Class 10

Chemistry

(Solutions)

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1.Chemical Reactions and Equations

LEVEL-1

1. The correct answer is(D)

The chemical reaction is the thermal decomposition of zinc carbonate ($ZnCO_3$) into zinc oxide (ZnO) and carbon dioxide (CO_2). This type of reaction is common for metal carbonates when heated.

The balanced chemical equation for this reaction is:

$$ZnCO_3 \rightarrow ZnO + CO_3$$

In this reaction, zinc carbonate decomposes upon heating to form solid zinc oxide and gaseous carbon dioxide.

Compounds which are reactants and products are,

1. Zinc carbonate ($ZnCO_3$): The reactant.
2. Zinc oxide (ZnO): One of the products.
3. Carbon dioxide (CO_2): The other product.

2. The correct answer is (C)

Complete combustion reaction of methane (CH_4) with oxygen (O_2) to produce carbon dioxide (CO_2) and water vapor (H_2O).

The balanced chemical equation for this reaction is:

$$CH_4\,(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

In the above equation, one molecule of methane reacts with two molecules of oxygen to produce one molecule of carbon dioxide and two molecules of water vapor.

3. The correct answer is (D)

The given unbalanced chemical equation is:

$$C_3H_8 + O_2 \rightarrow CO_2 + H_2O$$

To balance the given chemical equation, we need to ensure that the number of atoms of each element is the same on both sides of the equation.

Balance the carbon atoms:

$$C_3H_8 + O_2 \rightarrow 3CO_2 + H_2O$$

Balance the hydrogen atoms:

$$C_3H_8 + O_2 \rightarrow 3CO_2 + 4H_2O$$

Lastly, balance the oxygen atoms:

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

The balanced chemical equation is:

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

In above balanced equation, the number of moles of oxygen (O_2) required is 5 moles.

4. The correct answer is (A)

A chemical equation is a way of representing a chemical reaction using symbols and chemical formulae. It shows the reactants on the left-hand side and the products on the right-hand side, separated by an arrow (→). The chemical symbols represent the elements involved, while the chemical formulae represent the compounds or molecules involved in the reaction.

For example, in the chemical equation:

$$2Na(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2(g)$$

The symbols 'Na' and 'H' represent the elements sodium and hydrogen, while the formulae 'HCl' and 'NaCl' represent the compounds hydrogen chloride and sodium chloride. The subscripts and parentheses indicate the number of atoms or molecules and their physical states (s represents solid, aq represents aqueous, and g represents gas).

5. The correct answer is (B)

According to law of conservation of mass in a chemical change, the atoms are neither created nor destroyed. This law, formulated by Lavoisier in the 18th century, states that in any chemical reaction, the total mass of the reactants equals the total mass of the products.

During a chemical change, the atoms in the reactants are rearranged to form new substances, but the atoms themselves are conserved. They simply form new bonds and structures.

In opposite, a physical change involves a change in the state or physical properties of a substance without altering its chemical composition. While atoms are not created or destroyed in a physical change, the importance on conservation of mass is more prominent in chemical changes due to the rearrangement of atomic bonds.

6. The correct answer is (A)

In a chemical reaction, the substances that participate and undergo changes are called reactants. These reactants are present at the beginning of the reaction and are consumed as the reaction proceeds to form new substances known as products.

Reactants are very crucial for initiating chemical changes and are specifically chosen based on the chemical process and yield of product.

For example, in the combustion reaction of propane (C_3H_8) with oxygen (O_2) to produce carbon dioxide (CO_2) and water (H_2O), propane and oxygen are the reactants. They react together in a specific stoichiometric ratio to yield the products.

7. The correct answer is (B)

Products are the new substances formed as a result of a chemical change during a chemical reaction. They are the end products of the rearrangement of atoms or the breaking and forming of chemical bonds that occur when reactants undergo a chemical transformation.

In a chemical equation, products are written on the right-hand side of the arrow (→), that they are formed from the reactants on the left-hand side. For example, in the chemical equation:

$$2Na(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2(g)$$

The products are sodium chloride (NaCl) and hydrogen gas (H_2).

Products can be elements, compounds, or a combination of both, and they can exist in various physical states.

8. The correct answer is (D)

The sign of an arrow (→) in a chemical equation is read as "to yield" or "to produce." This explains about the direction in which reactants react to form products during a chemical reaction.

Option (A) "To yield" describes this directional transformation in a chemical equation. It explains the outcome of the reaction, where reactants interact and are transformed into products through chemical bonds breaking and forming.
Option (B) "To form" is also a valid interpretation and closely related to "to yield," as it highlights the creation or production of products from reactants.

9. The correct answer is (B)

A balanced chemical equation is a reaction in which the number of atoms of each element involved in the reaction is the same on both the reactant and product sides. This obey the law of conservation of mass and ensures that no atoms are created or destroyed during the chemical reaction.
In a balanced chemical equation, the coefficients are adjusted such a way that the total number of atoms of each element is equal on both sides of the arrow. This balance is crucial because it accurately represents the stoichiometry of the reaction.
For example, in the balanced equation for the combustion of methane,

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

the number of carbon, hydrogen, and oxygen atoms is the same on both sides of the arrow, ensuring that the equation is balanced and consistent with the law of conservation of mass.

10. The correct answer is (C)

In chemistry, a skeletal equation provides a basic representation of a chemical reaction by showing the reactants and products without indicating the correct proportions of each substance involved.
It serves as a preliminary form of the equation before the coefficients are adjusted to ensure that the law of conservation of mass is satisfied, meaning that the number of atoms of each element is the same on both sides of the equation.
Therefore, the correct term for an unbalanced equation is (C) skeletal equation.

11. The correct answer is (C)

The correct order of balancing equation is:
$Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$
Step-by-step process of balancing the equation,
1. Balance the lead (Pb) atoms first:
$Pb(NO_3)_2 \rightarrow PbO$
2. Balance the nitrogen (N) atoms:
$Pb(NO_3)_2 \rightarrow PbO + 2NO_2$
3. Balance the oxygen (O) atoms:
$Pb(NO_3)_2 \rightarrow PbO + 2NO_2 + 3O_2$
The balanced equation is,
$Pb(NO_3)_2 \rightarrow PbO + 2NO_2 + 3O_2$
By following the order of balancing lead (Pb) first, then nitrogen (N), and finally oxygen (O), we obtain the balanced chemical equation that represents the stoichiometric relationships between the reactants and products.

12. The correct answer is (A, D)

In the given chemical equation: $Al_2(SO_4)_3 + HNO_3 \rightarrow 2Al(NO_3)_3 + 3SO_2 + H_2O$
The elements that are balanced are (A) Al and (D) S.
Aluminum (Al):

On the reactant side, there are 2 aluminum atoms in $Al_2(SO_4)_3$.
On the product side, there are 2 aluminum atoms in $2Al(NO_3)_3$.
Hence, the aluminum atoms are balanced.
Sulphur (S):
On the reactant side, there are 3 sulphur atoms in $Al_2(SO_4)_3$.
On the product side, there are 3 sulphur atoms in $3SO_2$.
Hence, the sulphur atoms are balanced.
Remaining elements hydrogen (H), nitrogen (N), and oxygen (O) are not balanced in the given equation.

13. The correct answer is (B, D)

The balanced equations from the given options are (B) and (D).
(A) $Hg_2(NO_3)_2 \rightarrow 2Hg + 2NO_2 + O_2$
The above equation is not balanced for the elements nitrogen (N) and oxygen (O) atoms.
((B) $CuSO_4 + 4KI \rightarrow Cu_2I_2 + 2K_2SO_4 + I_2$
The above equation is balanced for all elements given copper (Cu), sulphur (S), potassium (K), and iodine (I).
(C) $SO_2 + Cl_2 + 2H_2O \rightarrow H_2SO_4 + HCl$
The above equation is not balanced for the hydrogen (H), sulphur (S), chlorine (Cl), and oxygen (O) atoms.
(D) $2NaOH + Cl_2 \rightarrow 2NaCl + NaClO + H_2O$
The above equation is balanced for all elements: sodium (Na), chlorine (Cl), hydrogen (H), and oxygen (O).
The balanced chemical equations are:
(B) $CuSO_4 + 4KI \rightarrow Cu_2I_2 + 2K_2SO_4 + I_2$
(D) $2NaOH + Cl_2 \rightarrow 2NaCl + NaClO + H_2O$

14. The correct answer is (A)

The correct balancing order of each element for the given equation:

$$K_2Cr_2O_7 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O + O_2$$

i) Balance chromium (Cr) first:

$$K_2Cr_2O_7 + H_2SO_4 \rightarrow Cr_2(SO_4)_3$$

ii) Balance hydrogen (H):

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 4H_2O$$

iii) Balance oxygen (O):

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 4H_2O + 3O_2$$

iv) Balance sulphur (S):

$$K_2Cr_2O_7 + 7H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 4H_2O + 3O_2 + 2K_2SO_4$$

v) Balance potassium (K):
The potassium atoms are already balanced in the equation.
The balanced equation is

$$K_2Cr_2O_7 + 7H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 4H_2O + 3O_2 + 2K_2SO_4$$

15. The correct answer is (D)

The statement that is not true is 'One gram of nitrogen and three grams of hydrogen combine to form two grams of ammonia.'
According to the balanced chemical equation:

$$N_2(g) + 3\ H_2(g) \rightleftharpoons 2\ NH_3(g)$$

The molar masses of the reactants and products are:

N_2 = 28.02 g/mol
H_2 = 2.02 g/mol
NH_3 = 17.03 g/mol
One mole of Nitrogen (28.02 g) and three moles of Hydrogen (3 × 2.02 = 6.06 g) combine to form two moles of NH_3 (2 × 17.03 = 34.06 g).
So, the masses of reactants and products are not in a 1:3:2 ratio. The correct mass ratio is
28.02 g N_2 + 6.06 g H_2 → 34.06 g NH_3.
The other statements (A), (B), and (C) are true and follow the laws of chemical combination.

16. The correct answer is (B)

The equation that represents a combination reaction is

$$C(s) + O_2(g) \rightarrow CO_2(g).$$

Combination reaction is a type of chemical reaction in which two or more reactants combine to form a single product. In this case, solid carbon (C) combines with gaseous oxygen (O_2) to produce carbon dioxide (CO_2) gas.
The given other options represent different types of reactions:
(A) $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ this reaction is a decomposition reaction, where a single reactant (silver chloride) breaks down into two or more products (silver and chlorine gas).
(C) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ is also a decomposition reaction, where calcium carbonate decomposes into calcium oxide and carbon dioxide gas.
(D) $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$ this reaction is a single displacement reaction, where an element (iron) displaces another element (copper) from a compound (copper sulphate) to form a new compound (iron sulphate) and the displaced element (copper).

17. The correct answer is (D)

The equation that is an example of a combination reaction in the given reactions is

$$2Mg + O_2 \rightarrow 2MgO$$

Combination reaction, is a type of chemical reaction where two or more reactants combine to form a single product. In this case, two atoms of magnesium (Mg) and one molecule of oxygen (O_2) combine to form two molecules of magnesium oxide (MgO).
The other equations represent different types of reactions:
(A) $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$ - This is an example of a single displacement reaction, where copper atoms displace silver ions from silver nitrate, forming copper nitrate and solid silver.
(B) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ - This is an example of a single displacement reaction, where zinc atoms displace copper ions from copper sulphate, forming zinc sulphate and solid copper.
(C) $CaCO_3 \rightarrow CaO + CO_2$ - This is an example of a decomposition reaction, where calcium carbonate decomposes into calcium oxide and carbon dioxide.

18. The correct answer is (C)

The equation that correctly represents the described reaction is

$$2Mg + O_2 \rightarrow 2MgO$$

Here, when a strip of magnesium ribbon is heated on a flame, it reacts with the oxygen present in the air to form magnesium oxide, which is a white substance. This reaction is an example of a combination reaction, where two reactants (magnesium and oxygen) combine to form a single product (magnesium oxide).
The balanced chemical equation for this reaction is:

$$2Mg + O_2 \rightarrow 2MgO$$

In this equation, two atoms of magnesium (2Mg) react with one molecule of oxygen gas (O_2) to produce two molecules of magnesium oxide (2MgO).
The other options are incorrect:
(A) $2Mg + 2O_2 \rightarrow 2MgO$ - This equation is unbalanced, as there are two oxygen molecules on the reactant side but only one oxygen molecule in MgO on the product side.
(B) $Mg + 2O_2 \rightarrow 2MgO_2$, (D) $Mg + O_2 \rightarrow MgO_2$ - This equation is incorrect because the product MgO_2 (magnesium dioxide) does not exist.

19. The correct answer is (C)

The equation that represents the correct word equation for the given reaction is
Potassium hydroxide + Sulphuric acid → Potassium sulphate + Water
In the above equation, the reactants (potassium hydroxide and sulphuric acid) are on the left side of the arrow, and the products (potassium sulphate and water) are on the right side of the arrow. The plus sign (+) indicates that the reactants are combined, and the arrow (→) indicates the direction of the reaction, with the reactants forming the products.
The other options are incorrect or incomplete:
(A) Potassium hydroxide − Sulphuric acid → Potassium sulphate − Water
This equation is incorrect because it uses the minus sign (-) instead of the plus sign (+).
(B) Potassium hydroxide + Sulphuric acid Potassium sulphate + Water
This equation is incomplete as it does not include the arrow (→) to indicate the direction of the reaction.
(D) Potassium hydroxide − Sulphuric acid Potassium sulphate − Water
This equation is incorrect because it uses the minus sign (-) instead of the plus sign (+) and it does not include the arrow (→) to indicate the direction of the reaction.

20. The correct answer is (A)

The equation that is an example of a single displacement reaction is

$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$$

A single displacement reaction is a type of chemical reaction in which an element from a compound is displaced by another element. In this equation, the iron (Fe) atoms displace the copper ions from copper sulphate ($CuSO_4$) solution, forming iron sulphate ($FeSO_4$) and solid copper (Cu) as the products.
The other equations represent different types of reactions:
(B) $CaCO_3 \rightarrow CaO + CO_2$ - This is an example of a decomposition reaction.
(C) $ZnCO_3 \rightarrow ZnO + CO_2$ - This is an of a decomposition reaction.
(D) $2Mg + O_2 \rightarrow 2MgO$ - This is an example of a combination reaction.

21. The correct answer is (D)

The balanced equation for the reaction of iron oxide (Fe_2O_3) with sulphuric acid (H_2SO_4) is:

$$Fe_2O_3 + 3H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 3H_2O$$

In this equation, one molecule of iron oxide reacts with three molecules of sulphuric acid to produce one molecule of iron (III) sulphate and three molecules of water.

The other options are incorrect:

(A) $Fe_2O_3 + H_2SO_4 \rightarrow 3Fe_2(SO_4)_3 + H_2O$

This equation is unbalanced, as the number of iron (Fe) atoms and sulphate (SO_4) ions is not balanced.

(B) $Fe_2O_3 + H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 3H_2O$

This equation is unbalanced, as the number of sulfuric acid (H_2SO_4) molecules is not balanced.

(C) $Fe_2O_3 + 3H_2SO_4 \rightarrow 3Fe_2(SO_4)_3 + H_2O$

This equation is unbalanced, as the products are incorrectly represented.

LEVEL-II

SINGLE ANSWER CORRECT QUESTIONS

1. The correct answer is (B)

The correct set of numbers of different elements in the balanced equation:

$$Al_2(CO_3)_3 + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2O + 3CO_2$$

	Al	C	O	H	S
(B)	2	2	5	2	1

After balancing the equation, the numbers of different elements present are:

Aluminum (Al): 2

Carbon (C): 2

Oxygen (O): 5

Hydrogen (H): 2

Sulphur (S): 1

The other options are incorrect:

(A) 2 2 5 2 2 - This option has an incorrect number of sulphur atoms (2 instead of 1).

(C) 2 5 2 2 2 - This option has incorrect numbers of carbon (5 instead of 2) and oxygen atoms (2 instead of 5).

(D) 2 5 2 5 2 - This option has incorrect numbers of carbon (5 instead of 2), oxygen (2 instead of 5), and hydrogen atoms (5 instead of 2).

2. The correct answer is (B)

The correct set of coefficients to balance the given equation

$$Mg + CO_2 \longrightarrow MgO + C$$

After balancing it, the coefficients are

	Mg	CO_2	MgO	C
(B)	2	1	2	1

The balanced equation is: $2Mg + CO_2 \rightarrow 2MgO + C$

In the above equation, two atoms of magnesium (2Mg) react with one molecule of carbon dioxide (CO_2) to produce two molecules of magnesium oxide (2MgO) and one atom of carbon (C).

The other options are incorrect:

(A) 1 1 1 1 - This set of coefficients does not balance the equation correctly.

(C) 1 2 1 2 - This set of coefficients also does not balance the equation, as there would be an imbalance in the number of carbon and oxygen atoms on the reactant and product sides.

(D) 2 2 1 1 - This set of coefficients does not balance the equation.

3. The correct answer is (B)

The correct set of coefficients to balance the given equation

$$KClO_3 \rightarrow KCl + O_2$$

After balancing, the coefficients are:

	$KClO_3$	KCl	O_2
(B)	2	2	3

The balanced equation is:

$$2KClO_3 \rightarrow 2KCl + 3O_2$$

In the above equation, two molecules of potassium chlorate decompose to produce two molecules of potassium chloride and three molecules of oxygen gas.

The other options are incorrect:

(A) 1 1 1 - This set of coefficients does not balance the equation correctly.

(C) 3 3 1 - This set of coefficients does not balance the equation, as there is an imbalance in the number of oxygen atoms on the reactant and product sides.

(D) 2 3 2 - This set of coefficients does not balance the equation.

4. The correct answer is (A)

The given equation explains about the combustion of benzene in the presence of oxygen to form carbon dioxide and water. This reaction is highly exothermic.

To balance the given equation

$$C_6H_6 + O_2 \rightarrow CO_2 + H_2O$$

Ensure the number of each type of atom are balanced on both sides of the equation.

Balanced equation:

$$2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O$$

Now in the given options provided:

(A) $2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O$; ΔH = −ve

Option (A) correctly balances the equation and indicates that the reaction is exothermic (ΔH = negative).

5. The correct answer is (C)

The balanced chemical equation for the reaction of iron (III) oxide with hydrogen gas to produce iron and water is $Fe_3O_4 + 4H_2 \rightarrow 3Fe + 4H_2O$

In this equation, the coefficients are:

Fe_3O_4: 1

H_2: 4

Fe: 3

H_2O: 4

The coefficients signify the number of atoms of each element is balanced on both sides of the equation. There are 3 iron atoms and 4 oxygen atoms on the reactant side, and the same number of iron and oxygen atoms on the product side.

6. The correct answer is (D)

The balanced chemical equation for the reaction of iron (II) sulphide with oxygen gas to produce iron (III) oxide and sulphur dioxide is:

$$4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$$

In this equation, the coefficients are:

FeS_2: 4
O_2: 11
Fe_2O_3: 2
SO_2: 8

On the reactant side, there are 4 iron atoms, 8 sulphur atoms, and 22 oxygen atoms. On the product side, there are 4 iron atoms, 8 sulphur atoms, and 22 oxygen atoms, ensuring the conservation of mass.

7. The correct answer is (D)

The balanced chemical equation for the reaction of copper with hot and concentrated sulfuric acid (H2SO4) to form copper sulphate, sulphur dioxide gas, and water is $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

In this balanced equation, the coefficients are:

Cu: 1
H_2SO_4: 2
$CuSO_4$: 1
SO_2: 1
H_2O: 2

On the reactant side, there is 1 copper atom (from Cu), 2 sulphur atoms (from 2 H_2SO_4), 4 hydrogen atoms (from 2 H_2SO_4), and 8 oxygen atoms (from 2 H_2SO_4). On the product side, there is 1 copper atom (from $CuSO_4$), 2 sulphur atoms (1 from $CuSO_4$ and 1 from SO_2), 4 hydrogen atoms (from 2 H_2O), and 8 oxygen atoms (4 from $CuSO_4$, 2 from SO_2, and 2 from 2 H_2O).

8. The correct answer is (C)

The correct answer to balance the chemical equation for the decomposition of lead nitrate crystals on strong heating is 2 2 2 1.

The balanced chemical equation can be written as:

$2Pb(NO_3)_2 \rightarrow 2PbO + 2NO_2 + O_2$

1. The lead (Pb) and nitrogen (N) atoms on the reactant side must be balanced with the product side.
2. For every two-lead nitrate molecule, there are two lead (Pb) atoms and four nitrogen (N) atoms.
3. On the product side, there are two lead oxide (PbO) molecules, accounting for two lead (Pb) atoms.
4. The two-nitrogen dioxide (NO_2) molecules account for two nitrogen (N) atoms, and one oxygen (O_2) molecule accounts for the remaining two oxygen atoms.

9. The correct answer is (B)

The answer for the given question is (B) 2 8 2 2 8 3.

The balanced chemical equation can be written as:

$$2K_2Cr_2O_7 + 8H_2SO_4 \rightarrow 2K_2SO_4 + 2Cr_2(SO_4)_3 + 8H_2O + 3O_2$$

1.The potassium (K), chromium (Cr), and sulphur (S) atoms on the reactant side must be balanced with the product side.

2.On the product side, there are two potassium sulphate (K_2SO_4) molecules, accounting four potassium (K) atoms.

3. The two-chromium sulphate ($Cr_2(SO_4)_3$) molecules account for four chromium (Cr) atoms and six sulphur (S) atoms.

4. The eight sulfuric acid (H_2SO_4) molecules on the reactant side provide the remaining eight sulphur (S) atoms and sixteen hydrogen (H) atoms.

5. The eight water (H_2O) molecules on the product side account for sixteen hydrogen (H) atoms.

6. The three oxygen (O_2) molecules on the product side balance the remaining oxygen atoms.

10. The correct answer is (C)

The correct answer is (C) 1 6 2 3 3.

The balanced chemical equation can be written as

$Al_2(CO_3)_3 + 6HNO_3 \rightarrow 2Al(NO_3)_3 + 3CO_2 + 3H_2O$

1. The aluminium (Al), carbon (C), nitrogen (N), and hydrogen (H) atoms on the reactant side must be balanced with the product side.

2. For every one aluminium carbonate ($Al_2(CO_3)_3$) molecule, there are two aluminium (Al) atoms and three carbon (C) atoms.

3. On the product side, there are two aluminium nitrates ($Al(NO_3)_3$) molecules, accounting for two aluminium (Al) atoms and six nitrogen (N) atoms.

4. The three carbon dioxide (CO_2) molecules account for three carbon (C) atoms.

5. The six nitric acid (HNO_3) molecules on the reactant side provide the six nitrogen (N) atoms and six hydrogen (H) atoms.

6. The three water (H_2O) molecules on the product side account for six hydrogen (H) atoms.

ONE OR MORE THAN ONE CORRECT ANSWER QUESTIONS

11. The correct answer is (D)

The balanced chemical equation is

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$$

The limitations provided on the arrow is (900 atm - 450°C, Fe) which indicate the physical conditions under which the reaction takes place, which are high pressure (900 atm) and high temperature (450°C), presence of an iron (Fe) catalyst.

A. The physical states of reactants (N_2 and H_2) and the product (NH_3) are correctly represented as gaseous (g).

B. The number of atoms and molecules are balanced on both sides of the equation.

C. The symbols and formulae of all the substances (N_2, H_2, and NH_3) are correct.

D. The physical conditions of high pressure, high temperature, and the presence of an iron catalyst are provided on the arrow, indicating that these conditions are necessary for the reaction to occur favourably

Therefore, the corrected limitations in the given balanced chemical equation are (D) Physical conditions of a reaction on the arrow.

12. The correct answer is (B, D)

The balanced chemical equation for the reaction is:

$$N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$$

Option (B) is correct because one molecule of nitrogen gas (N_2) and three molecules of hydrogen gas (H_2) reacts to form two molecules of ammonia (NH_3).

Option (D) is correct because the molar masses of the reactants and products are equal.

The molar mass of N_2 molecule is 28 g/mol, and the molar mass of H_2 molecule is 2 g/mol.

So, the total mass of reactants is 28 + (3 × 2) = 34 g/mol. The molar mass of NH_3 is 17 g/mol, and since two molecules of NH_3 are formed, the total mass of products is 2 × 17 = 34 g/mol.

Thus, the total mass of reactants (44 grams) is equal to the total mass of products (44 grams).

13. The correct answer is (B, C, D)

(B) $2\ Mg + O_2 \rightarrow 2\ MgO$

The above equation represents the formation of magnesium oxide (MgO) from the reaction between two atoms of magnesium (Mg) and one molecule of oxygen gas (O_2). The given coefficients on both sides are same.

(C) $2\ CO + O_2 \rightarrow 2\ CO_2$

The above equation represents the combustion of carbon monoxide (CO) in the presence of oxygen gas (O_2) to form carbon dioxide (CO_2). The given coefficients on both sides are same.

(D) $Fe + S \rightarrow FeS$

The above equation represents the formation of iron sulphide (FeS) from the reaction between one atom of iron (Fe) and one atom of sulphur (S). The coefficients are balanced, with one atom of each element on both sides of the equation.

Option (A), $H_2 + C_2 \rightarrow 2\ HC\lambda$, is not a balanced equation.

14. The correct answer is (A, B, C, D)

(A) $Na_2CO_3 + 2\ HCl \rightarrow 2\ NaCl + CO_2 + H_2O$

Above equation represents the reaction between sodium carbonate and hydrochloric acid, producing sodium chloride, carbon dioxide, and water.

(B) $Ca\ (OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

This equation represents the reaction between calcium hydroxide and carbon dioxide, forming calcium carbonate and water.

(C) $2\ H_2S + SO_2 \rightarrow 2\ H_2O + 3\ S$

This equation explains about the reaction between hydrogen sulphide and sulphur dioxide, producing water and elemental sulphur.

(D) $4\ NH_3 + 5\ O_2 \rightarrow 4\ NO + 6\ H_2O$

This equation represents the reaction between ammonia and oxygen, forming nitric oxide and water.

15. The correct answer is (A, B)

Among the given options, the balanced chemical equations are:

(A) $2\ K_2Cr_2O_7 + 8\ H_2SO_4 \rightarrow 2\ K_2SO_4 + 2\ Cr_2(SO_4)_3 + 8\ H_2O + 3\ O_2$

Above equation represents the reaction between potassium dichromate and sulfuric acid, producing potassium sulphate, chromium (III) sulphate, water, and oxygen.

(B) $3\ CaOCl_2 + 2\ NH_3 \rightarrow 3\ CaCl_2 + 3\ H_2O + N_2$

This equation represents the reaction between calcium hypochlorite and ammonia, forming calcium chloride, water, and nitrogen gas. Coefficients on both sides are equal.

Options (C) and (D) are unbalanced equations.

16. The correct answer is (A, B)

(A) $Ca_3(PO_4)_2 + 3\ SiO_2 + 5\ C \rightarrow 3\ CaSiO_3 + 2\ P + CO$

Above equation is not balanced because the number of atoms of each element is not equal on both sides. On the left side, there are 3 calcium atoms, 2 phosphorus atoms, 6 oxygen atoms, 3 silicon atoms, and 5 carbon atoms. On the right side, there are 3 calcium atoms, 2 phosphorus atoms, 9 oxygen atoms, 3 silicon atoms, and 1 carbon atom.

(B) $2\ NH_3 + Na \rightarrow 2\ NaNH_2 + 2\ H_2$

Above equation is not balanced because the number of atoms of each element is not equal on both sides. On the left side, there are 2 nitrogen atoms, 6 hydrogen atoms, and 1 sodium atom. On the right side, there are 2 nitrogen atoms, 6 hydrogen atoms, and 2 sodium atoms.

(C) $2\ Cr(OH)_3 + 6\ Na_2O_2 \rightarrow 2\ Na_2CrO_4 + 2\ NaOH + 2\ H_2O$

This equation is balanced, with the number of atoms of each element equal on both sides.

(D) $2\ CuFeS_2 + O_2 \rightarrow Cu_2S + 2\ FeS + SO_2$

This equation is balanced, with the number of atoms of each element equal on both sides.

LEVEL-I

Chemical Reactions

1. The correct answer is (D)

When a match stick burns, it undergoes a chemical reaction that produces new substances. The head of a match stick contains chemicals like sulphur, potassium chlorate, and phosphorus. When struck, the friction generates heat, causing these chemicals to react with oxygen in the air. This reaction produces new compounds.

$$S\ (\text{sulphur}) + O_2\ (\text{oxygen}) \rightarrow SO_2\ (\text{sulphur dioxide})$$

$$P\ (\text{phosphorus}) + O_2\ (\text{oxygen}) \rightarrow P_2O_5\ (\text{phosphorus pentoxide})$$

These reactions produce new chemical substances, releasing heat and light, which indicates a chemical change. It is a clear example of a chemical change.

The other options are examples of physical changes:

- (A) Glowing of an electric bulb involves the flow of electricity, which is a physical process.
- (B) Crushing an ice cube changes its shape but not its chemical structure.
- (C) Magnetizing an iron needle changes the alignment of its magnetic domains but not its chemical composition.

2. The correct answer is (A)

Burning of petrol or diesel is a chemical change.

A chemical change is a process that involves the rearrangement of atoms or molecules to form new substances with different chemical properties. When petrol or diesel is burned, it undergoes a combustion reaction with oxygen in the air. This reaction involves the breaking and rearrangement of chemical

bonds in the hydrocarbon molecules present in petrol or diesel, resulting in the formation of new compounds.

The general equation for the combustion of hydrocarbons (such as petrol or diesel) can be represented as:

Hydrocarbon + Oxygen → Carbon dioxide + Water + Energy (heat and light)

During this process, the original substances (petrol/diesel and oxygen) are chemically transformed into new substances (carbon dioxide, water vapor, and other combustion products).

3. The correct answer is (C)

When zinc hydroxide is heated, it decomposes to form zinc oxide. This is an example of a chemical change.

A chemical change involves a transformation that produces one or more new substances with different chemical properties and compositions. The heating of zinc hydroxide ($Zn(OH)_2$) causes it to break down into zinc oxide (ZnO) and water (H_2O). The chemical equation for this reaction is

$$Zn(OH)_2 \rightarrow ZnO + H_2O$$

In this process, zinc hydroxide, a compound, is decomposed by heat into zinc oxide, a different compound, and water. This decomposition reaction involves the breaking of chemical bonds in zinc hydroxide and the formation of new bonds in zinc oxide and water.

4. The correct answer is (A)

When a drop of water is poured over anhydrous copper sulphate ($CuSO_4$), it turns blue. This phenomenon is an example of a physical change.

A physical change is a process in which the physical properties of a substance, such as colour, state, or appearance, can be changed without changing its chemical composition and molecular structure.

In the case of anhydrous copper sulphate, the white colour is due to the absence of water molecules in its crystal structure. When water is added, the water molecules are absorbed into the crystal lattice, forming a hydrated compound called copper sulphate pentahydrate ($CuSO_4 \cdot 5H_2O$).

The hydration process can be represented by the following equation is

$$CuSO_4 \text{ (anhydrous)} + 5\ H_2O \rightarrow CuSO_4 \cdot 5H_2O \text{ (hydrated)}$$

5. The correct answer is (A)

When a small piece of quicklime (calcium oxide, CaO) is dropped into water, it undergoes an exothermic reaction, forming calcium hydroxide, $Ca(OH)_2$. The chemical equation for this reaction is:

$$CaO + H_2O \rightarrow Ca(OH)_2 + \text{Heat}$$

In this reaction, the water becomes very hot, and a milky white solution of calcium hydroxide is formed. The heat released during the reaction is responsible for the increase in temperature, causing the water to become hot.

This reaction explains about a change involving energy. It involves the release of energy in the form of heat, which is a characteristic of an exothermic reaction.

6. The correct answer is (D)

(i) Breaking of a glass pane is a physical change.

This statement is true. When a glass pane breaks, it undergoes a change in its physical state (from a solid structure to smaller pieces), but the chemical

composition and molecular structure of the glass remain unchanged (no new molecule is formed).

(ii) When a glass tumbler breaks into pieces, the arrangement of molecules in glass changes.

This statement is also true. Although the chemical composition and molecular structure of the glass remain unchanged, the arrangement of the molecules in the glass is altered when it breaks into pieces. The molecules in the solid glass structure are displaced from their original positions.

So, both statements (i) and (ii) are correct.

7. The correct answer is (B)

A chemical equation is defined as the symbolic representation of a chemical reaction using the chemical formulas of the reactants and products involved. It shows the reactants on the left side and the products on the right side, separated by an arrow symbol (→) that indicates the direction of the reaction.

For example, the chemical equation for the reaction between sodium and chlorine gas to form sodium chloride can be written as:

$$2\ Na + Cl_2 \rightarrow 2\ NaCl$$

In this equation, the reactants are sodium (Na) and chlorine gas (Cl_2), and the product is sodium chloride (NaCl). The coefficients (2) before the reactants and products ensure that the equation is balanced.

8. The correct answer is (A)

The balanced chemical equation represents the reaction between aluminum (Al) and solid sulphur (S_8), where aluminum sulphide (Al_2S_3) is formed as the product.

In order to balance the equation, ensure that the number of atoms of each element is equal on both sides of the equation.

On the reactant side:

- There are 16 aluminum atoms (16 Al)
- There are 24 sulphur atoms (3 × 8 atoms in S_8)

On the product side:

- There are 16 aluminum atoms (8 × 2 atoms in Al_2S_3)
- There are 24 sulphur atoms (8 × 3 atoms in Al_2S_3)

Therefore, the balanced equation $16\ Al + 3\ S_8 \rightarrow 8\ Al_2S_3$ accurately represents the reaction between aluminum and solid sulphur.

9. The correct answer is (C)

A balanced chemical equation is a representation of a chemical reaction where the number of atoms of each element is equal on both the reactant and product sides. This ensures that the total mass of the reactants is equal to the total mass of the products, in accordance with the law of conservation of mass.

The law of conservation of mass, formulated by Antoine Lavoisier, which states that in a chemical reaction, the total mass of the reactants is equal to the total mass of the products.

10. The correct answer is (B)

To balance the given chemical equation, apply the principles of mass conservation and charge balance.

$$2\ KMnO_4 + 5\ H_2SO_4 + 5\ FeSO_4 \rightarrow K_2SO_4 + 2\ MnSO_4 + 5\ Fe_2(SO_4)_3 + 4\ H_2O$$

- Potassium permanganate ($KMnO_4$) is the oxidizing agent, and ferrous sulphate ($FeSO_4$) is the reducing agent.
- 5 moles of sulfuric acid (H_2SO_4) are required to produce 5 moles of ferric sulphate ($Fe_2(SO_4)_3$) and release 4 moles of water (H_2O).
- The potassium ions (K^+) from $KMnO_4$ combine with the sulphate ions (SO_4^{2-}) from H_2SO_4 to form potassium sulphate (K_2SO_4).
- The manganese ions (Mn^{7+}) from $KMnO_4$ are reduced to manganese (II) ions (Mn^{2+}) and form manganese sulphate ($MnSO_4$).
- The iron ions (Fe^{2+}) from $FeSO_4$ are oxidized to iron (III) ions (Fe^{3+}) and form ferric sulphate ($Fe_2(SO_4)_3$).

Combination Reactions

11. The correct answer is (A)

In the process of forming ferric chloride ($FeCl_3$) on an iron surface due to the reaction between iron and chlorine, iron undergoes oxidation.

$$2Fe(s) + 3Cl_2(g) \rightarrow 2FeCl_3(s)$$

In this reaction, iron atoms lose electrons, and their oxidation state changes from 0 (in elemental iron) to +3 (in ferric chloride). This is the process of oxidation, where an element loses electrons.

The oxidation half-reaction is:

$$2Fe(s) \rightarrow 2Fe^{3+}(aq) + 6e^-$$

Chlorine gas gains electrons and is reduced, forming chloride ions:

$$3Cl_2(g) + 6e^- \rightarrow 6Cl^-(aq)$$

12. The correct answer is (D)

Ammonia is formed by the combination of nitrogen and hydrogen gases through the Haber process, which can be represented by the chemical equation

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

(A) Ammonia is the reduction product of N2:

In the formation of ammonia, nitrogen gains hydrogen atoms and is reduced. The nitrogen in N_2 has an oxidation state of 0, and in ammonia, nitrogen has an oxidation state of -3.

$$N_2(g) + 6e^- \rightarrow 2N^{3-}(aq)$$

(B) Ammonia is the oxidation product of H2:

In this process, hydrogen loses electrons as it combines with nitrogen. The hydrogen in has an oxidation state of 0, and in ammonia, hydrogen has an oxidation state of +1. Thus, hydrogen is oxidized.

$$6H^+(aq) + 6e^- \rightarrow 3H_2(g)$$

(C) Nitrogen acts as an oxidizing agent:

Nitrogen gains electrons (or hydrogen atoms) during the reaction, causing hydrogen to be oxidized. Therefore, nitrogen acts as an oxidizing agent.

13. The correct answer is (B)

During the combustion of methane, it reacts with oxygen to form carbon dioxide and water. The chemical equation for this reaction is:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

In this above reaction, oxygen gas undergoes a reduction process. In the combustion reaction, oxygen molecules gain electrons (through the formation of water and carbon dioxide) and thereby get reduced.

On the other hand, methane gets oxidized. During combustion, methane loses electrons as it reacts with oxygen to form carbon dioxide and water. Thus, methane acts as the reducing agent, not the oxidizing agent.

$$CH_4(g) + 2H_2O(l) \rightarrow CO_2(g) + 8H^+(aq) + 8e^-$$

14. The correct answer is (B)

A chemical combination reaction, occurs when two or more elements or compounds chemically react to form a single new product. In this type of reaction, the reactants combine their atoms or ions to create a new compound with a different chemical composition and properties.

Examples of chemical combination reactions include:

1. Formation of water from hydrogen and oxygen gases:

$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

2. Formation of carbon dioxide from carbon and oxygen:

$C(s) + O_2(g) \rightarrow CO_2(g)$

In these reactions, the reactants combine their atoms or ions to form a single new product, without any separate byproducts.

15. The correct answer is (D)

An element-compound combination reaction is a type of chemical reaction where an element reacts with a compound to form a new compound.

$$O_2 + 2H_2O \rightarrow 2H_2O_2$$

Here, oxygen an element, reacts with water , a compound, to form hydrogen peroxide a new compound. This reaction involves the combination of an element with a compound to create a different compound.

The other options are different types of combination reactions involving elements only:

- (A) $C + O_2 \rightarrow CO_2$ involves carbon and oxygen combining to form carbon dioxide.
- (B) $H_2 + O_2 \rightarrow 2H_2O$ involves hydrogen and oxygen combining to form water.
- (C) $H_2 + Cl_2 \rightarrow 2HCl$ involves hydrogen and chlorine combining to form hydrogen chloride.

16. The correct answer is (A)

2 moles of sulphur trioxide (SO_3) will be produced when 2 moles of sulphur (S) combine with 3 moles of oxygen (O_2).

The balanced chemical equation for the reaction is:

$$2S(s) + 3O_2(g) \rightarrow 2SO_3(g)$$

From the above balanced equation, we observe that:

- 2 moles of sulphur react with 3 moles of oxygen.
- 2 moles of sulphur trioxide are produced.

Given we have 2 moles of sulphur and 3 moles of oxygen available as reactants. Therefore, following the mole ratio from the balanced equation, 2 moles of sulphur trioxide will be produced.

The mole ratios are:

2 moles of S: 3 moles of O_2: 2 moles of SO_3

Therefore, when 2 moles of sulphur combine with 3 moles of oxygen, 2 moles of sulphur trioxide will be produced.

17. The correct answer is (C)

When copper is heated in air, it reacts with oxygen to form copper (II) oxide, which is black in colour. The chemical reaction can be represented as follows:

$2Cu$ (copper) + O_2 (oxygen) → $2CuO$ (copper (II) oxide)

In above reaction, copper reacts with oxygen from the air to form copper (II) oxide. The black colour of copper (II) oxide is a clear indication of this chemical change. This reaction occurs because copper has a strong affinity for oxygen, and the heat provides the necessary energy for the reaction to take place.

18. The correct answer is (B)

I. Burning of coal and II. Burning of magnesium are combination reactions.

In the burning of coal, carbon (C) from coal reacts with oxygen (O_2) from the air to form carbon dioxide (CO_2).

The chemical equation for this reaction is:

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

In the burning of magnesium, magnesium (Mg) reacts with oxygen (O_2) from the air to form magnesium oxide (MgO).

The chemical equation for this reaction is:

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

In both these processes, the reactants (coal/magnesium and oxygen) combine to form a new product (carbon dioxide or magnesium oxide).

19. The correct answer is (B)

The reaction $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ is not an example of a combination reaction. It is a single displacement reaction, where the zinc metal replaces the hydrogen in hydrochloric acid, forming zinc chloride and hydrogen gas.

In a combination reaction, two or more reactants combine to form a single product.

In this case, the reactants (zinc and hydrochloric acid) form two products (zinc chloride and hydrogen gas).

20. The correct answer is (B)

Assertion: Addition of water to a drum containing calcium oxide makes the drum hot if touched from outside.

This assertion is true. When water is added to calcium oxide (CaO), an exothermic reaction occurs, releasing some amount of heat. This heat makes the drum hot to the touch from the outside.

The chemical equation for the reaction is:

$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + heat$

Reason: Reaction between calcium oxide and water is an endothermic reaction.

The reason is false. The reaction between calcium oxide and water is an exothermic reaction, not an endothermic reaction. Exothermic reactions release heat to the surroundings, while endothermic reactions absorb heat from the surroundings.

So, option (B) is the correct answer: Both Assertion and Reason are true, but Reason is not the correct explanation of the Assertion.

Decomposition Reactions

21. The correct answer is (C)

The reaction $2KClO_3 \rightarrow 2KCl + 3O_2$ is a decomposition reaction, where a single compound (potassium chlorate, $KClO_3$) breaks down into two or more simpler substances (potassium chloride, KCl, and oxygen gas, O_2).

In a decomposition reaction, a complex compound is broken down into smaller and simpler compounds or elements, usually by the application of heat, light, or electricity.

22. The correct answer is (D)

A decomposition reaction is a type of chemical reaction where a single compound breaks down into two or more simpler substances. It is usually an endothermic process, that energy (heat) is absorbed from the surroundings to break the bonds in the reactant compound.

Examples:

$2NaHCO_3(s) \rightarrow Na_2CO_3(s) + H_2O(g) + CO_2(g)$ (endothermic)

$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$ (endothermic)

Therefore, decomposition reactions are mostly endothermic, and they involve the breakdown of one substance into simpler substances.

23. The correct answer is (C)

A decomposition reaction is one type of chemical reaction in which a single compound breaks down into two or more simpler substances.

Examples of decomposition reactions include:

1. Thermal decomposition of calcium carbonate (limestone) to form calcium oxide (quicklime) and carbon dioxide:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

2. Electrolytic decomposition of water into hydrogen and oxygen gases:

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$

In these reactions, a single reactant compound decomposes or breaks down into two or more simpler products, without any other reactant participating in the process.

24. The correct answer is (C)

A photochemical reaction is a type of chemical reaction that proceeds with the absorption of light energy, from the visible or ultraviolet region of the electromagnetic spectrum. The absorbed light energy provides necessary activation energy to initiate the chemical reaction.

Examples of photochemical reactions include:

- Photosynthesis in plants: $CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$ (light energy absorbed by chlorophyll)
- Photochemical smog formation: $NO_2 + h\nu \rightarrow NO + O$ ($h\nu$ represents light energy)
- Photodegradation of plastics and dyes.

25. The correct answer is (A)

Exothermic reactions are chemical reactions in which heat energy is released to the surroundings during the process. In these reactions, the products have lower energy than the reactants, and the excess energy is released in the form of heat.

Examples of exothermic reactions include:

1. Combustion of methane:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) + \text{heat}$$

2. Neutralization reaction between hydrochloric acid and sodium hydroxide:

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l) + \text{heat}$$

3. Thermite reaction:

$$2Al(s) + Fe_2O_3(s) \rightarrow Al_2O_3(s) + 2Fe(l) + \text{heat}$$

26. The correct answer is (A)

A promoter is a substance that improves the efficiency of a catalyst in a chemical reaction without itself participating in the reaction or undergoing any permanent chemical change.

Promoters enhance the catalytic activity of catalysts by facilitating the interaction between reactants and the catalyst surface, thereby increasing the rate of the reaction.

They do not change the nature of the reaction or the products formed but help in achieving higher reaction rates under specific conditions.

27. The correct answer is (D)

The reaction in which atoms of oxygen in the upper region of the atmosphere absorb ultraviolet rays from sunlight and combine to form ozone gas is a photochemical reaction.

The overall reaction can be represented as:

$$3O_2(g) + \text{UV light} \rightarrow 2O_3(g)$$

In this reaction, the absorption of ultraviolet light energy from the Sun provides the necessary activation energy to break the strong bonds in the oxygen molecules (O_2). The oxygen atoms then rearrange and combine to form ozone molecules (O_3).

Displacement Reactions

28. The correct answer is (D)

The given reaction:

$$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$$

is an example of a displacement reaction, also known as a single displacement reaction.

In a displacement reaction, one element in a compound is displaced or replaced by another element from a different compound. In this specific reaction, the aluminum (Al) atoms displace or replace the iron (Fe) atoms from iron (III) oxide (Fe_2O_3) to form aluminum oxide (Al_2O_3) and elemental iron (Fe).

29. The correct answer is (A)

The reaction A + B → AC + B represents a simple displacement reaction (A).

In a simple displacement reaction, a more reactive element displaces a less reactive element from a compound. The general equation for a simple displacement reaction is

$$A + BC \rightarrow AC + B$$

For example, if the reaction is:

$$2Na + 2HCl \rightarrow 2NaCl + H_2$$

30. The correct answer is (C)

In the given reaction, $CuSO_4(aq) + Zn(s) \rightarrow A(aq) + Cu(s)$, zinc metal (Zn) is displacing copper from copper sulphate solution ($CuSO_4(aq)$).

Since zinc is more reactive than copper, it can displace copper from its compound.

The balanced chemical equation for the reaction is:

$$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$$

31. The correct answer is (D)

In the first reaction, $2K + [I] \rightarrow 2KOH + H_2$, potassium metal (K) reacts with water ([I] represents water) to form potassium hydroxide (KOH) and hydrogen gas (H_2). Therefore, [I] represent $2H_2O$.

$$2K + 2H_2O \rightarrow 2KOH + H_2$$

In the second reaction, $Fe + 2HCl \rightarrow [II] + [III]$, iron metal (Fe) reacts with hydrochloric acid (HCl) to form iron (II) chloride ([II] represents $FeCl_2$) and hydrogen gas ([III] represents H_2).

$$Fe + 2HCl \rightarrow FeCl_2 + H_2$$

32. The correct answer is (C)

The information provided is:

1. Metal C displaces metal B from its oxide when heated, indicating that C is more reactive than B.
2. There is no reaction when metal C is heated with the oxide of metal A, implying that A is more reactive than C.

Based on these observations, we can arrange the metals A, B, and C in increasing order of reactivity as follows:

B (least reactive) < C < A (most reactive)

33. The correct answer is (C)

The reactivity series of metals determines which metal can displace another metal from its compound in a single displacement reaction. The more reactive metal will displace the less reactive metal from its compound.

The reactivity order of the given metals is: Fe > Zn > Cu > Pb > Au

Iron (Fe) and zinc (Zn) are more reactive than copper, so they can displace copper from copper sulphate solution according to the following reactions:

$$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$$
$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

Lead (Pb) is less reactive than copper, so it cannot displace copper from its compounds. Gold (Au) is the least reactive among the given options, making it unable to displace copper from copper sulphate solution.

34. The correct answer is (C)

4When an aqueous solution of barium nitrate ($Ba(NO_3)_2$) is mixed with an aqueous solution of sodium fluoride (NaF), double displacement reaction takes place. In this reaction, the cations (Ba^{2+} and Na^+) and anions (NO_3^- and F^-) exchange partners to form a precipitate of barium fluoride (BaF_2) and sodium nitrate ($NaNO_3$) in solution.

The balanced chemical equation for the reaction is:

$$Ba(NO_3)_2(aq) + 2NaF(aq) \rightarrow BaF_2(s) + 2NaNO_3(aq)$$

In this reaction, barium fluoride (BaF_2) is formed as a solid precipitate due to its low solubility in water.

35. The correct answer is (C)

In this activity, the reaction of aluminum foil with different solutions in the three test tubes will determine the production of hydrogen gas.

Test Tube X: Aluminum reacts with sulfuric acid to produce aluminum sulphate and hydrogen gas.

$$2Al(s) + 3H_2SO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$$

Test Tube Y: Aluminum reacts with copper sulphate solution, undergoing a single displacement reaction, where aluminum displaces copper from the solution, forming aluminum sulphate and copper metal. Hydrogen gas is also produced in this reaction.

$$2Al(s) + 3CuSO_4(aq) \rightarrow Al_2(SO_4)_3(aq) + 3Cu(s) + 3H_2(g)$$

Test Tube Z: Aluminum does not react with zinc sulphate solution because aluminum is less reactive than zinc. No hydrogen gas is produced in this test tube.

36. The correct answer is (A)

When zinc metal reacts with ferrous sulphate solution, a single displacement reaction occurs. In this reaction, zinc, being more reactive than iron, displaces iron from the ferrous sulphate solution, forming zinc sulphate and iron.

The balanced chemical equation for the reaction is:

$$Zn(s) + FeSO_4(aq) \rightarrow ZnSO_4(aq) + Fe(s)$$

In this reaction, zinc metal oxidizes to form zinc ions (Zn^{2+}) and release electrons, while iron ions (Fe^{2+}) in the ferrous sulphate solution are reduced by accepting these electrons, resulting in the formation of metallic iron.

37. The correct answer is (A)

$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu.$$

This equation represents a single displacement reaction, where a more reactive metal (iron, Fe) displaces a less reactive metal (copper, Cu) from its compound (copper sulphate, $CuSO_4$).

In a single displacement reaction, a more reactive element displaces a less reactive element from one of its compounds.

The general equation for a single displacement reaction is:

$$A + BC \rightarrow AC + B$$

Where A is a more reactive element than B, and BC is a compound containing B.

38. The correct answer is (D)

According to the given reactivity series, the metals are arranged in decreasing order of their reactivity as:

Magnesium > Zinc > Iron > Lead > Copper > Silver > Gold

In a single displacement reaction, a more reactive metal can displace a less reactive metal from its compound in an aqueous solution.

Since zinc (Zn) is more reactive than iron (Fe), it can displace iron from iron sulphate ($FeSO_4$) solution.

The balanced chemical equation for the reaction is:

$$Zn(s) + FeSO_4(aq) \rightarrow ZnSO_4(aq) + Fe(s)$$

39. The correct answer is(A)

When zinc metal reacts with copper sulphate solution, a single displacement reaction occurs. In this reaction, zinc, being more reactive than copper, displaces copper from the copper sulphate solution, forming zinc sulphate and metallic copper.

The balanced chemical equation for the reaction is:

$$Zn(s) + CuSO_4\ (aq) \rightarrow ZnSO_4\ (aq) + Cu(s)$$

In this reaction, zinc metal oxidizes to form zinc ions (Zn^{2+}) and release electrons, while copper ions (Cu^{2+}) in the copper sulphate solution are reduced by accepting these electrons, resulting in the formation of solid metallic copper.

40. The correct answer is (C)

The given reaction, $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$, is both a redox reaction and a displacement reaction.

It is a redox (reduction-oxidation) reaction because it involves the transfer of electrons. Aluminum is oxidized by losing electrons, while hydrogen ions (H^+) from hydrochloric acid are reduced by gaining electrons to form hydrogen gas.

It is also a displacement reaction because the more reactive aluminum metal displaces hydrogen from hydrochloric acid, forming aluminum chloride and releasing hydrogen gas.

Double Displacement

41. The correct answer is (C)

A double displacement reaction, also known as a double replacement reaction or metathesis reaction, is a type of chemical reaction where two compounds exchange ions or atoms, forming two new compounds. This reaction is called a "double displacement" or "double decomposition" because two compounds undergo displacement or decomposition simultaneously.

The general equation for a double displacement reaction is:

$$AB + CD \rightarrow AD + CB$$

Where A and C represent cations, and B and D represent anions.

42. The correct answer is (A)

A precipitation reaction is a type of double displacement reaction in which one of the products formed is an insoluble solid called a precipitate. The precipitate separates from the solution due to its low solubility in water.

The general equation for a precipitation reaction is:

$$AB\ (aq) + CD\ (aq) \rightarrow AD(s) + CB\ (aq)$$

Where AB and CD are the reactants (soluble ionic compounds), AD is the insoluble precipitate, and CB is the soluble compound remaining in the solution.

43. The correct answer is (C)

A neutralization reaction is a type of double displacement reaction where an acid and a base react to form a salt and water. In this reaction, one of the products formed is a soluble salt, and the other product is water.

The general equation for a neutralization reaction is:

$$Acid + Base \rightarrow Salt + Water$$

For example, when hydrochloric acid reacts with sodium hydroxide, the products formed are sodium chloride, a soluble salt, and water.

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

44. The correct answer is (C)

Based on the given information:

1. Compound A reacts with compound B to form compounds C and D.
2. Compound C can easily dissolve in compound D.
3. Compound D on electrolysis gives hydrogen and oxygen.

From point 3, we can infer that compound D is water (H_2O), as the electrolysis of water produces hydrogen and oxygen gases.

From point 2, we know that compound C is a salt, which can dissolve in water (compound D).

For an acid (A) and a base (B) to react and form a salt (C) and water (D), it must be a neutralization reaction.

The balanced chemical equation for a neutralization reaction is:

Acid (A) + Base (B) → Salt (C) + Water (D)

44. The correct answer is (D)

A double displacement reaction involves the exchange of ions between two compounds. In this process, the anions (negatively charged ions) of one molecule and the cations (positively charged ions) of another molecule switch partners, resulting in the formation of two new compounds. This type of reaction can be represented by the general equation:

$$AB + CD \rightarrow AD + CB$$

where A and C represent cations, and B and D represent anions. The ions rearrange themselves to form two new ionic compounds, AD and CB.

46. The correct answer is (D)

$$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2\ NaCl.$$

Above reaction is a double displacement reaction because it involves the exchange of ions between two ionic compounds. In this reaction, the barium cation (Ba^{2+}) from barium chloride and the sulphate anion (SO_4^{-2}) from sodium sulphate switch partners to form the new ionic compound barium sulphate. Simultaneously, the chloride anions (Cl^-) from barium chloride and the sodium cations (Na^+) from sodium sulphate combine to form sodium chloride.

47. The correct answer is (C)

$$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow 2\ NaCl(aq) + BaSO_4(s)$$

Above reaction represents a double displacement reaction, also known as a metathesis reaction. In this reaction, the sodium cation (Na^+) from sodium sulphate and the chloride anion (Cl^-) from barium chloride exchange partners, forming sodium chloride. Simultaneously, the barium cation (Ba^{2+}) from barium chloride and the sulphate anion (SO_4^{-2}) from sodium sulphate combine to form the insoluble precipitate barium sulphate.

Oxidation and Reduction Reactions

48. The correct answer is (B)

Oxidation is a chemical process in which an atom, molecule, or ion loses one or more electrons. When a substance undergoes oxidation, it loses electrons, increasing its oxidation state. Oxidation is one half of the redox (reduction-oxidation) reaction, with reduction being the complementary process where electrons are gained.

Oxidation occurs when an element or compound loses electrons, typically by transferring them to another substance.

49. The correct answer is (B)

In the reaction,

$$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s),$$

the zinc ion (Zn^{2+}) gains two electrons, resulting in the formation of solid zinc metal. This process is called reduction, which is the gain of electrons by a substance.

Reduction is the opposite of oxidation, where a substance loses electrons. In this reaction, the zinc ion acts as an oxidizing agent, accepting electrons from another species (not shown in the equation).

50. The correct answer is (A)

An oxidizing agent is a substance that has the ability to accept electrons from another substance during a chemical reaction. This process is called oxidation. When an oxidizing agent accepts electrons, it undergoes reduction itself, as the gain of electrons leads to a decrease in its oxidation state. Oxidizing agents are often used in redox (reduction-oxidation) reactions to facilitate the removal of electrons from other substances.

Common examples of oxidizing agents include oxygen gas (O_2), hydrogen peroxide (H_2O_2), potassium permanganate ($KMnO_4$), and potassium dichromate ($K_2Cr_2O_7$).

51. The correct answer is (C)

In the reaction between zinc and iodine to form zinc iodide, zinc is oxidized. The chemical equation can be represented as,

$$Zn\ (zinc) + I_2\ (iodine) \rightarrow ZnI_2\ (zinc\ iodide)$$

In this reaction, the zinc atom loses electrons to form zinc ions (Zn^{2+}), which is the process of oxidation. Specifically, zinc goes from an oxidation state of 0 to +2, indicating that it has lost two electrons.

52. The correct answer is (B)

In the given reaction,

$$H_2S(g) + Cl_2(g) \rightarrow S(s) + 2\ HCl(g)$$

hydrogen sulphide (H_2S) is being oxidized.

Oxidation is the process of losing electrons, resulting in an increase in the oxidation state of an element or compound. In this reaction, the sulphur atom in hydrogen sulphide (H_2S) loses two electrons, changing its oxidation state from -2 to 0, forming elemental sulphur (S).

The hydrogen atoms in H_2S remain unchanged, with an oxidation state of +1 before and after the reaction. The chlorine molecules (Cl_2) gain electrons, undergoing reduction to form hydrogen chloride (HCl).

53. The correct answer is (C)

In the given reaction, when zinc oxide (ZnO) is heated with carbon (C), the following reaction takes place:

$$ZnO(s) + C(s) \rightarrow Zn(s) + CO(g)$$

In above reaction, zinc oxide acts as the oxidizing agent, facilitating the oxidation of carbon to form carbon monoxide (CO). Simultaneously, the zinc ion (Zn^{2+}) in ZnO is reduced by gaining two electrons, resulting in the formation of elemental zinc (Zn).

The oxidation and reduction processes can be described as follows:

1. Oxidation: $C(s) \rightarrow CO(g) + 2e^-$ (Carbon is oxidized)
2. Reduction: $Zn2+(s) + 2e^- \rightarrow Zn(s)$ (Zinc ion is reduced)

54. The correct answer is (B)

When iron (Fe) is added to a copper sulphate (CuSO4) solution, the following reaction occurs:

$$Fe(s) + CuSO4(aq) \rightarrow FeSO4(aq) + Cu(s)$$

In this reaction, the copper ions (Cu^{2+}) present in the copper sulphate solution are reduced to metallic copper (Cu), which precipitates out as a solid. The reduction of copper ions is facilitated by the oxidation of iron atoms, where iron loses electrons and forms iron (II) sulphate (FeSO4).

The oxidation and reduction half-reactions can be represented as:

Reduction: Cu2+(aq) + 2e- → Cu(s)

Oxidation: Fe(s) → Fe2+(aq) + 2e-

55. The correct answer is (C)

The reaction between hydrogen gas (H_2) and bromine gas (Br_2) to form hydrogen bromide (HBr) involves both oxidation and reduction processes.

The balanced chemical equation is,

$$H_2 + Br_2 \rightarrow 2HBr$$

In this above reaction, the hydrogen molecule (H_2) is oxidized by losing electrons, and the bromine molecule (Br_2) is reduced by gaining electrons.

The oxidation half-reaction is: $H_2 \rightarrow 2H^+ + 2e^-$ (Hydrogen is oxidized)

The reduction half-reaction is: $Br_2 + 2e^- \rightarrow 2\ Br^-$ (Bromine is reduced)

56. The correct answer is (A)

In the given reaction,

$$2Ag + 2H_2SO_4 \rightarrow Ag_2SO_4 + SO_2 + 2\ H_2O$$

sulfuric acid (H_2SO_4) acts as the oxidizing agent.

An oxidizing agent is a substance that has the ability to accept electrons from another substance during a chemical reaction, causing the oxidation of that substance.

In this reaction, sulfuric acid oxidizes the silver atoms (Ag) by accepting electrons from them. The silver atoms lose electrons and are oxidized to form silver(I) sulphate (Ag_2SO_4). Simultaneously, the sulphur atom in sulfuric acid is reduced from an oxidation state of +6 to +4, forming sulphur dioxide (SO_2) gas.

57. The correct answer is (B)

In the given reaction, Zn (zinc) loses two electrons to form Zn^{2+} (zinc ion), which is the process of oxidation. On the other hand, Cu^{2+} (copper ion) gains two electrons to form Cu (copper metal), which is the process of reduction.

The reaction can be represented as:

Zn --> $Zn^{2+} + 2e^-$ (Oxidation)

$Cu^{2+} + 2e^-$---> Cu (Reduction)

58. The correct answer is (B)

In the below reaction, aluminum (Al) metal reduces ferric oxide (Fe_2O_3) to form aluminum oxide (Al_2O_3) and iron (Fe) metal. The reaction can be represented as:

$$2Al + Fe_2O_3 \text{ --> } Al_2O_3 + 2Fe$$

In this reaction, aluminum metal loses electrons and gets oxidized to aluminum oxide (Al_2O_3), while ferric oxide (Fe_2O_3) gains electrons and gets reduced to iron (Fe) metal.

$Fe_2O3 + 6e^-$ --> $2Fe + 3O^{2-}$ (Reduction of Fe_2O_3)

$2Al$ --> $2Al^{3+} + 6e^-$ (Oxidation of Al)

59. The correct answer is (C)

According to the electronic concept of oxidation and reduction reactions:

(A) Oxidation is the loss of electrons.

(B) The electron-donating species is called the reducing agent, not the oxidizing agent.

(C) Reduction is the gain of electrons.

(D) The electron-accepting species is called the oxidizing agent, not the reducing agent.

In a redox reaction, oxidation and reduction occur simultaneously. Oxidation involves the loss of electrons by a species, while reduction involves the gain of electrons by another species.

60. The correct answer is (B)

Rancidity is a process of oxidation that occurs in fats and oils, leading to the development of unpleasant odors and Flavors. Rancidity can be checked or prevented by using anti-oxidants.

Anti-oxidants are substances that inhibit the process of oxidation by reacting with free radicals and other reactive oxygen species. They help in preventing the onset of rancidity in foods containing fats and oils.

Some common anti-oxidants used to check rancidity include vitamins C and E.

61. The correct answer is (B)

In the given reaction equation for the rusting of iron,

$$4Fe + 3O_2 \text{ --> } 2Fe_2O_3$$

iron (Fe) undergoes oxidation to form iron (III) oxide (Fe_2O_3), which is rust.

Oxidation is the process of losing electrons, and in this reaction, each iron atom loses three electrons to form the Fe^{3+} ions in iron (III) oxide.

The equations for the oxidation half-reaction redox reaction are:

$4Fe \text{ --> } 4Fe^{3+} + 12e^-$ (Oxidation of iron)

$3O_2 + 12e^- \text{ --> } 6O^{2-}$ (Reduction of oxygen)

62. The correct answer is (C)

In a redox reaction, the substance that gets oxidized loses electrons, and the substance that gets reduced gains electrons.

- (A) In oxidation, oxygen is added to a substance or electrons are lost.
- (B) In reduction, hydrogen is added to a substance or electrons are gained.
- (C) An oxidizing agent is actually reduced; it gains electrons and causes another substance to be oxidized.
- (D) A reducing agent is oxidized; it loses electrons and causes another substance to be reduced.

63. The correct answer is (B)

In the reaction:

$$NaH + H_2O \rightarrow NaOH + H_2$$

Hydrogen undergoes both oxidation and reduction.

- In NaH (sodium hydride), hydrogen has an oxidation state of -1.
- In H_2O (water), hydrogen has an oxidation state of +1.
- In NaOH (sodium hydroxide), hydrogen has an oxidation state of +1.
- In H_2 (molecular hydrogen), hydrogen has an oxidation state of 0.

In the above reaction, the hydrogen in NaH (oxidation state -1) is oxidized to H2 (oxidation state 0), and the hydrogen in H_2O (oxidation state +1) is reduced to H_2 (oxidation state 0). Hence, hydrogen undergoes both oxidation and reduction.

64. The correct answer is (D)

$$Zn \rightarrow Zn^{2+} + 2e^-$$

In this reaction, zinc (Zn) undergoes oxidation by losing two electrons to form the zinc ion (Zn^{2+}). Oxidation is the process of losing electrons, and in this case, the zinc atom is oxidized by losing two electrons.

(A) $Zn^{2+} + 2e^- \rightarrow Zn$: In this reaction, the zinc ion is reduced by gaining two electrons to form zinc metal (Zn). This is the opposite of oxidation, which is reduction.

(B) $H_2 \rightarrow 2H^+ + 2e^-$: In this reaction, hydrogen gas is oxidized by losing two electrons to form two hydrogen ions. Zinc is not involved in this reaction.

(C) $2H^+ + 2e^- \rightarrow H_2$: This reaction represents the reduction of hydrogen ions by gaining two electrons to form hydrogen gas. Zinc is not involved in this reaction.

65. The correct answer is (C)

In a redox (reduction-oxidation) reaction, oxidation and reduction are invert processes that involve the transfer of electrons.

Oxidation: A species undergoes oxidation when it loses electrons. This process involves an increase in the oxidation state of the species.

Reduction: A species undergoes reduction when it gains electrons. This process involves a decrease in the oxidation state of the species.

66. The correct answer is (A)

When an apple is cut or exposed to air, the exposure to oxygen in the air causes an oxidation reaction to occur. This oxidation reaction involves enzymes present in the apple's tissues, particularly polyphenol oxidase enzymes, which catalyse the oxidation of phenolic compounds naturally present in the apple.

It results in the formation of brown-coloured compounds called quinones.

Oxygen (from air) + Phenolic compounds (in apple) → Quinones (brown-coloured compounds).

67. The correct answer is (A)

Rancidity term refers to the spoilage of fats and oils in foods, leading to an unpleasant taste and odor. Oxygen plays a critical role in this process by initiating oxidative reactions in unsaturated fats. These reactions involve the addition of oxygen atoms to the double bonds of fatty acids, leading to the formation of compounds such as peroxides and aldehydes, which are responsible for the rancid smell and taste.

68. The correct answer is (D)

Hydrogenation of oils is a process that involves the addition of hydrogen to unsaturated fatty acids present in oils, resulting in the formation of saturated fatty acids. This process is an example of a reduction reaction.

During hydrogenation, the double bonds present in the unsaturated fatty acid molecules are broken, and hydrogen atoms are added to these locations. The addition of hydrogen atoms to the unsaturated fatty acids results in the reduction of the molecules, converting them into saturated fatty acids.

Unsaturated fatty acid + H_2 → Saturated fatty acid

Exothermic and Endothermic Reactions

69. The correct answer is (D)

In an exothermic reaction, energy is released to the surroundings in the form of heat or light. The combustion of carbon with oxygen to produce carbon dioxide is an exothermic reaction, as it releases heat energy.

The balanced chemical equation for this reaction is:

$$C + O_2 \rightarrow CO_2 + \text{heat}$$

During this reaction, the strong bonds in the reactants (carbon and oxygen) are broken, and new, even stronger bonds are formed in the product (carbon dioxide). The excess energy released during the formation of these stronger bonds is given off as heat, making the overall reaction exothermic.

70. The correct answer is (C)

In the given reaction,

$$A + B \rightarrow C + D + \text{Heat}$$

the presence of "Heat" on the product side indicates that heat energy is released during this chemical process.

An exothermic reaction is a type of chemical reaction in which energy, typically in the form of heat, is released to the surroundings. The release of heat is a characteristic feature of exothermic reactions.

71. The correct answer is (A)

In an exothermic reaction, the energy required to break the bonds of the reactants is less than the energy released when new bonds are formed in the products.

Exothermic reactions often involve highly stable products compared to the reactants, allowing for a spontaneous release of energy during the reaction process.

An example of an exothermic reaction is the combustion of methane:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{energy}$$

72. The correct answer is (A)

The energy released or absorbed during a chemical reaction indicates whether it is exothermic or endothermic. In this case, since the reaction releases 182 kJ of energy, it is classified as an exothermic reaction. Exothermic reactions release energy in the form of heat to the surroundings. The presence of a catalyst (V_2O_5) suggests that it accelerates the reaction without being consumed itself.

73. The correct answer is (B)

Photosynthesis is the process by which plants and certain organisms convert carbon dioxide and water into glucose and oxygen, using sunlight as the energy source. This process is an endothermic reaction, that it requires an input of energy from an external source (sunlight) to proceed.

The overall equation for photosynthesis is:

$$6CO_2 + 6H_2O + \text{Sunlight energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

74. The correct answer is (A)

The given reaction,

$$CaCO_3(s) + \text{Heat} \rightarrow CaO(s) + CO_2(g)$$

represents the thermal decomposition of calcium carbonate (also known as limestone or chalk) into calcium oxide (quicklime) and carbon dioxide gas.

In this reaction, the word "Heat" is mentioned on the reactant side, indicating that heat energy needs to be supplied to the system for the reaction to occur. This implies that the reaction is endothermic.

Irreversible and Reversible Reactions

75. The correct answer is (B)

A reversible reaction is a chemical reaction where the products can react together to reform the original reactants under suitable conditions. This means that the reaction can proceed in both the forward and reverse directions. The symbol ⇌ is used to represent a reversible reaction.

For example, the formation of ammonia from nitrogen and hydrogen (H2):

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

76. The correct answer is (A)

An irreversible reaction is a chemical reaction where the products formed do not react with each other to retain the original reactants. In other words, once the reaction proceeds to form products, it does not reverse back to the reactants under the same conditions. Irreversible reactions typically proceed in one direction due to formation of stable compounds or the release of gases.

For example, the combustion of methane in the presence of oxygen to form carbon dioxide and water is an irreversible reaction:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

77. The correct answer is (D)

(A) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Above reaction involves the dissociation of phosphorus pentachloride gas into phosphorus trichloride gas and chlorine gas. The forward and reverse reactions can occur in a closed container, establishing an equilibrium.

(B) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

This reaction represents the thermal decomposition of calcium carbonate into calcium oxide and carbon dioxide gas. In a closed container, the reverse reaction can also occur, leading to an equilibrium state.

(C) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

This reaction involves the dissociation of hydrogen iodide gas into hydrogen gas and iodine gas. In a closed container, the forward and reverse reactions can occur, establishing an equilibrium.

All these reactions are reversible, the products can react to re-form the original reactants in a closed container.

78. The correct answer is (B)

Haber's process of ammonia synthesis, represented by the equation

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

is a reversible reaction.

Meaning that the forward and reverse reactions occur simultaneously, and the system reaches an equilibrium state where the rates of the forward and reverse reactions are equal.

The reaction is exothermic in the forward direction (synthesis of ammonia) and endothermic in the reverse direction (decomposition of ammonia). The reaction is reversible because it involves gaseous reactants and products, and the reaction can proceed in either direction depending on the conditions of temperature, pressure, and catalyst.

LEVEL-II
SINGLE CORRECT ANSWER QUESTIONS

1. The correct answer is (D)

The balanced chemical equation for the reaction between barium chloride ($BaCl_2$) and ammonium phosphate ($(NH_4)_3PO_4$) is:

$$3BaCl_2 + 2(NH_4)_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NH_4Cl$$

According to the given equation, 3 moles of $BaCl_2$ react with 2 moles of $(NH_4)_3PO_4$ to produce 6 moles of NH_4Cl (ammonium chloride).

Given:

- 3 moles of $BaCl_2$
- 2 moles of $(NH_4)_3PO_4$

From the balanced equation, observe that for every 3 moles of $BaCl_2$ reacting with 2 moles of $(NH_4)_3PO_4$, 6 moles of NH_4Cl are formed.

2. The correct answer is (B)

Hydrogen sulphide (H_2S) is the strongest reducing agent among the given options. A reducing agent is a substance that donates electrons in a redox (reduction-oxidation) reaction.

The strength of a reducing agent is determined by its ability to donate electrons and become oxidized. The order of reducing strength among the given options is:

$$H_2S > SnCl_2 > H_2SO_3 > HNO_3$$

Hydrogen sulphide is the strongest reducing agent because the sulphur atom can readily accept electrons and undergo oxidation to form sulphur containing compound.

3. The correct answer is (C)

$$Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$$

In this reaction, water acts as a reducing agent by donating electrons to chlorine and getting oxidized in the process.

Here, chlorine is the oxidizing agent, and hydrogen peroxide is the reducing agent. In the question asks for the reaction in which water acts as the reducing agent.

To understand this, we can break down the overall reaction into two half-reactions:

$Cl_2 + 2e^- \rightarrow 2Cl^-$ (Reduction half-reaction)

$H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$ (Oxidation half-reaction)

In the oxidation half-reaction, water is oxidized to oxygen gas by donating two electrons, thereby acting as a reducing agent.

4. The correct answer is (B)

A reducing agent is a substance that has the ability to donate electrons to another substance during a chemical reaction. The process of donating electrons is known as reduction, and the substance that receives the electrons is called the oxidizing agent.

The general equation for a reduction reaction can be represented as:

Reducing Agent → Oxidized Form + Electrons

5. The correct answer is (A)

In the given reaction

$$2Na + Cl_2 \rightarrow 2NaCl$$

sodium is oxidized, and chlorine acts as the oxidizing agent. Sodium atoms donate electrons to chlorine molecules, resulting in the formation of sodium cations (Na^+) and chloride anions (Cl^-). This process can be represented by the following half-reactions:

Oxidation half-reaction: $2Na \rightarrow 2Na^+ + 2e^-$

Reduction half-reaction: $Cl_2 + 2e^- \rightarrow 2Cl^-$

In this redox reaction, sodium atoms undergo oxidation by losing electrons, while chlorine molecules accept these electrons and undergo reduction.

6. The correct answer is (B)

To balance the given reaction: $Al(s) + aHCl(aq) \rightarrow AlCl_3(aq) + bH_2(g)$, we need to follow the rules of balancing chemical equations.

Step 1: Balance the aluminum (Al) and chlorine (Cl) atoms.

$$Al(s) + 3HCl(aq) \rightarrow AlCl_3(aq) + bH_2(g)$$

Step 2: Balance the hydrogen (H) atoms.

$$Al(s) + 3HCl(aq) \rightarrow AlCl_3(aq) + (3/2)\ H_2(g)$$

Therefore, the values of the constants a and b that balance the reaction are:

a = 3 (coefficient of HCl)

b = 3/2 (coefficient of H_2)

7. The correct answer is (C)

Column I represent the types of decomposition reactions, and Column II provides examples of these reactions.

(a) **Thermal decomposition reaction**: This type of reaction occurs when a compound breaks down into simpler substances upon heating or increasing temperature.

The example given in Column II, "$CaCO_3 \rightarrow CaO + CO_2$," represents the thermal decomposition of calcium carbonate into calcium oxide and carbon dioxide.

(b) **Electrolytic decomposition reaction**: This type of reaction involves the decomposition of a compound by passing an electric current through it. The example given in Column II, "$2H_2O \rightarrow 2H_2 + O_2$," represents the electrolytic decomposition of water into hydrogen and oxygen gases.

(c) **Photolytic decomposition reaction**: This type of reaction involves the decomposition of a compound by the absorption of light or photons. The example given in Column II, "$O_2(g) \rightarrow 2O(g)$," represents the photolytic decomposition of oxygen molecules into oxygen atoms upon exposure to ultraviolet (UV) radiation.

8. The correct answer is (A)

Elements are pure substances composed of only one type of atom and are homogeneous in nature. This means that all samples of a particular element have the same chemical composition and properties.

- (B) Compounds always contain two or more different elements is true because compounds are composed of atoms of different elements chemically bonded together in fixed proportions.

- (C) A mixture is not always heterogeneous is true because mixtures can be homogeneous (uniform composition throughout, like salt dissolved in water) or heterogeneous (non-uniform composition, like a mixture of sand and water).
- (D) Air is a heterogeneous mixture is true because it consists of a mixture of gases (mainly nitrogen, oxygen, carbon dioxide, etc.) with varying compositions at different locations or times.

9. The correct answer is (C)

To balance the given reaction: $C_6H_{14} + O_2 \rightarrow CO_2 + H_2O$, we need to follow the rules of balancing chemical equations.

Step 1: Balance the carbon (C) atoms.

$$C_6H_{14} + O_2 \rightarrow 6CO_2 + H_2O$$

Step 2: Balance the hydrogen (H) atoms.

$$C_6H_{14} + O_2 \rightarrow 6CO_2 + 7H_2O$$

Step 3: Balance the oxygen (O) atoms.

$$C_6H_{14} + 19O_2 \rightarrow 6CO_2 + 7H_2O$$

In this balanced reaction, the number of moles of O_2 required is 19 moles.

10. The correct answer is (C)

The loss of lustre on copper objects and the formation of a green coating, known as the "rust of copper," is a result of the reaction between copper and various atmospheric gases, including water vapor, oxygen, and carbon dioxide.

This process, known as corrosion, involves the following steps:

1. Copper reacts with oxygen to form copper (II) oxide (CuO):

$$2Cu + O_2 \rightarrow 2CuO$$

2. The copper (II) oxide then reacts with water vapor and carbon dioxide to form basic copper carbonate:

$$CuO + H_2O + CO_2 \rightarrow CuCO_3{\cdot}Cu\ (OH)_2$$

The basic copper carbonate, $CuCO_3{\cdot}Cu\ (OH)_2$, is the green coating that forms on the surface of copper objects, giving them a dull and tarnished appearance.

11. The correct answer is (B)

Rusting of iron is a chemical process that involves the combination of iron with oxygen and water from the atmosphere to form hydrated iron (III) oxide, which is commonly known as rust.

The overall reaction for the rusting of iron can be represented as:

$$4Fe(s) + 3O_2(g) + 2xH_2O(l) \rightarrow 2Fe_2O_3{\cdot}xH_2O(s)$$

In this reaction, iron combines with oxygen from the air and water to form hydrated iron (III) oxide ($2Fe_2O_3{\cdot}xH_2O$), which is the rust.

This process is termed a "combination reaction" because two or more reactants combine to form a single product.

12. The correct answer is (B)

The formula for rust, which is the reddish-brown coating that forms on iron and steel surfaces when exposed to oxygen and moisture, is $Fe_2O_3.xH_2O$.

Rust primarily consists of iron oxide (Fe_2O_3) in hydrated form (denoted as xH_2O, where x can vary depending on the specific conditions and type of rust).

14. The correct answer is (C)

In the given reaction: $PbO + C \rightarrow Pb + CO$, carbon (C) acts as a reducing agent.

A reducing agent is a substance that donates electrons in a chemical reaction, causing itself to undergo oxidation. In this reaction, carbon donates electrons to lead (II) oxide (PbO), reducing it to elemental lead (Pb).

The half-reactions involved are:

Oxidation half-reaction: $C + O^{2-} \rightarrow CO_2 + 4e^-$

Reduction half-reaction: $Pb^{2+} + 2e^- \rightarrow Pb$

15. The correct answer is (D)

When iron nails are placed in a solution of $CuSO_4$ (copper sulphate), a displacement reaction occurs.

- Iron (Fe) is more reactive than copper (Cu), so it displaces copper from copper sulphate ($CuSO_4$).
- The iron nails react with $CuSO_4$, causing copper metal (Cu) to be displaced from the solution.
- Simultaneously, iron ions (Fe^{2+}) are released into the solution, forming ferrous sulphate ($FeSO_4$).

The solution initially containing blue CuSO4 turns light green due to the presence of ferrous ions (Fe^{2+}).

$$Fe\ (s) + CuSO_4\ (aq) \rightarrow FeSO_4\ (aq) + Cu\ (s)$$

16. The correct answer is (A)

When zinc (Zn) reacts with sulfuric acid (H_2SO_4), the following reaction takes place:

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$$

In this reaction, zinc acts as a reducing agent, which means it donates electrons and undergoes oxidation. Specifically, the zinc atoms lose two electrons each, forming zinc ions (Zn^{2+}) in the solution.

The half-reaction for the oxidation of zinc is:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

17. The correct answer is (D)

The given compounds are products of corrosion of certain metals under specific conditions.

(A) $CuCO_3 \cdot xCu\ (OH)_2$, known as basic copper carbonate, is formed by the corrosion of copper in the presence of carbon dioxide and moisture.

(B) $Fe_2O_3 \cdot xH_2O$, known as hydrated iron (III) oxide or rust, is formed by the corrosion of iron in the presence of oxygen and moisture.

(C) Ag_2S, known as silver sulphide or silver tarnish, is formed by the corrosion of silver in the presence of sulphur-containing compounds like hydrogen sulphide.

18. The correct answer is (D)

The methods listed in the options (A), (B), and (C) are all commonly used techniques to prevent corrosion of metals.

(A) Painting the surface of the metal with a protective coating creates a barrier between the metal and the corrosive environment, preventing direct contact and inhibiting corrosion.

(B) Greasing the surface of the metal involves applying a thin layer of grease or oil, which acts as a barrier and prevents moisture and oxygen from reaching the metal surface, thereby preventing corrosion.

(C) Varnishing or coating the surface of the metal with a protective layer, such as a polymer or enamel coating, also creates a barrier that isolates the metal from the corrosive environment.

19. The correct answer is (B)

When packing fats and oils in food, it is important to prevent oxidation, which can lead to rancidity and spoilage. Nitrogen gas (N_2) is used because it is chemically inert and does not react easily with other substances, including fats and oils.

By replacing the air (which contains oxygen) with nitrogen during packaging, the exposure of fats and oils to oxygen is minimized. This helps to preserve the quality and freshness of the food products for a longer period, as oxidation reactions, which require oxygen, are significantly reduced or prevented altogether.

20. The correct answer is (C)

An equilibrium can be attained from either the forward or backward direction in a reversible chemical reaction. The direction from which the equilibrium is approached does not matter.

In a reversible reaction, both the forward and backward reactions occur simultaneously. Initially, if the reactants are present, the forward reaction will predominate, and the products will start to form. As the concentration of products increases, the backward reaction will also begin to occur, and the formation of reactants will start.

Eventually, a point is reached where the rates of the forward and backward reactions become equal, and the concentrations of reactants and products remain constant. This state is known as chemical equilibrium.

21. The correct answer is (A)

The reaction between an acid and a base to form a salt and water is known as a neutralization reaction. This type of reaction is characterized by the combination of a proton donor (acid) and a proton acceptor (base), resulting in the formation of an ionic compound (salt) and water.

For example, when hydrochloric acid (HCl) reacts with sodium hydroxide (NaOH), the products formed are sodium chloride (NaCl) and water (H2O). The overall reaction can be represented as:

$$HCl\ (acid) + NaOH\ (base) \rightarrow NaCl\ (salt) + H_2O\ (water)$$

22. The correct answer is (B)

1. Balance aluminum (Al):
   Left: a, right: c
   a = c
2. Balance chlorine (Cl):
   Left: 3a, right: 2d
   3a = 2d
3. Balance calcium (Ca):
   Left: b, right: d
   b = d
4. Balance hydroxide (OH):
   Left: 2b, right: 3c
   2b = 3c

From these above relationships, we can deduce:

a = c = 3

b = d = 3

2b = 3c, which is satisfied by b = 3 and c = 2

So, the correct answer is (B) a=2 b=3 c=2 d=3.

23. The correct answer is (C)

A reducing agent, is a substance that donates electrons to another species in a chemical reaction. By donating electrons, it causes the reduction of another substance while itself becoming oxidized.

This electron transfer is the basis of redox reactions. The ability to donate electrons makes reducing agents essential in various chemical processes, including industrial applications, biological systems, and laboratory synthesis.

Example equation:

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ (reduction)

$Zn \rightarrow Zn^{2+} + 2e^-$ (oxidation, Zn is the reducing agent)

ONE OR MORE THAN ONE CORRECT ANSWER QUESTIONS

24. The correct answer is (B)

$$2\ HgO \rightarrow 2\ Hg + O_2$$

This reaction represents a chemical decomposition, where a single compound breaks down into simpler substances. In this case, mercury (II) oxide decomposes into mercury metal and oxygen gas when heated. This is a classic example of thermal decomposition.

Option (A) is a combination reaction, (C) is also a decomposition but more complex, and (D) is a synthesis reaction.

25. The correct answer is (C)

Energy released or absorbed during a chemical reaction is measured in terms of joules or kilo joules (kJ). One joule is a small unit of energy, so kilo joules (1 kJ = 1,000 joules) are often used to quantify the energy changes in chemical reactions.

This unit is part of the International System of Units (SI) and is widely used in scientific contexts to express energy, heat, and work.

26. The correct answer is (A, B)

When calcium carbonate is heated, it undergoes thermal decomposition, breaking down into calcium oxide, also known as quicklime, and carbon dioxide gas. This reaction is an important industrial process used in the production of lime.

The chemical equation for this decomposition reaction is:

$$CaCO_3 \rightarrow CaO + CO_2$$

27. The correct answer is (C)

A negative catalyst, also known as an inhibitor, is a substance that decreases the rate of a chemical reaction or reduces the effectiveness of a catalyst.

Unlike positive catalysts that speed up reactions, negative catalysts slow them down. They work by interfering with the reaction mechanism, often by occupying active sites on a catalyst's surface or by reacting with intermediates in the reaction pathway.

Negative catalysts are useful in various applications, such as:

1. Controlling polymerization reactions.

2. Preventing unwanted reactions in stored chemicals.

28. The correct answer is (A)

Combination reactions involve the joining of two or more reactants to form a single product. The most common type of combination reaction involves elements combining with other elements to form a compound.

For example, the combination of hydrogen and oxygen to form water is a classic example:

$$2H_2 + O_2 \rightarrow 2H_2O$$

In this reaction, hydrogen and oxygen are elements that combine to form water a compound.

29. The correct answer is (D)

The mass of the substance changes in a chemical reaction. This statement is false, as it goes against the law of conservation of mass, which states that mass is neither created nor destroyed in a chemical reaction. The true statements for the given chemical reaction, 2H2(g) + O2(g) → 2H2O(g), are:

(A) Hydrogen burns in oxygen to form water.

(B) Two elements combine to form a single compound.

(C) The formation of water from hydrogen and oxygen is a combination reaction.

30. The correct answer is (D)

For the given reaction:

$$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$$

The false statement is 'It is a decomposition reaction'.

- (A) In this reaction, sulphur dioxide reacts with oxygen to form a new compound, sulphur trioxide, so this statement is true.

- (B) This reaction is indeed a combination reaction because two substances (SO_2 and O_2) combine to form a single product (SO_3), so this statement is true.

- (C) The new product formed is sulphur trioxide (SO_3), so this statement is true.

- (D) A decomposition reaction involves a single compound breaking down into two or more simpler substances, which is not the case here. Therefore, this statement is false.

31. The correct answer is (C)

Chemical changes involve the transformation of substances at the molecular level, often resulting in the formation of new substances with different properties. These changes are typically accompanied by energy changes, most commonly in the form of heat being released (exothermic) or absorbed (endothermic).

The other options are incorrect:

(A) Chemical changes are generally not easily reversible.

(B) The total mass remains constant in a chemical reaction (Law of Conservation of Mass).

(D) New substances are indeed formed in chemical changes.

32. The correct answer is (A, B, C)

(A)Burning of fuels: This involves the combustion of hydrocarbons, producing carbon dioxide and water. Example:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

(B) Burning of crackers: This involves rapid oxidation reactions, producing light, heat, and various gases.

(C) Strong heating of mercuric oxide: This causes thermal decomposition.

$$2HgO \rightarrow 2Hg + O_2$$

These processes involve the formation of new substances with different chemical properties from the original materials.

(D) Melting of ice is not a chemical change. It's a physical change where only the state of water changes from solid to liquid, without altering its chemical composition.

33. The correct answer is (A, B, C, D)

(A) Cooking of food involves various chemical reactions, altering the molecular structure of ingredients.

(B) Ripening of fruits includes complex biochemical processes, changing sugars, acids, and other compounds.

(C) Rusting of iron is an oxidation reaction:

$$4Fe + 3O_2 + 2H_2O \rightarrow 4Fe\ (OH)_3$$

(D) Blackening of silver-ware is typically due to the formation of silver sulphide:

$2Ag + H_2S \rightarrow Ag_2S + H_2$

Each of these processes involves the formation of new substances with different chemical properties from the original materials.

34. The correct answer is (C)

$$Hg\ (NO_3)_2 + 2Ag \rightarrow 2Ag\ (NO_3)_2 + Hg$$

This reaction is cannot happen because silver (Ag) is less reactive than mercury (Hg). In the reactivity series, mercury is above silver, meaning it's more likely to form compounds and less likely to be displaced by silver.

35. The correct answer is (A)

To determine which reaction occurs, we need to consider the reactivity series of metals.

Iron is more reactive than copper, so it will displace copper from its salt solution.

The reaction can be represented by the equation:

$$Fe + CuCl_2 \text{ -> } FeCl_2 + Cu$$

In this reaction, iron replaces copper in the solution, forming iron (II) chloride and solid copper.

The other options do not result in reactions because:

(B) Silver is less reactive than hydrogen.

(C) Copper is less reactive than lead.

(D) Zinc is more reactive than chromium, but chromium is above hydrogen in the reactivity series.

36. The correct answer is (A)

$$ZnSO_4 + Ag \rightarrow AgSO_4 + Zn.$$

Above reaction is not possible because silver is less reactive than zinc in the reactivity series of metals. Silver cannot displace zinc from its salt solution.

All other reactions given are possible:

(B) Lead is less reactive than calcium, so calcium can displace lead.

(C) Mercury is less reactive than magnesium, so magnesium can displace mercury.

(D) Copper is less reactive than iron, so iron can displace copper.

37. The correct answer is (B)

The correct answer is Chemical double displacement reaction.

The given equation '(A+B-) + (C+D-) → (A+D-) + (C+B-)' represents a double displacement reaction. In this type of reaction, two compounds exchange ions to form two new compounds.

The characteristic feature of a double displacement reaction is that two reactants exchange partners. In this case:

- A+ exchanges with C+
- B- exchanges with D-

This results in the formation of two new compounds (A+D-) and (C+B-).

38. The correct answer is (A)

In a chemical reaction, the Law of Conservation of Mass states that the mass of the reactants equals the mass of the products. This principle ensures that matter is neither created nor destroyed during a chemical reaction. For example, in the reaction:

$$2H_2 + O_2 \rightarrow 2H_2O$$

The mass of 4 grams of hydrogen and 32 grams of oxygen will result in 36 grams of water. This explains that the total mass of the reactants equals the total mass of the products.

39. The correct answer is (B, C)

Explosion reactions, combustion reactions are generally considered irreversible due to their rapid, exothermic nature and the dispersal of products. Once an explosion occurs, it's practically impossible to collect and recombine the products back into the original reactants.

The other options are not universally irreversible:

(A) Ionic reactions can often be reversed.

(D) Many precipitation reactions can be reversed by changing conditions like pH or temperature.

Explosion reactions, however, involve such rapid energy release and physical dispersion of products that reversal is not feasible.

40. The correct answer is (D)

$$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$$

This reaction between silver nitrate and sodium chloride forms silver chloride, which is insoluble in water and precipitates out of solution. The formation of a solid precipitate from aqueous reactants is the defining characteristic of a precipitation reaction.

41. The correct answer is (A)

$$2K_2\,MnO_4 + Cl_2 \rightarrow 2KMnO_4 + 2KCl$$

This reaction is an example of oxidation.

1. The oxidation state of manganese changes from +6 in K_2MnO_4 (potassium manganate) to +7 in $KMnO_4$ (potassium permanganate).
2. The increase in oxidation state indicates that manganese is being oxidized.
3. Chlorine (Cl_2) is acting as the oxidizing agent, accepting electrons and being reduced to chloride ions (Cl^-) in KCl.

42. The correct answer is (A)

In the reaction:

$$Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$$

Here, Cu(s) is oxidized to Cu^{2+}.

In this reaction, solid copper (Cu) loses two electrons to form Cu^{2+} ions, which means it is oxidized. Simultaneously, Ag^{+} ions in the aqueous solution gain electrons to form solid silver (Ag), indicating that Ag^{+} is reduced. Therefore, copper undergoes oxidation and silver ions undergo reduction.

43. The correct answer is (D)

$$2S_2O_3^{2-} \rightarrow S_4O_6^{2-}$$

This reaction involves the conversion of thiosulfate ions to tetrathionate ions. In this process, the oxidation state of sulphur remains unchanged:

- In $S_2O_3^{2-}$, sulphur has oxidation states of +6 and -2.
- In $S_4O_6^{2-}$, sulphur maintains the same oxidation states of +6 and -2.

Since there's no change in oxidation states, this reaction involves neither oxidation nor reduction.

44. The correct answer is (D)

In this reaction, chlorine (Cl_2) undergoes disproportionation, where it simultaneously oxidizes and reduces. However, the question asks about chlorine's overall role in the reaction.

Chlorine acts as an oxidizing agent because:

1. It accepts electrons from OH^-, oxidizing it to form $ClO3^-$ (chlorate ion).
2. The oxidation state of some chlorine atoms increases from 0 to +5 in $ClO3^-$.
3. While some chlorine is reduced to Cl^- (oxidation state -1), the key role is its ability to oxidize OH^-.

45. The correct answer is (A)

$$3Br_2 + 6CO_3^{2-} + 3H_2O \rightarrow 5Br^- + BrO_3^- + 6HCO_3$$

This reaction is a disproportionation of bromine, where bromine acts as both an oxidizing and reducing agent. However, the question asks about the overall change in bromine's oxidation state.

Bromine is oxidised:

1. In Br_2, bromine has an oxidation state of 0.
2. In the products, we have Br- (oxidation state -1) and BrO_3^- (oxidation state +5).
3. While some bromine is reduced to Br^-, one bromine atom is oxidized to BrO_3^-.

46. The correct answer is (D)

$$2Cu_2O(s) + Cu_2S(s) \longrightarrow 6Cu\ (s) + SO_2\ (g)$$

1. Cu_2O (copper(I) oxide) is actually oxidized, not reduced. The oxidation state of copper changes from +1 in Cu2O to 0 in Cu.
2. Cu_2S (copper(I) sulphide) is oxidized. The sulphur changes from an oxidation state of -2 in Cu_2S to +4 in SO_2.

The correct interpretation is:

- Cu_2O acts as the oxidizing agent (it's reduced)
- Cu_2S acts as the reducing agent (it's oxidized)

47. The correct answer is (A)

$$2FeCl_3 + H_2S \longrightarrow 2FeCl_2 + 2HCl + S$$

In this reaction:

1. $FeCl_3$ (iron (III) chloride) is reduced to FeCl2 (iron (II) chloride). The iron's oxidation state changes from +3 to +2.

2. H_2S (hydrogen sulphide) is oxidized to elemental sulphur (S). The sulphur's oxidation state changes from -2 to 0.

$FeCl_3$ accepts electrons from H_2S, thereby oxidizing it. This makes $FeCl_3$ the oxidizing agent in the reaction.

48. The correct answer is (D)

A redox (reduction-oxidation) reaction involves the transfer of electrons between reactants. In option (D), evaporation of H_2O (water) does not involve any change in oxidation states or the transfer of electrons. It is a physical process where liquid water changes into water vapor due to an increase in temperature, without altering the chemical composition of water molecules.

49. The correct answer is (B)

$$A\ (s) + CuSO_4\ (aq) \rightarrow ASO_4\ (aq) + Cu\ (s)$$

This reaction represents a single displacement reaction where a more reactive metal replaces a less reactive metal from its salt solution. For this reaction to occur spontaneously, 'A' must be more reactive than copper.

Among the given options:

(A) Ag (silver) is less reactive than copper

(B) Zn (zinc) is more reactive than copper

(C) Pb (lead) is less reactive than copper

(D) Fe (iron) is more reactive than copper, but less reactive than zinc

50. The correct answer is (D)

Redox reactions involve the transfer of electrons between substances, leading to changes in oxidation states.

In option (A):

$$CuO\ (solid) + H_2\ (gas) \rightarrow Cu\ (solid) + H_2O\ (gas)$$

Copper (II) oxide (CuO) is reduced to copper (Cu), and hydrogen (H_2) is oxidized to water (H_2O).

In option (B):

$$2K\ (solid) + F_2\ (gas) \rightarrow 2K+\ F-\ (solid)$$

Potassium (K) is oxidized to potassium ion (K^+), and fluorine (F_2) is reduced to fluoride ion (F^-).

COMPREHENSION TYPE QUESTIONS

51. The correct answer is (A)

In a redox reaction, the reducing agent is defined as the compound or molecule or atom that loses electrons and is oxidized. In the given reaction:

$$6Fe^{2+}(aq) + BrO^{3-}(aq) + 6H^+ \rightarrow 6Fe3^+(aq) + Br^-(aq) + 3H_2O(l)$$

The Fe^{2+} ions are oxidized to Fe^{3+} ions, meaning they lose electrons:

$$Fe^{2+} \longrightarrow Fe^{3+} + e^-$$

Since Fe^{2+} is losing electrons, it is acting as the reducing agent.

52. The correct answer is (B)

In the given reaction:

$$6Fe^{2+}(aq) + BrO^{3-}(aq) + 6H^+ \rightarrow 6Fe3^+(aq) + Br^-(aq) + 3H_2O(l)$$

BrO_3^- (bromate ion) is the oxidizing agent. An oxidizing agent is a substance that gains electrons and is reduced in a chemical reaction. In this reaction, BrO_3^- is reduced to Br^-, indicating it has gained electrons. Meanwhile, Fe^{2+} is oxidized to Fe^{3+}, indicating it has lost electrons. Therefore, BrO_3^- acts as the oxidizing agent by accepting electrons from Fe^{2+}.

53. The correct answer is (B)

Oxidation is a chemical process where a substance gains oxygen or loses electrons. It is often accompanied by the reduction of another substance, which loses oxygen or gains electrons. For example, in the reaction of magnesium with oxygen:

$$2Mg + O_2 \rightarrow 2MgO$$

Magnesium (Mg) gains oxygen to form magnesium oxide (MgO), and this process is called oxidation.

54. The correct answer is (C)

In chemical reactions, reduction refers to the process where a substance gains electrons or gains hydrogen atoms. This often involves the decrease in the oxidation state of the element being reduced. For example, in the reaction where hydrogen gas reduces copper (II) oxide to copper, hydrogen atoms are gained:

$$CuO + H_2 \rightarrow Cu + H_2O$$

Here, copper (II) oxide (CuO) is reduced to copper (Cu) by gaining hydrogen atoms from hydrogen gas (H_2).

STATEMENT TYPE QUESTIONS

55. The correct answer is (A)

Both statements are accurate. Zinc can indeed replace copper from copper sulphate solution, which is represented by the reaction:

$$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$$

This displacement reaction occurs because zinc is more reactive than copper, as stated in Statement-2. The reactivity difference explains why zinc can displace copper from its salt solution. Zinc's higher reactivity means it has a greater tendency to lose electrons and form positive ions compared to copper.

This allows zinc to reduce copper ions in solution, replacing them in the compound. Thus, Statement-2 correctly explains the phenomenon described in Statement-1.

56. The correct answer is (A)

Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

Statement–1 is true because in the reaction $Fe \rightarrow Fe^{2+} + 2e^-$, iron (Fe) loses electrons and is oxidized to Fe^{2+}. In this process, Fe acts as a reducing agent because it donates electrons to another substance, reducing it.

Statement–2 is also true because the loss of electrons is indeed called oxidation. This definition explains why Fe is acting as a reducing agent in Statement–1.

MATCHING TYPE QUESTIONS

57. The correct matching is:

(a) $Fe^+ \rightarrow Fe^{2+} + 1e^-$ matches with (S) oxidation reaction.

(b) $Fe^{2+} + 1e^- \rightarrow Fe^+$ matches with (Q) reduction reaction.

(c) $Fe^{3+} + Sn^{2+} \rightarrow Fe^{2+} + Sn^{4+}$ matches with (P) redox reaction.

(d) $HCl + NaOH \rightarrow NaCl + H_2O$ matches with (R) neutralization reaction.

Reasons:

(a) is oxidation as Fe^+ loses an electron.

(b) is reduction as Fe^{2+} gains an electron.

(c) is a redox reaction involving both oxidation (Sn^{2+} to Sn^{4+}) and reduction (Fe^{3+} to Fe^{2+}).

(d) is a neutralization reaction between an acid and a base.

58. The correct matching is:

(a) $2Na + Cl_2 \rightarrow 2NaCl$ matches with (R) combination reaction.

(b) $AB + CD \rightarrow AD + BC$ matches with (Q) Double displacement reaction.

(c) $CuSO_4 + Zn \rightarrow ZnSO_4 + Cu$ matches with (P) displacement reaction.

(d) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$ matches with (S) Double decomposition reaction.

Reason:

(a) is a combination reaction where two elements combine to form a compound.

(b) represents the general form of a double displacement reaction.

(c) is a single displacement reaction where Zn replaces Cu in the compound.

(d) is a double decomposition reaction, also known as double displacement.

INTEGER TYPE QUESTIONS

59. To balance this reaction, determine the correct coefficients for each compound.

1) First, balance potassium (K):

$$2KClO_3(s) \rightarrow bKCl(s) + cO_2(g)$$

2) Now, chlorine (Cl) is balanced:

$$2KClO_3(s) \rightarrow 2KCl(s) + cO_2(g)$$

3) Finally, oxygen (O) is balanced:

$$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$$

The balanced equation is:

$$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$$

Here, a = 2, b = 2, and c = 3.

The sum of a, b, and c is:

2 + 2 + 3 = 7

60.To determine the value of x + y, balance the equation:

1) Start with the unbalanced equation:

$$xNa + Cl_2 \rightarrow yNaCl$$

2) Balance sodium (Na):

$$2Na + Cl_2 \rightarrow yNaCl$$

3) Chlorine (Cl) is already balanced:

$$2Na + Cl_2 \rightarrow 2NaCl$$

The balanced equation is:

$$2Na + Cl_2 \rightarrow 2NaCl$$

Here, x = 2 and y = 2.

Therefore, the value of x + y is: 2 + 2 = 4.

2. Periodic Classification

Solutions

LEVEL – I

1. Correct option: (B) Transition element

The element with 18 electrons in the penultimate shell and 1 electron in the ultimate shell is copper (Cu). This makes it a transition element, as transition elements have partially filled d subshells in their valence shells. The penultimate shell of copper is the 4th shell, which has 18 electrons. The valence shell is the 5th shell, which has 1 electron. The 18 electrons in the penultimate shell are arranged in the 4s and 3d subshells. The 4s subshell is filled with 2 electrons, and the 3d subshell is filled with 10 electrons.

2. Correct option: (C) IA

The chloride of an element A gives a neutral solution in water because the chloride of a strong acid and a strong base is neutral.

The reaction is:

$2M + Cl_2 \rightarrow 2MCl$, $MCl + H_2O \rightarrow HCl + MOH$ (strong acid + strong base)

where M is the metal atom.

Therefore, the element A belongs to the first group.

3. Correct option: (A) C_2^{2-}, NO^+, CN^-, O_2^{2+}

Isoelectronic species have the same number of electrons. CO, CN^-, NO^+, and C_2^{2-} are isoelectronic because they all have 14 electrons.

4. Correct option: (D) For transition elements, the d-sub-shells are filled with electrons monotonically with an increase in atomic number

The d-subshells of transition elements are not filled with electrons monotonically with increasing atomic number. Instead, the filling of d-subshells is often interrupted by the filling of s-subshells, or the d-subshell may be left partially filled. This is because the stability of half-filled and full-filled electron configurations can override the general trend of filling subshells in order of increasing energy.

5. Correct option: (C) 1.96 Å, 0.72 Å

When an atom loses an electron, its ionic radius decreases. When an atom gains an electron, its ionic radius increases. Since the ionic radii of K+ and F- are nearly the same, the atomic radii of K and F must be 1.96 Å and 0.72 Å respectively.

6. Correct option: (D) O

Ionization potential is the energy required to remove an electron from an atom in its gaseous state. The ionization potential increases as the atomic number increases. This is because the electrons in an atom are held more tightly by the nucleus as the atomic number increases.

Among the choices, oxygen (O) has the highest atomic number (8), so it has the highest ionization potential.

7. Correct option: (B) 2x + 48

The fourth IP is very high, which means that it is difficult to remove an electron from the atom after it has lost three electrons. This suggests that the element is a metal from group 13 of the periodic table, such as aluminum (Al).

Aluminum forms a trivalent oxide, Al_2O_3. So, the molecular weight of the oxide will be 2 * atomic weight of Al + atomic weight of O = 2 * 13 + 16 = 48.

The generalized formula for the molecular weight of the oxide of any group 13 element:

Molecular weight = 2x + 3*16

8. Correct option: (B) Cl

Chlorine has the highest electron gain enthalpy among the halogens. This is because chlorine has the highest nuclear charge and the smallest atomic radius. The nuclear charge pulls the incoming electron more strongly in chlorine than in the other halogens. The smaller atomic radius also means that the incoming electron is closer to the nucleus, which makes it even easier to attract.

9. Correct option: (A) Al

Beryllium and aluminum have similar electronegativity because of the diagonal relationship. This relationship occurs between elements in the second and third periods of the periodic table that are diagonally adjacent to each other. The elements in these pairs have similar properties, such as electronegativity, ionic radii, and chemical reactivity, because they have similar effective nuclear charges.

10. Correct option: (B) I < II < III < IV

A. Element I: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ has the highest electron gain enthalpy because it has the most valence electrons (5). The valence electrons are the electrons that are furthest away from the nucleus, so they are the easiest to add an electron to.

B. Element II: $1s^2\ 2s^2\ 2p^3$ has the lowest electron gain enthalpy because it has the fewest valence electrons (3). The valence electrons are also closer to the nucleus in this case, making it more difficult to add an electron.

C. Element III: $1s^2\ 2s^2\ 2p^5$ has an intermediate electron gain enthalpy. This is because it has the same number of valence electrons as element I, but the valence electrons are closer to the nucleus.

D. Element IV: $1s^2\ 2s^2\ 2p^6\ 3s^1$ has an electron gain enthalpy that is slightly higher than element III. This is because the valence electron in this case is farther away from the nucleus than the valence electrons in element III.

11. Correct option: (B) 289

The electron affinity of iodine is the energy released when an electron is added to an iodine atom in the gas phase to form an iodide ion. The equation for this process is:

$I(g) + e^- \rightarrow I^-(g)$

The electron affinity of iodine is equal to the energy released when 1 mole of iodine atoms in vapor state are converted into I^- ions.

Electron affinity = -4.81×10^{-13} J/atom

Number of atoms in 1 mole of iodine = 6.022×10^{23}

Electron affinity (in kJ/mol) = -5×10^{-13} J/atom $\times$ 6.022×10^{23} atoms/mol / 1000 J/kJ

= -289 kJ/mol

Therefore, the electron affinity of iodine is -289 kJ/mol.

12. Correct option: (D) Their ionisation potential is low

Alkali metals are powerful reducing agents because they have low ionization potentials. This means that they can easily lose electrons to form positive ions. The positive ions are then attracted to negative ions, making alkali metals good at reducing other substances.

13. Correct option: (D) S

The oxides of elements in the same group become more basic as you go down the group. This is because the atoms become larger and the valence electrons are further away from the nucleus, making them easier to lose.

Since the oxide of S is basic, it follows that S is the most electropositive element.

14. Correct option: (A) I > II > III

As we move down the group basic nature of hydroxides of halogens increases so correct order of decreasing acid strength of hydroxides of halogens is I > II> III.

15. Correct option: (B) 53

The elements in group 17 are the halogens. The halogens have 7 valence electrons. When an atom of a halogen gains an electron, it becomes a halide ion with a negative charge.

The element in the third period with an atomic number of 17 is chlorine. The element in the fifth period with the same valence electron configuration is astatine, which has an atomic number of 53.

16. Correct option: (B) Mg, Ca, Sr

Dobereiner's triads are groups of three elements that have similar chemical properties. The atomic mass of the middle element is approximately the average of the atomic masses of the other two elements.

The triad of magnesium, calcium, and strontium is not a Dobereiner triad because the atomic mass of strontium is not the average of the atomic masses of magnesium and calcium.

17. Correct option: (D) All

Lothermeyer curve is a graph of atomic volume vs. atomic mass. The curve shows that the elements with large atomic volume and high electropositivity have peaks in the curve.

Alkali metals are highly electropositive elements and have large atomic volume. These properties are the reason why they occupied the peaks in the Lothermeyer curve.

18. Correct option: (B) An electron

Electrons are the outermost particles in an atom and can be removed with the least energy change compared to protons and neutrons, which are located in the nucleus. Alpha particles are also located in the nucleus and are more tightly bound than electrons.

19. Correct option: (A) Mg – Ba

Both magnesium (Mg) and barium (Ba) are in group 2 of the periodic table, which is the alkaline earth metals

20. Correct option: (D) (n-1) s^2 p^6 d^{1-10} ns^{1or2}

Transition elements are elements with valence electrons in the d orbitals. Their general electronic configuration is $(n-1)d^{1-10}\,ns^{1-2}$, where n is the period number of the element.

21. Correct option: (A) 6s, 4f, 5d, 6p

The energy levels in the 6th period of the periodic table are filled in the following order:

- 6s orbital
- 4f orbitals (14 electrons)
- 5d orbitals (10 electrons)
- 6p orbitals (6 electrons)

The 6s orbital is filled first because it is the outermost orbital. The 4f orbitals are filled next, even though they are lower in energy than the 5d orbitals. This is because the 4f orbitals are more shielded from the nucleus than the 5d orbitals. The 5d orbitals are filled after the 4f orbitals, and the 6p orbitals are filled last.

22. Correct option: (B) Z_{eff} increases down the group

- Screening effect increases down the group. This is because the number of electrons increases down the group, which increases the shielding effect.
- Zeff increases in a period. This is because the nuclear charge increases in a period, which increases the attractive force between the nucleus and the electrons.

Therefore, the statement that Zeff increases down the group is false.

23. Correct option: (A) Atomic size $\propto Z_{eff}$

Atomic size is inversely proportional to Zeff, not directly proportional.

24. Correct option: (D) $(n-1)d^6\,ns^2$

The outer shell electronic configuration of alkaline earth metals is ns^2.

Alkaline earth metals are the elements in group 2 of the periodic table. They have two valence electrons, which are located in the s orbital of the outermost shell.

25. Correct option: (B) 3, 11

Elements in the same group of the periodic table have the same chemical properties. The elements with atomic numbers 3 and 11 are both in group 1, the alkali metals. They both have one valence electron, which makes them very reactive and gives them similar chemical properties.

26. Correct option: (C) None of the above (30)

Sulfur has an atomic number of 16, so X^{2+} has an atomic number of Z = 16 - 2 = 14. The mass number of X^{2+} is the sum of its protons and neutrons, which is 14 + 16 = 30. So, the answer is 30.

27. Correct option: (D) All of the above

The second-period elements show different properties from their respective group due to their small size, absence of vacant d-orbitals, and high ionization energy or electronegativity.

The small size of the atoms means that the valence electrons are held more tightly by the nucleus, which makes them less reactive. The absence of vacant d-orbitals limits the number of bonds that they can form and their chemical reactivity. The high ionization energy or electronegativity makes it difficult to remove an electron from their atoms or to share electrons with other atoms.

28. Correct option: (D) Halogens have the highest I.E. in the respective period

Halogens do have high ionization energies, but they are not the highest in the respective period. The noble gases have the highest ionization energies in their respective periods.

29. Correct option: (D) CsF

The anion to cation size ratio is the size of the anion divided by the size of the cation. The smaller the anion to cation size ratio, the more tightly the ions are packed together.

In CsF, the cation (Cs^+) is the largest and the anion (F^-) is the smallest. This gives CsF the lowest anion to cation size ratio.

30. Correct option: (C) $r_P > r_S$

Atomic radius decreases as you go across a period from left to right. This is because the nuclear charge increases as you go across the period, which pulls the electrons closer to the nucleus.

In this case, phosphorus (P) is to the right of sulfur (S) on the periodic table. Therefore, phosphorus should have a smaller atomic radius than sulfur.

The other options are correct. Copper (Cu) has a larger atomic radius than zinc (Zn) because copper has a lower nuclear charge. Chlorine (Cl) has a larger atomic radius than fluorine (F) because chlorine has more electrons in its valence shell. Scandium (Sc) has a larger atomic radius than titanium (Ti) because scandium has a lower nuclear charge.

31. Correct option: (C) [Ne] $3s^2$ $3p^5$

Ionization energy is the energy required to remove an electron from an atom in its ground state. The higher the ionization energy, the more difficult it is to remove an electron.

In the given elements, [Ne] 3s2 3p5 has the highest ionization energy because it has the most tightly held electron. This is because the valence electron is in the 3p orbital, which is the outermost orbital and is furthest from the nucleus.

The other elements have fewer valence electrons or the valence electrons are in lower energy orbitals, which makes them easier to remove.

32. Correct option: (D) Alkali metals

Alkali metals have the lowest ionization energy in the periodic table. This is because they have only one valence electron, which is located in the outermost shell and is furthest

from the nucleus. The valence electron is therefore easily removed, requiring the least amount of energy.

The other options are incorrect. Alkaline earth metals have two valence electrons, which are held more tightly than the single valence electron of alkali metals. Halogens have seven valence electrons, which are also held more tightly than the single valence electron of alkali metals. Inert gases have a full valence shell, which makes it very difficult to remove any electrons.

33. Correct option: (A) Li^+

Among the given elements, Li^+ has the highest ionization potential. This is because Li^+ has the smallest size and the highest nuclear charge. The small size of the atom means that the valence electron is held more tightly by the nucleus. The high nuclear charge also pulls on the valence electron, making it even more difficult to remove.

- He has a lower ionization potential than Li^+ because He has a closed shell electronic configuration. A closed shell electronic configuration is one in which all of the orbitals in a particular shell are filled. This makes the electrons in the closed shell more tightly held by the nucleus, making it more difficult to remove them.
- Ne also has a closed shell electronic configuration, but its ionization potential is higher than He because Ne has more protons than He. The greater number of protons in Ne pulls on the electrons more strongly, making it more difficult to remove them.
- Mg^+ has a lower ionization potential than Li^+ because Mg^+ has a larger size than Li^+. The larger size of the atom means that the valence electron is held less tightly by the nucleus.

34. Correct option: (D) $M^{2+}(g) \rightarrow M^{3+}(g) + e^-$

The transition $M^{2+}(g) \rightarrow M^{3+}(g)$ involves maximum amount of energy. The removal of an electron from a dipositive cation requires the most energy because the effective nuclear charge per electron is the highest.

35. Correct option: (A) Na

The first ionization enthalpy is the energy required to remove the outermost electron from an atom in its ground state. The second ionization enthalpy is the energy required to remove a second electron from the same atom.

The difference between the first and second ionization enthalpies is greatest for Na because Na has a single valence electron. Once the first electron is removed, Na+ has a filled valence shell, which makes it much more difficult to remove a second electron.

The other elements have more than one valence electron, so the difference between their first and second ionization enthalpies is smaller.

36. Correct option: (C) I < Br < F < Cl increasing electron gain enthalpy (with negative sign)

The electron gain enthalpy is the energy released when an electron is added to an atom in its ground state. The order of electron gain enthalpy is generally reversed as you go across a period from left to right. This is because the nuclear charge increases as you go across the period, which pulls on the electrons more strongly, making it less likely for an electron to be added.

In the given options, F has the highest electron gain enthalpy, followed by Cl, Br, and I. Therefore, the order I < Br < F < Cl is not correct.

The correct order of electron gain enthalpy is:

F > Cl > Br > I

37. Correct option: (D) All of these

The magnitude of electron gain enthalpy (EGE) depends on atomic size, nuclear charge, and electronic configuration.

- Atomic Size: As atomic size increases, EGE becomes more negative.
- Nuclear Charge: As nuclear charge increases, EGE becomes less negative.
- Electronic Configuration: If electronic configuration is stable, EGE becomes more positive.

38. Correct option: (B) IE_1

In the halogens, the only property that does not increase from iodine to fluorine is ionization energy (IE1). Electronegativity, oxidizing power, and electron affinity all increase across this group due to the decreasing atomic size, resulting in stronger attraction for electrons and higher tendency to gain electrons, thereby enhancing these properties.

39. Correct option: (C) O → O^{2-}

O → O^{2-}

This involves the formation of an oxygen ion with an additional electron. Energy is needed to overcome the electrostatic repulsion between the incoming electron and the existing electrons in the oxygen atom, leading to an endothermic process.

40. Correct option: (C) In the 5th period, the orbitals 5s 4 f 4d 6 p are filled

A) Lithium and magnesium share similarities due to their comparable ion sizes and behaviors, demonstrating a diagonal relationship.

B) Elements in groups 1-2 and 13-17 are representative elements, as they showcase a wide range of chemical properties.

C) The correct order is 5s 4d 5p. Electrons fill orbitals in increasing order of energy. The 5s orbital is filled before the 4f and 4d orbitals in the 5th period, as per the Aufbau principle.

D) The general electronic configuration of f-block elements is $(n-2)f^{1-14}$ $(n-1)d^{0,1}$ ns^2, characterizing the filling of inner f-orbitals before outer d-orbitals and ns^2 orbitals.

41. Correct option: (C) Isotone with $_{18}A^{36}$

This is because the element has 20 protons (as indicated by its atomic number) and is isotone with Argon (18 protons, 20 neutrons) which has the same neutron number (20), making them isotones. The other options are incorrect because they do not accurately describe the properties of this element. It is not necessarily a transition element, and the information provided does not indicate its ionization energy (I.E) or oxide stability (M^2O).

42. Correct option: (C) Valency increase along with the group in term of oxygen

This statement is incorrect because valency doesn't necessarily increase uniformly with the group in terms of oxygen. Elements within the same group may have different valencies depending on their electron configurations and chemical properties.

43. Correct option: (A) Al^{3+}

The hydration energy of Mg^{2+} ions is lesser than that of Al^{3+} ions. This is because Al^{3+} ions have a higher charge, leading to stronger electrostatic attraction with water molecules during hydration. The other options, Ba^{2+}, Na^+, and K^+, have different cations with different charge and do not directly relate to the hydration energy of Mg^{2+} ions.

44. Correct option: (C) 1, 3, 5, 7, 5, 3, 1

As we move from Group I to Group VII of the periodic table, the valency of elements in their hydrides increases progressively from 1 to 7 and then decreases back to 1. This reflects the number of electrons needed to achieve a stable noble gas configuration by forming hydrides.

LEVEL – II

1. Correct option: (A) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$

The electronic configuration of the element with atomic number 43 is [Kr] $4d^5\ 5s^2$. The element just above it in the same group has one fewer electron, which will be in the 4d orbital.

Option (B) is incorrect because it has a 4p electron, which is not possible for an element in the same group as technetium (Tc). Option (C) is incorrect because it has a $3d^6$ electron, which is not possible for an element with an atomic number of 42. Option (D) is incorrect because it has a $4s^1\ 4p^6$ electron configuration, which is not possible for an element in the same group as technetium.

2. Correct option: (D) Zn

Zinc (Zn) remains in the same position in the periodic table's d-block, even if the Aufbau principle is not followed. The other elements (Ca, Ge, Sn) would change their positions if electron filling were not based on increasing atomic number, altering their block classifications.

3. Correct option: (C) (A) is true but (R) is false

Zinc (Zn) is indeed a transition element, as it is located in the d-block. However, the reason provided in (R) is not correct. Transition elements are defined by their partially filled d-orbitals, but Zn has a completely filled d-orbital (3d10), so (R) is false.

4. Correct option: (A) Transition elements are placed from 3rd to 6th period

Transition elements are placed in the d-block and span the 3rd to 7th period of the periodic table, not just the 3rd to 6th period.

5. Correct option: (C) [Rn] $7s^2\ 5f^7\ 6d^1$

The electronic configurations of A, B, and D are consistent with the f-block, d-block, and p-block, respectively. However, option C represents an element with electron filling that is inconsistent with any specific block, making it the outlier.

6. Correct option: (C) Lawrencium

Lawrencium belongs to Group 3 of the periodic table (the lanthanide series) and, when undergoing alpha decay, retains its position in the same group. The other options belong to different groups and do not exhibit this behavior.

7. Correct option: (B) 27

If an orbital could contain 3 electrons, then the number of groups in the periodic table would be 27. This is because there are 27 different sets of 3 quantum numbers that can be assigned to electrons. The current periodic table has 18 groups, which is based on the fact that an orbital can contain up to 2 electrons.

The other options are incorrect:

- (A) 20: This is the number of groups in the current periodic table.
- (C) 36: This is the number of orbitals in the first 4 shells.
- (D) 42: This is the number of orbitals in the first 5 shells.

8. Correct option: (C) Zr and Y have the same radius

Lanthanide Contraction refers to the decrease in atomic and ionic radii that occurs as you move across the 4f series of lanthanide elements. This contraction can result in elements in the same period, like Zirconium (Zr) and Yttrium (Y), having similar radii. The other options are not directly related to Lanthanide Contraction and involve different chemical properties.

9. Correct option: (A) The imperfect shielding on outer electrons by 4f-electrons from the nuclear charge

Lanthanide contraction is the decrease in the atomic radius of the lanthanide elements from lanthanum (La) to lutetium (Lu). This is due to the imperfect shielding of the 4f

electrons by the 5d and 6s electrons. The 4f electrons are not as effective at shielding the nuclear charge as the 5d and 6s electrons, so the valence electrons are pulled in closer to the nucleus.

The other options are incorrect:

- (B) The appreciable shielding on outer electrons by 4f-electrons from the nuclear charge: This would cause the atomic radius to increase, not decrease.
- (C) The appreciable shielding on outer electrons by 5d-electrons from the nuclear charge: The 5d electrons are not as effective at shielding the nuclear charge as the 4f electrons, so they would actually cause the atomic radius to decrease further.
- (D) The same effective nuclear charge from Ce to Lu: This would not cause any change in the atomic radius

10. Correct option: (B) 0.41 Å

The ionic radius of Si^{4+} can be calculated based on their isoelectronic nature, where both have the same number of electrons. Since O^{2-} has a radius of 1Å, Si^{4+}, having a significantly lower electron count, will have a smaller radius. Therefore, the correct answer is: B) 0.41 Å

The other options are larger than what is expected for Si^{4+} due to the reduction in electron count.

11. Correct option: (B) $1s^2\ 2s^2\ 2p^6\ 3s^1$

In this electronic configuration, the maximum difference in ionization energies occurs when the first electron is removed from the 3s orbital, followed by the second ionization involving the removal of an electron from the $2p^6$ filled subshell, resulting in a substantial energy difference.

12. Correct option: (B) 'Y' would have an ionisation potential in between those of 'X' and 'Z'

'Y' would have an ionization potential in between those of 'X' and 'Z'.

This statement is accurate because, within a group (Group 1 - alkali metals), ionization energy tends to decrease from top to bottom. Therefore, it is likely that the ionization energy of element Y (with an atomic number of 37) falls between those of element X (with an atomic number of 19) and element Z (with an atomic number of 55).

13. Correct option: (C) Nature of bonding between A & C is covalent

The nature of bonding between A and C is ionic, not covalent. This is because C is an alkali metal and has one valence electron. Alkali metals are very electropositive and tend to form ionic bonds with non-metals. Therefore, the bonding between A and C is ionic, not covalent.

The other options are correct:

- EN of A is maximum in the period because electronegativity increases from left to right across a period.
- IP of C is minimum in the period because ionization energy increases from left to right across a period.
- B has zero valency because it has one electron less than C, which is an alkali metal and has one valence electron.

14. Correct option: (A) a-S, b-R, c-Q, d-P

a) $ns^2\ np^6$ corresponds to S because it represents a noble gas configuration with a full outer shell, resulting in high ionization potentials.

b) $ns^2\ np^4$ corresponds to R because it represents an outer electron configuration with partial filling of the p-subshell, resulting in moderate ionization potentials.

c) $ns^2\ np^5$ corresponds to Q because it represents an outer electron configuration with partial filling of the p-subshell, resulting in moderate ionization potentials.

d) ns^2 np^3 corresponds to P because it represents an outer electron configuration with fewer electrons in the p-subshell, resulting in relatively low ionization potentials.

15. Correct option: (D) All are correct

The given values of successive ionization energies suggest that element A is a metal with an intermediate electronegativity, likely located in the 3rd period of the periodic table. This implies that A could form an oxide with the formula A_2O_3 and exhibit amphoteric properties, meaning it can react with both acids and bases.

Therefore, all three statements – A) Element 'A' may be metal, B) Formula of oxide of A may be A_2O_3, and C) Oxide of element A may be amphoteric – are correct.

16. Correct option: (A) $I.E._1$ (Cu) < $I.E._1$ (Zn) but $I.E._2$ (Cu) > $I.E._2$ (Zn)

This relation aligns with the periodic trend where ionization energy (I.E.1) tends to increase across a period from left to right, making Zn have a higher I.E.1 than Cu. However, I.E.2 decreases across a period, leading to Cu having a higher I.E.2 than Zn. Other options do not accurately represent this trend.

17. Correct option: (D) All

- IPF: The ionization energy of fluorine is greater than the ionization energy of chlorine.
- EAF: The electron affinity of fluorine is greater than the electron affinity of chlorine.
- IPF + IPCl > EAF + EACl: The sum of the ionization energies of fluorine and chlorine is greater than the sum of the electron affinities of fluorine and chlorine.
- IPF - IPCl > EAF - EACl: The difference between the ionization energies of fluorine and chlorine is greater than the difference between the electron affinities of fluorine and chlorine.

This is because fluorine is a more electronegative element than chlorine. Electronegativity is the tendency of an atom to attract electrons. Therefore, fluorine is more likely to attract an electron than chlorine.

18. Correct option: (C) Electrostatic repulsion outweighs the energy gain due to coupling

Formation of oxide or sulfide ions involves adding electrons to oxygen or sulfur atoms, which results in significant electrostatic repulsion due to the negative charges of the added electrons. This repulsion can outweigh the energy gain from electron coupling, making the process endothermic.

19. Correct option: (A) 2X – EA – IP = 0

Electron affinity: The amount of energy released when an electron is added to the outermost shell of one mole of an isolated gaseous atom in its lower energy state.

Ionisation energy: Minimum energy required to remove most loosly held outer most shell e – in ground state from an isolated gaseous atom is known as ionisation potential.

$X = \frac{IP+EA}{2}$

$2X – IP – EA = 0$

20. Correct option: (A) |I.E. of process (II)| = |E.A. of process (I)|

The ionization energy of process (II), $Cl^- \rightarrow Cl$, is equal to the electron affinity of process (I), $Cl \rightarrow Cl^-$. This is because process (II) is the reverse of process (I), and reversing a physical process also reverses the sign of the energy involved. Gaining an electron (E.A. of Cl) releases energy, which is the same amount of energy required to remove that electron (I.E. of Cl^-).

21. Correct option: (C) EA of B^+ is 250 $Kcalmol^-$

The electron affinity (EA) of an ion refers to the energy released when an electron is added to it. Since B has a higher EA (90 $Kcalmol^{-1}$) than A (70 $Kcalmol^{-1}$), EA of B+ should be less than EA of A+, not 250 $Kcalmol^{-1}$

22. Correct option: (D) $BaSO_4$

The solubility of sulfates in water decreases as the atomic number of the metal ion increases. This is because the sulfate ion is a large ion, and it is more difficult for the metal ion to attract it as the atomic number increases.

Therefore, BaSO4, which has the highest atomic number, is the least soluble in water.

23. Correct option: (A) Li^+, Mg^{+2}

Both Li^+ and Mg^{+2} have the same charge-to-size ratio (2+ charge for Li^+ and 2^+ charge for Mg^{+2}), which results in similar polarizing power. The other options involve ions with different charge-to-size ratios, leading to different polarizing powers.

24. Correct option: (B) $H_2O < CuO < CaO < CO_2$

The acidic strength of an oxide increases as the oxidation state of the metal increases. The oxidation state of carbon in CO_2 is +4, the oxidation state of copper in CuO is +2, the oxidation state of calcium in CaO is +2, and the oxidation state of hydrogen in H_2O is +1.

Therefore, CO_2 is the least acidic oxide, followed by H2O, CuO, and CaO.

25. Correct option: (D) Bi_2O_3

Amphoteric oxides are oxides that can react both as an acid and as a base.

N_2O_3 is a neutral oxide.

P_2O_3 is an acidic oxide.

Sb_2O_3 is an amphoteric oxide.

Bi_2O_3 is an amphoteric oxide.

26. Correct option: (B) Al_2Cl_6, N_2H_4

A) KHF_2, B_2H_6

- KHF_2 does not readily accept electron pairs.
- B_2H_6 (diborane) has empty p-orbitals that can accept electron pairs.

B) Al_2Cl_6, N_2H_4

- Al_2Cl_6 (aluminum hexachloride) can accept electron pairs due to the empty p-orbitals in aluminum.
- N_2H_4 (hydrazine) has lone pairs and can donate electron pairs.

C) N_2O, H_3PO_2

- N_2O doesn't have a readily available site to accept electron pairs.
- H_3PO_2 (phosphinous acid) can donate an electron pair but does not have a readily available site to accept electron pairs.

D) CrO_5, H_2O_2

- CrO_5 does not readily accept electron pairs.
- H_2O_2 (hydrogen peroxide) has lone pairs but is typically seen as a Lewis base rather than an acid.

27. Correct option: (B) K_2O

When K_2O reacts with CO_2, it forms K_2CO_3 (potassium carbonate)

$K_2O + CO_2 \rightarrow K_2CO_3$

28. Correct option: (D) CO_3^{-2}, H_2CO_3 and HCO_3^-

When CO_2 is dissolved in water, it reacts with water to form carbonic acid (H_2CO_3):

$3CO_2 + H_2O \rightleftharpoons H_2CO_3$

Carbonic acid (H_2CO_3) can further dissociate to form bicarbonate (HCO_3^-):

$H_2CO_3 \rightleftharpoons H + HCO_3^-$

Bicarbonate (HCO_3^-)can further dissociate to form carbonate (CO_3^{2-}):

$HCO_3^- \rightleftharpoons H + CO_3^{2-}$

So, in an aqueous solution of CO_2, we have:

- H_2CO_3 (carbonic acid)
- HCO_3^- (bicarbonate ion)

- CO_3^{2-} (carbonate ion)

29. Correct option: (D) PbO_2 – only acidic

A) $B(OH)_3$ - acidic: Accurate, as boric acid is weakly acidic.

B) *SnO*, *PbO* - amphoteric: SnO is amphoteric, and PbO is more basic but can exhibit some acidic character.

C) GeO_2 - acidic: Accurate, germanium dioxide is acidic.

D) PbO_2 – only acidic: This is the incorrect match since PbO_2 can behave as both an acid and a base, making it amphoteric.

30. Correct option: (B) II is a d-block element

A) I is a strong reducing agent: True. [Xe]$6s^1$ (Cesium), as an alkali metal, is indeed a strong reducing agent.

B) II is a d-block element: Technically true. While [Xe]$4f^{14}5d^16s^2$ (Lutetium) is part of the lanthanide series in the f-block, the presence of an electron in the 5d orbital allows it to be considered as a d-block element. However, if one strictly defines d-block elements based on the principal quantum number, then Lutetium, with its primary configuration in the 4f orbital, would not be classified as such.

C) III has high Electron gain enthalpy: True. [Ar]$4s^23d^{10}4p^5$ **(**Bromine) has a high tendency to gain an electron, resulting in high electron gain enthalpy.

D) IV shows variable oxidation states: True. [Ar]$3d^74s^2$ (Cobalt) can exhibit multiple oxidation states, primarily +2 and +3.

31. Correct option: (A) 1st and 17th groups

The most predominantly electrovalent (ionic) compounds are typically formed between elements with a large difference in electronegativity. Electronegativity tends to increase from left to right across a period and decrease from top to bottom within a group in the periodic table.

1st and 17th groups:

- The 1st group elements are alkali metals with low electronegativity.
- The 17th group elements are halogens with high electronegativity.

The combination of elements from the 1st and 17th groups often results in predominantly electrovalent (ionic) compounds because of the significant difference in electronegativity.

32. Correct option: (C) Be, Mg, Ca

Newlands' Octave Rule states that every eighth element has similar properties when elements are arranged in order of increasing atomic weights, mainly applicable up to calcium.

A) Na, K, Rb: These don't follow the octave sequence; there are elements in between.

B) F, Cl, Br: Don't follow the octave sequence; there are elements between them.

C) Be, Mg, Ca: These are alkaline earth metals, and their properties roughly align with Newlands' observation.

D) B, Al, Ga: Don't strictly follow the octave rule.

33. Correct option: (D) Less than p-electrons

The screening effect, also known as the shielding effect, describes how the inner (core) electrons shield the outer (valence) electrons from the positive charge of the nucleus.

The d-electrons, which are found in the transition metals, have a poorer shielding effect than the s and p electrons, because they are more diffused and are farther away from the nucleus compared to the s and p electrons of the same shell.

34. Correct option: (C) 4s < 3d & D) 4s > 3p

For potassium (K) with the atomic number 19, the electronic configuration is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$.

Based on the Aufbau principle and the general energy levels of atomic orbitals:

1. The 4s orbital fills before the 3d orbital. So, 4s has lower energy than 3d.
2. The 4s orbital fills after the 3p orbital. So, 4s has higher energy than 3p.

35. Correct option: (B) 17

Bromine has an atomic number of 35, so it has 35 electrons. The electronic configuration of bromine is: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$

To determine the number of electrons in all p-orbitals of bromine:

- $2p^6$: 6 electrons
- $3p^6$: 6 electrons
- $4p^5$: 5 electrons

Adding up all these electrons:

6 + 6 + 5 = 17 electrons

36. Correct option: (D) $(n-2)f^{1-14}\ (n-1)d^{1-10}\ ns^2$ = d - block elements

D) $(n-2)f^{1-16}\ (n-1)d^{1-10}\ ns^2$ = d - block elements, because it incorrectly suggests that the f-orbitals can accommodate 16 electrons, which is not the case.

37. Correct option: (D) II, III and IV

I) Bi^{3+} is more stable than Bi^{5+}: This statement is correct. Bismuth (Bi) in the +3 oxidation state is more stable due to the inert pair effect.

II) Mn shows + 8 oxidation state: This statement is incorrect. The highest oxidation state for manganese (Mn) is +7, as seen in MnO4−MnO4− (permanganate ion).

III) The oxidation state of an element is always less than or equal to its group number: This statement is not universally true. For example, the group number of iron (Fe) is 8, but its common oxidation states are +2 and +3.

IV) s-block elements show variable oxidation states: This statement is incorrect. s-block elements (alkali and alkaline earth metals) primarily exhibit +1 (for alkali metals) and +2 (for alkaline earth metals) oxidation states.

38. Correct options: (C) I.P. of the Be is more than 'B'

A) Not universally true. High ionization energy doesn't always mean high electron affinity.

B) Incorrect. The 2nd I.P. is always higher than the 1st.

C) Correct. Beryllium (Be) has a higher ionization potential than boron (B) due to a fully filled 2s subshell.

D) Correct. The 2nd I.P. of nitrogen is higher than oxygen because removing an electron from a half-filled p-subshell in nitrogen is harder.

39. Correct option: (B) Si

The significant jump is between IP4 and IP5, suggesting that the fifth electron is being removed from a new energy level.

This indicates that the element has 4 electrons in its outermost shell, pointing towards a group 14 element.

40. Correct option: (A) Metalic, Vanderwaal and covalent radii respectively

1. Van der Waals radius is the largest. It represents the half of the distance between two non-bonded atoms in a solid state.
2. Metallic radius is in between. It's half the distance between two adjacent atoms in a metal lattice.
3. Covalent radius is the smallest. It's half the distance between two atoms that are bonded together in a molecule.

41. Correct option: (A) Above Ne

The ionization potential (I.P.) generally increases across a period. In the second period (Li - Ne), Na (11) is located after Ne (10). As Ne typically has a higher ionization potential, Na's ionization potential would be below Ne.

42. Correct option: (C) K > Ca > Mg > Li

A) Represents lanthanide contraction; the order is correct.

B) Considering ion sizes, the order is correct.

C) *K>Ca>Mg>Li* is incorrect as atomic size increases from Li to K but slightly decreases from K to Ca.

D) Given the nature of anions and parent atoms, the order is correct.

43. Correct option: (B) x > y > z

As we move down the periodic table within a group, the increase in atomic size becomes less pronounced. This is due to increasing nuclear charge that somewhat counterbalances the addition of a new energy level, leading to a smaller increment in size.

44. Correct option: (A) O – F

O−F bond is the least polar due to less electronegativity difference between oxygen and fluorine atoms.

45. Correct option: (C) 0.5 Å

$r_c + r_f = 1.33$ Å

$r_{Si} + r_f = 1.54$ Å

Adding both equations,

$r_c + r_{Si} + 2\ r_f = 2.87$ Å

given, $r_c + r_{Si} = 1.87$ Å

$1.87 + 2\ r_f = 2.87$ Å

$2\ r_f = 1.00$ Å

$r_f = 0.5$ Å

46. Correct option: (A) Paulling scale (E.N. of F – atom) < Mulliken scale (E.N. of F – atom)

A) Different electronegativity scales can provide varying values. This statement's correctness is questionable due to the unconventional ordering.

B) & C) While electronegativity is constant, electron distribution changes in different molecules. The statement gives a relative comparison and seems correct.

D) Electronegativity increases with positive oxidation state, making this statement likely correct.

47. Correct option: (B) Electronegativity increases means metallic character increases

A) Higher ionization energy generally correlates with higher electronegativity.

B) Increased electronegativity indicates more non-metallic character, not metallic. This statement is incorrect.

C) Elements with low ionization energies are better reducing agents.

D) Chlorine has a higher electron affinity than fluorine.

48. Correct option: (D) Bi_2O_3

A) SeO_2 - Selenium dioxide: Typically, non-metal oxides are acidic rather than basic.

B) Al_2O_3 - Aluminum oxide: Generally amphoteric, meaning it can exhibit both acidic and basic properties depending on the conditions.

C) Sb_2O_3 - Antimony(III) oxide: Can also be amphoteric, displaying both acidic and basic properties.

D) Bi_2O_3 - Bismuth(III) oxide: Bismuth oxides tend to be weakly basic.

49. Correct option: (C) As_4O_6

Between phosphorus (P) and arsenic (As) in the periodic table, arsenic is lower and thus its oxides would generally be less acidic than the oxides of phosphorus. Among the

arsenic oxides, As_4O_6 will be less acidic compared to As_4O_{10} due to its lesser oxygen content, which contributes to acidity.

50. Correct option: (D) Ti^{4+}

Explanation:

Ionic radius tends to decrease with an increase in positive charge (or removal of electrons) for cations of the same period, because of the increase in the nuclear charge felt by the remaining electrons, causing them to be pulled closer to the nucleus.

51. Correct option: (A) 69 % Mg^{+} , 31 % Mg^{+2}

Energy absorbed in the ionisation of 1 mole of Mg to $Mg^{+}(g)$ =750 kJ

Energy left unused =1200−750 =450 kJ

% of $Mg^{+}(g)$ converted into $Mg^{2+}(g)$

$$=1450450\times100=31\%$$

Hence, the % of Mg^{+}=100−31=69%

52. Correct option: (C) O > F > N > C

- Carbon (C) for its second ionization will remove an electron from a $2p$ orbital.
- Nitrogen (N) has a half-filled $2p$ configuration after the first ionization, which is stable.
- Oxygen (O) after the first ionization also has a half-filled $2p$ configuration.
- Fluorine (F) after the first ionization is one electron away from a full $2p$ shell.

Considering the stability of half-filled configurations:

- Oxygen's second ionization will require the most energy, followed by fluorine (due to its near-full shell), then nitrogen (due to its stability), and lastly carbon.

53. Correct option: (B) A_2

The electronic configuration of element A is $1s^2\ 2s^2\ 2p^5$. This configuration indicates that the atom is one electron short of achieving a full outer shell (noble gas configuration).

Atoms strive to achieve a full outer shell either by gaining, losing, or sharing electrons. In this case, element A will try to gain one electron to achieve a full outer shell.

Since element A needs only one electron, when it reacts with another atom of element A, they can share their unpaired electrons to form a single covalent bond.

54. Correct option: (A) Same

The ionization potential (or ionization energy) is the energy required to remove an electron from an atom or ion in its gaseous state. It depends on the effective nuclear charge and the distance of the outermost electron from the nucleus.

Isotopes of an element have the same number of protons (and hence the same atomic number) but different numbers of neutrons. This difference in neutrons doesn't significantly impact the effective nuclear charge felt by the valence electrons or their distance from the nucleus.

55. Correct option: (C) Element in which valence electron resides in (n-1)d subshell – VI A group

A) 14th group elements have the configuration $ns^2\ np^2$. This is correct.

B) 17th group elements have one unpaired p-electron in their valence shell. This is correct.

C) 16th group elements (old VI A) have valence electrons in s and p orbitals, not d. This is incorrect.

D) II A group elements like Calcium have the configuration $[Ar]4s^2$. This is correct.

56. Correct option: (C) X_8O_{11}

Let the compound be X_nO_p

since X_nO_p is neutral

$$\frac{n}{4}x+2+\frac{3n}{4}x+3-2p=0$$

$n\left[\frac{2+9}{4}\right] = 2p$

$$\frac{n}{p} = \frac{8}{11}$$

57. Correct option: (A) 2.582Å

1. Volume of one atom: Divide the atomic weight by Avogadro's number and multiply by the density

Volume per atom = $\frac{26.98\ g/mol}{6.022 \times 10^{23}\ atoms/mol} \times 2.7\ g/cc$ = 1.661 x 10^{-23} cm^3

2. Radius from volume: Since we assume a spherical atom, use the formula for the volume of a sphere:

$\frac{4}{3}$ x π x r^3 = 1.661 x 10^{-23} cm^3

$$r = (3 * 1.661 \times 10^{-23}\ cm^3 / 4\pi)^{1/3} = 1.582 \times 10^{-8}\ cm$$

3. Convert to Angstroms: Multiply by 10^{10} to get the answer in Angstrom

Radius = 1.582 x 10^{-8} cm x 10^{10} Å/cm = 2.582 Å

58. Correct option: (A) 2.53

△ = EC−H - √EH−H×EC−C

= 98.8 - [(104.2)×(83.1)]12 = 5.75 kcal.

(XC−H) E.N of C-H bond = 0.218 √△ = 0.218 (5.75)12

= 0.43

XC = 0.43 + XH = 0.43 + 2.1 = 2.53

59. Correct option: (B) II IE_1 520 IE_2 7300

Alkali metals have the lowest IE_1 values in their respective periods. Also, there is a large jump between IE_1 and IE_2 due to the removal of the second electron from the noble gas core. Therefore, out of the given IE_1 and IE_2 values, (II) belongs to alkali metals.

60. Correct option: (B) 8

$_{29}^{63}X + H^1 \rightarrow 6_0n^1 + 2_1H^1 + X$

64=6+4+2+A

⇒A=52

29+1=30=0+2+2+z

⇒z=26

element X should be iron in group

LEVEL-III

PREVIOUS YEAR QUESTIONS

1. Correct option: (B) -349, -333, -325 and -296

The electron gain enthalpy trend follows the order Cl > F > Br > I, with values of -349, -328, -325, and -295 kJ/mol, respectively. Fluorine exhibits lower electron gain enthalpy than chlorine because of its exceptionally small size, leading to strong inter-electronic repulsion during the addition of an extra electron. This is attributed to the high electron density, making the process less favorable. Bromine, with a slightly larger size than fluorine, has a marginally lower electron gain enthalpy due to a reduced effective nuclear charge on the incoming electron.

2. Correct option: (A) Fe

Electronic configuration of

Mn^{+2} is $[Ar]3d^5$, Co^{+2} is $[Ar]3d^7$, Fe^{+2} is $[Ar]3d^6$, and Ni^{+2} is $[Ar]3d^8$

Among all these elements Fe^{+2} is most likely to lose an electron very easily so as to attain the half-filled d-subshell stability.

3. Correct option: (C) c < b < a < d < e

Moving across a period leads to a consistent decrease in size, whereas progressing down a group results in a regular increase in size.

Br > Cl > C > O > F

4. Correct option: (D) 3, diamagnetic

The CN^- ion, with 14 electrons akin to nitrogen molecules, adopts a diamagnetic character according to Molecular Orbital Theory (MOT). The electron arrangement, featuring paired electrons in σ and π orbitals, results in a bond order (B.O.) of 3, indicating a stable, diamagnetic state.

B.O. $= \frac{1}{2}[Nb - Na]$

$= \frac{1}{2}[8-2]$

$=3.$

5. Correct option: (A) Cl, S and Li

As we move down a group, electron affinity decreases due to the increasing size of atoms. Consequently, electron affinity follows the trend of S > Se and Li > Na, leading to greater energy release. However, an exception occurs in the case of F and Cl, where Cl > F. This exception is attributed to the small size of the F atom, causing incoming electrons to experience high electron-electron repulsion, resulting in greater energy release in Cl.

6. Correct option: (B) -349, -333, -325 and -296

The electron gain enthalpy trend follows the order Cl > F > Br > I, with values of -349, -328, -325, and -295 kJ/mol, respectively. Fluorine exhibits lower electron gain enthalpy than chlorine because of its exceptionally small size, leading to strong inter-electronic repulsion during the addition of an extra electron. This is attributed to the high electron density, making the process less favorable. Bromine, with a slightly larger size than fluorine, has a marginally lower electron gain enthalpy due to a reduced effective nuclear charge on the incoming electron.

7. Correct option: (D) Atomic radius

Generally, when moving across a period, the atomic radius tends to decrease, while ionization enthalpy, electron gain enthalpy, and electronegativity increase. This trend is attributed to the rise in effective nuclear charge (Zeff) along the period.

8. Correct option: (A) in the stratosphere, CFCs release chlorine-free radicals (Cl) which reacts with O_3 to give chlorine dioxide radicals.

In the upper stratosphere, UV radiation is absorbed, leading to the decomposition of certain compounds and the generation of highly reactive chlorine-free radicals. These radicals then interact with ozone (O_3) in the ozone layer, resulting in the release of oxygen and chlorine monoxide. Subsequent reactions involving chlorine monoxide and O_3 produce molecular oxygen (O_2) along with active chlorine radicals.

$$CF_2Cl_2(g) + h\nu \rightarrow \dot{C}F_2Cl(g) + \dot{C}l(g)$$

$$\dot{C}l(g) + O_3(g) \rightarrow Cl\dot{O}(g) + O_2(g)$$

$$Cl\dot{O}(g) + O_3(g) \rightarrow 2O_2(g) + \dot{C}l(g)$$

9. Correct option: (D) X < Y < Z

As we progress from left to right in the periodic table, the acidic character of oxides tends to increase, parallel to the increase in the atomic number of the elements X, Y, and Z, where the order of acidic character is X < Y < Z, and the order of atomic number is X < Y < Z.

10. Correct option: (D) $Na_{(g)} \rightarrow Na^{-}_{(g)} + e^{-}$

Only option (d) represents an exothermic process, where energy is released. All other options involve gaining electrons against repulsion, which requires energy and are thus endothermic.

11. Correct option: (B) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

Ionic radius is essentially the distance from the nucleus of an ion to the outermost electron cloud, representing the effective span between the nucleus and the outer shell containing the electron. When a cation is formed through the loss of one or more electrons, its radius becomes smaller than that of the original atom. Conversely, when an anion is formed by gaining one or more electrons, its radius increases compared to that of the original atom.

12. Correct option: (C) Electronegativity and atomic radius

As we progress down a group, electronegativity tends to decrease, while atomic radius tends to increase.

13. Correct option: (D) K

K = 2, 8, 8, 1

Following the removal of one electron, the second electron needs to be removed from a shell that possesses an octet configuration. This results in a substantial disparity between the first and second ionization energies.

14. Correct option: (C) both form basic carbonates

A) Li and Mg both form nitrides (Li_3N_2 and Mg_3N_2) when directly combined with N_2.

B) Both Li and Mg undergo thermal decomposition of their hydroxides and nitrates to yield oxides, as illustrated by the reactions $LiNO_3 \rightarrow Li_2O + NO_2 + O_2$ and $Mg(NO_3)_2 \rightarrow MgO + NO_2 + O_2$.

C) magnesium forms basic carbonates ($MgCO_3{\cdot}Mg(OH)_2{\cdot}xH_2O$), whereas lithium forms normal carbonates, lacking the formation of basic carbonates.

D) Despite being in the same group, Li and Mg exhibit differences in solubility; their bicarbonates are more soluble in water than carbonates, a contrast to the high solubility of alkali metal carbonates.

15. Correct option: (B) acid and base

Zinc oxide (ZnO) when reacting with Na_2O acts as an acid while with CO_2 it acts as base.

$$ZnO + Na_2O \rightarrow Na_2ZnO_2$$

Acid Base Salt

$$ZnO + CO_2 \rightarrow ZnCO_3$$

Base Acid Salt

16. Correct option: (D) Sc

The high ionization energy of Scandium (Sc) is attributed to the limited shielding effectiveness of its d-orbital, resulting in a higher effective nuclear charge (Zeff).

17. Correct option: (C) 1.71, 1.40 and 1.36

For isoelectronic species, size of anion increases as negative charge increases. Thus the correct order is $N^{3-} > O^{2-} > F^{-}$

18. Correct option: (A) −5.1 eV

$Na\,(g) \rightarrow Na^{+}(g) + e$; IE = 5.1Ev

$Na^{+}(g) + e \rightarrow Na\,(g)$; EA = -IE = -5.1 eV

19. Correct option: (C) $Ca^{+2} > K^{+} > Cl^{-} > S^{-2}$

In isoelectronic species, the shielding constant (σ) is consistent for all. The size is inversely proportional to the effective nuclear charge (Z_{eff}), where $Z_{eff}=(Z-\sigma)$, with Z being the atomic number. For the given species, the atomic numbers are in the order Ca > K > Cl > S. Consequently, the ionic radii follow the order Ca^{2+}, K^{+}, Cl^{-}, and S^{2-}.

20. Correct option: (A) $O^{2-} > F^{-} > Na^{+} > Al^{3+}$

All the provided species have the same electron count, making them isoelectronic. In isoelectronic species, the anion with a higher negative charge tends to be larger in size, while the cation with a higher positive charge tends to be smaller. Therefore, the correct size order is $O^{2-} > F^{-} > Na^{+} > Mg^{2+} > Al^{3+}$.

21. Correct option: (C) Cl > F > Br > I

As we progress down a group in the Periodic Table, the atomic size increases, resulting in a decrease in the force of attraction for the added electron, and, consequently, a decrease in electron gain enthalpy. The observed order for electron gain enthalpy is Cl > F > Br > I. The lower electron gain enthalpy for fluorine compared to chlorine is attributed to the relatively greater effectiveness of the 2p-electron in the small fluorine atom to repel the additional electron entering the atom, as opposed to the 3p-electrons in the larger chlorine atom.

22. Correct option: (A) $Al_2O_3 < MgO < Na_2O < K_2O$

While moving from left to right in periodic table, basic character of oxide of elements will decrease.

$Al_2O_3 < MgO < Na_2O$

And while descending in the group, basic character of corresponding oxides increases.

$Na_2O < K_2O$

23. Correct option: (C) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

A) Down the group, the stability of the +2 oxidation state increases, thereby augmenting the oxidizing power of MO_2 (where M = C, Si, Sn, Pb).

B) Moving down a group enhances the stability of the conjugate base, leading to an increase in the acidity of HX (where X is a halogen).

C) In hydrides of 15th group elements, the basic strength diminishes down the group due to increased electron density in NH_3, attributed to the small size of nitrogen, resulting in a higher tendency to donate a lone pair. The correct order is $SbH_3 < AsH_3 < PH_3 < NH_3$.

D) Progressing along a period results in an increase in effective nuclear charge, leading to a rise in ionization energy.

24. Correct option: (B) $Na^{+} > Li^{+} > Mg^{2+} > Be^{2+}$

Ionic radius is essentially the distance from the nucleus of an ion to the outermost electron cloud, representing the effective span between the nucleus and the outer shell containing the electron. When a cation is formed through the loss of one or more electrons, its radius becomes smaller than that of the original atom.

While magnesium (Mg) is in the 3rd period and lithium (Li) is in the 2nd period, Mg^{2+} has a smaller ionic radius compared to Li^{+}. This is attributed to the significantly high effective nuclear charge in Mg^{2+} due to its higher positive charge, causing a substantial reduction in the size of Mg^{2+} ions.

25. Correct option: (C) $K^{+} < Ca^{2+} < Mg^{2+} < Be^{2+}$

The polarizing power of a cation is determined by its charge-to-size ratio. According to Fajan's rule, the ability of a cation to distort the electron cloud of a negative ion (anion) is known as its polarizing power, and the anion's capability to undergo distortion is termed its polarizability.

Down a group, both the size of the atom and its cation increase, while along a period, the size generally decreases. Consequently:

$Be^{2+} < Mg^{2+} < Ca^{2+}$ (Size of ion increases)

Additionally, the size of K^{+} is greater than that of Ca^{2+}.

Therefore, the increasing order of the polarizing power of the cationic species is

$K^{+} < Ca^{2+} < Mg^{2+} < Be^{2+}$.

26. Correct option: (B) HNO_3 is a stronger acid than HNO_2

+5 +3

HNO_3 is stronger than HNO_2. The more the oxidation state of Nitrogen, the more is the acidic character.

27. Correct option: (C) K^+, Cl^-, Ca^+, Sc^{+3}

Isoelectronic species are those that contain the same number of electrons.

In option (a), N^{3-},O^{2-},F^-, and S^{2-} are not isoelectronic.

In option (b), Li^+,Na^+,Mg^{2+}, and Ca^{2+} are not isoelectronic.

In option (c), as K^+,Cl^-,Ca^{2+}, and Sc^{3+} are isoelectronic species, each containing 18 electrons.

In option (d), Ba^{2+},Sr^{2+},K^+, and S^{2-} are not isoelectronic.

28. Correct option: (B) B< S < P < F

Within a group, ionization potential (IP) tends to decrease as we move down, while ionization energy increases across a period from left to right.

In the case of phosphorus (P) with the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^3$ and sulfur (S) with the configuration $1s^2 2s^2 2p^6 3s^2 3p^4$, P exhibits a stable half-filled electronic configuration, providing a greater stability compared to S. Consequently, the ionization enthalpy of P is higher than that of S.

29. Correct option: (B) In alkali metals the reactivity increases but in the halogens, it decreases with an increase in atomic number down the group

In alkali metals, reactivity increases as we move down the group due to the expanding size, leading to a greater ability to lose electrons.

Conversely, in halogens, reactivity decreases for several reasons:

a) A decrease in electronegativity down the group.

b) Diminished oxidizing ability resulting from the inert pair effect.

c) An increase in atomic radius down the group, which reduces the attraction of valence electrons from other atoms, thereby lowering reactivity.

30. Correct option: (D) SnO_2

A) Calcium oxide (CaO) exhibits basic properties as it reacts with water to form a strong base, calcium hydroxide ($Ca(OH)_2$):

$CaO+H_2O \rightarrow Ca(OH)_2$

B) Carbon dioxide (CO_2) is acidic, dissolving in water to produce unstable carbonic acid:

$H_2O+CO_2 \rightarrow H_2CO_3$

C) Silica (SiO_2) is insoluble in water and behaves as a very weak acid.

D) Tin(IV) oxide (SnO_2) is amphoteric, reacting with both acids and bases:

$SnO_2+2H_2SO_4 \rightarrow Sn(SO_4)_2+2H_2O$

$SnO_2+2KOH \rightarrow K_2SnO_3+H_2O$

31. Correct option: (B) HF > HCl > HBr > HI

The thermal stability of hydrogen halides decreases as we move down the group, primarily attributed to a reduction in bond strength.

32. Correct option: (B) B < C < N < O – increasing first ionization energy

While the general trend is an increase in ionization enthalpy from left to right across a period, with the order B < C < N < O, nitrogen (N) deviates due to its stable half-filled electronic configuration ($1s^2 2s^2 2p^3$). Consequently, the ionization enthalpy of nitrogen is greater than that of oxygen (O). Therefore, the correct order of increasing first ionization enthalpy is B < C < O < N.

33. Correct option: (C) Zr and Hf have about the same radius

The phenomenon of lanthanoid contraction results in elements belonging to the same group in the 2nd and 3rd transition series having nearly identical atomic radii. As an example, both 40Zr (160 pm) and 72Hf (159 pm) exhibit almost the same radii due to this contraction.

34. Correct option: (A) O^{-2}

Li^+ and B^{3+} are isoelectronic, each containing 2 electrons. Similarly, O^{2-} and F^- are isoelectronic with 10 electrons each. Anions with a higher negative charge tend to have a larger ionic radius, as the increased electron repulsion surpasses the nuclear charge, causing the ion to expand in size. Therefore, the ionic radius of O^{2-} is maximum.

35. Correct option: (C) K^+, Ca^{2+}, Sc^{3+}, Cl^-

A) K^+ (19-1 = 18e), Cl^- (17+1 = 18e), Mg^{2+} (12-2 = 10e), Sc^{3+} (21-3 = 18e)

B) Na^+ (11-1 = 10e), Ca^{2+} (20-2 = 18e), Sc^{3+} (21-3 = 18e), F^- (9+1 = 10e)

C) K^+ (19-1 = 18e), Ca^{2+} (20-2 = 18e), Sc^{3+} (21-3 = 18e), Cl^- (17+1 = 18e)

D) Na^+ (11-1 = 10e), Mg^{2+} (12-2 = 10e), Al^{3+} (13-3 = 10e), Cl^- (17+1 = 18e)

36. Correct option: (A) Hydration enthalpy

Fluorine is a potent oxidizing agent because of its low heat of dissociation and high heat of hydration, compensating for its comparatively lower electron affinity.

37. Correct option: (A) O^- ion will tend to resist the addition of another electron

The addition of second electron in an atom or ion is always endothermic.

38. Correct option: (A) Atomic radii

The inadequate shielding effect of 4f electrons results in the outer electrons experiencing a stronger attractive force from the nucleus. This enhanced attraction leads to a contraction in the size of elements, a phenomenon referred to as lanthanide contraction.

39. Correct option: (A) s > p > d >f

The order of orbital penetration into the electron clouds of other orbitals is s > p > d > f. As a result, an electron in an s-orbital experiences the highest effective nuclear charge, followed by a p-orbital, and so forth. Ionization energy rises with increased penetration power, establishing the order of screening effect as s > p > d > f.

40. Correct option: (B) High ionization potential

An electronegative element tends to accept electrons rather than lose them, leading to a high ionization potential.

41. Correct option: (A) Increases one by one from IA to VIIA

Because oxygen is a more electronegative element, it tends to form oxides with elements from Group IA to VIIA in their highest oxidation states. The maximum oxidation state is equal to the number of valence electrons in the outermost shell, and it increases by one within any period.

42. Correct option: (B) Be^-

The outer electronic configurations of the ions are Li^- (most stable with $2s^2$), Be^- (least stable with $2s^2\ 2p^1$), B^- (stable with $2s^2\ 2p^2$), and C^- (more stable due to the half-filled configuration $2s^2\ 2p^3$).

Be^-, derived from the most stable Be ($2s^2$ configuration) by adding one electron, is the least stable and consistently exhibits a tendency to lose one electron to attain the stable $2s^2$ configuration.

43. Correct option: (B) Be > Mg > Ca

The first ionization potential decreases down a group due to the increase in size and increases across a period as the size decreases.

44. Correct option: (B) $F^{-} < O^{2-} < N^{3-}$

A) N < Be < B: All three elements are in the same period. Moving along the period, atomic size decreases due to an increase in nuclear charge. The order of radii is: Be < B < N.

B) $F^{-} < O^{2-} < N^{3-}$: These species are isoelectronic with 10 electrons each. As the negative charge increases, the atomic size also increases. The order of radii is: $F^{-} < O^{2-} < N^{3-}$.

C) $Na^{+} < Li^{+} < K^{+}$: These elements are in the same group, and as we move down the group, the atomic radii increase due to the addition of shells. The order of radii is: $Li^{+} < Na^{+} < K^{+}$.

D) $Fe^{3+} < Fe^{2+} < Fe^{4+}$: As the positive charge on the ion increases, the atomic radius decreases because the number of electrons decreases. The order of radii is: $Fe^{4+} < Fe^{3+} < Fe^{2+}$.

45. Correct option: (D) $P^{3+} > P^{5+}$

A) Inaccurate. Ti^{4+} and Mn^{7+} are isoelectronic with 18 electrons each. The size decreases with an increase in positive charge for these species.

B) Incorrect. These ions belong to two different isotopes of the same element. Since the proton number is the same in isotopes, the nuclear attraction is also the same, and thus, their radii must be equal.

C) Incorrect. K^{+} and Cl^{-} are isoelectronic, but K^{+} has more protons, making it smaller in size than Cl^{-}.

D) Correct. For different ions of the same element, the size decreases with an increase in positive charge.

46. Correct option: (B) The second ionisation potential of Mg is greater than the second ionization potential Na

Explanation:

A) The electronic configuration of Al is $1s^2 2s^2 2p^6 3s^2 3p^1$, while Mg has a stable electronic configuration with a fully filled subshell in the valence shell ($1s^2 2s^2 2p^6 3s^2$). Consequently, the first ionization energy of Al is less than the first ionization potential of Mg.

B) Upon ionization, Mg forms Mg^{+} with the electronic configuration $1s^2 2s^2 2p^6 3s^1$, and Na forms Na^{+} with the configuration $1s^2 2s^2 2p^6$. As Na^{+} has a completely filled valence subshell, the second ionization potential of Mg is less than the second ionization potential of Na.

C) Comparing Mg ($1s^2 2s^2 2p^6 3s^2$) and Na ($1s^2 2s^2 2p^6 3s^1$), the stable electronic configuration of Mg results in a lower first ionization potential than that of Na.

D) Finally, considering Mg^{2+} ($1s^2 2s^2 2p^6$) and Al^{2+} ($1s^2 2s^2 2p^6 3s^1$), the stable electronic configuration of Mg^{2+} leads to a greater third ionization potential compared to Al^{2+}.

47. Correct option: (D) Stable oxide M_2O

An element with an atomic weight of 39 has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$.

Identified as potassium (K), it forms a monovalent cation, K^{+}, with a valency of 1.

48. Correct option: (B) Be → Be^{+}

The ionization energy is highest for the process Be ⟶ Be^{2+}. Smaller atoms require more energy to remove an electron, i.e., exhibit higher ionization energy. In the case of beryllium (Be), with an atomic number of 4, the conversion to Be^{2+} involves the release of two electrons from the tightly bound 2s subshell, demanding a substantial amount of ionization energy.

49. Correct option: (A) $Hg^{+1} > Hg^{+2}$

A cation with a higher positive charge will exhibit a smaller radius due to the increased attraction of electrons to the nucleus.

50. Correct option: (D) II, III and IV

Isoelectronic structures have the same number of electrons.

A) CH_3^+ has 6 + 3 × 1 − 1 = 8 electrons.

B) (ii) H_3O^+ has 3 × 1 + 8 − 1 = 10 electrons.

C) (iii) NH_3 has 7 + 3 × 1 = 10 electrons.

D) (iv) CH_3^- has 6 + 3 × 1 + 1 = 10 electrons.

In H_3O^+, NH_3, and CH_3^-, the number of electrons is the same; therefore, these are isoelectronic structures.

51. Correct option: (C) I^-

Bromine (Br) is positioned above Iodine (I) in the periodic table, making Br smaller than I^+, I, and I^-. Among I^+, I, and I^-, I^- experiences the lowest effective nuclear charge, resulting in the largest size. Therefore, the largest species among the given options is I^-.

52. Correct option: (B) Absence of bond polarity

The cyanide ion (CN^-) is a strong nucleophile due to the availability of lone pairs, allowing it to donate to a suitable electrophile. N_2 on the other hand has a symmetrical electron distribution, and the presence of three multiple bonds between two nitrogen atoms makes it highly unreactive and inert, attributed to a high bond dissociation energy. Additionally, the covalent bond is non-polar, as it occurs between two identical atoms.

53. Correct option: (C) $1s^2, 2s^22p^6, 3s^23p^64s^1$

In options a and b, the last electron enters in d subshell and the elements are chromium and copper respectively.

54. Correct option: (C) [Ne] $3s^23p^3$

The configuration [Ne]$3s^23p^3$ will exhibit the highest ionization energy among the given options. This is because a half-filled electronic configuration is stable, requiring a substantial amount of energy to remove an electron from it. Option d is incorrect because n=4 indicates a larger size compared to n=3, and it is known that as the size of an atom increases, ionization energy decreases.

55. Correct option: (C) ns^2np^5

Within the configuration ns^2np^5, only one electron is needed to achieve a fully filled electronic configuration. Hence, it is the most electronegative element among the provided elements.

56. Correct Answer: 101

57. Correct Answer: 9

The given ionization enthalpy values provide clues to the element's position in the periodic table and its chemical properties. Element n+2 with a surprisingly low first ionization enthalpy (496 kJ/mol) suggests it's an alkali metal. Alkali metals typically reside in the second period, and their low first ionization enthalpies reflect their ease of losing an electron. Combining this information with the ionization enthalpy values, we can deduce that element n+2 is Sodium (Na) with a first ionization enthalpy of 496 kJ/mol.

n+2 = 11

n = 9

3. Acids, Bases and Salts

Solutions

LEVEL – I

1. Correct Option: C) Sour

Explanation:

Acidus is a Latin adjective meaning "sour, tart, or acidic." It's the root of the English word "acid."

2. Correct Option: D) Ascorbic acid

Explanation:

Ascorbic acid is also known as Vitamin C. Lemons are famous for their Vitamin C content, not the other acids listed.

3. Correct Option: B) Sulphuric acid

Explanation:

Sulphuric acid is a strong acid. It dissociates completely in water, producing a high concentration of hydrogen ions, making it a strong electrolyte.

4. Correct Option: B) Acetic acid

Explanation:

Acetic acid is the weak acid among the options. Sulphuric, nitric, and hydrochloric acids are all strong acids, meaning they virtually dissociate completely in water, whereas acetic acid only partially dissociates, leading to a lower concentration of hydrogen ions and weaker acidic character.

5. Correct Option: D) HCl

Explanation:

Mineral acids hail from inorganic sources like salts. Palmitic and acetic are organic (living things), carbonic can be both.

6. Correct Option: B) Red

Explanation:

In response to acidic conditions, blue litmus undergoes a chromic shift, resulting in a distinct red coloration.

7. Correct Option: B) Red

Explanation:

In an acidic medium, methyl orange exhibits a pronounced chromic shift, transitioning from its original orange hue to a distinct red colour.

8. Correct Option: B) basicity of an acid

Explanation:

The basicity of an acid refers to the number of hydrogen ions (H^+) it can furnish when it dissolves in water.

A) Acidity of base refers to H^+ concentration in a basic solution.

C) & D) classify acids by source, not H^+ donation capacity.

9. Correct Option: C) Organic acid

Explanation:

They're naturally found in living things and contain carbon, unlike mineral acids from inorganic sources. Citrusy citric acid or tangy lactic acid for examples.

10. Correct Option: C) Sodium hydroxide

Explanation:

A) Na_2CO_3 (washing soda): While exhibiting limited basicity under specific conditions, the carbonate (CO_3^{2-}) lacks the inherent electron density of strong bases for readily accepting H^+ ions.

B) CH_3COOH (acetic acid): Primarily exhibiting acidic properties due to its ability to donate H^+ through the carboxylic acid group (-COOH), it lacks the structural features for significant base behaviour.

C) NaOH (sodium hydroxide): The presence of the hydroxide (OH^-) group, a potent electron donor, readily facilitates H^+ acceptance, classifying it as a strong base due to its high proton affinity.

D) HCl (hydrochloric acid): Possessing a readily available H^+ for donation, its structure lacks the characteristics necessary for accepting H^+ ions, solidifying its acidic nature.

11. Correct Option: C) Sodium hydroxide

Explanation:

A) NaOH (Strong Base): $NaOH \rightarrow Na^+(aq) + OH^-(aq)$

All OH^- ions are deployed, ready for action.

B) Ammonium hydroxide (NH_4OH) (Weak Base): $NH_4OH + H_2O \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

Only partial dissociation, resulting in a smaller OH^- contingent.

C) Calcium hydroxide ($Ca(OH)_2$) (Moderately Strong Base): $Ca(OH)_2 \rightleftharpoons Ca^{2+}(aq) + 2OH^-(aq)$

Not all $Ca(OH)_2$ molecules dissociate, limiting the OH^- presence.

D) Nitric acid (HNO_3) (Strong Acid): $HNO_3 \rightarrow H^+(aq) + NO_3^-(aq)$

HNO_3 releases H^+ ions, not OH^-, making it an acidic adversary.

12. Correct Option: C) NH_4OH

Explanation:

While strong bases like NaOH and KOH readily dissociate into their constituent ions (Na^+/K^+ and OH^-), NH_4OH undergoes a partial equilibrium reaction with water:

$$NH_4OH + H_2O \rightleftharpoons NH_4^+ + OH^-$$

This equilibrium favours the undissociated NH_4OH molecules, resulting in a lower concentration of free OH^- ions compared to strong bases.

13. Correct Option: D) Pink

Explanation:

Phenolphthalein acts as a pH indicator, remaining colourless in acidic solutions and magically transforming to a vibrant pink in basic environments

14. Correct Option: A) 2

Explanation:

H_2SO_4, also known as sulphuric acid, is a dibasic acid, meaning it can donate two hydrogen ions (H^+) per molecule when dissolved in water. Basicity refers to the number of H^+ ions an acid donates, not its overall strength.

15. Correct Option: C) 3

Explanation:

When $Al(OH)_3$ reacts with any acid, $Al(OH)_3 + 3HCl \rightarrow AlCl_3 + 3H_2O$

It requires 3 acidic H^+ ions to neutralize therefore its acidity is 3.

16. Correct Option: A) NaOH

Explanation:

Monoacidic bases can accept only one hydrogen ion (H^+) per molecule.

NaOH readily accepts one H^+ ion from its dissociation in water,

$$NaOH + H_2O \rightarrow Na^+ + OH^-.$$

$Ba(OH)_2$, $Mg(OH)_2$, and $Al(OH)_3$ are diacidic or polyacidic bases, meaning they can accept two or more H^+ ions per molecule.

For example, $Ba(OH)_2$ can accept two H^+ ions:

$$Ba(OH)_2 + 2H^+ \rightarrow Ba^{2+} + 2H_2O.$$

17. Correct Option: A) $Ba(OH)_2$

Explanation:

Diacidic bases can accept two hydrogen ions (H^+) per molecule.

$Ba(OH)_2$ readily accepts two H^+ ions through its dissociation in water,

$$Ba(OH)_2 + 2H^+ \rightarrow Ba^{2+} + 2H_2O.$$

NH_4OH and KOH are monoacidic bases like NaOH, accepting only one H^+ ion each.

$Al(OH)_3$ is a triacidic base, capable of accepting three H^+ ions due to its three OH^- groups.

18. Correct Option: D) HNO_3

Explanation:

Arrhenius acids dissociate in water to release hydrogen ions (H^+).

$$HNO_3 + H_2O \rightarrow H^+ + NO_3^-.$$

CO_2 and SO_3 are acid anhydrides, reacting with water to form acids but not directly releasing H^+ ions.

$FeCl_3$ is a salt, not an acid, as it doesn't dissociate to release H^+ ions.

19. Correct Option: C) NaCl

Explanation:

Ionization refers to the separation of a molecule into charged ions in solution.

While HCl, CH_3COOH, and HBr are polar covalent molecules that dissociate into ions in water (e.g., $HCl \rightarrow H^+ + Cl^-$), NaCl is an ionic compound already composed of Na^+ and Cl^- ions. Thus, it doesn't need further ionization in water.

20. Correct Option: B) $H_2PO_2^-$

Explanation:

$H_2PO_2^-$ (hydrogen phosphite ion) lacks a conjugate base. It's already donated its two acidic protons, leaving no remaining H^+ to accept and form a conjugate pair. Unlike $H_2PO_4^-$ (dihydrogen phosphate) and $H_2PO_3^-$ (hydrogen phosphite), which can accept protons ($H_2PO_4^- + H^+ \rightleftharpoons H_3PO_4$, $H_2PO_3^- + H^+ \rightleftharpoons H_3PO_3$), $H_2PO_2^-$ remains proton-less.

21. Correct Option: D) Both Lewis and Bronsted neutralization

Explanation:

This reaction exhibits characteristics of both:

1. Bronsted: H_3O^+ donates a proton (H^+) to OH^-, fulfilling the Bronsted acid-base definition.
2. Lewis: H_3O^+ acts as a Lewis acid (electron acceptor) by accepting a lone pair from OH^-, a Lewis base (electron donor).

22. Correct Option: D) $H_2PO_4^-, PO_4^{-3}$

Explanation:

A conjugate acid-base pair differs by one proton (H^+).

In A, B, and C, the difference between the two species is indeed one H^+:

A) HSO_4^- gains H^+ to become SO_4^{2-}.

B) $H_2PO_4^-$ gains H^+ to become HPO_4^{2-}.

C) $H_2PO_4^-$ loses H^+ to become H_3PO_4.

However, in D, $H_2PO_4^-$ and PO_4^{3-} differ by two protons. $H_2PO_4^-$ would need to lose two H^+ to reach PO_4^{3-}, not just one.

23. Correct Option: A) OH^-

Explanation:

The conjugate acid of a base differs by one proton (H^+).

The oxide ion (O^{2-}) accepts a proton to become the hydroxide ion (OH^-): $O^{2-} + H^+ \rightarrow OH^-$.

24. Correct Option: D) HN_3

Explanation:

Bases accept protons (H^+). While the other options readily accept protons due to their lone pairs (N_2H_4, NH_2OH, $(CH_3)_3N$), HN_3 acts as an acid itself, containing a readily donatable proton on the nitrogen atom.

25. Correct Option: C) BF_3

Explanation:

1. Brønsted-Lowry acids accept lone pairs and donate protons (H^+). BF_3 readily accepts a lone pair from a Lewis base like F^- ($F^- + BF_3 \rightarrow BF_4^-$).
2. Arrhenius acids dissociate in water to release H^+ ions. BF_3 doesn't have any H^+ to release and doesn't dissociate in water, preventing it from exhibiting Arrhenius behaviour.

26. Correct Option: C) Arrhenius theory

Explanation:

According to Arrhenius theory, an acid is a substance that dissociates in water to release hydrogen ions (H^+). Sulphur trioxide (SO_3) readily dissolves in water and dissociates into H^+ and SO_4^{2-} ions, making it an acid based on the Arrhenius definition.

The other options:

A) Lewis theory: While SO_3 can act as a Lewis acid by accepting electron pairs, it doesn't directly release H^+ in water, so it wouldn't be classified as an acid.

B) Usanovich theory: This lesser-known theory focuses on the interaction of protons and electrons, but it's not as widely used as the other options and wouldn't be the primary explanation for SO_3 being an acid.

D) Bronsted-Lowry theory: This theory expands on Arrhenius by stating that an acid is a proton donor. SO_3 can indeed donate a proton to water, making it an acid according to Bronsted-Lowry, but the Arrhenius definition is more specific about H^+ release in water.

27. Correct Option: D) Acetylene

Explanation:

The strongest conjugate base results from acetylene since it is the weakest acid among formic acid, benzoic acid, acetic acid, and acetylene.

The order of strength of acids is given below:

Acetylene < Acetic Acid < Formic Acid < Benzoic Acid

28. Correct Option: A) HNO_3

Explanation:

Amphoteric substances can act as both acids and bases, depending on the solution they're in. They have the ability to both donate and accept protons (H^+).

A) HCO_3^- (bicarbonate ion) can act as a base by accepting protons ($HCO_3^- + H^+ \rightleftharpoons H_2CO_3$), or as an acid by donating its own proton ($HCO_3^- \rightleftharpoons CO_2 + OH^-$)

B) H_2O (water) can act as an acid (donating H^+ to strong bases) and a base (accepting H^+ from strong acids).

C) NH_3 (ammonia) can act as a base by accepting H^+ (forming NH_4^+), but under specific conditions, it can also donate H^+ to even stronger bases like lithium amide.

HNO_3, however, is a strong acid and doesn't readily accept protons due to its highly electronegative nitrate group (NO_3^-) already possessing a strong hold on its protons. Therefore, it primarily remains an acid regardless of the solution it's in.

29. Correct Option: A) Weak

Explanation:

Strong electrolytes readily ionize in water, completely dissociating into ions.

Acetic acid is a weak acid compared to water. While it can still conduct electricity due to some dissociation, it does so to a much lesser extent compared to water.

When a strong electrolyte dissolves in acetic acid, the weakness of the solvent hinders its ability to fully ionize the electrolyte. The strong electrolyte experiences partial or incomplete dissociation, leading to a weaker overall ionization compared to its behaviour in water.

30. Correct Option: C) Benzene

Explanation:

In water, ammonia, and aqueous NaOH, HCl readily donates its proton (H^+), showcasing its acidic prowess. It forms H_3O^+ in water, NH_4^+ in ammonia, and Cl^- and Na^+ in aqueous NaOH. Benzene, however, is a relatively non-polar molecule with weak electron-accepting abilities, unlike the other options. Therefore, HCl remains stubbornly intact in benzene, failing to exhibit its acidic nature.

31. Correct Option: D) Acetic acid

Explanation:

Amphiprotic solvents are those which can donate and accept a proton.

A) $2H_2O \rightleftharpoons H_3O^+ + OH^-$. Here one water molecule is accepting a proton and the other is donating, so water is amphiprotic.

C) Alcohol also can accept and donate a proton.

D) Acetic acid can only donate a proton but can't accept.

32. Correct Option: B) NH_3

Explanation:

Brønsted-Lowry theory defines acids as proton (H^+) donors.

A) $HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$ (sulphate ion readily donates a proton)

C) BCl_3 acts as a Lewis acid, accepting a lone pair from Cl^- to form BCl_4^- (indirectly involving proton transfer)

D) $HNO_3 \rightleftharpoons H^+ + NO_3^-$ (nitric acid readily dissociates)

However, NH_3 prefers the role of a Brønsted-Lowry base, readily accepting protons to form the ammonium ion: $NH_3 + H^+ \rightleftharpoons NH_4^+$.

33. Correct Option: B) Lewis acid

Explanation:

A Lewis acid is a species that accepts a lone pair of electrons.

Reaction analysis: In this reaction, $AlCl_3$ accepts a lone pair of electrons from the chloride ion (Cl^-) to form the $AlCl_4^-$ complex ion. The empty 3p orbital on the aluminium atom acts as the electron pair acceptor.

Other options:

A) Lewis base: Cl^- donates the lone pair, making it the Lewis base in this scenario.

A) Lewis salt: There's no formation of a salt in this reaction, which typically involves the complete transfer of electrons to form ions.

B) Arrhenius acid: $AlCl_3$ doesn't release H^+ in aqueous solution, so it doesn't fit the Arrhenius definition of an acid.

34. Correct Option: C) Dative bond

Explanation:

Dative bond: This type of bond involves one atom donating a lone pair of electrons to another atom, without sharing electrons equally. In a Lewis acid-base reaction, the base (electron donor) donates its lone pair to the empty orbital of the acid (electron acceptor), forming a dative bond.

Other options:

A) Covalent bond: While some covalent character may exist due to electron sharing, the emphasis is on the lone pair donation from the base, making it primarily a dative bond.

B) Ionic bond: This typically involves complete electron transfer, which isn't the case in Lewis

acid-base interactions.

D) H-bond: While hydrogen bonds can be present in the resulting molecule, they wouldn't be the primary bond formed in the actual neutralization reaction.

35. Correct Option: D) Lewis theory cannot explain relative strength of acids and bases

Explanation:

Lewis theory: While it effectively defines acids and bases based on electron accepting/donating abilities, it lacks a quantitative measure for comparing their strengths. It tells you "who's who" in the acid-base world but doesn't provide a scale for "how strong" they are.

Arrhenius and Bronsted-Lowry theories: Both offer frameworks for comparing acid/base strengths. Arrhenius uses the degree of dissociation in water, while Bronsted-Lowry considers the pKa value (proton dissociation constant).

36. Correct Option: C) N_2

Explanation:

Lewis acids are electron acceptors, seeking to fill empty orbitals with lone pairs from other molecules.

N_2 possesses an empty π* antibonding orbital, making it eager to accept electron pairs. This empty orbital can readily interact with lone pairs on other molecules, engaging in electron pair sharing.

O, F, and O_2 primarily act as Lewis bases due to their lone pairs. They are electron donors, not acceptors.

37. Correct Option: C) Lewis theory

Explanation:

A) Arrhenius theory: This theory defines acids and bases in terms of their dissociation in water and release of H^+ ions. It isn't directly applicable to gaseous phase neutralization, where water isn't present.

B) Bronsted-Lowry theory: This theory expands on Arrhenius by defining acids as proton donors and bases as proton acceptors. While it can explain some gaseous phase reactions involving proton transfer, it isn't as general as Lewis theory.

C) Lewis theory: This theory defines acids and bases based on their electron accepting and donating abilities, respectively. It broadly encompasses both proton transfer and other types of electron pair interactions. This makes it the most suitable framework for explaining gaseous phase neutralization reactions, which can involve various kinds of electron pair sharing and acceptance between molecules.

D) Bohr's theory: This theory focuses on electron structure and doesn't directly address acid-base behaviour.

38. Correct Option: A) The presence of three oxygen atoms, e^- density will increase on sulphur atom in the SO_3

Explanation:

Lewis acid strength is directly related to the electron-accepting ability of the central atom. The easier it is for the atom to accept a lone pair of electrons, the stronger the Lewis acid.

In SO_3, the three oxygen atoms surrounding the sulphur atom are highly electronegative. They draw electron density towards themselves, creating a partial positive charge on the sulphur atom. This increases the sulphur's ability to accept lone pairs from other molecules, making SO_3 a stronger Lewis acid compared to SO_2.

SO_2 has only two oxygen atoms, resulting in a less positive charge on the sulphur atom and a weaker electron-accepting ability compared to SO_3.

Therefore, the presence of three oxygen atoms in SO_3 plays a crucial role in enhancing its Lewis acidic strength by increasing the electron density deficit on the sulphur atom.

39. Correct Option: C) NBr_3 has more acidic strength than NI_3

Explanation:

In the context of Lewis acids, the electronegativity of the surrounding atoms plays a crucial role in determining their acidic strength. The more electronegative the surrounding atoms, the greater their ability to pull electron density away from the central N atom, creating a partial positive charge that makes it more receptive to accepting lone pairs (acting as a Lewis acid).

A) NI_3 vs. NF_3: Fluorine is more electronegative than iodine, so NF_3 will have a higher partial positive charge on the N atom, making it more acidic than NI_3.

B) NI_3 vs. NBr_3: Bromine is more electronegative than iodine, so NBr_3 will have a higher partial positive charge on the N atom, making it more acidic than NI_3.

D) NCl_3 vs. NBr_3: Chlorine is more electronegative than bromine, so NCl_3 will have a higher partial positive charge on the N atom, making it more acidic than NBr_3.

Therefore, based on the electronegativity trend, option C) NBr_3 has more acidic strength than NI_3 is the correct choice.

40. Correct Option: D) It does not acts as Lewis acid

Explanation:

A) While $BaCl_2$ is indeed an ionic compound, it doesn't directly influence its Lewis acid behaviour.

B) Barium (Ba) in $BaCl_2$ possesses filled electronic orbitals in its outermost shell ($6s^2$). There are no readily available empty orbitals for accepting lone pairs, a key characteristic of Lewis acids.

C) This statement refers to Beryllium (Be), not Barium. Additionally, while back bonding can occur in some Be compounds, it's not relevant to the Lewis acid nature of $BaCl_2$.

D) Both Ba and Cl in $BaCl_2$ have complete octets in their stable ionic configuration. They neither lack electron pairs nor possess empty orbitals to readily accept them, eliminating the possibility of Lewis acid behaviour.

41. Correct Option: D) $\sqrt{K_W}$

Explanation:

At any temperature, the proton concentration of pure water is approximately $\sqrt{K_W} \approx 10^{-7}$ M, reflecting its equilibrium relationship with hydroxide ions via the ionic product of water (K_w).

42. Correct Option: B) 0

Explanation:

In a 1 M HCl solution, the complete dissociation of HCl occurs as

$HCl\ (aq) \rightarrow H^+(aq) + Cl^-(aq)$

$pH = -\log[H^+]$

$= -\log[1]$

$pH = 0$

43. Correct Option: D) Decreased by 10 times

Explanation:

1. Initial H^+ concentration:

$$pH = -\log_{10}[H^+]$$
$$4 = -\log_{10}[H^+]$$
$$[H^+] = 10^{-4}\ M$$

2. Desired H^+ concentration for pH 5.0:

$$pH\ 5.0 = -\log_{10}[H^+]_{new}$$
$$[H^+]_{new} = 10^{-5}\ M$$

3. Change in H^+ concentration:

$$\frac{[H^+]_{new}}{[H^+]_{initial}} = \frac{10^{-5}\,\text{M}}{10^{-4}\,\text{M}} = \frac{1}{10}$$

44. Correct Option: D) Decreased by 100 times

Explanation:

Initial $[H^+] = 10^{-2}$ M (pH 2). Doubling pH requires two one-unit increases, leading to:

Final $[H^+] = 10^{-2} \times 10^{-1} \times 10^{-1} = 10^{-4}$ M (decreased by 100 times).

45. Correct Option: A) 9

Explanation:

Initial pH = 6, meaning $[H^+] = 10^{-6}$ M (because, pH = $-\log_{10}[H^+]$).

H^+ concentration decreases by 1000 times, so the new $[H^+] = 10^{-6}$ M / 1000 = 10^{-9} M.

new pH = $-\log_{10}(10^{-9}) = 9$.

46. Correct Option: A) $FeSO_4.\ (NH_4)2SO_4.\ 6H_2O$

Explanation:

A mixed salt contains two different metal cations or two different non-metal anions.

A) $FeSO_4.\ (NH_4)2SO_4.\ 6H_2O$ contains both Fe^{2+} and NH_4^+ cations, along with SO_4^{2-} anions.

B) CH_3COONH_4 (ammonium acetate) is a single salt, not a mixed salt. It has NH_4^+ as the cation and CH_3COO^- as the anion.

C) Na_2KPO_4 (sodium potassium phosphate) is also a single salt. It has Na^+ and K^+ cations, along with PO_4^{3-} anions.

D) $K_4[Fe(CN)_6]$ (potassium hexacyanoferrate) is a complex salt, not a mixed salt. It has the complex anion $[Fe(CN)_6]^{4-}$ and K^+ cations.

47. Correct Option: B) POP has the property of setting into a hard mass on wetting with water

Explanation:

A) Normal salts are neutral in nature. So, A is wrong.

C) Salt is an electrovalent or ionic compound formed by the partial or complete replacement of the replaceable (ionisable) hydrogen atoms (i.e.,H+ ions) of an acid by a metallic ion or ammonium ion. So, C is wrong.

48. Correct Option: A, B and C

Explanation:

A phenomenon where a compound loses its water of crystallization on exposure to dry air is called efflorescency. Substances showing efflorescency are called efflorescent substances. Here except the option D, everything is an efflorescent substance because of presence of water of crystallization in them.

49. Correct Option: C) Sodium hydroxide

Explanation:

A) Silver chloride (AgCl):

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$, equilibrium heavily favours the solid

B) Calcium sulphite ($CaSO_3$): It exhibits limited solubility, forming a slightly cloudy solution

$$CaSO_3(s) \rightleftharpoons Ca^{2+}(aq) + SO_3^{2-}(aq)$$, equilibrium favours the solid

C) Sodium hydroxide (NaOH):

$$NaOH(s) \rightarrow Na^+(aq) + OH^-(aq)$$

D) Sodium aluminate ($NaAlO_2$): This complex salt forms a colloidal suspension rather than a true solution

$$NaAlO_2(s) + H_2O \rightleftharpoons Na^+(aq) + AlO_2(OH)^-(aq) + H_2O$$

50. Correct Option: D) $CaOCl_2$

Explanation:

The reaction refers to the chlorination of calcium hydroxide ($Ca(OH)_2$) to form calcium oxychloride ($CaOCl_2$), also known as bleaching powder.

$$2Ca(OH)_2(s) + Cl_2(g) \rightarrow CaOCl_2(s) + CaCl_2(s) + H_2O(l)$$

51. Correct Option: C) Neutral salt

Explanation:

Strong acids and strong bases completely dissociate in water, meaning their ions readily separate and remain free in the solution. When a strong acid reacts with a strong base, the resulting salt is formed by the combination of their respective ions. Since both strong acids and strong bases have no inherent acidic or basic properties due to complete dissociation, their conjugate ions formed in the salt also lack these properties. Therefore, the resulting salt exhibits neither acidic nor basic character, making it a neutral salt.

52. Correct Option: B) Basic salt

Explanation:

Washing soda, also known as sodium carbonate (Na_2CO_3), is formed from the reaction of a strong base (sodium hydroxide, NaOH) with a weak acid (carbonic acid, H_2CO_3).

$$2NaOH(aq) + H_2CO_3(aq) \rightarrow Na_2CO_3(aq) + 2H_2O(l)$$

In this reaction, the strong base completely dissociates in water, while the weak acid partially dissociates. The resulting salt, Na_2CO_3, contains the conjugate base of the weak acid (CO_3^{2-}) which retains some of its basic properties.

53. Correct Option: D) Carbon dioxide

Explanation:

When you mix baking powder with a liquid and an acidic ingredient, the following reaction occurs:

$$2NaHCO_3(aq) + H^+(aq) \rightarrow Na_2CO_3(aq) + CO_2(g) + H_2O(l)$$

54. Correct Option: C) 1/ 2

Explanation:

Plaster of Paris, also known as calcium sulphate hemihydrate ($CaSO_4 \cdot \frac{1}{2}H_2O$), contains one molecule of water of crystallization per two formula units of calcium sulphate. This translates to half a water molecule per formula unit, making option C the correct choice.

55. Correct Option: D) Calcium oxide and carbon dioxide

Explanation:

When calcium carbonate ($CaCO_3$) is heated, it undergoes a decomposition reaction:

$$CaCO_3\ (s) \rightarrow CaO\ (s) + CO_2\ (g)$$

This reaction essentially breaks down the calcium carbonate molecule, releasing calcium oxide (CaO, also known as lime) as a solid and carbon dioxide (CO_2) as a gas. So, both products, calcium oxide and carbon dioxide, are formed during the heating process.

56. Correct Option: A) O_2

Explanation:

The chlor-alkali process produces chlorine gas (Cl_2), hydrogen gas (H_2), and sodium hydroxide (NaOH), but it does not yield oxygen gas (O_2) as a product.

$$2NaCl(aq) + 2H_2O(l) \rightarrow Cl_2(g) + H_2(g) + 2NaOH(aq)$$

LEVEL – II
SINGLE OPTION CORRECT TYPE

1. Correct Option: B) Bitter

Explanation:

Bases generally taste bitter, though not all and intensity varies. Strong ones have a soapy/astringent bitterness, while weak ones might be subtle or even slightly sweet

2. Correct Option: A) Neutralization

Explanation:

Neutralization refers to the chemical reaction between an acid and a base, where they react completely to form a salt and water. The resulting solution becomes neutral (neither acidic nor basic) due to the complete consumption of both the acidic and basic properties.

3. Correct Option: C) Citric acid

Explanation:

Citric acid is used for making sour drinks such as Limca.

4. Correct Option: C) Inorganic acids

Explanation:

A) Organic acids: These acids come from living organisms or their remains, such as citric acid (found in fruits) and acetic acid (vinegar).

B) Strong acids: This refers to the strength of an acid in terms of its ability to dissociate in water and donate protons. While some inorganic acids like sulfuric acid are strong, not all are, and strength isn't directly related to their origin.

C) Inorganic acids: These are acids derived from mineral sources, meaning they originate from non-living materials found in the Earth's crust. Examples include sulfuric acid (H_2SO_4), hydrochloric acid (HCl), and nitric acid (HNO_3).

D) Weak acids: Similar to strong acids, this categorizes acids based on their dissociation degree. Again, some inorganic acids are weak, but it's not a defining characteristic based on their source.

5. Correct Option: C) Sulphuric acid

Explanation:

Sulphuric acid is often referred to as the "king of chemicals" because of its widespread industrial use and importance in various chemical processes.

6. Correct Option: (C)

Explanation:

A triacidic base is a base that can accept three protons (H^+ ions). This means it has three hydroxide (OH^-) groups available for donation.

NaOH (sodium hydroxide) and $Ca(OH)_2$ (calcium hydroxide) are monobasic bases, meaning they have only one OH^- group and can accept only one H^+ ion.

$Al(OH)_3$ (aluminium hydroxide), however, has three OH^- groups and can therefore accept three H^+ ions, qualifying it as a triacidic base.

$Mg(OH)_2$ (magnesium hydroxide) is a dibasic base, meaning it has two OH^- groups and can accept two H^+ ions.

7. Correct Option: A) HCl

Explanation:

A conjugate base is formed when an acid donates a proton (H^+). In this case, the Cl^- ion is formed when HCl loses its H^+:

$$HCl\ (aq) \rightarrow H^+\ (aq) + Cl^-\ (aq)$$

8. Correct Option: C) Na_2O

Explanation:

A) CO_2 (carbon dioxide): Although not a strong acid, CO_2 readily dissolves in water to form carbonic acid (H_2CO_3), making it acidic.

$$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$$

B) SO_3 (sulphur trioxide): Similar to CO_2, SO_3 reacts vigorously with water to form sulphurous acid (H_2SO_3), exhibiting acidic properties.

$$SO_3(g) + H_2O(l) \rightarrow H_2SO_3(aq)$$

C) Na_2O (sodium oxide): Unlike the other options, Na_2O reacts with water to form sodium hydroxide (NaOH), a strong base. This reaction consumes H^+ ions from water, making the resulting solution slightly alkaline, not acidic.

$$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq) + H^+(aq)$$

D) N_2O_5 (dinitrogen pentoxide): This potent oxidizing agent also undergoes hydrolysis in water, forming nitric acid (HNO_3) and establishing its acidic character.

$$N_2O_5(s) + H_2O(l) \rightarrow 2HNO_3(aq)$$

9. Correct Option: A) ClO^-

Explanation:

Brønsted basicity is inversely proportional to the stability of the conjugate acid. In other words, a weaker conjugate acid indicates a stronger Brønsted base.

In this case, all the given anions are derived from the same acid, $HClO_4$ (perchloric acid). When they accept a proton (H^+), they form their respective conjugate acids:

- $ClO^- + H^+ \rightarrow HClO$ (hypochlorous acid)
- $ClO_2^- + H^+ \rightarrow HClO_2$ (chlorous acid)
- $ClO_3^- + H^+ \rightarrow HClO_3$ (chloric acid)
- $ClO_4^- + H^+ \rightarrow HClO_4$ (perchloric acid)

Among these conjugate acids, hypochlorous acid (HClO) is the weakest, thereby making ClO^- the strongest Bronsted base.

10. Correct Option: D) I < iii < ii < iv

Explanation:

HSO_3F is a super acid. Its acidic strength is greater than any given species.

The pK_a value of other species are

$HCO_3^- \rightarrow 10.25$

$H_3O^+ \rightarrow -1.74$

$HSO_4^- \rightarrow 1.92$

Lesser the pK_a value higher will be its acidic strength. Hence, sequence of acidic strength will be $HSO_3F > H_3O^+ > HSO_4^- > HCO_3^-$

11. Correct Option: C) $C_5H_5NH_2$ (P^{Kb} = 3.2)

Explanation:

P^{Kb} is the negative logarithm of the base dissociation constant (Kb). Higher P^{Kb} values indicate weaker bases because they dissociate less readily in water.

Comparing the P^{Kb} values:

- $C_5H_5NH_2$ (P^{Kb} = 3.2) has the highest P^{Kb} value, making it the weakest base.
- NH_4OH (P^{Kb} = 4.8) has a lower P^{Kb} value than $C_5H_5NH_2$, indicating it's a stronger base.
- $C_6H_5NH_2$ (P^{Kb} = 9.4) and C_9H_7N (P^{Kb} = 9.2) have even higher P^{Kb} values, making them even weaker bases than $C_5H_5NH_2$.

12. Correct Option: C) 1 and 4 are correct

Explanation:

1) In a neutral medium, $[H^+]$ = $[OH^-]$ = 1 x 10^{-7} mol L^{-1} at 25^oC. So, pH = -log $[H^+]$ = 7 at 25^oC is correct.
As the temperature increases, K_w increases. For example, pH range is from 0 to 14 at 25^oC and K_w = 10^{-14} and neutral pH point for pure water is 7.
But pH range is 0 to 13 at 80^oC and K_w = 10^{-13} and neutral pH is 6.5 for pure water at this temperature. So, pH of water decreases with rise in temperature but will still remain neutral.
2) When base is added $[OH^-]$ of water increases and pH increases. So, the given option is wrong.
3) When acid is added $[H^+]$ of water increases and pH decreases. So, the given option is wrong.
4) pH + pOH = pK_w,
Where K_w = -log (K_w) and K_w is the ion product constant of water. For pure water at 25^oC, K_w is approximately 1 x 10^{-14}, and at this temperature, the pH of pure water is 7.
So, at any temperature, the sum of the pH and pOH in water is constant and equal to the pK_w value for that temperature.

13. Correct Option: A) Temperature

Explanation:
B) Nature of the solution: Obviously, an acidic solution will have a different pH than a basic or neutral one.
C) Degree of dissociation of acid (or) base: The extent to which an acid or base releases its ions affects the concentration of H^+ and OH^- ions, directly impacting pH.
D) Volume of the solution: While not as significant as other factors, changing the volume of a solution (without altering the amount of acid/base) can slightly affect the concentration of ions and consequently, the pH.

14. Correct Option: (D)

Explanation:
K_a, the acid dissociation constant, measures the extent of an acid's dissociation in water. A low K_a value indicates limited dissociation, meaning only a small fraction of the acid molecules ionize into H^+ and conjugate base. Conversely, a high K_a value signifies strong dissociation, with a significant portion of the acid molecules releasing protons.
Therefore, a weak acid is characterized by a low K_a, reflecting its limited tendency to dissociate and release H^+ ions.
Examining the other options
A) A very high K_a indicates a strong acid, not a weak one.
B) P^{Ka} (the negative logarithm of K_a) is inversely related to K_a. So, a low K_a translates to a high P^{Ka}, not a very low one.
C) Alpha (α) represents the degree of dissociation, which is typically low for weak acids due to their limited ionization.

15. Correct Option: D) $(NH_4)_2CO_3$

Explanation:
A) Na_2CO_3 (sodium carbonate):
This salt is derived from a weak acid (H_2CO_3) and a strong base (NaOH). However, the carbonate ion (CO_3^{2-}) has a weak ability to pull protons from water, meaning the solution remains slightly alkaline due to excess OH^- ions.
B) $AlCl_3$ (aluminium chloride):
This salt is formed from a strong acid (HCl) and a strong base ($Al(OH)_3$). Therefore, the solution has no excess H^+ or OH^- ions and will be neutral (pH ≈ 7).
C) NaCl (sodium chloride):

This salt is derived from strong acid (HCl) and strong base (NaOH). Similar to $AlCl_3$, it has no excess ions and will have a neutral pH.

D) $(NH_4)_2CO_3$ (ammonium carbonate):

This salt is formed from a weak acid (H_2CO_3) and a weak base (NH_4OH). While the carbonate ion contributes some alkalinity, the ammonium ion (NH_4^+) can react with water to release H^+ ions. This reaction, known as hydrolysis, can tip the balance slightly towards acidity, potentially resulting in a pH below 7.

16. Correct Option: C) salt of strong base and strong acid

Explanation:

Strong acids and bases completely dissociate in water, meaning they break down entirely into their respective ions. For example, HCl (strong acid) dissociates into H^+ and Cl^-, and NaOH (strong base) dissociates into Na^+ and OH^-. When a salt is formed from a strong acid and a strong base, both components are already completely dissociated. Therefore, the resulting salt simply exists as separate ions in the solution, achieving 100% ionization.

Examining the other options

A) Salt of strong acid and weak base: The weak base only partially dissociates in water, meaning some of its molecules remain intact. This incomplete dissociation translates to an incomplete ionization of the salt as well.

B) Salt of strong base and weak acid: Similar to option A, the weak acid's partial dissociation hinders the complete ionization of the salt.

D) Both A & B: As explained, neither salts of strong acids with weak bases nor salts of strong bases with weak acids achieve complete ionization due to the incomplete dissociation of one of the components.

17. Correct Option: B) Sodium carbonate decahydrate

Explanation:

The chemical name of washing soda is sodium carbonate decahydrate, often represented as $Na_2CO_3 \cdot 10H_2O$

18. Correct Option: D) Both (B) & (C)

Explanation:

A) Dehydration of NaCl solution: This doesn't directly produce sodium hydroxide. Dehydration focuses on removing water molecules, and while a concentrated NaCl solution can be dehydrated to form anhydrous NaCl, it wouldn't yield NaOH.

B) Electrolysis of aqueous NaCl solution: This is one method for producing sodium hydroxide, known as the chlor-alkali process. The overall reaction involves the electrolytic decomposition of brine (NaCl solution):

$$2NaCl(aq) + 2H_2O(l) \rightarrow Cl_2(g) + H_2(g) + 2NaOH(aq)$$

As you can see, this process generates chlorine gas (Cl_2), hydrogen gas (H_2), and the desired sodium hydroxide (NaOH) solution.

C) Chlor-alkali process: This encompasses option B and refers specifically to the electrolytic production of chlorine, hydrogen, and sodium hydroxide from brine.

D) Both (B) & (C): This is the correct answer. Electrolysis of aqueous NaCl solution, particularly via the chlor-alkali process, is the primary method for industrial production of sodium hydroxide

19. Correct Option: A) 2

Explanation:

Gypsum, also known as calcium sulphate dihydrate, has the chemical formula $CaSO_4 \cdot 2H_2O$. This indicates that one molecule of calcium sulphate ($CaSO_4$) combines with two molecules of water ($2H_2O$) as water of crystallization.

4. CHEMICAL BONDING

LEVEL – I

1. Correct Option: A) Chemical bond

Explanation:

Chemical bond encompasses the various forces (covalent, ionic, metallic) that bind atoms in molecules, crystals, and other structures

B) Valence: While valence refers to the number of electrons in an atom's outermost shell that can participate in bonding, it's not the force itself but rather a property that influences bonding behaviour.

C) Dative bond: This is a specific type of covalent bond where one atom donates both electrons in the bond, but it doesn't encompass all types of chemical bonds.

D) Hydrogen bond: This is a type of intermolecular force, not a true chemical bond, and it primarily occurs between molecules with hydrogen atoms bonded to highly electronegative atoms like oxygen or nitrogen.

2. Correct Option: B) Exothermic

Explanation:

When atoms form chemical bonds, they share electrons, creating a more stable configuration compared to their individual states. This process usually releases energy, making the overall reaction exothermic. The released energy can be in the form of heat, light, or sound

3. Correct Option: A) Potential energy of system decreases

Explanation:

When atoms form chemical bonds, they typically achieve a more stable electronic configuration by sharing electrons. This creates a lower energy state compared to the individual atoms, resulting in a decrease in the system's potential energy.

4. Correct Option: A) Kessel and Lewis

Explanation:

The significance of the stable octet in the formation of chemical bonds was first recognized by Gilbert N. Lewis and Kossel independently.

5. Correct Option: A) Valence electrons

Explanation:

B) Core electrons: These are electrons located closer to the nucleus in inner energy levels and are tightly bound to the nucleus. They are not directly involved in chemical bonding.

C) Lone pair electrons: These are valence electrons that are not involved in bonding with other atoms. While they are part of the valence shell, they are not the defining characteristic of all valence electrons.

D) Bonding electrons: These are valence electrons that are involved in forming chemical bonds with other atoms. Again, while they are a subset of valence electrons, they don't encompass all of them.

6. Correct Option: B) Core electrons

Explanation:

A) Valence electrons: These are the electrons present in the outermost energy level, not the inner one. They are crucial for chemical bonding and determining an atom's reactivity.

C) Lone pair electrons: These are valence electrons that are not involved in bonding with other atoms. While they are part of the valence shell, they are not the defining characteristic of all valence electrons.

D) Bonding electrons: These are valence electrons that are involved in forming chemical bonds with other atoms. Again, while they are a subset of valence electrons, they don't encompass all of them.

7. Correct Option: C) Cl^- ion

Explanation:

A) Cl atom: A neutral chlorine atom has 7 valence electrons. While it can achieve an octet by sharing electrons with other atoms, it wouldn't have a complete octet in its isolated state as a Lewis dot structure.

B) Cl^+ ion: This ion loses an electron, resulting in 6 valence electrons. An octet is not achieved, making it unstable as a Lewis dot structure.

C) Cl^- ion: By gaining an electron, chlorine gains a total of 8 valence electrons (7 from the atom + 1 gained), achieving a stable octet configuration. This is represented by a chlorine atom surrounded by 8 dots (7 dots representing its own electrons and 1 dot representing the gained electron).

D) Cl^{2+} ion: This ion loses an electron, leaving it with 6 valence electrons, similar to Cl^+ ion and lacking a stable octet.

8. Correct Option: C) He

Explanation:

A) Li: Loses 1 electron to achieve a stable octet configuration (filled outer shell), but lacks the exceptional stability of He's filled first energy level.

B) Mg: Similar to Li, loses 2 electrons for an octet configuration, gaining stability but not reaching He's unique level.

C) He: Despite only 2 electrons (not 8), its filled first energy level orbital provides unmatched stability due to optimal electron-proton ratio and minimal shielding.

D) Ne: Achieves a stable octet with 8 electrons, following the rule, but its stability doesn't surpass He's unique first-level configuration.

9. Correct Option: B) 4, 6, 7

Explanation:

Carbon (C): 4 - It's in Group 14 of the periodic table, meaning it has 4 electrons in its outermost shell.

Sulfur (S): 6 - Located in Group 16, so it has 6 valence electrons.

Fluorine (F): 7 - As the first element in Group 17 (the halogens), it has 7 valence electrons.

10. Correct Option: D) Both A and B.

Explanation:

A) CH_4: Carbon in CH_4 forms four covalent bonds with each hydrogen atom, achieving an octet configuration. So, it obeys the octet rule.

B) CO_2: Each oxygen atom in CO_2 forms a double bond with the central carbon, sharing two electrons each. Carbon forms four single bonds with each oxygen, sharing one electron each. This also satisfies the octet rule for all atoms.

C) BF_3: Boron in BF_3 forms three covalent bonds with each fluorine atom, sharing one electron each. This fulfills the octet rule for boron and fluorine.

D) Both A and B: As explained above, both CH_4 and CO_2 obey the octet rule.

11. Correct Option: A) BCl_3

Explanation:

A contracted octet molecule has fewer than 8 electrons around its central atom due to an inability to fulfill the usual octet rule.

A. BCl_3: Boron has 3 valence electrons, forms 3 bonds (sharing 1 each), resulting in 6 electrons (contracted octet).
B. CH_4: Carbon has 4, forms 4 bonds, gets 8 electrons (octet rule).

C. SF_6: Sulfur has 6, forms 6 bonds, gets 12 electrons (exceeds octet).
D. PCl_5: Phosphorus has 5, forms 5 bonds, gets 10 electrons (exceeds octet).

12. Correct Option: D) CH_4

Explanation:

A super octet molecule has more than 8 electrons around its central atom

A. PCl_5: Phosphorus (central atom) has 5 valence electrons, forms 5 bonds, acquires 10 electrons (super octet, violates octet rule).
B. IF_7: Iodine has 7, forms 7 bonds, gets 14 electrons (super octet).
C. SF_6: Sulfur has 6, forms 6 bonds, gets 12 electrons (super octet).
D. CH_4: Carbon has 4, forms 4 bonds, gets 8 electrons (octet rule satisfied).

13. Correct Option: D) Formal charge

Explanation:

The formal charge is the difference between the number of valence electrons in an isolated atom and the number of electrons assigned to that atom in the Lewis structure.

A) Atomic number: This signifies the total number of protons, not related to electron distribution in molecules.

B) Oxidation number: While related to charge, it focuses on electron gain/loss during reactions, not electron distribution within a molecule.

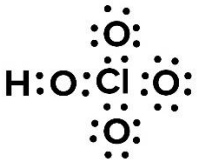
C) Valency: This indicates the combining capacity of an atom, not the specific electron allocation in a Lewis structure.

14. Correct Option: (C)

Explanation:

Formal Charge = V – N - $\frac{B}{2}$

Where, V = the number of valence electrons present in the neutral atom of the compound.
N = the number of non – bonding valence electrons in the same atom.
B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.



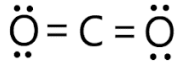
Formal Charge of Cl= 7 – 0 - $\frac{14}{2}$ = 0

15. Correct Option: A) 0

Explanation:

Formal Charge = V – N - $\frac{B}{2}$

Where, V = the number of valence electrons present in the neutral atom of the compound.
N = the number of non – bonding valence electrons in the same atom.
B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.



Formal Charge of C= 4 – 0 - $\frac{8}{2}$ = 0

16. Correct Option: A) Valency

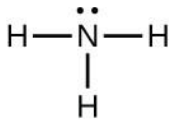
Explanation:

Valency describes the combining capacity of an atom, meaning the number of bonds it can form with other atoms.

B) Chemical bond: This refers to the actual force connecting atoms, not the potential for forming bonds.
C) Oxidation state: This captures the hypothetical charge an atom would have if all its bonds were ionic, not its bonding capacity.
D) Oxidation number: Similar to oxidation state, it focuses on the theoretical charge, not the inherent tendency to form bonds.

17. Correct Option: B) 3

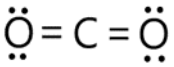
Explanation:
Nitrogen has 5 valence electrons ($2s^2 2p^3$). In NH_3, N forms covalent bonds with 3 hydrogen atoms. It shares one electron with each H, resulting in 3 shared electrons. This fulfils the octet rule (8 electrons for stability).



ammonia

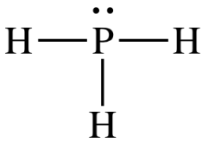
18. Correct Option: C) 4

Explanation:
In the CO_2 molecule, carbon (C) forms two double bonds with oxygen (O). Each double bond involves the sharing of two electron pairs. Therefore, the valency of carbon in CO_2 is 4.



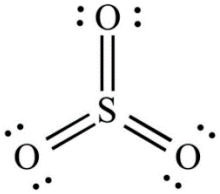
19. Correct Option: D) 3

Explanation:
In the PH_3 molecule, phosphorus (P) forms three covalent bonds with hydrogen (H). Each covalent bond involves the sharing of one electron pair. Therefore, the valency of phosphorus in PH_3 is 3.



20. Correct Option: B) 6

Explanation:
In the SO_3 molecule, sulphur (S) forms six covalent bonds with oxygen (O). Each covalent bond involves the sharing of one electron pair. Therefore, the valency of sulphur in SO_3 is 6.



LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Correct Option: B) 16

Explanation:

Element 20 (calcium) has 2 valence electrons and seeks stability through the octet rule (8 outer shell electrons).

A) 11 (Sodium, Na): Sodium typically loses one electron to achieve a stable electron configuration.

B) 16 (Sulphur, S): Sulphur typically gains two electrons to achieve a stable electron configuration.

C) 18 (Argon, Ar): Argon is a noble gas with a stable electron configuration.

D) 10 (Neon, Ne): Neon is a noble gas with a stable electron configuration.

Therefore, atom with atomic number 20 is most likely to combine chemically with the atom of atomic number 16.

2. Correct Option: B) F^+

Explanation:

F has 7 electrons in its valence shell. Thus, to attain stability, it should have lost one electron.

3. Correct Option: A) XY_2

Explanation:

Given, X....[Ar] $4s^2$

X is a metal and forms X^{+2} ion

Y....[Ne] $3s^23p^5$

Y is a non-metal and forms Y^- ion

X^{+2} Y^-

Applying criss-cross method the formula is written as XY_2

4. Correct Option: C) H_2O, NH_3, CO, CH_4

Explanation:

A: PCl_5 violates octet rule (P has 10 electrons).

B: BH_3 violates octet rule (B has 6 electrons).

C: All atoms in H_2O, NH_3, CO, and CH_4 achieve 8 electrons (octet rule).

D: CO_2 and $AlCl_3$ violate octet rule (O with 7, Al with 6 electrons).

5. Correct Option: D) IF_7

Explanation:

A. Carbon (C) in CH_4: 4 valence electrons + 4 shared with H (1 each) = 8 electrons (octet rule satisfied).

B. Carbon (C) in CO_2: 4 valence electrons + 4 shared with O (2 each) = 8 electrons (octet rule satisfied).

C. Nitrogen (N) in NF_3: 5 valence electrons + 3 shared with F (1 each) = 8 electrons (octet rule satisfied).

D. Iodine (I) in IF_7: 7 valence electrons + 7 shared with F (1 each) = 14 electrons (significantly exceeds octet rule).

6. Correct Option: C) 14

Explanation:

Xenon (Xe): It has 8 valence electrons in the outermost shell ($5s^25p^6$).

Fluorine (F): Each F atom has 7 valence electrons ($1s^22s^22p^5$).

1. Shared electrons: XeF_4 has 4 F atoms, and each forms a single bond with Xe (1 shared electron/bond). So, Xe shares 4 electrons (4 x 1).

2. Remaining valence electrons in Xe: 8 valence electrons - 4 shared electrons = 4 electrons.

3. Lone pairs in Xe: Divide remaining electrons by 2 (as each lone pair has 2 electrons): 4 electrons / 2 = 2 lone pairs.
4. Lone pairs in F: Each F has 3 lone pairs (7 valence electrons - 1 shared electron = 6 electrons / 2 = 3 lone pairs).
Total lone pairs: 2 lone pairs (Xe) + 3 lone pairs x 4(4 x F) = 14 lone pairs.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

7. Correct Option: A, B , C

Explanation:
Formal Charge = V – N - B/2
Where, V = the number of valence electrons present in the neutral atom of the compound.
N = the number of non – bonding valence electrons in the same atom.
B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.
Number 2 O atom: FC = 6 – 4 – 4/2 = 0
Number 1 O atom: FC = 6 – 2 – 6/2 = 1
Number 3 O atom: FC = 6 – 6 – 2/2 = -1

8. Correct Option: A, B, C

Explanation:
Expanded octet refers to a central atom having more than 8 electrons around it due to exceptions to the octet rule. Among the given options:
A) SF_6: Sulphur (central atom) has 6 valence electrons and shares 6 electrons with fluorine (1 each), resulting in 12 electrons around it (expanded octet).
B) PCl_5: Phosphorus has 5 valence electrons and shares 1 with each chlorine, leading to 10 electrons around it (expanded octet).
C) IF_7: Iodine has 7 valence electrons and shares 1 with each fluorine, giving it 14 electrons around it (expanded octet).
D) $BeCl_2$: Beryllium only has 2 valence electrons and shares 1 with each chlorine, totaling 4 electrons (expanded octet rule not satisfied).

9. Correct Option: C, D

Explanation:
Duplet configuration means having only two valence electrons in the outermost shell.
A) H^-: Hydrogen gains one electron, resulting in two (duplet) around it.
B) H_2: Each hydrogen shares one electron with the other, both achieving two (duplet) electrons.
C) Li^{2+}: Lithium typically loses 1 electron to achieve a duplet configuration. In this case, Li^{2+} has lost 2 valence electrons, leaving it with one valence electron.
D) Be^{3+}: Beryllium loses three electrons, leaving it with no valence electrons (not a duplet).

10. Correct Option: B) By sharing of electrons

Explanation:
Octet rule states that atoms tend to have 8 electrons in their outermost shell (valence shell) to achieve stability.
A. Electron transfer: While transferring electrons can sometimes lead to an octet configuration (e.g., ionic bonding), it's not the primary path due to potential energy imbalances created.
C. Gaining electrons: This can contribute to octet configuration, but it depends on the initial electron count and may not always be applicable.
D. Not sharing electrons: This wouldn't fulfill the octet rule, as electrons remain isolated and don't contribute to filling the outer shell.

COMPREHENSION TYPE

Passage-I:

11. Correct Option: B) 0, 3, 2, 0

Explanation:

Helium (He): Helium has a full outer shell with 2 electrons ($1s^2$). It doesn't need to gain or lose electrons to achieve stability, so its valency is 0.

Phosphorus (P): Phosphorus has 5 valence electrons ($3s^23p^3$). It can gain 3 electrons to achieve an octet configuration (P^{3-}), resulting in a valency of 3. Alternatively, it can share 5 electrons with other atoms ($3s^23p^2$), also achieving an octet, leading to a valency of 5.

Sulphur (S): Sulfur has 6 valence electrons ($3s^23p^4$). It can gain 2 electrons (S^{2-}) or share 4 electrons ($3s^23p^2$) to reach an octet, resulting in valencies of 2 or 4, respectively. Additionally, it can participate in hypervalent bonding, sharing 6 electrons ($3s^23d^2$), leading to a valency of 6.

Neon (Ne): Similar to helium, neon has a full outer shell with 8 electrons ($1s^22s^22p^6$). It doesn't need to gain or lose electrons, so its valency is 0.

12. Correct Option: B) 1, 7

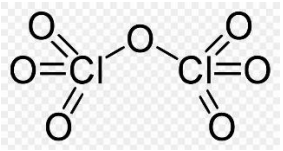
Explanation:

A) Cl in HCl:

In HCl, chlorine shares one electron with hydrogen to form a covalent bond. Its valence electrons are $3s^23p^5$, and sharing one electron allows it to achieve a stable octet configuration. Therefore, valency of Cl in HCl is 1

B) Cl in Cl_2O_7:

In the Cl_2O_7 molecule, each chlorine (Cl) forms 3 double bonds and 1 single bond with oxygen (O). Valency of Cl in Cl_2O_7 = 3 x 2 + 1 = 7



13. Correct Option: C) 3

Explanation:

Al has 3 valence electrons ($3s^23p$). It shares 1 electron each with 3 oxygen atoms in Al_2O_3. Sharing 3 electrons allows Al to reach a stable octet (8 total electrons).

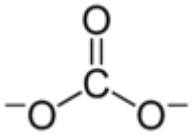
Passage-II:

14. Correct Option: D) 0

Explanation:

Formal Charge = V – N - $\frac{B}{2}$

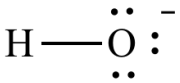
Where, V = the number of valence electrons present in the neutral atom of the compound.
N = the number of non – bonding valence electrons in the same atom.
B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.



Formal Charge of C = 4 – 0 - $\frac{8}{2}$ = 0

15. Correct Option: C) -1

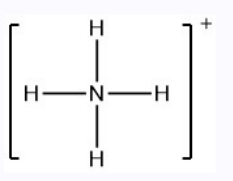
Explanation:



Formal Charge of O = 6 – 6 – $\frac{2}{2}$ = -1

16. Correct Option: A) 1

Explanation:



Formal Charge of N = 5 – 0 - $\frac{8}{2}$ = 1

MATRIX MATCHING TYPE

17. a) R b) S c) Q d) P

18. Correct Option: a)-R), b)-S), c)-Q), d)-P)

Explanation:

a) Boron (B): 3

- It has the electron configuration: $1s^2 2s^2 2p^1$.
- Valence electrons are those in the outermost shell, so in this case, the single electron in the 2p orbital is considered the valence electron.

b) Calcium (Ca): 2

- Its electron configuration is: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$.
- The outermost shell (4s) contains two valence electrons.

c) Bromine (Br): 7

- Its configuration is: $1s^2 2s^2 2p^6 3s^2 3p^5$.
- Similar to boron, the seven valence electrons reside in the outermost shell (3p), with five electrons contributing to the valence count.

d) Krypton (Kr): 8

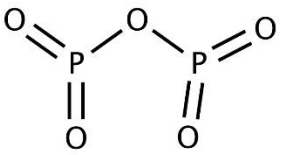
- Its configuration is: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$.
- Although it has electrons in the 4d orbital, the valence electrons are considered those in the outermost 4p orbital, which has a full eight electrons, making Kr a noble gas.

INTEGER TYPE

19. Correct Option: 5

Explanation:

In the P_2O_5 molecule, each P forms 2 double bonds and 1 single bond with O. Each double bond involves the sharing of two electron pairs. And one single bond involves the sharing of 1 electron pair. Valency of P in P_2O_5 = 2 x 2 + 1 = 5

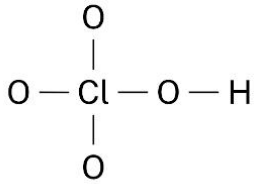


20. Correct Option: 3

Explanation:

Formal Charge = V – N - $\frac{B}{2}$

Where, V = the number of valence electrons present in the neutral atom of the compound.
N = the number of non – bonding valence electrons in the same atom.
B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.



Formal Charge of Cl = 7 – 0 – $\frac{8}{2}$ = 3

Chemical Bond

LEVEL – I

1. Correct Option: A) Transfer of electron

Explanation:

Ionic bonds are formed by the complete transfer of electrons between atoms. One atom loses electrons and becomes a positively charged ion (cation), while another atom gains electrons and becomes a negatively charged ion (anion).

B) Sharing of electrons defines covalent bonds, where atoms share electrons to achieve a stable octet configuration.

C) While loss of electrons is involved in ionic bond formation, it's not the complete picture. Both loss and gain of electrons are necessary for the ionic bond to form.

D) Gain of electrons alone can occur in other types of bonding, not just ionic.

2. Correct Option: B) Electrostatic force

Explanation:

Ionic compounds consist of oppositely charged ions held together by strong electrostatic forces of attraction. These forces arise due to the Coulombic interaction between positive and negative charges.

A) Van der Waals forces are much weaker intermolecular forces that exist between neutral atoms or molecules due to temporary fluctuations in electron distribution. While they exist to some extent in ionic compounds, they are not the primary force holding them together.

C) Hydrogen bonding involves specific interactions between polar molecules with hydrogen atoms bonded to highly electronegative elements like oxygen, nitrogen, or fluorine. It's not relevant in ionic compounds, which lack these functionalities.

D) Repulsive forces occur between like charges and cannot form a stable structure like an ionic compound.

3. Correct Option: A) Electro negative element

Explanation:

A) Electropositive elements: These tend to lose electrons easily and become positively charged ions (cations). They have low electronegativity, meaning they don't attract electrons strongly.

B) Electronegative elements: These tend to gain electrons readily and become negatively charged ions (anions). They have high electronegativity, meaning they have a strong pull towards electrons.

C) Non-metals: While most non-metals are electronegative and gain electrons, some can exhibit varying behavior depending on the context. For example, hydrogen can sometimes lose an electron, and halogens may share electrons in covalent bonds.

4. Correct Option: B) Electropositive elements

Explanation:

A) Electropositive elements: These tend to lose electrons easily and become positively charged ions (cations). They have low electronegativity, meaning they don't attract electrons strongly.

B) Electronegative elements: These tend to gain electrons readily and become negatively charged ions (anions). They have high electronegativity, meaning they have a strong pull towards electrons.

C) Non-metals: While most non-metals are electronegative and gain electrons, some can exhibit varying behavior depending on the context. For example, hydrogen can sometimes lose an electron, and halogens may share electrons in covalent bonds

5. Correct Option: B) loss of 2 electrons

Explanation:

Magnesium (Mg) is an electropositive element, meaning it tends to lose electrons to achieve stability. It has 2 valence electrons in its outermost shell ($3s^2$), and losing these 2 electrons would give it the stable electronic configuration of a noble gas (neon) with a full outer shell (8 electrons).

A) & C) Gaining electrons wouldn't be favorable for Mg as it already has 2 valence electrons and wouldn't achieve a full outer shell with 3 or 4 electrons.

D) Losing 1 electron wouldn't result in a stable configuration either. Mg would have 1 remaining valence electron, leaving it positively charged but not achieving the same stability as losing both and resembling a noble gas.

6. Correct Option: D) Ionic bond

Explanation:

Potassium (K) is an electropositive element, readily losing its single valence electron to become K^+. Chlorine (Cl) is an electronegative element, readily gaining an electron to become Cl^-. This complete transfer of electrons from K to Cl defines an ionic bond, not a covalent or dative bond.

7. Correct Option: C) Electro valency

Explanation:

A. Valency: This refers to the combining power of an atom, indicating how many bonds it can form. It doesn't necessarily specify whether electron transfer or sharing occurs.
B. Covalency: This term describes the formation of covalent bonds through electron sharing, not applicable to ionic bonds.
C. Electro valency: This specifically refers to the number of electrons an atom gains or loses when forming an ionic bond. This transfer of electrons allows each atom to achieve a stable outer shell configuration.
D. Atomic number: This is the number of protons in an atom's nucleus and doesn't change during bond formation.

8. Correct Option: A) +1

Explanation:

Potassium (K) is an electropositive element with one valence electron in its outermost shell ($4s^1$). In KCl, K atoms readily lose their single valence electron to achieve a stable octet configuration similar to a noble gas (argon) with a full outer shell (8 electrons). Losing one electron results in a positively charged potassium ion (K^+).

Electrovalency represents the number of electrons lost or gained. Since K loses one electron, its electrovalency in KCl is +1.

9. Correct Option: B) K and O_2

Explanation:

A. Na and Ca: Both are metals and tend to lose electrons, making them less likely to form an ionic bond where one loses and the other gains.
B. K and O_2: Potassium (K) is a highly electropositive metal with a low ionization energy, readily losing one electron. Oxygen (O) is an electronegative non-metal with a high electron affinity, readily gaining two electrons. This electron transfer scenario creates oppositely charged ions, leading to ionic bonding.
C. O_2 and Cl_2: Both are non-metals, so sharing electrons in a covalent bond is more likely than complete transfer.
D. Br_2 and I_2: Both are halogens, forming covalent bonds with each other.

10. Correct Option: B) High charge on ions, large cation and small anion

Explanation:

High charge on ions: Greater charge difference leads to stronger electrostatic attraction, making the ionic bond more stable.

Large cation and small anion: This size difference allows for closer packing of the ions, maximizing electrostatic attraction and minimizing repulsive forces.

11. Correct Option: B) Non direction

Explanation:

Ionic bonds are formed due to the electrostatic attraction between oppositely charged ions. This attraction exists in all directions around each ion, not just in specific directions like covalent bonds.

12. Correct Option: D) They exhibit space isomerism

Explanation:

A. High melting and boiling points: Ionic compounds have strong electrostatic forces between oppositely charged ions, requiring high energy to overcome, leading to high melting and boiling points.
B. Good conductors in molten state: When molten, ionic compounds dissociate into free ions, enabling them to conduct electricity.
C. Exist as solids: Ionic compounds typically form stable crystal lattices due to the strong electrostatic forces, resulting in solid states.
D. Space isomerism: This type of isomerism arises from different spatial arrangements of atoms within a molecule. However, ionic compounds lack individual molecules and instead have repeating crystal structures. The arrangement of ions in these structures is determined by the minimization of repulsion and maximization of attraction, not by specific spatial relationships like in molecules.

13. Correct Option: B) Fused NaCl

Explanation:

A. Crystalline NaCl: Despite having charged ions arranged in a regular lattice, these ions are fixed in their positions and cannot move freely. This lack of mobile charges prevents conduction in solid NaCl.
B. Fused NaCl: When heated to a high temperature, NaCl melts, forming a molten state. In this state, the ions become free to move, allowing them to carry current and making the solution a good conductor.
C. Solid KCl: Similar to crystalline NaCl, the ions in solid KCl are fixed in their positions, leading to no electrical conductivity.
D. Diamond: Diamond is a covalent network solid where electrons are tightly shared between carbon atoms. This lack of free mobile charges also prevents electrical conduction.

14. Correct Option: A) Good conductors

Explanation:

Ionic compounds in their solid state are not good conductors of electricity. This is because the ions are fixed in their crystal lattice and cannot move freely. However, when ionic compounds are fused (melted) or dissolved in water (aqueous state), the situation changes. In these states, the ionic compounds dissociate into free ions. These ions are no longer fixed in place and can move freely. The freely moving ions can carry electric current, making the fused or aqueous solution a good conductor.

15. Correct Option: C) Solution state

Explanation:

Ionic compounds are typically solids in their pure state. In this solid state, the ions are fixed in a crystal lattice and cannot move freely. For a reaction to occur, the reactants need to collide with enough energy (activation energy) for the reaction to proceed. In a solid state, with fixed ions, collisions are less frequent and the activation energy is high, making reactions less likely. However, when ionic compounds are dissolved in water, they dissociate into free ions. These free ions can move around freely in the solution, increasing the frequency of collisions and lowering the activation energy. This makes ionic reactions much more likely to occur in solution

16. Correct Option: D) RbCl

Explanation:

- Non-directional bonds: These bonds lack a specific directionality and exist equally in all directions around the atom.
- RbCl: This is an ionic compound formed by the complete transfer of electrons from rubidium (Rb) to chlorine (Cl). The resulting oppositely charged ions (Rb^+ and Cl^-) attract each other due to electrostatic forces, which act in all directions around each ion.
- NCl_3, $BeCl_2$, and BCl_3: These are covalent compounds where atoms share electrons to achieve stable configurations. The sharing of electrons occurs in specific orbitals and leads to directional bonds with well-defined geometries (trigonal pyramidal for NCl_3, linear for $BeCl_2$, and trigonal planar for BCl_3).

17. Correct Option: A) Lattice energy/enthalpy

Explanation:

A. Lattice energy: This term precisely refers to the energy needed to break all the ionic bonds in one mole of a solid ionic compound, separating it into individual gaseous ions.
B. Electrostatic force: While electrostatic forces are the primary attraction between ions in an ionic compound, "electrostatic force" doesn't directly describe the energy required for separation.
C. Nuclear energy: This refers to energy within the nucleus of an atom, unrelated to the separation of ions in a compound.
D. Ionic energy: While this term might seem relevant, it's not a widely used or recognized term for lattice energy.

18. Correct Option: D) Charge and size of the ions

Explanation:

A. Charge on the ions: Higher charges lead to stronger electrostatic attraction between ions, resulting in higher lattice energy.
B. Size of the ions: Smaller ions can pack closer together, minimizing repulsion and maximizing attraction, leading to higher lattice energy.
C. Packing of the ions: The arrangement of ions in the crystal lattice also impacts the overall attraction and repulsion forces, influencing lattice energy.

19. Correct Option: C) Oppositely charged ions

Explanation:

Electrovalent compounds are formed by the complete transfer of electrons between atoms. This transfer results in the formation of positively charged cations and negatively charged anions. These oppositely charged ions are attracted to each other by strong electrostatic forces.

20. Correct Option: D) Al_2O_3 > CaO > $MgBr_2$ > NaCl

Explanation:

Lattice energy: Increases with higher charge on ions and smaller ion size.

Al_2O_3: Highly charged Al^{3+} cation and small O^{2-} anion lead to the highest lattice energy.

CaO: Smaller Ca^{2+} compared to Mg^{2+} and the same O^{2-} contribute to a higher lattice energy than $MgBr_2$.

$MgBr_2$: Larger Br^- anion reduces the lattice energy compared to CaO.

NaCl: Largest anion (Cl^-) among the options results in the lowest lattice energy.

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Correct Option: B) Fluorine

Explanation:

Potassium is an electropositive element with one valence electron it tends to lose easily. Among the options, fluorine is the most electronegative element, readily gaining an electron to achieve a stable configuration. This strong difference in electronegativity facilitates the complete transfer of the electron from potassium to fluorine, forming an ionic bond. While other halogens (chlorine, bromine, iodine) are also electronegative, their larger sizes compared to fluorine lead to slightly weaker attractions with potassium and potentially some covalent character in the bond.

2. Correct Option: A) LiCl > $BeCl_2$ > BCl_3 > CCl_4

Explanation:

Ionic character: Increases with decreasing electronegativity difference between the elements involved.

- LiCl: Li has the lowest electronegativity, leading to the largest difference with Cl and the most ionic character.
- $BeCl_2$: Be is slightly more electronegative than Li, resulting in a slightly smaller difference and less ionic character compared to LiCl.
- BCl_3: B is even more electronegative than Be, leading to a further decrease in electronegativity difference and a trend towards more covalent character.
- CCl_4: C is the most electronegative among the cations, resulting in the smallest difference with Cl and the least ionic character, bordering on covalent.

3. Correct Option: D) CaI_2

Explanation:

Ionic character decreases with increasing size and decreasing electronegativity of the anion. Iodine (I) in CaI_2 is the largest anion among the options, resulting in weaker electrostatic attraction with the Ca^{2+} cation. The electronegativity of I is also lower compared to F, Br, and Cl. This smaller electronegativity difference reduces the driving force for complete electron transfer, leading to a more covalent character in the bond.

4. Correct Option: D) Anhydrous and aqueous

Explanation:

- Anhydrous HCl: In this state, HCl exists as a gas. Gases generally lack sufficient mobile ions to conduct electricity effectively. Therefore, anhydrous HCl is a bad conductor.

- Aqueous HCl: When dissolved in water, HCl dissociates into H^+ and Cl^- ions. These freely moving ions can carry electric current, making aqueous HCl a good conductor.

5. Correct Option: A) Li

Explanation:

Electrovalent compounds are formed due to the complete transfer of electrons.

Lithium (Li) has the lowest ionization energy among the options, meaning it requires the least energy to lose its single valence electron. Losing this electron readily gives it a stable configuration similar to helium, making it highly favorable for Li to form ionic bonds.

While Na, Be, and Mg also tend to lose electrons, their higher ionization energies compared to Li make them slightly less inclined to form purely ionic bonds

6. Correct Option: A) X^+Y^-

Explanation:

The compound formed between a strongly electropositive element (X) and a strongly electronegative element (Y) with both being univalent (having a charge of +1 and -1 respectively) would be an ionic compound. In an ionic compound, electrons are transferred from the electropositive element to the electronegative element.

So, the correct representation for the compound formed would be X^+Y^-

7. Correct Option: A) A_2B_3

Explanation:

Element A needs to lose 3 electrons to achieve a stable octet configuration (similar to noble gases).

Element B needs to gain 2 electrons to achieve a stable octet configuration.

To minimize the number of ions formed, A loses 3 electrons while B gains 2 electrons per molecule. This leads to the formation of 2 A cations (A^{2+}) and 3 B anions (B^-) per molecule.

Combining these ions in the simplest ratio gives the formula A_2B_3.

8. Correct Option: A) Ionic

Explanation:

Cations are positively charged ions, and anions are negatively charged ions. These oppositely charged ions are attracted to each other due to electrostatic forces. When these ions pack together in a crystal lattice, they form an ionic compound.

Metallic compounds are composed of metal atoms, covalent compounds involve sharing electrons between atoms, and dipole-dipole interactions occur between neutral molecules.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Correct Option: A & D

Explanation:

Ionic bonds form between atoms when one atom (electron acceptor) gains one or more electrons from another atom (electron donor), resulting in oppositely charged ions attracted to each other by electrostatic forces.

A) High electron affinity: Electron acceptors need high electron affinity to readily attract and stabilize the donated electrons.

B) Less size of anion: Anion size has little bearing on ionic bond formation as long as the charges can interact effectively.

C) Low charge of anion: A low charge on the anion would reduce the electrostatic attraction with the cation, weakening the bond.

D) High electronegativity: High electronegativity reflects a strong attraction for electrons, making the atom a good electron acceptor.

10. Correct Option: A& B

Explanation:

Electrovalency refers to the number of electrons gained or lost by an atom to form an ion.

In ionic compounds, metals lose electrons (positive electrovalency) and non-metals gain electrons (negative electrovalency).

A. KCl: Potassium (K) loses 1 electron (electrovalency +1), and Chlorine (Cl) gains 1 electron (electrovalency -1).
B. NaCl: Sodium (Na) loses 1 electron (electrovalency +1), and Chlorine (Cl) gains 1 electron (electrovalency -1).
C. MgO: Magnesium (Mg) loses 2 electrons (electrovalency +2), and Oxygen (O) gains 2 electrons (electrovalency -2).
D. CaO: Calcium (Ca) loses 2 electrons (electrovalency +2), and Oxygen (O) gains 2 electrons (electrovalency -2).

11. Correct Option: A, B, C & D

Explanation:

A. They have high melting point and boiling point: Correct. The strong electrostatic forces between oppositely charged ions require a lot of energy to overcome, resulting in high melting and boiling points.
B. Ionic bonds are non-directional: Correct. Ionic bonds arise from the attraction between opposite charges, not specific orbital overlaps, making them non-directional.
C. Ionic compounds are hard and brittle: Correct. The rigid structure of ionic crystals and strong ionic bonds lead to hardness, but also make them susceptible to shattering rather than bending, making them brittle.
D. Ionic compounds exist as solids: Correct. The strong ionic forces favor tightly packed arrangements, resulting in solid-state structures.

12. Correct Option: A, B, C, D

Explanation:

Each statement represents the ionization process of a metal atom, where it loses electrons to achieve a stable, filled outer electron shell configuration. This electron loss results in the formation of a positively charged ion (cation).

COMPREHENSION TYPE

Passage – I:

13. Correct Option: D) X_2Y

Explanation:

Since element X is univalent (loses one electron) and element Y is bivalent (gains two electrons), two atoms of X will combine with one atom of Y to balance the charges.

This gives us the formula X_2Y, where the subscripts indicate the number of atoms of each element present in the compound.

14. Correct Option: C) 35

Explanation:

A) 10: This is neon (Ne), a noble gas with a full outer shell and very low electronegativity. It won't readily accept an electron from sodium, making an ionic bond unlikely.

B) 34: This is selenium (Se), which has some electronegativity but not enough to form a strong ionic bond with sodium. It's more likely to form a covalent bond.

C) 35: This is chlorine (Cl), which has a high electronegativity and readily accepts an electron from sodium. This forms the strong ionic compound sodium chloride (NaCl).

D) 37: This is rubidium (Rb), another metal with lower electronegativity than chlorine. While it can form an ionic compound with sodium, it wouldn't be as strong as with chlorine.

15. Correct Option: C) Higher electronegativity

Explanation:

In an ionic bond, one atom loses electrons and becomes a positively charged cation. This atom typically has lower electronegativity as it readily gives up electrons.

The other atom gains electrons and becomes a negatively charged anion. This atom has higher electronegativity as it attracts and holds electrons strongly.

Passage – II:

16. Correct Option: C) II < IV < III < I

Explanation:

The ionic character of a bond depends on the electronegativity difference between the participating atoms. Higher the difference, greater the ionic character.

- LiF: Lithium (Li) has low electronegativity (0.98), while Fluorine (F) has the highest (3.98). This large difference (3) leads to significant electron transfer, making LiF highly ionic.
- ClF_3: Chlorine (Cl) has moderate electronegativity (3.19), but the three Fluorines still create a considerable difference (around 2.81). ClF_3 is relatively ionic but less than LiF.
- K_2O: Potassium (K) has low electronegativity (0.82), and Oxygen (O) has moderate (3.44). The difference (2.62) suggests moderate ionic character.
- SO_2: Sulfur (S) has moderate electronegativity (2.58), and Oxygen (O) also has moderate (3.44). The smaller difference (around 0.86) makes SO_2 less ionic than the others.

17. Correct Option: D) II, III and IV

Explanation:

I) AgBr: Silver bromide (AgBr) is generally considered insoluble in water due to its strong ionic bond and high lattice energy.

II) $BaSO_4$: Barium sulfate ($BaSO_4$) is also highly insoluble in water due to its strong ionic bond and large size of the sulfate ion.

III) KOH: Potassium hydroxide (KOH) is a highly soluble ionic compound that readily dissolves in water to form potassium and hydroxide ions.

IV) $Ca_3(PO_4)_2$: Calcium phosphate ($Ca_3(PO_4)_2$) has low solubility in water due to its strong ionic bonds and large size of the phosphate ion.

18. Correct Option: D) Both (b) and (c)

Explanation:

For solubility in water, the trend should be decreasing as we move down a group (from Ca to Sr to B) or across a period (from F to Cl to Br). This is because the larger size of the ions or increased attraction between the ions and the smaller radius of the F^- ion make it harder for the ionic compound to dissolve in water.

The correct order for solubility in water is:

$BF_2 > SrF_2 > CaF_2$

KF > NaF > LiF

INTEGER TYPE

19. Correct Option: 4

Explanation:

In ionic bonds, the maximum electrovalence is determined by the number of valence electrons the atom can lose to achieve a stable, noble gas configuration. While some elements like aluminum can lose 3 electrons, the maximum electrovalence is typically +4, achieved by elements like beryllium and titanium.

20. Correct Option: 1

Explanation:

Chlorine initially has 7 valence electrons (in its outer shell). For stability, it aims to achieve a full outer shell like a noble gas, which typically requires 8 electrons. In NaCl, sodium loses 1 electron, which is then gained by chlorine, resulting in both sodium (Na^+) and chlorine (Cl^-) having stable 8-electron outer shells.

Covalent Bond and Coordinate Covalent Bond

LEVEL – I

1. Correct Option: B) Covalent bond

Explanation:

A) Ionic bond: Ionic bonds involve the complete transfer of electrons from one atom to another, not equal sharing.

B) Covalent bond: Covalent bonds are formed by the sharing of electron pairs between atoms, usually resulting in a stable electron configuration for both.

C) Dative bond: This is a type of covalent bond where one atom donates both electrons in the shared pair. While related, it's not the general definition of a bond formed by equal sharing.

D) Coordinate covalent bond: This is another term for a dative bond.

2. Correct Option: C) H_2O

Explanation:

NaCl, KCl, and $MgCl_2$: These compounds are all ionic compounds. They form through the transfer of electrons, not sharing, resulting in positively charged cations (Na^+, K^+, Mg^{2+}) and negatively charged anions (Cl^-).

H_2O: This is a covalent compound. Each hydrogen atom shares one electron with the oxygen atom, forming a stable configuration for all three atoms.

3. Correct Option: D) The number of electrons shared by an atom.

Explanation:

A) The number of electrons are accepted by an atom: This describes electronegativity, not covalency.

B) The number of electrons are donated by an atom: This applies to dative covalent bonds, but not the broader concept of covalency.

C) The number of electrons transferred by an atom: This definition applies to ionic bonds, not covalent bonds.

D) The number of electrons shared by an atom: This accurately describes the essence of covalency, where atoms pool their electrons to form stable configurations.

4. Correct Option: C) 4

Explanation:

Covalent is defined as the number of bonds an element forms with other elements in a molecule.

In a Carbon dioxide molecule, each Oxygen shares electrons with carbon, forming two bonds each with Carbon. As the total number of bonds formed with carbon is , hence, the covalency of carbon in carbon dioxide is 4.

5. Correct Option: C) 2

Explanation:

An oxygen molecule (O_2) consists of two oxygen atoms joined by a double bond. A double bond is a type of covalent bond where two pairs of electrons are shared between the atoms.

6. Correct Option: B) 3

Explanation:

Nitrogen molecules (N_2) are held together by a triple covalent bond. This means that each nitrogen atom shares three pairs of electrons with the other nitrogen atom.

7. Correct Option: C) 3

Explanation:

Elements in Group IVA (also known as Group 14) have 4 valence electrons in their outermost shell. To achieve a stable octet configuration (8 valence electrons), they can share electrons with other atoms. However, sharing electrons comes with a stability limit. Sharing

more than 3 electrons with another atom can lead to repulsion between the electron clouds, making the bond unstable. Therefore, for two IVA group elements to bond with each other, the maximum number of shared electron pairs is 3, resulting in a triple bond.

8. Correct Option: D) F-F

Explanation:

A nonpolar covalent bond occurs when the electronegativity difference between the two atoms is very small or nonexistent. In this case, the electrons are shared equally between the atoms.

Among the options provided, the bond between two fluorine atoms (F-F) is nonpolar because fluorine atoms have similar electronegativities, so the electron pair is shared equally.

9. Correct Option: B) Low melting point

Explanation:

A) High solubility in water: This depends on the polarity of the covalent compound. Polar covalent compounds generally dissolve in water, while nonpolar ones do not.

B) Low melting point: Covalent bonds are weaker than ionic bonds due to the sharing of electrons instead of complete transfer. This results in lower melting and boiling points for covalent compounds.

C) High electrical conductivity: Covalent compounds generally do not have free ions and are poor conductors of electricity.

D) High boiling point: Similar to melting point, covalent compounds tend to have lower boiling points due to weaker bonds compared to ionic compounds.

10. Correct Option: C) They show ionic reactions

Explanation:

A) They may exhibit isomerism: Isomerism is the phenomenon where different molecules have the same chemical formula but different structures. This can occur in covalent compounds due to the way the atoms are arranged.

B) They have low melting and boiling points: As mentioned earlier, covalent bonds are weaker than ionic bonds, leading to lower melting and boiling points for covalent compounds.

D) They show molecular reactions: Covalent compounds react by breaking and forming new covalent bonds between molecules. This is known as a molecular reaction.

11. Correct Option: B) Absence of free electrons

Explanation:

Diamond is a bad conductor of electricity because it lacks free electrons. In diamond, each carbon atom forms strong covalent bonds with four neighboring carbon atoms, creating a rigid three-dimensional network structure known as a tetrahedral arrangement. This arrangement locks the valence electrons in place, preventing them from moving freely and conducting electricity.

12. Correct Option: A) Small cation and large anion

Explanation:

Fajans' rule: ionic bonds favor small cations and large anions. This maximizes charge density and polarization, strengthening electrostatic attraction and electron transfer.

B) Small cation and small anion: Repulsion

C) Large cation and large anion: Weaker attraction

D) Large cation and small anion: Less effective electron transfer

13. Correct Option: B) $SiCl_4$

Explanation:

Factors affecting covalent character:

1. Electronegativity difference: Smaller difference indicates more sharing of electrons, leading to more covalent character.
2. Size of cation: Smaller cation leads to higher charge density and greater polarization of the anion, increasing covalent character.

A. NaCl: High electronegativity difference and large Na^+ cation suggest strong ionic character.
B. $SiCl_4$: Smaller electronegativity difference and smaller Si^{4+} cation compared to Na^+ imply more covalent character.
C. $AlCl_3$: Smaller electronegativity difference than NaCl but larger Al^{3+} cation than Si^{4+} leads to less covalent character than $SiCl_4$.
D. $AgCl_2$: High electronegativity difference and large Ag^+ cation indicate strong ionic character.

14. Correct Option: A) $KCl < CaCl_2 < AlCl_3 < SnCl_4$

Explanation:

Covalent character generally increases with:

1. Decreasing electronegativity difference: Smaller difference between the electronegativity of the cation and anion leads to more sharing of electrons and a more covalent bond.
2. Smaller cation size: Smaller cation creates a more concentrated positive charge, attracting the shared electrons more strongly and increasing covalent character.

15. Correct Option: A) Ni^{3+}

Explanation:

Polarizing power of an ion is determined by its charge-to-radius ratio. Higher the ratio, greater the ability of the ion to distort the electron cloud of surrounding anions, leading to greater polarizability.

16. Correct Option: A) Maximum polarization is done by a cation of high charge.

Explanation:

Maximum polarization is done by a cation of high charge

According to Fajans rules, polarization will be favoured by small size of cation, large size of anion and high charge of cation and anion.

17. Correct Option: C) $PbCl_2$

Explanation:

Factors affecting covalent character:

1. Electronegativity difference: Smaller difference indicates more sharing of electrons and a more covalent bond.
2. Size of cation: Smaller cation creates a more concentrated positive charge, attracting the shared electrons more strongly and increasing covalent character.

18. Correct Option: C) Coordinate covalent bond

Explanation:

A. Ionic bond: This type of bond involves the complete transfer of electrons from one atom to another, not just sharing.
B. Covalent bond: This is the sharing of electrons equally between two atoms.
C. Metallic bond: This is a type of bonding where electrons are delocalized across a metal lattice, not shared between specific atoms.
D. Coordinate covalent bond (dative covalent bond): This type of bond involves the sharing of an electron pair donated by only one atom, with the other atom providing an empty orbital.

19. Correct Option: D) Coordinate Covalent bond

Explanation:

- NH_3 (ammonia) acts as a Lewis base: It has a lone pair of electrons on the nitrogen atom that it can donate.
- BF_3 (boron trifluoride) acts as a Lewis acid: It has an empty orbital on the boron atom that can accept electron density.

In the formation of their adduct, NH_3 donates its lone pair to the empty orbital of BF_3, resulting in a coordinate covalent bond. This bond forms readily due to the strong attraction between the electron-rich NH_3 and the electron-deficient BF_3.

20. Correct Option: B) Electrons of the atom are shared between two atoms

Explanation:

A) Electrons are equally shared by the atoms: This is a characteristic of a regular covalent bond, not a coordinate covalent bond.

C) Hydrogen bond is formed: Hydrogen bonds can be present alongside coordinate covalent bonds, but they are not directly related to the electron sharing mechanism.

D) Electrons are not shared by the atoms: This is incorrect. The definition of a coordinate covalent bond involves the sharing of an electron pair, even though both electrons in the pair come from one atom.

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Correct Option: D) $BeCl_2$

Explanation:

- Electronegativity difference: Smaller difference indicates less complete electron transfer, leading to more covalent character. Be has high electronegativity (1.57), closer to Cl (3.16) compared to Mg, Sr, and S.
- Cation size: Smaller cations have higher effective nuclear charge, attracting electrons more tightly and reducing ionic character. Be^{2+} is the smallest cation among the options.

Therefore, the combination of smaller electronegativity difference and smaller cation size makes $BeCl_2$ the least ionic among the given compounds.

2. Correct Option: D) CaI_2

Explanation:

As per Fajan's rule as the size of anion increases covalent character increase.

Order of anions size: $F^- < Cl^- < Br^- < I^-$

So, order of covalent character is $CaF_2 < CaCl_2 < CaBr_2 < CaI_2$

3. Correct Option: B) Fluorine

Explanation:

As per Fajan's rule as the size of anion increases covalent character increase. Since fluorine is smaller among all, so the answer is fluorine.

4. Correct Option: D) MgS

Explanation:

Factors affecting solubility:

1. Hydration energy: This is the energy released when ions are surrounded by water molecules. Larger ions interact more favourably with water, leading to higher hydration energy and generally better solubility.
2. Lattice energy: This is the energy required to break apart the crystal lattice of a solid ionic compound. It depends on the charges of the ions and the distance between them. Higher charges and smaller distances lead to higher lattice energy and lower solubility.

A. NaCl: Na^+ is larger than Mg^{2+}, and Cl^- is smaller than S^{2-}. This gives Na^+ a potential advantage in hydration energy but also means a tighter lattice in NaCl due to smaller ions.
B. Na_2S: Both Na^+ and S^{2-} are larger than Cl^-, potentially favoring hydration energy, but the lattice has two S^{2-} interacting with smaller Na^+, again suggesting a tighter structure.
C. $MgCl_2$: Mg^{2+} is smaller than Na^+, which could hinder its hydration. Cl^- is smaller than S^{2-}, potentially contributing to a tighter lattice.
D. MgS: Both Mg^{2+} and S^{2-} are larger ions, which could improve hydration energy. However, the interaction between a small Mg^{2+} and a large S^{2-} in the lattice can still be strong, potentially leading to higher lattice energy.

5. Correct Option: D) 4

Explanation:

Covalency refers to the number of covalent bonds an atom forms with other atoms to achieve a stable electron configuration.

6. Correct Option: A) Covalent bond

Explanation:

When two non-metals come together, they are more likely to share electrons rather than transfer them completely. This sharing of electrons results in a covalent bond.

B) Ionic bond: Ionic bonds typically form between a metal and a non-metal, involving the complete transfer of electrons from the metal to the non-metal.

C) Dative bond: While a type of covalent bond, it's a specific scenario where one atom donates both electrons in the shared pair. It's not the general term for bonds between non-metals.

D) Metallic bond: Metallic bonds form between metal atoms through a delocalized sea of electrons, not between individual atoms like in the case of non-metals.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

7. Correct Option: A, B, D

Explanation:

A & B: Smaller Li^+ ion creates stronger polarization of Cl^- leading to weaker ionic bonds in LiCl, resulting in lower melting point and higher solubility in organic solvents.

C: Smaller size and higher charge density of Li^+ make LiCl slightly more ionic than NaCl. However, the difference is minor.

D: Ionic conductivity relies on mobile ions. Smaller and more tightly packed Li^+ ions in LiCl make it less conductive than NaCl with larger and more loosely packed Na^+ ions.

8. Correct Option: A, D

Explanation:

A. H_3O^+ (hydronium ion): In this ion, a water molecule (H_2O) donates a lone pair of electrons from its oxygen atom to an empty orbital on a hydrogen ion (H^+), forming a coordinate covalent bond between the oxygen and the hydrogen.
B. BF_3 (boron trifluoride): This molecule does not contain any coordinate covalent bonds. It has three covalent bonds between boron and fluorine atoms.
C. H_2O (water): This molecule only has regular covalent bonds between hydrogen and oxygen atoms.
D. NH_4^+ (ammonium ion): An ammonia molecule (NH_3) donates a lone pair of electrons from its nitrogen atom to an empty orbital on a hydrogen ion (H^+), forming a coordinate covalent bond between the nitrogen and the hydrogen.

9. Correct Option: C) Lewis base

Explanation:

A. Lewis acids: These are electron pair acceptors and tend to have empty orbitals. They do not donate electrons.
B. Donors: This is a generic term that can apply to any substance that donates something, including electrons. However, it doesn't specifically refer to electron pair donors in the context of Lewis theory.
C. Lewis bases: These are electron pair donors and have lone pairs of electrons or negative charges, making them capable of donating electrons to Lewis acids to form coordinate covalent bonds.
D. Acceptor: This term is similar to Lewis acid, but it's not as specific in the context of Lewis theory.

10. Correct Option: A, B

Explanation:

A. Ionic bond: The NH_4^+ (ammonium) ion and the Cl^- (chloride) ion are held together by an ionic bond.
B. Covalent bonds: Within the NH_4^+ ion, covalent bonds exist between the nitrogen atom and each of the four hydrogen atoms. These bonds involve the sharing of electrons between the atoms.

11. Correct Option: A, D

Explanation:

A nonpolar bond is a bond in which the electrons are shared equally between the two atoms. This results in no positive or negative end to the bond.

Electronegativity is a measure of an atom's ability to attract electrons in a bond. The greater the difference in electronegativity between two atoms, the more polar the bond will be.

- CCl_4 All four atoms are identical (carbon and chlorine), so there is no difference in electronegativity and the electrons are shared equally.
- P_4 All four atoms are identical (phosphorus), so there is no difference in electronegativity and the electrons are shared equally.

12. Correct Option: A) Lewis acid

Explanation:

A. Lewis acids: These are electron pair acceptors and possess empty orbitals to accommodate electron pairs donated by Lewis bases. They do not donate electrons themselves.
B. Donar: This is a general term for any substance that gives something, including electrons. But it's not specific to the context of Lewis theory and electron pair donation.
C. Lewis base: These are electron pair donors with lone pairs of electrons or negative charges, capable of donating electrons to Lewis acids to form coordinate covalent bonds.
D. Acceptor: Similar to Lewis acid, but less specific within Lewis theory.

13. Correct Option: D) Both b and C

Explanation:

B) The ligand has no lone pair of electrons hence it is called a Lewis base: In a coordinate covalent bond, the ligand is the electron donor and must have a lone pair to donate. It cannot be a Lewis base without a lone pair.

C) The metal ion has empty orbitals hence it accepts a lone pair of electrons from the ligand: While some Lewis acids do have empty orbitals, it's not a strict requirement for accepting an electron pair. Lewis acids can also accept electrons into higher-energy orbitals or delocalize the donated electrons throughout their structure.

COMPREHENSION TYPE

Passage – I:

14. Correct Option: B) HF > H_2O > SO_2 > NH_3

Explanation:

Polarity is a measure of the separation of electric charge within a molecule. A molecule is polar if it has a positive end and a negative end. The polarity of a molecule is determined by the difference in electronegativity between the atoms in the molecule. Electronegativity is a measure of an atom's ability to attract electrons towards itself in a bond.

The greater the difference in electronegativity between two atoms, the more polar the bond between them will be. And the more polar bonds a molecule has, the more polar the molecule will be.

In the given options, HF has the largest electronegativity difference, followed by H_2O, SO_2, and NH_3. Therefore, the correct order of decreasing polarity is HF > H_2O > SO_2 > NH_3.

15. Correct Option: D) Fajan's rule

Explanation:

Fajan's rule states that smaller cations and larger anions favor covalent character due to increased polarization of the anion by the cation.

- In $AlCl_3$, the Cl^- anion is larger than the F^- anion in AlF_3. This makes the electron cloud of Cl^- easier to polarize by the smaller Al^{3+} cation in $AlCl_3$, leading to more covalent character.
- AlF_3 has both a smaller anion (F^-) and a smaller cation (Al^{3+}) compared to $AlCl_3$. This minimizes polarization and strengthens the ionic character of the bond.

16. Correct Option: A) AlF_3

Explanation:

- Electronegativity difference: Fluorine has the highest electronegativity among halogens, leading to the largest electronegativity difference with aluminium in AlF_3, promoting an ionic bond.
- Cation size: While all halides have larger anions than the Al^{3+} cation, fluorine, being the smallest halogen, minimizes potential polarization effect in AlF_3, further strengthening its ionic character.

MATRIX MATCHING TYPE

17. Correct Option: a)-Q), b)-P), c)-R), d)-S)

Explanation:

a) Diamond is a pure covalent network solid where carbon atoms share electrons to form strong covalent bonds.

b) Table salt (NaCl) is an ionic compound where Na^+ and Cl^- ions are held together by strong electrostatic forces due to complete electron transfer.

c) Ammonium ion is a polyatomic ion where N forms covalent bonds with H atoms. Additionally, N donates a lone pair to an H^+ ion, forming a dative covalent bond.

d) Ammonium chloride combines characteristics of the individual ions: ionic bond between NH_4^+ and Cl^-, covalent bonds within NH_4^+, and a dative bond between N and H^+.

18. Correct Option: a)-Q), b)-S), c)-P), d)-R)

Explanation:

a) AlF_3 is an ionic compound due to the high electronegativity difference between Al and F, resulting in complete electron transfer from Al to F and electrostatic attraction between ions.

b) CCl_4 is a covalent molecule as all atoms share electrons. Its tetrahedral structure and weak intermolecular forces lead to a liquid state at room temperature.

c) Graphite is a layered structure with weak van der Waals forces between sheets, making it slippery and a good lubricant.
d) BCl_3 has an empty orbital and can accept an electron pair from a Lewis base, acting as a Lewis acid in various reactions.

INTEGER TYPE

19. Correct Option: 2

Explanation:

In hydrogen peroxide, each oxygen atom is bonded to one hydrogen atom on one side and to another oxygen atom on the other side. Each bond between oxygen and hydrogen involves the sharing of one electron pair, meaning each oxygen atom participates in two covalent bonds. Since each oxygen atom has six valence electrons, sharing one electron pair in each bond fulfils the octet rule for both oxygen atoms, achieving stability

20. Correct Option: 3

Explanation:

Each nitrogen atom has five valence electrons in its outer shell. To achieve the stable octet configuration (eight electrons in the outer shell), each nitrogen atom shares three electrons with the other nitrogen atom. Each shared pair of electrons forms a covalent bond, resulting in a triple bond (N≡N).

Valence Bond Theory (VBT)

LEVEL – I

1. Correct Option: B) Decreases the bond length

Explanation:

When atoms form bonds, they share electrons in their outermost orbitals. This sharing attracts the positively charged nuclei of the atoms closer together. This attraction overcomes the opposing force of electron-electron repulsion between the shared electrons and the inner electrons of the atoms. As a result, the internuclear distance (bond length) decreases compared to the distance between the isolated atoms.

2. Correct Option: A) P and P orbitals

Explanation:

Stronger bonds result from greater overlap between the orbitals of bonding atoms. P orbitals have more complex shapes than S orbitals, allowing for more extensive overlap and stronger interactions. When P orbitals overlap end-to-end (head-on), the overlap is maximized, leading to a particularly strong bond.

3. Correct Option: B) Orbitals overlap will not occur

Explanation:

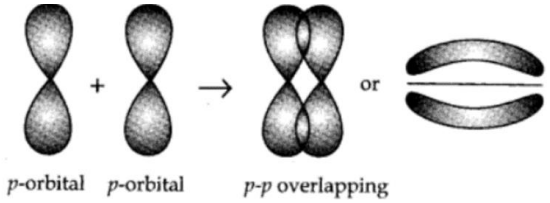
The Pauli Exclusion Principle states that two electrons within an atom cannot occupy the same quantum state, which includes having the same spin. In other words, electrons with the same spin cannot pair up.

For a diatomic molecule to form, orbitals containing unpaired electrons on individual atoms must overlap, allowing sharing of electron density and bonding. Since both atoms have electrons with the same spin, there are no unpaired electrons available for overlap, thus preventing bond formation.

4. Correct Option: D) P-P orbitals side wise manner

Explanation:

• Pi (π) bonds: Formed by the sideways or lateral overlapping of p orbitals. The p orbitals don't overlap end-to-end, but their lobes interact sideways, creating electron density above and below the plane of the atoms. This characterizes the pi bond.



• Sigma (σ) bonds: Formed by the axial or end-to-end overlapping of atomic orbitals. This overlap can occur between s-s orbitals, s-p orbitals, or p-p orbitals (where they overlap directly along their internuclear axis in end-to-end manner). This head-on overlap creates a cylindrical electron density distribution around the bond axis, which defines the sigma bond.

5. Correct Option: B) sp - sp

Explanation:

As we know that hybrid orbital always gives better overlap than pure orbitals because of the same size and equivalent energy. Out of various hybrid orbitals above the most efficient overlapping is shown by the sp-sp hybrid orbitals as there are 50% s and 50% p.

Since greater the s-character, smaller is the hybridized orbital and more is the tendency for overlapping. Also, more is the bond energy, smaller is the bond length.

6. Correct Option: B) BF_3

Explanation:

A. $BeCl_2$ (sp): Only 2 valence electrons, forms 2 bonds using unhybridized orbitals.
B. BF_3 (sp^2): 3 valence electrons, forms 3 bonds by combining 1s and 2p orbitals into trigonal planar sp^2 hybrids.
C. NH_3 (sp^3): 5 valence electrons, forms 3 bonds and has 1 lone pair, achieved by combining 1s and 3p orbitals into tetrahedral sp^3 hybrids.
D. XeF_3 (sp^3d): 8 valence electrons, forms 3 bonds and has 2 lone pairs, requiring the involvement of 1s, 3p, and 1d orbital in sp^3d hybridization.

7. Correct Option: D) BF_3

Explanation:

IF_3 has bent-T shape

PCl_3 and NH_3 has pyramidal geometry

BF_3 has trigonal planar geometry. BF_3 does not have any lone pair, the three F atoms are arranged around central Boron at a bond angle of 120°

8. Correct Option: B) Trigonal planar, sp^2

Explanation:

BF_3 has trigonal planar geometry. BF_3 does not have any lone pair, the three F atoms are arranged around central Boron at a bond angle of 120°

BF_3 - Boron trifluoride (BF_3) undergoes sp^2 hybridization. Boron (B) has three valence electrons, and in BF_3, it forms three sigma bonds with three fluorine (F) atoms. To accommodate the three sigma bonds, boron undergoes sp^2 hybridization.

9. Correct Option: B) BF_3, AlF_3

Explanation:

A) BF_3, NF_3: Although both have three central atoms bonded to three other atoms, NF_3 has a pyramidal geometry due to the presence of a lone pair on the nitrogen atom.

B) BF_3, AlF_3: Both BF_3 and AlF_3 have three electron domains (three bonding atoms) around the central atom, leading to the same trigonal planar geometry without any lone pairs influencing the molecular shape.

C) BeF_2, H_2O: BeF_2 has a linear geometry with two fluorine atoms bonded to beryllium in a straight line. H_2O has a bent or V-shaped geometry due to the presence of two lone pairs on the oxygen atom.

D) BCl_3, PCl_3: Both have a trigonal pyramidal geometry, but not the same as the trigonal planar geometry of BF_3 and AlF_3. This is because they have a lone pair on the central atom (B or P).

10. Correct Option: A) 25% s- character

Explanation:

Sp^3 hybridization involves combining one s orbital and three p orbitals to form four sp^3 hybrid orbitals. Each sp^3 hybrid orbital contains:

- 25% s character: This comes from the contribution of the single s orbital.
- 75% p character: This comes from the contribution of the three p orbitals.

11. Correct Option: B) SF_4

Explanation:

A. SO_4^{2-}: Sulphur in SO_4^{2-} has four single bonds and one negative charge, indicating four electron domains (bonding atoms and lone pairs). This necessitates sp^3 hybridization for optimal orbital overlap and bonding.

B. SF_4: In SF_4, sulphur forms four single bonds with fluorine atoms. However, it also has one lone pair of electrons. Due to the presence of this lone pair, sulphur undergoes sp^3d hybridization to accommodate the five electron domains (four bonds and one lone pair).

C. SF_2: Sulphur in SF_2 has two single bonds and two lone pairs, resulting in four electron domains. Therefore, sp^3 hybridization allows for efficient bonding with the fluorine atoms.

D. SCl_2: Similar to SF_2, sulphur in SCl_2 forms two single bonds and has two lone pairs, requiring sp^3 hybridization for the four electron domains.

12. Correct Option: A) NH_3

Explanation:

A. NH_3 (ammonia): Nitrogen has 5 valence electrons, forming 3 bonds with H atoms and having 1 lone pair. This leads to 4 electron domains (3 bonds + 1 lone pair) and necessitates sp^3 hybridization, resulting in a trigonal pyramidal shape.

B. SiF_4 (silicon tetrafluoride): Silicon has 4 valence electrons, forming 4 bonds with F atoms. This results in 4 electron domains (all bonds) and sp^3 hybridization, leading to a tetrahedral shape.

C. H_2O (water): Oxygen has 6 valence electrons, forming 2 bonds with H atoms and having 2 lone pairs. This creates 4 electron domains (2 bonds + 2 lone pairs) and necessitates sp^3 hybridization, leading to a bent or V-shaped geometry.

D. BF_3 (boron trifluoride): Boron has 3 valence electrons, forming 3 bonds with F atoms. This results in 3 electron domains (all bonds) and sp^2 hybridization, leading to a trigonal planar shape.

13. Correct Option: B) $sp^3 > sp^2 > sp$

Explanation:

Hybrid orbital size is primarily determined by the contribution of s and p orbitals:

- s orbitals are smaller and more spherical than p orbitals.
- As the s character in a hybrid orbital increases, its size decreases.

In this specific case:

- sp^3 orbitals: Contain 25% s character and 75% p character. They are the largest due to the lowest s character contribution.
- sp^2 orbitals: Contain 33.3% s character and 66.6% p character. They are smaller than sp^3 orbitals due to the higher s character.
- sp orbitals: Contain 50% s character and 50% p character. They are the smallest due to the highest s character contribution.

14. Correct Option: D) Increases progressively

Explanation:

As hybridization changes, the shapes of the hybrid orbitals and the angles between them also change.

- Sp^3: This has tetrahedral geometry with four hybrid orbitals arranged at 109.5° angles for maximum orbital overlap and minimum repulsion.
- Sp^2: This has trigonal planar geometry with three hybrid orbitals arranged at 120° angles, a bit closer due to the influence of more directional p orbitals.
- Sp: This has linear geometry with two hybrid orbitals aligned at 180°, closest due to the dominance of p orbitals with minimal s character.

15. Correct Option: A) sp^3d

Explanation:

The number of sigma bonds formed by the phosphorus atom is five and no lone pair is attached to the central atom so the steric number of PCl_5 is 5

For steric number 5, the type of hybridization is sp^3d

16. Correct Option: B) Octahedral

Explanation:

- sp^3d^2 hybridization: This combines one s orbital, three p orbitals, and two d orbitals to form six hybrid orbitals.
- These six hybrid orbitals are arranged in an octahedral geometry, with bond angles of 90° between each orbital.
- Compounds like SF_6 and XeF_6 exhibit sp^3d^2 hybridization and an octahedral shape.

17. Correct Option: A) sp^3d^2 hybridization

Explanation:

A. sp^3d^2 hybridization: Involves six hybrid orbitals arranged in an octahedral geometry. This arrangement leads to bond angles of 90° between each pair of bonding orbitals.

B. sp^3d hybridization: has four hybrid orbitals, but they form a trigonal prism geometry, resulting in bond angles of less than 90°.

C. dsp^2 hybridization: Has four hybrid orbitals in a trigonal planar geometry, leading to bond angles of 120°.

D. dsp^3 hybridization: Has five hybrid orbitals in a trigonal bipyramidal geometry, with bond angles ranging from 90° to 120°.

18. Correct Option: D) Octahedral

Explanation:

Hybridization = $\frac{1}{2}(s + v - q)$

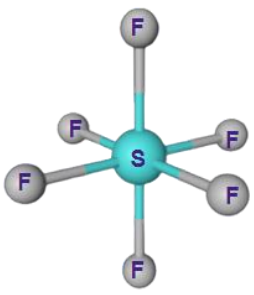
where, s = number of monovalent surrounding atoms

v = number of valence electrons on the central atom

q = charge (with sign)

In case of SF_6, s = 6, v = 6, q = 0 Hence, x = 6

∴ Hybridization becomes sp^3d^2 and geometry is octahedral.



19. Correct Option: D) Pentagonal bipyramidal

Explanation:

Hybridization (x) = $\frac{1}{2}(s + v - q)$

In IF_7,

v = Total number of valence electron pairs=7

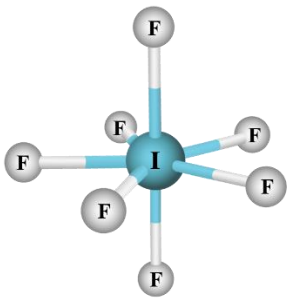
s = Total number of bonding electron pairs=7

q = Total number of lone pairs=0

x = 7

So hybridisation is sp^3d^3.

Shape is pentagonal bipyramidal



20. Correct Option: D) 4,3

Explanation:

Steric number = Number of bonded atoms + Number of lone pairs

NH_4^+:

- Steric number = 4 (bonding atoms) + 0 (lone pairs) = 4
- Since the steric number is 4, the hybridization is sp^3, which involves 4 hybrid orbitals.

NO_3^-:

- Steric number = 3 (bonding atoms) + 1 (lone pair) = 4
- However, in the case of NO_3^-, the presence of a lone pair doesn't contribute to the hybridization state. Only the bonding atoms determine the hybridization.
- Therefore, the hybridization is sp^2 (trigonal planar) using 3 hybrid orbitals.

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Correct Option: C) SO_4^{2-} and BF_4

Explanation:

Isostructural species have the same arrangement of atoms in space, even though they might differ in the types of atoms involved. Here's why the other options are not iso-structural:

A) CH_3^- and CH_3^+: Both have tetrahedral geometry, but the extra electron in CH_3^- changes the electron density distribution and bond lengths.

B) NH_4^+ and NH_3: NH_4^+ has a tetrahedral geometry while NH_3 is pyramidal due to the lone pair on nitrogen.

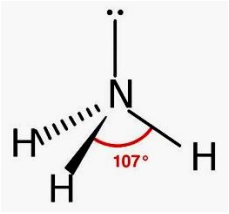
D) NH_4^+ and BF_4^-: NH_4^+ has a tetrahedral geometry while BF_4^- is square planar due to different hybridization and no lone pairs.

2. Correct Option: C) NH_3 & H_2O

Explanation:

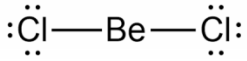
A. NH_3 (Ammonia):

- Has a trigonal pyramidal shape due to sp3 hybridization around the nitrogen atom with one lone pair.
- Non-linear due to the presence of the lone pair affecting the bond angles.



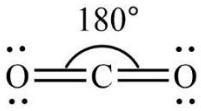
B. $BeCl_2$ (Beryllium chloride):

- Has a linear shape due to the absence of lone pairs and sp hybridization around the beryllium atom.
- Linear and does not deviate from a straight line.



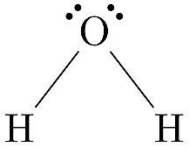
C. CO_2 (Carbon dioxide):

- Has a linear shape due to the absence of lone pairs and sp^2 hybridization around the carbon atom.
- Linear and does not deviate from a straight line.



D. H_2O (Water):

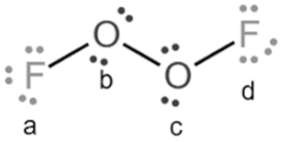
- Has a bent or V-shaped shape due to sp^3 hybridization around the oxygen atom with two lone pairs.
- Non-linear due to the presence of the lone pairs affecting the bond angles.



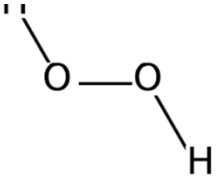
3. Correct Option: B) H_2O_2

Explanation:

O_2F_2: Both oxygen atoms have 2 lone pairs and 1 bonding pair each, leading to a bent or V-shaped structure with a bond angle of approximately 103°. This shape is known as an "open book" or "bent" structure.



H_2O_2: Both oxygen atoms have 2 lone pairs and 1 bonding pair each, resulting in a similar bent or V-shaped structure with a bond angle of around 105°. This shape also resembles an "open book" or "bent" configuration.

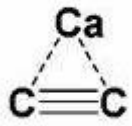


All the other options have a linear shape due to the absence of lone pairs and sp hybridization.

4. Correct Option: B) One sigma and two pi bonds

Explanation:

CaC_2 is an ionic compound, not a covalent molecule. It exists as Ca^{2+} and C_2^{-2} ions. Within the C_2^{-2} ion, the carbon atoms form a triple bond. A triple bond consists of one sigma bond and two pi bonds.



5. Correct Option: C) 109°

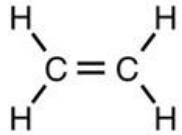
Explanation:

In BH_4^-, the boron atom (B) undergoes sp^3 hybridization to accommodate its four electron domains (one bond with each H and one lone pair). sp^3 hybridization involves mixing one 2s orbital and three 2p orbitals to form four tetrahedral hybrid orbitals. Due to the tetrahedral arrangement of these hybrid orbitals, the H-B-H bond angle in BH_4^- is 109.5°, which is close to the ideal angle for sp^3 hybridization.

6. Correct Option: B) sp^2 hybridization

Explanation:

Each carbon atom forms three sigma (σ) bonds: two with hydrogen atoms and one with the other carbon atom. To accommodate these bonds, each carbon undergoes sp^2 hybridization. This involves mixing one 2s orbital with two 2p orbitals, resulting in three sp2 hybrid orbitals arranged in a trigonal planar geometry (120° apart). The remaining 2p orbital on each carbon remains unhybridized and forms a pi (π) bond with the other carbon atom, contributing to the double bond character of ethylene.



ONE OR MORE THAN ONE OPTION CORRECT TYPE

7. Correct Option: D) Shape of molecule depends upon type of hybridization

Explanation:

A) Hybridization involves orbitals of comparable energies, not necessarily nearly equal. Small energy differences may exist but shouldn't significantly hinder mixing.

B) Hybrid orbitals are similar in energy but not identical in all respects. They inherit some characteristics from the original atomic orbitals they mix from.

C) Hybrid orbitals can form sigma (σ) and pi (π) bonds, not just sigma bonds. Sigma bonds arise from head-on overlap, while pi bonds result from sideways overlap of hybrid orbitals.

8. Correct Option: D) $SnCl_2$

Explanation:

Isostructural molecules have the same arrangement of atoms in space, even though they might differ in the types of atoms involved.

CO_2, SO_2, and NO_2: All three molecules have a linear geometry with a central atom (C, S, or N) bonded to two terminal atoms (O). This common arrangement makes them iso-structural.

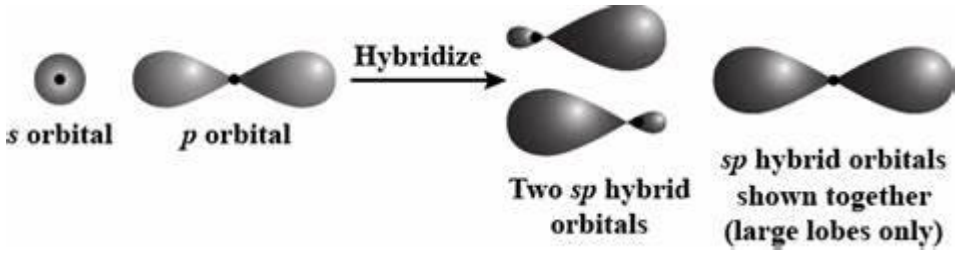
C_2H_2: This molecule has a linear geometry with a triple bond between the two carbon atoms. Each carbon is also bonded to a hydrogen atom, resulting in a different spatial arrangement compared to the other options.

$SnCl_2$: While $SnCl_2$ is linear, the central tin atom (Sn) has sp^2 hybridization, resulting in a bent (V-shaped) geometry due to the presence of one lone pair. This differs from the linear structure of CO_2, SO_2, and NO_2.

9. Correct Option: B) Two equivalent hybrid orbitals at 180°

Explanation:

Hybridizing one s orbital and one p orbital results in sp hybridization. This involves mixing one 2s orbital with one 2p orbital (usually the 2px orbital). The mixing creates two sp hybrid orbitals with equal contributions from the s and p orbitals. These sp orbitals are not mutually perpendicular (like in sp3 hybridization with three orbitals) or arranged in a plane (like in sp2 hybridization with three orbitals). Instead, they are oriented at 180° to each other, forming a linear geometry.



10. Correct Option: B) sp^2

Explanation:

Number of sigma bond around the Sulphur atom in SO_2 = 2 (one with each oxygen atom)

Sulphur atom has 6 valence electrons, out of which 4 of them are shared with two oxygen atoms in SO_2.

So, Number of lone pairs of electrons on a Sulphur atom in $SO_2 = \frac{6-4}{2} = 1$

Hence, steric number (X) for SO_2 = 2 + 1 = 3 and it is already known to you that for X = 3, hybridization is sp^2

11. Correct Option: B) S-S sigma bond is stronger than P-P sigma bond

Explanation:

A) Hybrid orbitals can form only sigma bonds: Hybrid orbitals can form both sigma (σ) and pi (π) bonds, depending on their overlap patterns. Sigma bonds result from head-on overlap, while pi bonds arise from sideways overlap.

C) Double bonds cause less repulsion than single bonds: Double bonds involve more electron density concentrated in the bonding region, leading to greater repulsion compared to single bonds where electron density is more spread out.

D) The number of hybrid orbitals formed are twice the number of orbitals participating in hybridization: Depending on the specific hybridization involved, the number of hybrid orbitals can be equal to or less than twice the number of participating orbitals. For example, sp hybridization involves one s and one p orbital, resulting in two sp hybrid orbitals.

12. Correct Option: A) Cl_2

Explanation:

P-P overlapping refers to the interaction between unpaired electrons in p orbitals of the same atom. This overlap leads to the formation of a sigma (σ) bond.

A. Cl_2: Both chlorine atoms have an unpaired electron in their 3p orbitals. These 3p orbitals overlap sideways to form a pi (π) bond, resulting in the diatomic chlorine molecule.
B. HCl: The hydrogen atom only has one 1s orbital, and the chlorine atom has an unpaired electron in its 3p orbital. They form a sigma (σ) bond by overlapping their s orbitals, not p orbitals.
C. H_2O: The oxygen atom has two unpaired electrons in its 2p orbitals, but they form sigma bonds with the hydrogen atoms (s-p overlap), not pi bonds.
D. NH_3: The nitrogen atom has an unpaired electron in a 2p orbital, but it forms sigma bonds with the hydrogen atoms (s-p overlap) and uses its remaining 2p orbitals to hold lone pairs, not for pi bonds.

COMPREHENSION TYPE

Passage – I:

13. Correct Option: B) sp^2

Explanation:

Hybridization (x) = $\frac{1}{2}$(s + v − q)

where, s = number of monovalent surrounding atoms

v = number of valence electrons on the central atom

q = charge (with sign)

In case of BH_3, s = 3, v = 3, q = 0 Hence, x = 3

∴ Hybridization becomes sp^2

14. Correct Option: C) sp

Explanation:

Hybridization (x) = $\frac{1}{2}$(s + v − q)

where, s = number of monovalent surrounding atoms

v = number of valence electrons on the central atom

q = charge (with sign)

In case of NO_2^+, s = 3, v = 2, q = -1 Hence, x = 2

∴ Hybridization becomes sp

15. Correct Option: A) sp^2, sp

Explanation:

Hybridization (x) = $\frac{1}{2}$(s + v − q)

where, s = number of monovalent surrounding atoms

v = number of valence electrons on the central atom

q = charge (with sign)

In case of CO_3^{2-}, s = 0, v = 4, q = -2 Hence, x = 3

∴ Hybridization becomes sp^2

In case of CO_2, s = 0, v = 4, q = 0 Hence, x = 2

∴ Hybridization becomes sp

16. Correct Option: D) sp^3d^3

Explanation:

A) sp^3

The angle between the sp3 hybrid orbitals is 109.28°

B) sp^3d

Three hybrid orbitals lie in the horizontal plane inclined at an angle of 120° to each other, known as the equatorial orbitals. The remaining two orbitals lie in the vertical plane at 90 degrees plane of the equatorial orbitals, known as axial orbitals.

C) sp^3d^2

They are inclined at an angle of 90 degrees to one another.

D) sp^3d^3

The angle between every five bonds in the X-Y plane is 72°. Whereas the angle between the bond along Z-axis and X-Y plane is 90°.

MATRIX MATCHING TYPE

17. Correct Option: a-R, b-S, c-P, d-Q

Explanation:

a. Ethane (CH_3CH_3): Each carbon forms 4 single bonds using sp^3 hybridization (1s orbital + 3p orbitals).

b. Ethylene ($CH_2=CH_2$): Each carbon forms 3 sigma bonds (using sp^2 hybridization) and a pi bond (using an unhybridized p orbital).
c. Acetylene (HC≡CH): Each carbon forms 1 sigma bond and 2 pi bonds, requiring sp hybridization (1s + 1p).
d. Benzene (C_6H_6): Each carbon forms 3 sigma bonds and 1 pi bond, employing sp^2 hybridization.

18. Correct Option: a-R, b-P, c-S, d-Q

Explanation:

a) BCl_3: Each Cl forms a single bond with B, totaling 3 electron domains (bonds).
b) NH_3: Each H forms a single bond with N, and N has 1 lone pair, totaling 4 electron domains.
c) PCl_5: Each Cl forms a single bond with P, totaling 5 electron domains.
d) XeF_4: Each F forms a single bond with Xe, and Xe has 2 lone pairs, totaling 6 electron domains.

INTEGER TYPE

19. Correct Option: 4

Explanation:

Each carbon atom in acetylene undergoes sp hybridization. This involves mixing one 2s and one 2p orbital from each carbon atom, resulting in two sp hybrid orbitals per carbon atom. Since there are two carbon atoms in acetylene, the total number of hybrid orbitals formed is 4 (2 sp orbitals per carbon atom * 2 carbon atoms).

20. Correct Option: 3

Explanation:

In SO_2, the sulphur atom (S) uses a total of 3 of its orbitals for bonding:

- 1 s orbital: This orbital hybridizes with 3 p orbitals to form four sp^3 hybrid orbitals.
- 2 p orbitals: Two of the four sp^3 hybrid orbitals participate in sigma bonding with the oxygen atoms.

VSEPR Theory & Bond Parameters

LEVEL – I

1. Correct Option: B) B

Explanation:

Given that the compound X has a bond angle of 120 degrees, it suggests a trigonal planar geometry, which indeed requires a central atom with three valence electrons. Among the options provided, boron (B) is the element with three valence electrons, consistent with the trigonal planar geometry.

2. Correct Option: C) 1:3

Explanation:

A bond angle of 109.28° corresponds to sp3 hybridization. In sp3 hybridization, one 2s orbital and three 2p orbitals from the same atom hybridize to form four sp3 hybrid orbitals. The contribution of each orbital to the hybrid orbital is not equal. The s orbital contributes 25% to the sp3 hybrid orbital, while each p orbital contributes 75%. Therefore, the ratio of s character to p character is:

25% s character / 75% p character = 1:3

3. Correct Option: D) ICl_3

Explanation:

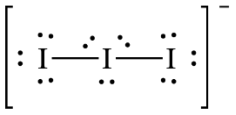
A. XeO_3: The central xenon atom has 1 lone pairs and 3 bonding pairs (double bonds with oxygen)

B. SF_4: The central sulfur atom has 1 lone pair and 4 bonding pairs (single bonds with fluorine).
C. PCl_3: The central phosphorus atom has 1 lone pair and 3 bonding pairs (single bonds with chlorine).
D. ICl_3: The central iodine atom has 2 lone pairs and 3 bonding pairs (single bonds with chlorine).

4. Correct Option: A) Linear

Explanation:

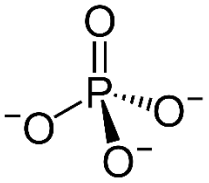
Each iodine atom (I) forms a single covalent bond with the central iodine atom, resulting in two I-I bonds. The central iodine atom also has one negative charge, which corresponds to a lone pair of electrons. To minimize electron-electron repulsion, the lone pair occupies an equatorial position, forming a linear arrangement of the three iodine atoms: I - I - I.



5. Correct Option: B) Tetrahedral

Explanation:

The central phosphorus atom (P) forms four single covalent bonds with four oxygen atoms (O). No lone pairs are present on the phosphorus atom. Based on the VSEPR theory (Valence Shell Electron Pair Repulsion), the four electron domains (four bonding pairs) around the phosphorus atom arrange themselves in a tetrahedral geometry to minimize electron-electron repulsion.



6. Correct Option: D) NH_3, PH_3

Explanation:

A) BF_3, NH_3:

- BF_3 has a trigonal planar geometry due to sp^2 hybridization and three bonding domains (three fluorine atoms).
- NH_3 has a trigonal pyramidal geometry due to sp^3 hybridization and four electron domains (three hydrogen atoms and one lone pair).

B) H_2O, C_2H_2:

- H_2O has a bent geometry due to sp^3 hybridization and two lone pairs.
- C_2H_2 has a linear geometry due to sp hybridization and one sigma and two pi bonds.

C) CO_2, SO_2:

- CO_2 has a linear geometry due to sp hybridization and two double bonds.
- SO_2 has a bent geometry due to sp^2 hybridization and one lone pair.

D) NH_3, PH_3:

- Both NH_3 and PH_3 have trigonal pyramidal geometry due to sp^3 hybridization and four electron domains (three hydrogen atoms and one lone pair).

7. Correct Option: D) SO_3

Explanation:

SO_2: Sulfur atom has sp^2 hybridization, using one s and two p orbitals to form three sp^2 hybrid orbitals. One lone pair remains on the sulfur atom.

BF_3: Boron atom has sp^2 hybridization, using one s and two p orbitals to form three sp^2 hybrid orbitals. No lone pairs are present.

B_2H_6: Each boron atom has sp^3 hybridization, using one s and three p orbitals to form four sp^3 hybrid orbitals. Each boron atom is bonded to two hydrogen atoms.
SO_3: Sulfur atom has sp^3d hybridization, using one s, three p orbitals, and one d orbital to form five sp^3d hybrid orbitals. One lone pair remains on the sulfur atom.

8. Correct Option: (D)

Explanation:

A. BF_3: This molecule has a trigonal planar geometry with sp^2 hybridization around the central boron atom. This results in bond angles of **120°** between the boron and fluorine atoms.
B. CH_4: This molecule has a tetrahedral geometry with sp^3 hybridization around the central carbon atom. This results in bond angles of **109.5°** between the carbon and hydrogen atoms.
C. CO_2: This molecule has a linear geometry with sp hybridization around the central carbon atom. This results in bond angles of **180°** between the carbon and oxygen atoms.
D. SO_2: This molecule has a bent geometry with sp^2 hybridization around the central sulphur atom. This results in bond angles of **~119°** between the sulphur and oxygen atoms.

9. Correct Option: B) 1

Explanation:

1. Central Atom: Xe (Xenon)
2. Bonding:
    - One double bond with oxygen (O) forming Xe-O bond.
    - Four bonds with fluorine (F) forming Xe-F single bonds.
3. Total valence electrons in Xe**:** 8
4. Electrons used in bonding: 6 (2 with O + 4 with F)
5. Remaining valence electrons: 2
6. Lone pairs: Since Xe needs to satisfy the octet rule, it forms one lone pair with the remaining 2 electrons.

10. Correct Option: B) Pyramidal

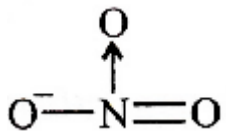
Explanation:

According to the VSEPR model, molecules arrange their electron domains (bonding pairs and lone pairs) to minimize electron-electron repulsion. With 4 electron domains, the most stable geometry for ClO_3^- is trigonal pyramidal, where the three Cl-O bonds form a triangle and the lone pair occupies one corner above the plane, creating a pyramidal shape.

11. Correct Option: D) 4, 0

Explanation:

Due to the presence of one negative charge, number of valence electrons= 5+ 1=6
One *O*-atom forms two bonds (= bond) and two *O*-atom are shared with two electrons of *N*-atom
Thus, 3*O*-atoms are shared with 8 electrons of *N*-atom.
Number of bond pairs (or shared pairs) =4
Number of lone pairs =0



12. Correct Option: D) Different with 1,0 and 2 lone pair of electrons respectively

Explanation:

SF_4: Sulfur (S) in SF_4 has 1 lone pair and 4 bonding pairs, leading to an sp^3d hybridization and a see-saw shape.

CF_4: Carbon (C) in CF_4 has 4 bonding pairs and no lone pairs, resulting in sp^3 hybridization and a tetrahedral shape.

XeF_4: Xenon (Xe) in XeF_4 has 2 lone pairs and 4 bonding pairs, leading to sp^3d^2 hybridization and a square planar shape.

13. Correct Option: D) C_2H_2

Explanation:

Factors influencing C-C bond length:

1. Bond order: A higher bond order corresponds to a shorter bond length due to stronger electron-nucleus attraction.
2. Hybridization: sp hybridization creates shorter bonds compared to sp^2 and sp^3.

A. C_2H_6 (ethane): All C-C bonds are single (bond order 1) and sp^3 hybridized.

B. C_2H_4 (ethylene): One C-C bond is double (bond order 2) and sp^2 hybridized, while the other is single.

C. C_6H_6 (benzene): All C-C bonds are part of a delocalized pi system with resonance, effectively increasing bond order. They are sp^2 hybridized.

D. C_2H_2 (acetylene): Both C-C bonds are triple (bond order 3) and sp hybridized.

14. Correct Option: D) $CO < CO_2 < CO_3^{-2}$

Explanation:

The bond length in a molecule depends primarily on two factors:

1. Bond order: Higher bond order generally corresponds to shorter bond length due to stronger electron-nucleus attraction.
2. Negative charge: The presence of a negative charge delocalizes electron density across the molecule, effectively increasing the size of the electron cloud and leading to longer bonds.

- CO: This molecule has a triple bond between C and O, resulting in the highest bond order (3) and the shortest bond length among the three options.
- CO_2: This molecule has a double bond between C and each O, leading to a bond order of 2 and a bond length longer than CO but shorter than CO_3^{-2}.
- CO_3^{-2}: This ion has a single bond between C and each O, with a negative charge delocalized across the molecule. The combination of lower bond order and negative charge leads to the longest bond length among the three.

15. Correct Option : (D)

16. Correct Option: D) HI

Explanation:

As the size of halogen atom(x) increases, the H-x bond length increases, and hence H-X bond strength decreases.

H-x bond energy decreases from H-F to H-I

H-I has least bond energy...

Bond energy order: H-F > H-Cl > H-Br > H-I

17. Correct Option: C) sp - s

Explanation:

The highest bond energy arises from the smallest overlap, which occurs with sp hybridization.

- smaller orbital size: sp orbitals are more concentrated compared to sp^2, sp^3, or sp^3d, leading to greater overlap with a standard s orbital.

- better directional overlap: sp orbitals have a more linear shape, maximizing overlap with an s orbital along the internuclear axis.

18. Correct Option: A) $F_2 > Cl_2 > Br_2 > I_2$

Explanation:

Bond energy typically increases with increasing electronegativity difference between the bonded atoms. Fluorine has the highest electronegativity among halogens, leading to the strongest F-F bond in F_2. As we move down the halogen group, electronegativity decreases, resulting in a weaker bond. Additionally, atomic size increases, creating greater orbital overlap and consequently lower bond energy. Therefore, Cl_2 will have a weaker bond than F_2, Br_2 weaker than Cl_2, and I_2 the weakest among them

19. Correct Option: B) O-H

Explanation:

Electronegativity difference: Oxygen (O) is significantly more electronegative than carbon (C), sulfur (S), and chlorine (Cl). This larger difference leads to stronger ionic character and a stronger bond.

Bond polarity: Due to the high electronegativity difference, O-H is the most polar bond, with stronger electrostatic attraction between the atoms.

20. Correct Option: A) NO_2^+

Explanation:

1. Hybridization: NO_2^+ has sp hybridization on the central nitrogen, while NO_2, NO_2^-, and NO_3^- have sp^2 hybridization. sp orbitals are more linear, leading to larger bond angles.
2. Lone pairs: NO_2^- and NO_3^- have lone pairs on the central atom, repelling bonding pairs and decreasing the bond angle. NO_2^+ has no lone pairs, further increasing the bond angle potential.

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Correct Option: D) CO_2

Explanation:

A. CH_4: The carbon atom in CH_4 is sp^3 hybridized, leading to a tetrahedral geometry with bond angles of 109.5°.

B. H_2O: Oxygen in H_2O is sp^3 hybridized, but two lone pairs repel bonding electron pairs, resulting in a bent shape and bond angles of 104.5°.

C. NH_3: Nitrogen in NH_3 is sp^3 hybridized with one lone pair. The lone pair repulsion reduces the bond angle to 107° compared to CH_4.

D. CO_2: Carbon in CO_2 is sp^2 hybridized, forming a linear molecule with a 180° bond angle. No lone pairs are present to cause repulsion.

2. Correct Option: A) NH_3

Explanation:

Key factors for a pyramidal shape:

1. Central atom with three bonded atoms: All the listed molecules have three bonded atoms.
2. Presence of a lone pair: This is crucial for the pyramidal shape. In NH_3, the nitrogen atom has one lone pair along with three hydrogen atoms attached.

Only NH_3 has the combination of a lone pair and sp^3 hybridization that generates the characteristic pyramidal shape.

3. Correct Option: A) S - S

Explanation:

Factors affecting bond strength:

- Atom size: Smaller atoms can form closer and stronger bonds due to less electron-electron repulsion. However, this is not the sole factor.
- Electronegativity: Higher electronegativity difference between atoms leads to greater attraction and a stronger bond.

While oxygen is indeed smaller than sulphur, its significantly higher electronegativity outweighs the slight size advantage in terms of overall bond strength. As we move down the group to Se and Te, their larger sizes weaken the bond energy due to the reduced orbital overlap and weaker electrostatic interactions.

4. Correct Option: D) XeO_4

Explanation:

For a molecule to be planar, it needs to fulfil two key requirements:

1. Central atom hybridization: The central atom in the molecule must be hybridized in a way that generates orbitals capable of forming planar bonds. This typically means sp^2 or dsp^2 hybridization, where the hybridized orbitals lie in the same plane.
2. No lone pairs around the central atom: Lone pairs occupy space and repel bonding pairs, pushing them out of the plane and disrupting planarity. Therefore, the central atom should not have any lone pairs for a truly planar structure.

A) XeF_4:	C) XeO_2F_2
Valence electrons of Xe= 8 Bounded electrons = 4 Lone pairs = (8-4)/2 = 2	Valence electrons of Xe= 8 Bounded electrons = 6 Lone pairs = (8-6)/2 = 1
B) $XeOF_2$	D) XeO_4
Valence electrons of Xe= 8 Bounded electrons = 4 Lone pairs = (8-4)/2 = 2	Valence electrons of Xe= 8 Bounded electrons = 8 Lone pairs = (8-8)/2 = 0

5. Correct Option: A) XeF_2, IF_2^-

Explanation:

Hybridization = $\frac{\text{Valence e}- + \text{Monovalent atoms} + \text{Anion} - \text{Cation}}{2}$

A) XeF_2, IF_2^-

XeF_2 Hybridization = (8+2)/2 = 5 (sp^3d)

IF_2^- Hybridization = (7+2+1)/2 = 5 (sp^3d)

B) NH_3, BF_3

NH_3 Hybridization = (5+3)/2 = 4 (sp^3)

BF_3 Hybridization = (3+3)/2 = 3 (sp^2)

C) CO_3^{2-}, SO_3^2

CO_3^{2-} Hybridization = (4+0+2)/2 = 3 (sp^2)

SO_3^{2-} Hybridization = (6+0+2)/2 = 4 (sp^3)

D) PCl_5, ICl_5

PCl_5 Hybridization = (5+5)/2 = 5 (sp^3d)

ICl_5 Hybridization = (7+5)/2 = 6 (sp^3d^2)

6. Correct Option: A) Three

Explanation:

In an octahedral molecule, the central atom M is surrounded by six identical atoms X arranged at the corners of a cube. Each X atom forms a bond with the central atom M at an angle of 90°. However, there are three specific X-M-X bond angles that are exactly 180°. These angles lie along the axes of the cube, where two X atoms are directly opposite each other on the same axis.

ONE OR MORE THAN ONE ANSWER TYPE

7. Correct Option: C, D

Explanation:

C) Slightly more than NH_3: This is because both molecules have a trigonal pyramidal geometry due to sp^3 hybridization of the central atom (P in PH_3 and N in NH_3). However, the larger size of the phosphorus atom in PH_3 compared to nitrogen in NH_3 leads to slightly less repulsion between the hydrogen atoms, resulting in a slightly larger bond angle in PH_3 (around 93.3°) compared to NH_3 (around 107°).

D) Much more than PF_3: This is due to the difference in hybridization between the central atoms. PH_3 has sp^3 hybridization, while PF_3 has sp^3d^2 hybridization. The inclusion of d orbitals in PF_3 expands the valence shell, leading to a larger bond angle (around 101.3°) compared to PH_3.

8. Correct Option: (B, C) None are always true

Explanation:

A) EN of central atom:

Decreasing EN generally leads to increased bond angle, but not always. Lower EN means less attraction to shared electrons, potentially reducing repulsion and allowing wider angles. However, other factors like size and hybridization play a role.

B) Size of central atom:

Increasing size can sometimes lead to increased bond angle, but again, not always. Larger atoms have more space to accommodate electron density, potentially reducing repulsion and allowing wider angles. However, hybridization and the nature of atom B also influence bond angles.

C) EN of atom B:

The effect of decreasing EN of atom B on bond angle is not consistent. Lower EN in B means less attraction towards shared electrons, potentially leading to less repulsion and possibly wider bond angles. However, this can be mitigated by other factors like the EN difference between A and B and the hybridization of the central atom.

D) EN of atom B:

Similar to C, the effect of decreasing EN of B is not always consistent on bond angle.

9. Correct Option: B, C, D

Explanation:

A) XeF_4: While XeF_4 has a central Xe atom surrounded by four atoms, it adopts a square planar geometry with bond angles of 90° due to sp^3d^2 hybridization and the absence of lone pairs.

B) XeF_2: With only two fluorine atoms, XeF_2 adopts a linear geometry with a bond angle of 180°. Square planar geometry requires four atoms around the central atom.

C) $XeOF_4$: This molecule has a tetrahedral geometry due to sp^3 hybridization on the Xe atom and the presence of four lone pairs (two lone pairs on Xe and two on the oxygen atoms). Square planar geometry requires no lone pairs.

D) CH_3Cl: The carbon atom in CH_3Cl has sp^3 hybridization, resulting in a tetrahedral geometry with bond angles of approximately 109.5°. Square planar geometry requires a central atom with sp^2d^2 hybridization.

10. Correct Option: A, B

Explanation:

Nitrogen remains sp^3 hybridized even after forming the adduct. Boron changes from sp^2 in BF_3 to sp^3 in the adduct due to accepting a lone pair from NH_3. This changes its shape from trigonal planar to tetrahedral.

COMPREHENSION TYPE

Passage – I:

11. Correct Option: B) IV<III<II<I

Explanation:

As the electronegativity of central atom increases the bond angle increases due to repulsion between bond pair and bond pair as bond pairs are more close to the central atom.

12. Correct Option: D) IV<II<III<I

Explanation:

I (NH_3): Nitrogen is sp^3 hybridized with one lone pair, resulting in a trigonal pyramidal geometry and a bond angle of ~107°.

II (PH_3): Phosphorus is sp^3 hybridized with one lone pair, similar to NH_3, but the larger P atom leads to less repulsion and a slightly wider bond angle of ~93.3°.

III (AsH_3): Arsenic is sp^3 hybridized with one lone pair, similar to NH_3 and PH_3, but the even larger As atom results in an even wider bond angle of ~91.8°.

IV (SbH_3): Antimony is sp^3 hybridized with one lone pair, but the significantly larger Sb atom causes the widest bond angle of ~90°.

13. Correct Option: A) OSF_2

Explanation:

Factors affecting the X-O-X bond angle:

1. Size of the halogen atom (X): Smaller size allows for closer packing and less steric hindrance, leading to a smaller bond angle.
2. Lone pairs: Their absence minimizes repulsion between electron pairs, allowing for a smaller bond angle.

- OSF_2: Fluorine is the smallest halogen, and there are no lone pairs on oxygen, leading to the smallest bond angle.
- $OSCl_2$, $OSBr_2$, OSI_2: All three have larger halogens than fluorine, increasing the bond angle. Additionally, $OSBr_2$ and OSI_2 have lone pairs on oxygen, further widening the bond angle.

Passage – II:

14. Correct Option: D) PI_3

Explanation:

Larger halogens like iodine (I) create more steric hindrance compared to smaller ones like fluorine (F), leading to wider bond angles. Therefore, as the halogen size increases down the group from F to I, the repulsion between the lone pairs and bonding pairs increases, resulting in the largest bond angle in PI_3.

15. Correct Option: C) XeF_4

Explanation:

a) BrF_3 has T-shape but perfect T but little distorted due to repulsion between lone pair and bond pair.

b) ClF_3 has T-shape but perfect T but little distorted due to repulsion between lone pair and bond pair.

c) XeF_4 square planar and lone pairs and above and below the plane. Therefore, no distortion.
d) XeF_6 has distorted octahedral shape.

16. Correct Option: B) F-N-F angle is NF_3 is smaller than H-N-H angle in NH_3

Explanation:
Both NH_3 and NF_3 have a pyramidal shape due to sp^3 hybridization of the central atom (N). However, fluorine (F) is more electronegative than hydrogen (H). This means F atoms pull electron density closer towards themselves, reducing the electron cloud size around the nitrogen atom. Smaller electron clouds experience less repulsion between each other, leading to a smaller bond angle in NF_3 compared to NH_3.

MATRIX MATCHING TYPE

17. Correct Option: a-Q, b-P, c-R, d-S

Explanation:
BrF_3: The central bromine atom (Br) in BrF_3 has one lone pair and sp^3d hybridization, resulting in a T-shaped geometry.
$[PCl_4]^+$: The phosphorus atom (P) in the positively charged tetrahedral ion $[PCl_4]+$ lacks lone pairs and has sp^3 hybridization, leading to a square planar geometry.
XeF_4: The xenon atom (Xe) in XeF_4 has no lone pairs and sp^3d^2 hybridization, resulting in a regular tetrahedral geometry.
XeO_3: The xenon atom (Xe) in XeO_3 has three lone pairs and sp^3d hybridization, causing the molecule to adopt a pyramidal shape with one lone pair occupying the apex.

18. Correct Option: a-R, b-Q, c-P, d-S

Explanation:
a) ClF_3: Chlorine has one lone pair, resulting in sp^3d hybridization and a T-shaped geometry.
b) NF_3: Nitrogen has one lone pair, leading to sp^3 hybridization and a pyramidal shape.
c) BF_3: Boron has no lone pairs, and sp^2 hybridization gives it a trigonal planar geometry.
d) SO_2: Sulfur has one lone pair, resulting in sp^2 hybridization and a bent (V-shaped) geometry.

INTEGER TYPE

19. Correct Option: 3

Explanation:
In the BCl_3 molecule, the central boron atom (B) forms bonds with three chlorine atoms (Cl). Each bond involves a shared pair of electrons. Therefore, there are 3 bonding pairs in BCl_3.

20. Correct Option: 3

Explanation:
$BeCl_2$: Beryllium in $BeCl_2$ has sp hybridization, leading to a linear geometry with two Cl atoms bonded in a straight line.
CO_2: Carbon in CO_2 also has sp hybridization, resulting in a linear geometry with two oxygen atoms bonded to the carbon.
HCN: The carbon atom in HCN has sp hybridization, forming a linear molecule with one hydrogen and one nitrogen atom bonded linearly.
SO_2: Sulfur in SO_2 has sp^2 hybridization, forming a bent (V-shaped) geometry due to the presence of one lone pair.
H_2O: Oxygen in H_2O has sp^3 hybridization, leading to a bent (V-shaped) geometry with two lone pairs causing repulsion.

5. Metals and Non-Metals

Solutions

LEVEL – I

1. Correct option: (B) Cations

Metals are highly electropositive. Thus they tend to lose one electron from their valence shell to gain electronic configuration of Noble gas. Hence they form cations.

2. Correct option: (C) Malleability

Malleability is a physical property of metals which defines the ability of metals to be beaten into sheets without breaking.

Tenacity is the ability of metal to resist the effect of tensile forces.

Ductility is a physical property for which metal can be drawn into thin wire.

Porosing is a defect that can leave machinery to a wide range of vulnerable problems.

3. Correct option: (C) Graphite

Graphite is a crystalline allotrope of carbon.

It has layered structure which are held by weak Van der Waals force of attraction. The layers are composed of planar hexagonal rings of carbon. It has free electrons in its structure as the fourth valence electron of carbon atom remains unpaired.

Hence Graphite is a good conductor of heat & electricity due to presence of free electrons

4. Correct option: (B) Positive

Metals can donate electrons and form positive ions. Thus metals have electropositive valency

5. Correct option: (B) Negative

Non-metals can gain electrons and form negative ions. Thus non-metals have electronegative valency.

6. Correct option: (C) Al

Ductility is a physical property for which metal can be drawn into thin wire.

The metal which can be drawn into thin wire more easily is more ductile.

Metals are commonly more ductile.

Aluminium is a ductile metal due to it's low metal density.

7. Correct option: (C) Hydrogen

Metals are placed above hydrogen in reactivity series and displace it during reaction with acids.

8. Correct option: (B) Acidic

Non-metals oxides (SO_2 , CO_2 etc.) reacts with water to produce acidic compounds.

Example – $SO_2 + H_2O \rightarrow H_2SO_3$

Hence non-metals oxides are acidic in nature

9. Correct option: (C) Basic

Metal oxides (BaO, MgO etc.) reacts with water to produce basic compounds.

Example : $MgO + H_2O \rightarrow Mg(OH)_2$

So metals oxides are basic in nature.

10. Correct option: (C) Hydrogen

Hydrogen gas is liberated when metals reacts with acid as metals are placed above hydrogen in reactivity series.

Example : $Zn(s) + HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

11. Correct option: (C) Both 'A' and 'B'

Metals are good conductor of heat because their particles are closely packed. Hence the vibrations of heat wave passed so quickly.

Metals also contain a large number of free electrons. Thus they are also good conductor of electricity.

12. Correct option: (A) Non-metal

Graphite is a allotrope of carbon with multiple hexagonal layers.

Hence it is a non-metal

It has free electrons in its structure as the fourth valence electron of carbon atom remains unpaired.

Hence Graphite is a good conductor of electricity due to presence of free electrons

13. Correct option: (C) Form acidic oxides

Malleability is a physical property of metals which defines the ability of metals to be beaten into sheets without breaking.

Ductility is a physical property of metal for which metal can be drawn into thin wire.

Metals contain a large number of free electrons. Thus they are also good conductor of electricity.

When metals oxides (CaO, BaO etc.) react with dilute acid, they form salt and water. Hence they are basic oxides.

14. Correct option: (B) Homogeneous mixture

Homogeneous mixture is a mixture in which composition is uniform throughout the mixture.

Heterogeneous mixture is a mixture in which composition is not uniform and contains smaller parts.

Brass is a mixture of copper and zinc. As the composition remains same throughout the mixture, it is a homogeneous mixture.

15. Correct option: (C) Conc.HCl : Conc.HNO_3 = 3 : 1

Aqua regia is mixture of Conc. HCl and Conc. HNO_3 at the ratio of 3:1.

It is a red-orange colour fuming liquid which is used to dissolve noble metals like gold and platinum.

16. Correct option: (B) (ii) and (iii)

In HCl and CCl_4 covalent bond is formed by mutual sharing of electron pairs. Hence they are covalent compounds.

But in KCl and NaCl ionic bond is formed by donating and accepting of electrons. Hence they are ionic compounds.

17. Correct option: (B) Electrical conductivity in solid state

Ionic compounds are highly soluble in water because their hydration energy is greater than lattice energy.

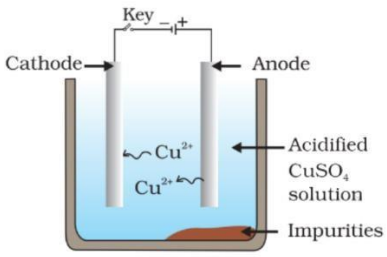
Ionic compounds break into cation and anion in molten or solution state. Thus they conduct electricity in molten or solution state due to availability of free ions.

But in solid state they can't conduct electricity due to absence of free ions.
Ionic bonds have high lattice energy. Hence they have high melting and boiling point.
So electrical conductivity in solid state is not exhibited by ionic compounds.

18. Correct option: (C) (ii) and (iv)

Au and Ag are placed below in the reactivity series. They are highly non reactive as they do not lose electrons more easily.
Thus Au and Ag exists in their native state in Nature.
But Cu and Zn are highly reactive. And they react with other elements.

19. Correct option: (C)



Electrolytic refining is used for extraction and purification of metals which are obtained by refining methods.
Cathode is negative where positive ions are attached.
Anode is positive which deposit positive ions.

20. Correct option: (D) $CuSO_4$+Fe

In reactivity series, the metal placed above is regarded as more reactive.
And a more reactive metal can displace less reactive metal.
Fe placed above Cu in reactivity series. So it can replace Cu and displacement reaction will occur.
But Mg and Zn placed above Fe in reactivity series. Hence reaction isn't possible.
Similarly Mg placed above Pb and reaction doesn't occur.

21. Correct option: (A) Deposited on cathode

Electrolytic refining is used for extraction and purification of metals which are obtained by refining methods.
For Zinc
Anode: Impure zinc
Cathode: Pure zinc
Electrolyte: Zinc sulphate solution
On passing current, zinc from the electrolyte is deposited at the cathode.

22. Correct option: (C) A homogenous mixture

An alloy is a mixture of two or more metal or a metal and a nonmetal.
Example – brass (Cu +Zn)
Alloy is a homogeneous mixture as it's composition is uniform throughout the whole mixture.

23. Correct option: (D) (i) and (ii)

Electrolysis is a process that leads to chemical change when current passes through an electrolyte.
Sodium and calcium are obtained by electrolysis of their chlorides in molten state as they are highly reactive.
But iron and copper are moderately reactive thus they can't be obtained by electrolysis of their chlorides.

24. Correct option: (C) Zinc

Iron in presence of air and moisture gets rusted. The rusting can be prevented by a coating with thin layer of zinc. This process of coating zinc on iron is called galvanisation. Hence zinc is the right answer.

25. Correct option: (B) $CuCO_3$

Copper is moderately reactive metal. When it is exposed to air it reacts with gases like carbon dioxide and vapour. This reaction leads to form copper carbonate.
It makes the shiny brown surface dull and green.
$2Cu(s) + CO_2(g) + H_2O(g) \rightarrow CuCO_3(s) + Cu(OH)_2(s)$
Since sulphur is not present in the air. It cannot react with sulphur to make $CuSO_4$.
Copper doesn't react with nitrogen. Thus $Cu(NO3)_2$ is not formed.
Copper first reacts with oxygen to form CuO. But further it reacts with carbon dioxide to form copper carbonate. So $CuCO_3$ is the correct answer.

26. Correct option: (C) Ag_2S

Silver articles become black on prolonged exposure to air due to thin coating of silver sulphide bring formed on the surface of silver.
Silver is not affected by dry or moist air.
But in the presence of hydrogen sulphide in the surroundings silver is affected . It reacts with the silver to form black coloured silver sulphide.
$2Ag(s) + H_2S(g) \rightarrow Ag_2S(s) + H_2(g)$

27. Correct option: (A) (i) and (ii)

Electrolytic refining is a process in which the extraction and purification of metals is done.
The impure metal is used as an anode and the pure metal is used as a cathode.
Copper and Gold is refined electrolytically as they are less reactive.
Sodium and potassium are highly reactive metals and are extracted by the electrolytic reduction of molten chloride salts.

28. Correct option: (A) decreasing order of reactivity

Reactivity series is a list of metals arranged in decreasing order of their reactivity. Most reactive metals are at the top while the least reactive metals at the bottom.
Hence option A is correct.

29. Correct option: (B) chemical; bauxite

Bayer's method for the extraction of Alumina from bauxite and gangue.
The difference in chemical affinity between bauxite and gangue is utilized for concentrating the ore.

30. Correct option: (A) oxygen

Non-metals are the species which are electronegative and have the tendency to gain electron to form negative ions.
Oxygen is the most abundant non-metal in the earth's crust.

31. Correct option: (A) iron forms ferrous sulphate solution

Iron is more reactive than copper. When nail (iron) is dipped in copper sulphate solution, it displaces copper from copper sulphate and forms greenish blue ferrous sulphate. Thus the solution turns light green colour.
$CuSO_4(aq) + Fe(s) \rightarrow FeSO_4(aq) + Cu(s)$

32. Correct option: (A) $CuCO_3.Cu(OH)_2$

Copper reacts with oxygen and vapour in the air to form copper oxide.

Copper oxide then combines with carbon dioxide to make copper carbonate, which gives it a green colour.
The green material is a mixture of copper hydroxide and copper carbonate.

33. Correct option: (D) Covering with glass

Corrosion is a process in which refined metals are converted to their more stable oxides.
It can be prevented if a barrier between metal and oxygen can be created.
Anodizing is process used to increased the natural oxide.
Alloying is a process in which two or more metals are combined together.
Painting , Anodizing and alloying can prevent corrosion.
But covering with glass can't prevent metal to get touch from oxygen. So it will not be able to prevent corrosion.

LEVEL - II

1. Correct option: (D) Carbon (Diamond)

Diamond is an allotrope of carbon.
In diamond each carbon is covalently bonded with other four carbons. This arrangement makes a tetrahedral crystal in which the atoms are closely packed and make a rigid structure. Thus diamond is the hardest substance.

2. Correct option: (D) Phosphorus

In metals, there are free held electrons which creates a sea of electrons.
Because of these free electrons metals are good conductors of electricity.
But non-metal crystal doesn't contain any free electrons, hence they are not good conductors of electricity.

3. Correct option: (D) Magnesium

Substances that have one type of atoms is called pure substance.
Duralumin → Al + Cu + Mg
Magnalium → Mg + Al
Bell metal → Cu + Sn
Magnesium → element → pure substance

4. Correct option: (D) Aluminium

In metals, there are free held electrons which creates a sea of electrons.
Because of these free electrons metals are good conductors of electricity.
But non-metal crystal doesn't contain any free electrons, hence they are not good conductors of electricity.
As sulphur, diamond (carbon) , Phosphorus are non-metals and are not good conductors of electricity.
But Aluminium is metal and one of the best conductor of electricity.

5. Correct option: (A) Diamond

Diamond is an allotrope of carbon.
In diamond each carbon is covalently bonded with other four carbons. This arrangement makes a tetrahedral crystal in which the atoms are closely packed and make a rigid structure. Thus diamond is the hardest substance.

6. Correct option: (D) Magnesium

Magnesium is present in chlorophyll.
It is located at the centre of chlorophyll.
Magnesium usually helps for the proper functioning of chlorophyll and also stimulates some enzymes to carry photosynthesis .

7. Correct option: (B) Iron

Haemoglobin is a metal complex.

The heme group of haemoglobin contains iron as the central metal atom.

This heme group is responsible for the binding and release of the oxygen atoms

8. Correct option: (C) Graphite

Graphite has layered structure which are held by weak Van der Waals force of attraction. The layers are composed of planar hexagonal rings of carbon. For these lars graphite is a slippery substance and used in pencil.

9. Correct option: (A) Free electron

Graphite is an allotrope of carbon. It has layered structure which are held by weak Van der Waals force of attraction. The layers are composed of planar hexagonal rings of carbon. It has free electrons in its structure as the fourth valence electron of carbon atom remains unpaired.

Hence Graphite is a good conductor of heat & electricity due to presence of free electrons.

10. Correct option: (A) Phosphorous

Red phosphorus is an allotrope of phosphorus which is actually used in match sticks.

It has low ignition temperature. Hence it can be burnt easily. Thus it is used in match sticks.

Ignition temperature of a substance is the minimum temperature that is required to start or cause combustion.

11. Correct option: (C) Gold, Mercury

An amalgam is a homogeneous mixture of metal and mercury.

If in alloy one of the components is mercury then it is called and amalgam.

Example *:* gold amalgam → Au + Hg → gold + mercury

12. Correct option: (A) isolation

An ore is full with many undesired earthly substances that are called gangue.

The process of removing unwanted materials such as clay, sand etc. from the ore is termed as zone refining, benefaction or dressing.

Whereas isolation is used for conversion of oxide of the ore.

13. Correct option: (C) *zinc with iron*

Iron in presence of air and moisture gets rusted. The rusting can be prevented by a coating with thin layer of zinc. This process of coating zinc on iron is called galvanisation

14. Correct option: (A) alloy of mercury

An amalgam is a homogeneous mixture of metal and mercury.

If in alloy one of the components is mercury then it is called and amalgam.

Example – (Na + Hg) is called sodium amalgam.

15. Correct option: (B) Cu

Fe , Mg and Zn are placed above Cu in reactivity series. Thus they can react with dilute HCl and produce H_2 gas.

$Fe + 2HCl\ (dil.) \rightarrow FeCl_2 + H_2 \uparrow$

$Zn + 2HCl\ (dil.) \rightarrow ZnCl_2 + H_2\uparrow$

$Mg + 2HCl\ (dil.) \rightarrow MgCl_2 + H_2\uparrow$

But Cu is less reactive and cannot react with dilute HCl

Cu + 2HCl (dil.) → No reaction

16. Correct option: (A) Gold , Platinum, B) Gold, Silver

Silver, gold and platinum come downwards in the reactivity series. They are less reactive and called inert metal. Even if in high temperature these elements do not react with oxygen.

But Zinc , lead and copper come upwards in the reactivity series. Hence they are more reactive and can react with oxygen.

17. Correct option: (D) electrolytic reduction

The oxides of highly reactive metals like K, Na cannot be oxidised by carbon. Because they have high affinity for oxygen then carbon.

Thus carbon is unable to remove oxygen from metal oxides.

Thus oxides of highly reactive metals are reduced by a process called electrolytic reduction.

Electrolytic reduction is a process where the metal ions comes in the touch of cathode, accepts electrons from it and get reduced .

18. Correct option: (B) ores

A mineral is called an ore if a metal is profitably extracted from it.

Through metallurgy metals are extracted from ore.

19. Correct option: (B) refining

The process used in purifying of metals is called refining.

Removal of impurities from metal ores are called enrichment of ore.

Cracking is a process in which the minerals are dissolved in acid and gets simplified

Roasting is process in which ore is converted into its oxide by heating it below its melting point in presence of excess air.

20. Correct option: (C) aluminothermic

Thermite reaction is a highly exothermic reaction which involves reaction between iron oxide with aluminium to produce molten iron.

This is also known as aluminothermic process.

Hall's process is the process for smelting aluminium.

Bayer's process is used to produce alumina from refined bauxite.

Lead chamber is used to produce sulfuric acids industries

21. Correct option: (C) calcinations

The process in which carbonate ores are converted to oxides is known as calcination.

In this process the ore is heated to high temperature in presence of absence of air.

22. Correct option: (A) carbon

Alloy is a homogeneous mixture of two or more metals or a metal and a nonmetal.

Carbon is non-metal which forms an alloy.

Example of carbon alloy – Carbon steel

23. Correct option: (C) zinc is more reactive than hydrogen

Zinc is situated above hydrogen in the reactivity series. So zinc is more reactive than hydrogen.

Hence it can react with HCl and replace hydrogen to produce H_2 gas.

$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

24. Correct option: (D) collectors; froth stabilizers

Froth floatation is the method employed in sulphide ores to remove the gangue.
To enhance the mineral particle non-wettability, fatty acids, xanthates, pine oils, etc are used as collectors.
For the stabilization of froth, cresols, aniline, etc can be used as froth stabilizers.

25. Correct option: (C) depressants

Froth floatation is the method employed in sulphide ores to remove the gangue.
It Is possible to separate two sulphide ores by adjusting the proportion of oil to water or by using depressants.

26. Correct option: (A) an alloy

Alloy is a homogeneous mixture of two or more metals or a metal and a nonmetal.
Minerals is a substance that is formed in the earth.
A mineral is called an ore if a metal is profitably extracted from it.
Flux is a chemical that is added to molten metal to connect with impurities that may then be removed.

27. Correct option: (B) hydrogen

Hydrogen is placed in the reactivity series. Because it has one electron in its valence shell. It can gain one electron to become a negative ion (H^-) and acts like a nonmetal.
And also can donate one electron to become a positive ion (H^+) and acts like a metal.

28. Correct option: (C) Ag

Corrosion is process which converts a refined metal into a more chemically stable oxide.
The metals that are less reactive does not corrode when exposed to air.
As silver (Ag) comes downwards in the reactivity series, it is less reactive and does not corrode.
Whereas Cu, Al and Fe come upwards in the reactivity series and are highly reactive. Thus corrosion occurrs.

29. Correct option: (B) reduction

The process of removal of unwanted materials from the ore is known as ore concentration.
Ore concentration can be accomplished through a variety of methods, including hydraulic washing, magnetic separation, froth flotation.
Hydraulic washing is used when the impurities are lighter and ore particles are heavier.
Froth floatation is the method employed in sulphide ores to remove the gangue.
Magnetic separation is a process in which solid ores are separated by a magnet.

30. Correct option: (B) Magnesium

Magnesium comes above zinc in reactivity series. Whereas Lead, copper and silver comes below zinc in reactivity series.
Thus Magnesium is more reactive than zinc and it can replace zinc in zinc sulphate solution.

$Mg(s) + ZnSO_4(aq) \rightarrow MgSO_4(aq) + Zn(s)$

6. General Organic Chemistry

Solutions

LEVEL – I

1. Correct option: (B)

$H_2N-CO-NH_2$(urea/carbamide) is considered an organic compound. Organic compounds are primarily composed of carbon and hydrogen atoms, often with other elements like oxygen, nitrogen, sulfur, and more. In this case, $H_2N-CO-NH_2$ is a compound that contains carbon and hydrogen atoms.

2. Correct option: (D)

Friedrich Wohler is famous for synthesizing urea from ammonium cyanate in 1828, marking a pivotal moment in the history of chemistry and demonstrating that organic compounds could be synthesized from inorganic substances.

3. Correct option: (D)

Friedrich Wohler synthesized urea from ammonium cyanate in 1828, and this is considered a landmark achievement in the history of chemistry, as it demonstrated that organic compounds could be artificially produced from inorganic materials.

4. Correct option: (A)

In the early days of chemistry, organic compounds were defined as those compounds that contained carbon, and urea contains carbon atoms in its chemical structure. So, it is an organic compound.

5. Correct option: (C)

For the first time in 1845, German chemist Hermann Kolbe synthesized acetic acid from inorganic compounds.

6. Correct option: (D)

Organic molecules are composed of carbon and hydrogen atoms, often with other elements like oxygen, nitrogen, sulfur, and more. Based on this definition, the following options are organic molecules:

Proteins (composed of carbon, hydrogen, oxygen, nitrogen, and sometimes sulfur) **Rubber** (composed of carbon and hydrogen atoms, typically in the form of long hydrocarbon chains)

7. Correct option: (A)

Organic compounds are defined by the presence of carbon atoms in their chemical structure. While organic compounds can also contain other elements like hydrogen, oxygen, nitrogen, and others, carbon is the key element that distinguishes them from inorganic compounds.

8. Correct option: (B)

Hydrogen is the second most abundant element in organic compounds, as it frequently forms covalent bonds with carbon in organic molecules.

9. Correct option: (D)

Compounds that contain carbon and hydrogen only are called Hydrocarbons.

10. Correct option: (D)

The radioactive isotope of carbon widely used in carbon dating to determine the age of fossils is C-14 (Carbon-14).

11. Correct option: (C)

Carbon has the unique ability to form long chains and rings with other carbon atoms, creating a vast variety of organic compounds. This property is known as catenation, and

it allows for the formation of a wide range of organic molecules, contributing to the diversity of carbon compounds.

12. Correct option: (C)

Carbon has the strongest catenation ability among the listed elements. Carbon is well-known for its ability to form long chains and rings with other carbon atoms, resulting in the vast diversity of organic compounds. While other elements like silicon, sulfur, and phosphorus can also form chains, they are not as versatile or extensive in their catenation as carbon.

13. Correct option: (C)

Carbon-nitrogen triple bonds are found in compounds like cyanides, where carbon is bonded to a nitrogen atom with a triple covalent bond. Each carbon and nitrogen atom will share three electrons. As a result, six electrons are shared, and a triple covalent bond is formed.

14. Correct option: (B)

Different compounds having the same molecular formula are called **Isomers**.

15. Correct option: (C) .

C_2H_6 is a saturated compound. Saturated hydrocarbons have only single covalent bonds between carbon atoms, and ethane (C_2H_6) is a saturated hydrocarbon consisting of single carbon-carbon and carbon-hydrogen bonds.

16. Correct option: (B)

C_3H_6 is an unsaturated compound. Unsaturated hydrocarbons have at least one or more carbon-carbon double or triple bonds. In this case, C_3H_6 (propene) has a carbon-carbon double bond, making it an unsaturated compound.

17. Correct option: (A)

Cyclohexane is an alicyclic compound. Alicyclic compounds are organic compounds that have carbon atoms arranged in a closed ring structure, and cyclohexane consists of a ring of six carbon atoms. It is a saturated alicyclic hydrocarbon with single bonds between all carbon atoms in the ring.

18. Correct options: (A, B, C.)

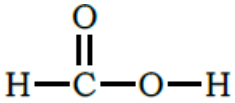
The non-benzenoid aromatic compounds are those which do not contain any benzene ring in their structure. So, options A, B, C are non-Benzenoid compounds.

19. Correct option: (B)

$H_3\underset{1}{C}-\underset{2}{C}\equiv\underset{3}{C}-\underset{4}{C}H_3$ in this compound 1st,4th carbon atom produce 4 sp^3 hybrid orbitals.

20. Correct option: (B)

Kekule structures are similar to Lewis Structures, but instead of covalent bonds being represented by electron dots, the two shared electrons are shown by a line.



LEVEL – II

1. Correct option: (C)

$$\underset{1}{C}H_3\text{-}\underset{2}{C}H=\underset{3}{C}H\text{-}\underset{4}{C}H_2\text{-}\underset{5}{C}H_3$$

A carbon atom attached to a double bond is sp^2 hybridized. Here C_2 is attached to double bond it is sp^2 hybridized and C_4 is sp^3 hybridized.

2. Correct option: (A)

A carbon atom attached to a double bond is sp^2 hybridized. A carbon atom attached to two double bonds or one triple bond is sp hybridized.

So here C_2 carbon atom is sp^2 hybridized. C_4 Carbon atom is sp^3 hybridized.

C_5 Carbon atom is sp hybridized.

3. Correct option: (B)

$(CH_3-CH_2-CH_2-CH_3)$ It can be written in form of C_4H_{10}. Here all C atoms are sp^3 hybridized so each gives 4 hybrid orbitals i.e. $4\times4=16$ in total. All hydrogen molecules can be calculated as $10\times1=10$.

Ratio can be written as $=10:16$
$=5:8$

4. Correct option: (C)

Neopentane (also known as 2, 2-dimethylpropane) has the molecular formula C_4H_{12}. All bonds in neopentane are sigma bonds. Bonds between C-C are4, bonds between C-H are 12 in total 4+12=16.

5. Correct option: (C)

The first homologue of an alkyne is C_2H_2, which is ethyne.

6. Correct option: (D)

The compounds in a particular homologous series differ by a $-CH_2$ unit.

7. Correct option: (B)

Given formula is $C_nH_{2n+1}OH$ place n=1 then we get C_2H_5OH (ethyl alcohol).

ONE OR MORE THAN ONE OPTION CORRECT TYPE

8. Correct options: (A, B, D)

$C_1-C_2\rightarrow2\pi,1\sigma$, $C_3-C_2\rightarrow0\pi,1\sigma$, $C_3-C_4\rightarrow1\sigma,1\pi$ bonds are present.

9. Correct options: (B, C)

10. Correct options: (A, B, D)

Compounds which obey huckels rule $(4n+2)\pi$ electrons and have a conjugated electron system are called as aromatic compounds.

-6π electrons.

-2π electrons

-2π electrons.

11. Correct options: (A, C)

COMPREHENSION TYPE

Passage – I:

12. Correct option: (A)

Primary carbon is those which are attached to one carbon atom only. So C_1 carbon 1^0 atom because it is attached to one carbon atom only. C_2 Carbon 1^0 atom because it is attached to one carbon atom only.

13. Correct option: (C)

Primary carbon are which is attached to one carbon atom only, secondary carbon are those which are attached to two carbon atoms, tertiary carbon are those which is attached to three carbon atoms, and quaternary carbon are those which is attached to three carbon atoms.

In the given compound carbons 1 & 2 are 1^0 and 4^0.

14. Correct option: (C)

Tertiary carbon is those which is attached to three carbon atoms. In the given question only C_3 fulfill the above statement.

15. Correct option: (B)

Tertiary carbon is those which are attached to three carbon atoms. Carbons 2 and 4 are tertiary carbon atoms.

MATRIX MATCHING TYPE

19. Correct Matching is: a-P; b-R; c-S; d-Q

a) C_2H_6 - In ethane, each carbon atom forms sigma bonds with three hydrogen atoms and a sigma bond with the other carbon atom. These sigma bonds are formed by the overlap of atomic orbitals. To accommodate this bonding, carbon atoms in ethane undergo sp^3 hybridization, which means that one s orbital and three p orbitals on each carbon atom combine to form four sp^3 hybrid orbitals. These four sp^3 hybrid orbitals then overlap with the hydrogen 1s orbitals to form four sigma bonds.

Correct option is P.

b) C_2H_2 - Acetylene has a linear molecular structure, and the carbon atoms are bonded together by a triple bond. In acetylene, each carbon atom forms two sigma (σ) bonds and one pi (π) bond. To accommodate this bonding arrangement, carbon atoms in acetylene use sp hybridization.

Correct option is R.

c) C_3H_6 - C3H6 can represent two different isomers Propene (or Propylene). Molecular Structure: $CH_2-CH_2-CH_3$

In propene, the carbon-carbon double bond (C=C) is present. The carbon atoms participating in the double bond are sp2 hybridized. The carbon atom bonded to only one other carbon and one hydrogen is sp3 hybridized.

Cyclopropane- In Cyclopropane, all three carbon atoms are bonded to each other in a ring. Each carbon atom in Cyclopropane is sp^3 hybridized. So, the hybridization of carbon atoms in propene is sp^2, while in Cyclopropane, it is sp^3.

Correct option is S.

d) The molecular formula C_7H_8 suggests that the compound likely contains a benzene ring, which is a six-membered ring of carbon atoms with alternating single and double bonds. The carbon atoms in a benzene ring are sp^2 hybridized. So, in C_7H_8, the carbon atoms are sp^2 hybridized due to the presence of the benzene ring.

Correct option is Q.

7. Solutions

Solutions

LEVEL- I

1. Correct option: (A)

When a solution can still dissolve more solute due to having less solute than its maximum amount, the solution is called unsaturated solution.

2. Correct option: (C)

This statement is incorrect because the solubility of gases generally decreases with an increase in temperature.

3. Correct option: (C)

A solution which remains in contact with undissolved solute is termed as **Saturated solution**.

A saturated solution is a solution in which no more solute can be dissolved at a given temperature and pressure, and any additional solute added will remain undissolved.

4. Correct options: (A, B, C)

The solubility of a gas in water can be influenced by the nature of the gas, the temperature of the solution, and the pressure of the gas above the solution.

5. Correct option: (A)

To find the solubility of NaOH in water, calculate the concentration of NaOH in grams per millilitre (g/mL).

Given:

Mass of NaOH = 10 grams

Volume of water = 100 mL

$$Solubility = \frac{Mass\,of\ NaOH}{Volume\,of\ water} \times 100$$

$$Solubility = \frac{10g}{100ml} \times 100$$

$$Solubility = 10g$$

6. Correct option: (B)

The temperature decreases, the solubility of the sugar solution also decreases. When the solubility decreases, the excess dissolved solute starts to crystallize out.

At 50°C: Solubility of the sugar solution is 20.

At 25°C: Solubility of the sugar solution is 10.

So, when the temperature drops from 50°C to 25°C, the excess solute will start to crystallize out until the solubility matches the new equilibrium solubility at 25°C.

The difference in solubility between 50°C and 25°C is $20-10=10$ grams per 100 grams of water.

This means that for every 100 grams of water, 10 grams of sugar will crystallize out as a solid when the temperature drops from 50°C to 25°C.

Therefore, the amount of solute that will crystallize out when the temperature drops from 50°C to 25°C is 10 grams.

7. Correct option: (B)

To form a saturated solution, the solubility limit needs to be reached. In this case, the solubility of KCl at 80°C is 50 g per 100 g of water.

If we have 200 g of water, the amount of KCl needed to form a saturated solution can be calculated using proportions.

Solubility of KCl at 80°C is 50 g per 100 g of water.

We can set up a proportion to find out how much KCl is needed for 200 g of water:

$$\frac{50g\ KCl}{100g\ water} = \frac{Xg\ KCl}{200g\ water}$$

Solving for X, the amount of KCl needed:

$$X = \frac{50 \times 200}{100} = 100g\ KCl$$

8. Correct option: (C)

To determine how much of the KNO_3 does not dissolve, we compare the amount of KNO_3 added with its solubility at 35°C.

Given:

Mass of KNO_3 added = 115 g

Solubility of KNO_3 at 35°C = 55 g per 100 g of water

At 35°C, the solubility of KNO_3 is 55 g per 100 g of water.

Therefore, for 100 g of water, the maximum amount of KNO_3 that can dissolve is 55 g.

However, 115 g of KNO_3 were added, which exceeds the solubility limit.The amount of KNO_3 that does not dissolve is the excess amount added over the solubility limit, which is 115g−55g=60g.

9. Correct option: (D)

Pressure typically affects the solubility of gases in liquids but has a negligible effect on the solubility of solids in liquids.

10. Correct option: (C)

Henry's law states that the solubility of a gas in a liquid is directly proportional to the partial pressure of that gas above the liquid, provided that the temperature remains constant.

In this case, when the pressure is doubled (from 1 atm to 2 atm), the amount of gas dissolved also doubles (from 0.05 mole to 0.1 mole), which is in accordance with Henry's law.

11. Correct option: (A)

The positive sign indicates that the dissolution process is endothermic, meaning it absorbs heat from the surroundings as the solute dissolves. Therefore, the correct answer is +ve (positive).

12. Correct option: (A)

Solubility of gases. It depends on Temperature and pressure.

13. Correct option: (D)

Henry's constant (KH) is indeed a function of the nature of the gas as well as the solvent and temperature. It represents the solubility of a gas in a particular solvent at a specific temperature.

14. Correct option: (A)

High pressure helps to force more carbon dioxide into solution, and low temperature enhances the solubility of gases in liquids according to Henry's law.

15. Correct option: (D)

The correct statement among the options is:

D) Pressure does not have a significant effect on the solubility of solids in liquids.

Changes in pressure typically affect the solubility of gases in liquids (Henry's law) rather than solids in liquids. Solids and liquids are generally considered to be incompressible, so option A is incorrect.

Solubility of NaCl (a polar substance) typically decreases in nonpolar solvents, so option B is incorrect.
Solubility of ether (a nonpolar substance) typically increases in nonpolar solvents, not in water, so option C is also incorrect.

16. Correct option: (B)

When dissolution is endothermic, meaning it absorbs heat, increasing the temperature typically increases solubility. This is because the endothermic process requires more energy, which is provided by the higher temperature, thereby allowing more solute to dissolve.

17. Correct option: (B)

Henry's law states that the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid when the temperature is constant. The Henry's constant (KH) represents the proportionality constant in this relationship.

18. Correct option: (A)

Henry's constant (KH) is a measure of the solubility of a gas in a liquid. The higher the value of KH, the more soluble the gas is in the liquid at that particular temperature.

19. Correct option: (C)

Henry's law states that the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid, at a constant temperature. According to Henry's law, the Henry's constant (KH) typically decreases with an increase in temperature, so option C is incorrect.

20. Correct option: (B)

The correct statement among the options is:

B) The solubility of gas O_2 increases with an increase in temperature.

According to Henry's Law, the solubility of a gas in a liquid generally increases with an increase in temperature.

LEVEL-II

1. Correct option: (C)

Alloys are homogeneous mixtures of two or more metals or a metal and a non-metal where the atoms of one element are uniformly dispersed among the atoms of another. This forms a solid solution where the solvent is typically a metal and the solute can be another metal or non-metal.

2. Correct option: (D)

When you add more solute to a saturated solution, it increases the concentration of the solute beyond what the solvent can hold at that temperature and pressure. This causes the solution to become unsaturated because there is now room for more solute to dissolve.

3. Correct option: (A)

The solubility of NaCl is typically affected by temperature changes. In general, as the temperature increases, the solubility of most solids in liquids also increases.
Given that the solubility of NaCl is 10 at 25°C, and we know that as temperature increases, the solubility generally increases, we can expect that the solubility at 35°C will be higher than 10.

4. Correct option: (C)

If the solubility of sugar is 20 grams at 40°C, and 80 grams of sugar have already been dissolved, then the solution is saturated. This means that the solution has reached its maximum capacity to dissolve sugar at that temperature.

Now, if 80 grams of sugar are dissolved, and the solubility limit is 20 grams, it means that 60 grams of sugar are in excess (80 grams dissolved - 20 grams solubility limit).

5. Correct option: (A)

A) Aquatic species are more comfortable in cold water rather than in warm water.

This statement is generally true. Many aquatic species, especially those adapted to cooler environments, are more comfortable in cold water because it holds more dissolved oxygen and can support more life.

6. Correct option: (C)

Air is primarily composed of various gases such as nitrogen, oxygen, carbon dioxide, and others. These gases are uniformly mixed in the atmosphere, making air a gas-gas solution.

7. Correct option: (D)

The phrase "of hydrogen over palladium" suggests a scenario where hydrogen is being absorbed or adsorbed onto the surface of palladium. This process, known as hydrogen absorption or adsorption onto palladium, is an example of: **Gas in solid.**

Palladium serves as the solid material onto which the gas (hydrogen) is being absorbed or adsorbed.

ONE ORE MORE THAN ONE OPTION CORRECT TYPE

8. Correct option: (A, D)

A) Solubility increases with a decrease in temperature, then ΔH solution is negative.

When solubility increases with a decrease in temperature, it indicates that the dissolution process is exothermic, leading to a negative ΔH solution.

B) Solubility increases with an increase in temperature, then ΔH solution is positive.This statement is generally true. When solubility increases with an increase in temperature, it indicates that the dissolution process is endothermic, leading to a positive ΔH solution.

C) If the hydration energy is greater than the lattice energy, then the ΔH solution is negative.

This statement is generally true. A negative ΔH solution typically indicates an exothermic process, where the energy released during hydration (solvent molecules surrounding the solute) exceeds the energy required to break the solute-solute interactions (lattice energy).

D) If the hydration energy is less than the lattice energy, then the ΔH solution is positive.

This statement is generally true. A positive ΔH solution typically indicates an endothermic process, where the energy required to break the solute-solute interactions (lattice energy) exceeds the energy released during hydration (solvent molecules surrounding the solute).

9. Correct option: (A, B, D)

A) At constant temperature, the solubility of a gas in a liquid is directly proportional to the pressure of the gas.

This statement is a simplified version of Henry's Law, which states that the solubility of a gas in a liquid is directly proportional to the partial pressure of that gas above the liquid, at a constant temperature. So, statement A is correct.

B) The partial pressure of the gas in the vapor phase (P) is directly proportional to the mole fraction of the gas (x) in the solution.

This statement is also consistent with Raoult's Law, which states that the partial pressure of a gas in the vapor phase is directly proportional to its mole fraction in the solution. So, statement B is correct.

C) Scuba divers use O_2 +He instead of $O_2 + N_2$.

This statement seems to be incomplete or contains a typo. Scuba divers typically use a mixture of oxygen and nitrogen, not oxygen and helium. Helium is used in diving gas

mixtures to reduce the effects of nitrogen narcosis and to decrease the risk of decompression sickness. Therefore, statement C is incorrect based on the provided information.

D) N_2 gas is more soluble at 293 K than 303 K.

In general, gases are more soluble at lower temperatures. So, it's more likely that N_2 gas is more soluble at 293 K than at 303 K. Therefore, statement D seems correct.

10. Correct option: (A, C)

A) **Nature of the solute**: Different solutes have different solubilities in a given solvent. The chemical structure, polarity, and interactions between solute and solvent molecules play a significant role in determining solubility.

B) **Temperature**: In general, the solubility of solids in liquids tends to increase with an increase in temperature for most substances. This is because higher temperatures provide more energy for solute particles to overcome intermolecular forces and dissolve in the solvent.

C) **Pressure**: Pressure typically has little effect on the solubility of solid solutes in liquid solvents under normal conditions. Unlike gases, which show significant changes in solubility with pressure, solids and liquids are relatively incompressible, so pressure variations usually do not affect their solubility to a significant extent.

D) **Volume of liquid**: The volume of the liquid does not directly affect the solubility of a solid in a liquid. Solubility is more dependent on factors like temperature, pressure, and the nature of the solute and solvent.

11. Correct option: (A, B)

A) **Temperature**: Generally, the solubility of liquids in gases decreases with an increase in temperature. Higher temperatures provide more energy to the gas molecules, allowing them to escape from the liquid phase more easily.

B) **Pressure**: The solubility of liquids in gases often increases with an increase in pressure. This relationship is described by Henry's Law, which states that the solubility of a gas in a liquid is directly proportional to the pressure of the gas above the liquid, at a constant temperature.

C) **Volume of liquid**: The volume of the liquid generally does not directly affect the solubility of a liquid in a gas. Solubility is primarily influenced by temperature and pressure.

D) **Mass of the gas**: The mass of the gas does not directly affect the solubility of a liquid in a gas. Solubility is primarily determined by temperature and pressure, as described by Henry's Law.

12. Correct option: (B)

Formation of a solution from a solute and a solvent is a physical process.

13. Correct option: (C)

The true solution among the options provided is **Sugar in water.**

In this case, sugar dissolves completely in water, forming a homogeneous mixture where the sugar molecules are evenly distributed throughout the water, creating a true solution.

COMPREHENSITON TYPE

Passage-I:

14. Correct option: (A)

According to Henry's law

$p_{N_2} = K_H . X_{N_2}$

$\Rightarrow X_{N_2} = \frac{0.987}{76800} = 1.29 \times 10^{-5}$

$Moles of H_2O in 1L = \frac{1000}{18} = 55.56$

Let us take the moles of nitrogen as "n".

$Total\ moles = n + 55.56$

$\therefore X_{N_2} = \frac{n}{n + 55.56} = 1.29 \times 10^{-5}$

$\Rightarrow n\left(1 - 1.29 \times 10^{-5}\right) - 55.56 \times 1.29 \times 10^{-5} = 0$

$\Rightarrow n = 0.7 \times 10^{-3} = 0.7 mil\ \lim oles$

15. Correct option: (A)

$Solubility\ of\ H_2S = 0.195 \frac{moles}{Kg}$

$1kg\ of\ solvent\ contains = \frac{1000}{18} = 55.55 moles$

$\therefore x_{H_2S} = \frac{n_{H_2S}}{n_{H_2S} + n_{H_2O}} = \frac{0.195}{0.195 + 55.55} = \frac{0.195}{55.745} = 0.0035$

Pressure at STP 0.987 bar =1 atm.

$P_{H_2S} = K_H \times X_{H_2S}$

$\therefore K_H = \frac{0.987 bar}{0.0035} = 282 bar$

16. Correct option: (B)

According to Henry's law,

$\frac{P_{O_2}}{P_{N_2}} = \frac{K_H(O_2) X_{O_2}}{K_H(N_2) X_{N_2}} \frac{X_{O_2}}{X_{N_2}}$

$\frac{K_H(O_2) X_{O_2}}{K_H(N_2) X_{N_2}} \frac{X_{O_2}}{X_{N_2}} = \frac{0.2}{0.8} \times \frac{6.6 \times 10^7}{3.3 \times 10^7} = \frac{1}{2}$

17. Correct option: (B)

$Pressure\ of\ CO_2 = 25 atm = 2.5 \times 1.013 \times 10^{13} P_a = 2.533125 \times 10^5 P_a$

$Given K_H = 1.67 \times 10^8 P_a$

According to Henny's law,

$P = (K_H)(X)$

$X = \frac{P}{K_H} = \frac{2.53 \times 10^5}{1.67 \times 10^8} = 1.52 \times 10^{-3}$

Given 500ml of soda water

Volume of water =500ml

Density of water =1g/ml

Use formula mass =volume × density

$Molar\ mass\ of\ water\ (H_2O) = 18g\,mol^{-1}$

$$Number\ of\ moles = \frac{mass}{molarmass}$$

$$Number\ of\ mass\ of\ water = \frac{500}{18} = 27.78\,mole\ of\ water$$

$$mole\ function, x = \frac{n_{CO_2}}{n_{CO_2} + n_{H_2O}} \approx \frac{n_{CO_2}}{n_{H_2O}} = 1.52\times10^{-3} = \frac{n_{CO_2}}{27.78}$$

$$n_{CO_2} = 0.042 moles$$

$$molar\ mass\ of\ CO_2 = 12 + 2(16) = 44\frac{g}{moles}$$

$$mass = (molar\ mass)(number\ of\ moles) = (0.042)(44)$$

$$mass = 1.8 grams$$

Quality of CO_2 is 1.8grams.

MATRIX MATCHING TYPE

18. Correct Matching is: a) Stainless steel - Q) Solid in solid
b) Sugar in water - P) Solid in liquid
c) Camphor in air - S) Solid in gas
d) Chloroform mixed with nitrogen gas - R) Liquid in gas

INTEGER TYPE

19. Correct Answer is:
Henry's law states that the amount of dissolved gas is proportional to its partial pressure in the gas phase. The proportionality factor is called the Henry's law constant.

Given $K_H = 4.27\times10^5 mmH$
$P = 760 mmHg$

If X is the molar fraction/Solubility then
According to Henry's law,

$$p = K_H X$$

$$760 = 4.27\times10^5 \times X$$

$$X = \frac{760}{4.27\times10^5} = 178\times10^{-5}$$

20. Correct Answer is:

$$P_{N_2} = K_H X_{N_2}$$

$$X_{N_2} = \frac{3}{4.27\times10^5} = 7.02$$

Vapour Pressure of Liquid Solution

LEVEL-I

1. Correct option: (C)
The vapor pressure of a solution depends on the vapor pressure of the individual components and the mole fraction of each component in the solution.
In the case of an aqueous solution of methanol in water, methanol is the solute and water are the solvent. Methanol has a lower boiling point and higher vapor pressure compared to water.

When methanol is added to water, the vapor pressure of the resulting solution will be higher than that of pure water. This is because methanol molecules escape into the vapor phase more readily than water molecules due to weaker intermolecular forces.

2. Correct option: (B)

Raoult's law states that the vapor pressure of a solvent above a solution is directly proportional to the mole fraction of the solvent in the solution. This law is applicable to ideal solutions, which are solutions that follow certain assumptions including:

The interactions between solute-solute, solvent-solvent, and solute-solvent particles are similar.

The solution behaves like an ideal gas.

Given this background, Raoult's law is generally applicable to:

B) Solutions containing volatile solvents only

It is most commonly used with solutions where the solute is non-volatile or significantly less volatile than the solvent, allowing the solvent's vapor pressure to dominate the behaviour of the solution.

3. Correct option: (B)

Vapor pressure generally increases with increasing temperature and decreasing boiling point. A substance with a lower boiling point typically has a higher vapor pressure at a given temperature compared to substances with higher boiling points.

Here are the boiling points given:

$C_6H_6\left(Benzene\right):80°C$

$CH_3OH\left(Methanol\right):65°C$

$C_6H_5NH_2\left(Aniline\right):184°C$

$C_6H_5NO_2\left(Nitrobenzene\right):212°C$

Among the given options, CH_3OH (methanol) has the lowest boiling point at 65°C. Therefore, it will have the highest vapor pressure at room temperature, which is typically around 20-25°C.

4. Correct option: (C)

Vapour pressure is same because it doesn't depend on the volume it depends only on Temperature.

5. Correct option: (B)

Weak intermolecular forces of attraction in a liquid typically result in lower boiling points and higher vapor pressures. So, among the options:

B) High vapor pressure

This is the characteristic that best characterizes weak intermolecular forces of attraction in a liquid. High vapor pressure indicates that molecules are escaping from the liquid phase into the vapor phase readily, which occurs more readily when intermolecular forces are weak.

6. Correct option: (B)

Raoult's Law states that the vapor pressure of a solution is directly proportional to the mole fraction of each component in the solution.

The total vapor pressure of the solution ($P_{solution}$) can be calculated using the mole fraction of each component and the vapor pressures of the pure components.

Given:

Vapor pressure of pure A $A\left(P_A^0\right)=100mm$

Vapor pressure of pure B $B\left(P_B^0\right)=150mm$

Moles of A = 2

Moles of B = 3

calculate the mole fractions of $A\left(X_A\right)$ and $B\left(X_B\right)$ in the solution:

$$X_A=\frac{moles\,of\ A}{total\,moles}=\frac{2}{2+3}=\frac{2}{5}$$

$$X_B=\frac{moles\,of\ B}{total\,moles}=\frac{3}{2+3}=\frac{3}{5}$$

$$P_{solution}=X_A\cdot P_A^0+X_B\cdot P_B^0$$

$$P_{solution}=(\frac{2}{5}\times 100mmHg)+(\frac{3}{5}\times 150mmHg)$$

$$P_{solution}=40mmHg+90mmHg=130mmHg$$

7. Correct option: (A)

$P_B^0=957mm$

$P_T^0=379.5mm$

According to Roults law,

Normal pressure p=760mm.

$$P=P_B^0X_B+P_T^0X_T$$

$$760=957X_B^0+379.5\left(1-X_B\right)$$

$$X_B^0=0.66$$

$$X_T^0=1-X_B^0=0.34$$

$$X_T^0=0.34$$

8. Correct option: (D)

According to Raoult's law, $p=p^0\times x$

p=partial vapour pressure of pure component A.

p^0=vapour pressure of component A

x=mole fraction of component A.

Substitute values in the above expression.

$$X_B=\frac{P^0-P}{P^0}=\frac{0.80-0.50}{0.80}=0.375$$

Hence, the mole fraction of component B is 0.375.

9. Correct option: (A)

$PA=\ 40mm\ of\ Hg$

$PT=32\ mm\ of\ Hg$

By Roults law,

$$P_T=P_A\times X_A$$

$$32=X_A\times 40$$

$$X_A=\frac{32}{40}$$

$$X_A=0.8$$

10. Correct option: (A)

$p = P_B^0 \chi_B$

$\chi_B = \frac{167}{268} = 0.623$

$\chi_{Solute} = 1 - \chi_B = 0.377$

11. Correct option: (A)

Mole fraction of liquid can be written as $\chi_l = \frac{0.2}{0.2 + 4.0} = 0.048$

Mole fraction of water can be written as $\chi_w = 1.0 - 0.048 = 0.952$

$0.210 = P_t^0 \chi_l + P_w^0 \chi_w$

$0.210 = P_l^0 \times 0.048 + 0.198 \times 0.952$

$P_l^0 = \frac{0.210 - 0.198 \times 0.952}{0.048} = 0.448 bar$

12. Correct option: (A)

According to Roults law,

$P_A = P_A^0 x_A$

Vapour pressure of pure ethyl alcohol $(p_A^O) = 40 mm of\ Hg$

Mole fraction of ethyl alcohol= $(x_A) = 0.80$

Vapour pressure of ethyl alcohol in solution= $(p_A) = 40 \times 0.80 = 32 mmHg$

13. Correct option: (C)

14. Correct option: (B)

Let X_A denote mole fractions of Ether.

$P_{solution} = P_s = P_A^0 \cdot x_A$

$P_{solution} = 640 \times 0.5 + 280 \times 0.5$

$P_{solution} = 460 mm$

Mole fraction of ether in vapour state = $\frac{x_A \cdot P_A^0}{P_s} = \frac{0.5 \times 640}{460}$

15. Correct option: (D)

The liquid pair of acetone-chloroform shows a positive deviation from Raoult's law.

16. Correct option: (D)

Carbon tetrachloride (CCl_4) and chloroform ($CHCl_3$) have different molecular structures and types of intermolecular forces. The difference in molecular structure and the presence of dipole moments in chloroform can lead to deviations from Raoult's law, making this mixture likely to be non-ideal.

17. Correct option: (B)

A mixture of chloroform and acetone shows a negative deviation from Raoult's law.

$\Delta V_{mixture} < 0 or V_t < (30 + 50) ml$

18. Correct option: (B)

Positive deviation from Raoult's Law occurs when the actual vapor pressure of the components in the solution is greater than what is expected from Raoult's Law. This typically happens when the intermolecular forces between unlike molecules are weaker than those between like molecules, leading to an increased vapor pressure.

Out of the given options, the pair of solutions that exhibit positive deviation from Raoult's Law are:

B) $CCl_4 + CHCl_3$

19. Correct option: (A)

Raoult's law is not applicable to solutions containing non-volatile solutes or solutes that dissociate into ions in solution. Among the options provided, the only one that fits this criterion is:

A) 1M NaCl

When NaCl dissolves in water, it dissociates into Na^+ and Cl^- ions, making it a non-volatile solute and rendering Raoult's law inapplicable.

20. Correct option: (A)

Raoult's law is obeyed for entire concentration range and temperatures

The enthalpy change of mixing is zero $\Delta H_{mix} = 0$

The volume of mixing is zero. \ Triangle $V_{mix} = 0$

But the entropy of mixing is not equal to zero $\Delta S_{mix} \neq 0$

LEVEL-II

1. Correct option: (B)

In a mixture of components, A and B that exhibits negative deviation from Raoult's law, the interactions between unlike molecules (A-B) are weaker than the interactions between like molecules (A-A and B-B). This scenario leads to the formation of a solution with a lower vapor pressure than predicted by Raoult's law.

2. Correct option: (B)

If liquid A and B form ideal solutions, it means that there are no intermolecular forces between unlike molecules, and the solution obeys Raoult's law. In such cases:

B) The enthalpy of mixing is zero

In ideal solutions, the enthalpy of mixing is indeed zero because there are no significant intermolecular forces between the solute and solvent molecules.

3. Correct option: (C)

$P_{total} = 254 - 119\chi_A$

For, $P_A^0 : put\, \chi_A = 1 \Rightarrow P_A^0 = 135 mmHg$

$P_B^0 : put\, \chi_B = 0 \Rightarrow P_B^0 = 254 mmHg$

4. Correct option: (D)

2:3 molar ratio and its total $(2+3=5)$ so the mole fractions will be $\frac{2}{5} = 0.4$ and $\frac{3}{5} = 0.4$ respectively for toluene and benzene respectively.

$Partial\ pressure\ of\ toluene = mole\ fraction \times vapour\ pressure = 0.4 \times 28 bar = 11.2\ bar$

$partial\ pressure\ of\ benzene = 0.6 \times 95 bar = 57\ bar$

so total pressure $11.2 + 57 = 68.2\ bar$

5. Correct option: (B)

$p_A^0 : p_B^0 = 1:2$

$p_A : \chi_A \times p_A^0 = A \times \chi_A$

$p_B : \chi_B \times p_B^0 = 2A \times 2\chi_A$

$P_{Total} = A\chi_A + 4A\chi_A = 5A\chi_A$

$\chi_A(vapour\ phase) = \frac{P_A}{P_{total}}$

$\chi_A(vapour\ phase) = \frac{A\chi_A}{5A\chi_A} = \frac{1}{5} = 0.2$

6. Correct option: (A)

The molar masses of heptane and octane are 100 g/mol and 114 g/mol.

25g of heptane corresponds to $\frac{25}{100} = 0.25$

35g of octane corresponds to $\frac{35}{114} = 0.31 moles$

Mole fraction of heptane can be written as $\chi = \frac{0.25}{0.25+0.32} = \frac{0.25}{0.57} = 0.43$

Mole fraction of octane can be written as $\chi = 1 - 0.43 = 0.57$

Partial pressure of heptane $p = 0.43 \times 105 = 45.15 kPa$

Partial pressure of octane $p' = 0.57 \times 46 = 26.22 kPa$

$P = p + p'$

Total Vapour pressure of solution $P = 45.15 + 26.22 kPa$

$P = 71.37 kPa$

7. Correct option: (B)

8. Correct option: (C)

Given $P^0_{methyl\ alcohol} = 0.158 bar$

$P_{solution} = 0.095 bar$

According to Raoult's law, $P = \chi_1 P^\circ$ (Mole fraction of ethyl alcohol)

$0.095 = \chi_1 \times 0.158$

$\chi_1 = \frac{0.095}{0.158} = 0.601$

9. Correct option: (C)

In vapour phase $P_A = \gamma_A P_{total}$

$P_A = 0.4 \times 600 = 240$

$P_A = P_A^0 \times X_A$

$240 = P_A^0 \times 0.60$

In liquid phase $P_A^0 = \frac{240}{0.60}$

$P_A^0 = 400$

$P_A^0 = 400 torr$

$$P_{total} = P_A^0 X_A + P_B^0 X_B$$
$$600 = 400 \times 0.6 + P_B^0 (1 - 0.6)$$
$$\therefore P_B^0 = \frac{600 - 240}{0.4} = 900 torr$$
$$P_A^0 = 400 torr, P_B^0 = 900 torr$$

10. Correct option: (C)

For total miscible liquids,

$$P_{total} = Mol.fraction of A \times p_A^0 + Mol.fraction of B \times p_B^0$$

No. of moles of A = $= \frac{28}{140}$

Liquid B is water, its mass is (100-28) =72

No. of moles of B = $= \frac{72}{18}$

Total number of moles = 0.2 + 4.0 = 4.2

As we know $P_{total} = 160 mm$

$$p_B^0 = 150 mm$$
$$160 = \frac{0.2}{4.2} \times p_B^0 + \frac{4.0}{4.2} \times 150$$
$$p_A^0 = 17.15 \times \frac{4.2}{0.2} = 360.15 mm$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct option: (C, D)

A) **Benzene + Toluene:**

These substances are relatively similar in structure and have comparable intermolecular forces. As a result, they form an almost ideal solution. Their interactions are close to what is expected in an ideal solution, making this pair a near-ideal solution.

B) **Chloroform + Benzene:**

Chloroform and benzene have different chemical structures and likely exhibit different types of intermolecular interactions. Their solution is likely to be non-ideal due to the dissimilarity in molecular structure and polarity.

C) **Water + Hydrochloric Acid:**

Water and hydrochloric acid form an ionic solution due to the dissociation of hydrochloric acid into ions (H+ and Cl-) in water. The formation of ions significantly affects the properties of the solution, making it a non-ideal solution.

D) **Ethyl Alcohol + Water:**

Ethyl alcohol and water form hydrogen bonds with each other due to their polar nature. However, the formation of hydrogen bonds between the two molecules is not as strong as the hydrogen bonds formed within water molecules. This makes the solution deviate from ideality, especially at higher concentrations of alcohol.

So, the non-ideal solutions from the given options are:

C) Water + Hydrochloric Acid

D) Ethyl Alcohol + Water

12. Correct option: (D)

Benzene and toluene are non-polar, operating intermolecular forces are almost similar. Hence, they form an ideal solution.

13. Correct option: (B)

Given,

$W_{H_2O} = 100g, Mw_{H_2O} = 18g\ mol^{-1}$

$W_{urea} = 10g, Mw_{urea} = 60g\ mol^{-1}$

$W_{sucrose} = 2g, Mw_{sucrose} = 342g\ mol^{-1}$

$P^0_{H_2O} = 0.3atm$

$$Amount\ fraction\ of\ solvent = \frac{Amount\ of\ solvent}{Amount\ of\ (solvent\ +\ solute)}$$

$$Amount\ fraction\ of\ solvent = \frac{\frac{100}{18}}{\frac{100}{18}+\frac{1}{60}+\frac{2}{342}} = \frac{5.556}{5.577} = 0.99$$

According to Roults law,

$P_A = P^0_A \chi_A = 0.3 \times 0.99 = 0.29atm$

COMPREHENSION TYPE

Passage-I:

14. Correct option: (C)

The total moles of the solution:

$n_{total} = n_A + n_B = 2 + 3 = 5moles$

Mole fractions of A, B is

$$X_A = \frac{n_A}{n_{total}} = \frac{2}{5} = 0.4$$

$$X_A = \frac{n_B}{n_{total}} = \frac{3}{5} = 0.6$$

According to Dalton's law of partial pressures, the total vapor pressure (P_{total}) of the solution is the sum of the partial pressures contributed by each component:

$P_{total} = P_A \times X_A + P_B \times X_B$

$P_{total} = (120\,mmHg \times 0.4) + (180\,mmHg \times 0.6)$

$P_{total} = 48\,mmHg + 108\,mmHg$

$P_{total} = 156\,mmHg$

15. Correct option: (D)

According to Raoult's law, $p_A = p^0_A X_A$

Vapour pressure of pure ethyl alcohol $(p^0_A) = 40mm of\ Hg$

Mole fraction of ethyl alcohol $(X_A) = 0.60$

Vapour pressure of ethyl alcohol in solution $(p_A) = 40 \times 0.60 = 24$

16. Correct option: (C)

Let us consider mole fraction of A $= \chi$

mole fraction of B= $= 1 - \chi$

Partial pressure of A $= 0.6\chi$

Partial pressure of B $= 0.2(1 - \chi)$

$0.6\chi = 0.2(1-\chi)$

$0.6\chi = 0.2 - 0.2\chi$

$0.8\chi = 0.2$

$\chi = 0.25$

17. Correct Matching is: a) Q b) PS c) PS d) R

a) $C_2H_5OH+H_2O$ - This combination represents ethyl alcohol (C_2H_5OH) mixed with water (H_2O), which typically forms an ideal solution obeying Raoult's law (Option P).

b) $C_2H_5Br + C_2H_5I$ - This combination represents ethyl bromide (C_2H_5Br) mixed with ethyl iodide (C_2H_5I). Since the molecules are similar, they are likely to form an almost ideal solution, obeying Raoult's law (Option P).

c) $P_A = P_A^0\chi_A$ - This is the expression for Raoult's law, which describes the behaviour of ideal solutions (Option S)

d) $H_2O+H_2SO_4$ - This combination represents water (H_2O) mixed with sulfuric acid (H_2SO_4), which typically forms a non-ideal solution with negative deviation from Raoult's law (Option R).

INTEGER TYPE

18. Correct Answer is:

Molar mass of benzene is $C_6H_6 = 78gmol^{-1}$

$$n(benzene) = \frac{75}{78} = 0.96$$

Molar mass of Toluene is $C_6H_5 - CH_3 = 92gmol^{-1}$

$$n(Toluene) = \frac{46}{92} = 0.5$$

$$x(Benzene) = \frac{0.96}{0.96+0.5} = \frac{0.96}{1.46} = 0.657$$

$$p(Benzene) = x(Benzene) \times P^{\circ}(Benzene)$$

$$p(Benzene) = 0.657 \times 75$$

$$p(Benzene) = 49.27$$

19. Correct Answer is:

$$P^{\circ} - P_S = P^{\circ} \times Mole\ fraction\ of\ solute$$

$$10 = P^{\circ} \times 0.2$$

$$20 = P^{\circ} \times \chi_2$$

$$\chi_2 = 0.4$$

$$\chi_1 = 1 - 0.4 = 0.6$$

20. Correct Answer is:

$$P_A = 520mmHg$$

$$P_B = 1000mmHg$$

$$P_C = 760mmHg$$

$$P_T = X_A P_A + X_B P_B$$

P_T=Total vapour pressure.

P_A, P_B=Vapour pressure of A, B.

X_A, X_B =Mole fraction of A, B.

$$P_T = X_A P_A + X_B P_B$$

$$760 = 520X_A + 1000(1 - X_A)$$

$$760 = 520X_A + 1000 - 1000X_A$$

$$240 = 480X_A$$

$$X_A = \frac{1}{2} = 0.5 = 50\%$$

Solutions & Colligative Properties

LEVEL-I

1. Correct option: (A)

Only Osmotic pressure is a colligative property.

'Relative lowering of vapour pressure of the solvent' is a colligative property but not only vapour pressure. Similar argument for boiling point and freezing point.

2. Correct option: (B)

Since on increasing the number of moles the molality also increases which is directly proportional to an elevation in boiling point.

Thus, the boiling point of the 2M of glucose will be greater than 1M glucose.

3. Correct option: (A, D)

$$For sucrose: \frac{3.42 \text{ g}}{342.3 \text{ g/mol}} = 0.01 \text{ mol}$$

$$For glucose: \frac{9g}{180.16\, g/mole} = 0.05\, mol$$

$$For urea(6g): \frac{6\,g}{60.06\, g/mol} = 0.1\, mol$$

$$For urea(0.6)\, g: \frac{0.6\, g}{60.06\, g/mol} = 0.01\, mol$$

4. Correct option: (B)

Equation can be written as $P_{solution} = X_{solvent} \times P_{solvent}$

$P_{solution}$ =vapor pressure of the solution.

$X_{solvent}$ =mole fraction of the solvent in the solution.

$P_{solvent}$ =vapor pressure of the pure solvent.

Given $P_{solution}$ =750 mmHg (glucose solution)

$P_{solvent}$ =760 mmHg (water).

Rearrange Raoult's law to solve for glucose $X_{solvent}$

$$X_{glucose} = \frac{P_{solution}}{P_{solvent}}$$

$$X_{glucose} = \frac{750 mmHg}{760 mmHg}$$

$$X_{glucose} = 0.9868$$

The mole fraction of glucose is:

$X_{glucose} = 1 - X_{water}$

$X_{glucose} = 1 - 0.9868$

$X_{glucose} = 0.013$

5. Correct option: (C)

Among the options provided, "C) Boiling point" is not a colligative property.

The colligative properties are those that depend on the number of solute particles in a solution, not on the nature of the solute particles.

6. Correct option: (B)

Colligative properties are used for the determination of B**) Molar mass**.

Colligative properties depend only on the number of solute particles in a solution, not on the nature of the solute particles themselves. Therefore, they can be used to determine the molar mass of the solute. The extent of the colligative property change is directly proportional to the concentration of the solute particles in the solution.

7. Correct option: (C)

The RLVP is directly proportional to the molality of the solution, which is calculated as:

$$Molality = \frac{moles\,of\ solute}{mass\,of\ solvent\,in\,kg}$$

Sucrose ($C_{12}H_{22}O_{11}$) with a molar mass of approximately $342.3 g/mol$

Urea ($(CO(NH_2)_2)$) with a molar mass of approximately $60.06 g/mol$

For Sucrose solution:

$Mass\ of\ sucrose\ = 3.42\ g$

$$Molesofsucrose = \frac{3.42\,\text{g}}{342.3\,\text{g/mol}} \approx 0.01\,\text{mol}$$

$Mass\ of\ water\ = 1000\ g\ (1\ kg)$

$$Molality\ of\ sucrose\ solution\ = \frac{0.01\,\text{mol}}{1\,\text{kg}} = 0.01\,\text{m}$$

For Urea solution:

$Mass\ of\ urea\ = 0.6\ g$

$$Molesofurea = \frac{0.6\,\text{g}}{60.06\,\text{g/mol}} \approx 0.01\,\text{mol}$$

$Mass\ of\ water\ = 1000\ g\ (1\ kg)$

$$Molality\ of\ urea\ solution\ = \frac{0.01\,\text{mol}}{1\,\text{kg}} = 0.01\,\text{m}$$

Both solutions have the same molality.

8. Correct option: (B)

$$Molality = \frac{w}{m \times W} \times 1000$$

where, w = mass of solute in grams;

W = mass of solvent in grams.

$$1.0 = \frac{w}{m \times W} \times 1000$$

$$\frac{1.0}{1000} = \frac{w}{m \times W}$$

$$\frac{w}{m \times W} = 0.001$$

Applying Raoult's law for dilute solution,

$$\frac{P_0 - P_s}{P_0} = \frac{w}{m \times W} \times M \, (M = 18)$$

$$\frac{760 - P_s}{760} = 0.001 \times 18$$

$$P_s = 760 - 760 \times 0.001 \times 18$$

$$P_s = 760 - 13.68 = 746.32$$

9. Correct option: (A)

Colligative properties are affected by the concentration of solute particles in a solution, regardless of the chemical nature of the solute. These properties include vapor pressure lowering, boiling point elevation, freezing point depression, and osmotic pressure. The more solute particles present, the greater the effect on these properties.

10. Correct option: (C)

Initial vapor pressure (vapor pressure of pure water) , P_0 = 20 mm

Mass of sucrose = 68.4 g

Molar mass of sucrose = 342 g/mol

$$number\ of\ moles\ of\ sucrose\ ,\ n_1 = \frac{68.4}{342} = 0.2\ mol$$

$$mass\ of\ water = 500g$$

$$molar\ mass\ of\ water = 18\frac{g}{mol}$$

$$number\ of\ moles\ of\ water\ ,\ n_2 = \frac{500}{18} = 27.78\ mol$$

$$Lowering\ of\ vapor\ pressure = \frac{\Delta P}{P_0} = mole\ fraction\ of\ solute$$

$$\frac{\Delta P}{P_0} = \frac{n_1}{(n_1 + n_2)}$$

$$\frac{\Delta P}{20} = 0.2/(27.78 + 0.2) = \frac{0.2}{27.98} = 0.143HG$$

11. Correct option: (A)

The vapour pressure (V.P.) of a dilute solution of non-volatile solute is P and the V.P. of pure solvent is P_0, the lowering of the V.P. is +ve. Lowering of vapor pressure is always positive.

12. Correct option: (D)

The lowering of vapor pressure is directly proportional to the number of solute particles in the solution, regardless of the type of solute (as long as it is non-volatile and doesn't dissociate significantly).

Therefore, the option where the most significant lowering of vapor pressure occurs is: D) 1 mole of urea.

13. Correct option: (C)

$\frac{P_0 - 21.85}{21.85} = \frac{30 \times 18}{90 \times m} \rightarrow I$

$weight(solvent) = 90 + 18 = 108g$

$\frac{P_0 - 22.15}{22.15} = \frac{30 \times 18}{108 \times m} \rightarrow II$

$From EQ - I$

$P^0 m - 21.85m = 21.85 \times 6 = 131.1$

$From EQ - 2$

$P^0 m - 22.15m = 22.15 \times 6 = 110.75$

$EQ - 1 \& 2$

$0.30m = 20.35$

$m = \frac{20.35}{0.30} = 67.83$

14. Correct option: (B)

$Mass\ of\ glucose\ =\ 10\ g$

$Number\ of\ moles\ of\ glucose\ =\ 0.0556$

$Mass\ of\ sucrose\ =\ 20\ g$

$Number\ of\ moles\ of\ sucrose\ =\ 0.0585$

$Mass\ of\ water\ =\ 100\ g$

$Number\ of\ moles\ of\ water\ =\ 5.556$

$Total\ number\ of\ moles\ =\ 5.556 + 0.0585 + 0.0556 = 5.67$

$Mole\ fraction\ of\ water = \frac{5.556}{5.67}$

$Vapour\ pressure\ of\ solution\ =\ Mole\ fraction\ of\ water \times P_0$

$Vapour\ pressure\ of\ solution\ = \frac{5.556}{5.67} \times 30 = 29.39mm$

15. Correct option: (A)

Applying Raoult's law,

$\frac{P_A^0 - P_S}{P_S} = \frac{w_B \times M_A}{w_A \times M_B} = \frac{\frac{w_B}{M_B}}{\frac{w_A}{M_A}}$ (Number of moles of solute per mole of benzene)

w_A =mass of A (Solvent)

w_B =mass of B (Solute)

M_A =molar mass of A

M_B =molar mass of B

$\frac{n_B}{n_A} = \frac{P_A^0 - P_S}{P_s} = \frac{(268 - 167)}{167} = 0.6047$

16. Correct option: (D)

The formula for boiling point elevation:

$\Delta T_b = i \times K_b \times m$

$\Delta T_b = Boilingpo\,\text{int}\,elevation$

$i = Vanthoffsfactor$

$K_b = ebullioscopic\ constant$

$m = molaritysolution$

$\Delta T_b = i \times K_b \times m$

$0.125^0 C = 1 \times 2.25 K\,kg / mol \times m$

$$m = \frac{0.125^0 C}{2.25 K\,kg / mol} \approx 0.0556 mol / kg$$

17. Correct option: (D)

The boiling point of a solution depends on the concentration of solute particles in the solvent. According to Raoult's Law, the boiling point elevation is directly proportional to the molality of the solute particles in the solution.

Given the options:

A) **Pure water** - Pure water has no solute particles, so it will have the lowest boiling point among the options.

B) **3.42 grams of sucrose in water** - This option presents a solution with sucrose dissolved in water. Sucrose is a non-volatile solute, and its presence in the solution will elevate the boiling point compared to pure water.

C) **0.6 grams of urea** - Urea is also a non-volatile solute. Dissolving urea in water will elevate the boiling point compared to pure water.

D) **18 grams of glucose** - Glucose, like sucrose and urea, is a non-volatile solute. Dissolving glucose in water will elevate the boiling point compared to pure water.

Comparing the options, the solution with the maximum boiling point would be the one with the highest concentration of solute particles. Therefore, option D) 18 grams of glucose would result in the maximum boiling point among the given options.

18. Correct option: (B)

Comparing the options:

Option A) 0.1 mole of urea

Option B) 0.2 mole of urea

Option C) 0.2 mole of sucrose

Option D) 0.1 mole of sucrose

Among these options, the solution with the highest number of moles of solute will produce the maximum elevation in boiling point.

Therefore, option B) 0.2 mole of urea will result in the maximum elevation in boiling point.

19. Correct option: (C)

Elevation in boiling point is a type of colligative property.

Colligative properties depend on the number of solute particles present in a solution, not on the identity of the solute particles. These properties include boiling point elevation, freezing point depression, vapor pressure lowering, and osmotic pressure.

20. Correct option: (B)

Given

$W_{solute} = 18g$

$W_{solvent} = 1kg$

$k_b = 0.52kgmol^{-1}$

First, we calculate elevation in the boiling point of solution.

$$\Delta T_b = \frac{K_b \times W_{solute}}{M_{w_{solute}} \times W_{solvent}}$$

$$\Delta T_b = \frac{0.52 \times 18}{180 \times 1} = 0.052K$$

water boils at 373.15 K at 1 atm pressure, therefore the boiling point of solution will be,

$T_B = T_b + \Delta T_b = 373.15 + 0.052 = 373.202K$

LEVEL-II

1. Correct option: (D)

The statement "The relative lowering of vapor pressure is equal to the mole fraction of the non-volatile solute" is known as Raoult's Law.

Raoult's Law describes the behaviour of ideal solutions and states that the vapor pressure of a solvent above a solution is directly proportional to the mole fraction of the solvent in the solution.

2. Correct option: (D)

$$\frac{P_\circ - P_s}{P_\circ} = \frac{5}{100} = 0.05 = X_B$$

$$0.05 = \frac{n_B}{n_B + n_A} = \frac{n_B}{n_B + \frac{171}{18}}$$

$n_B = 0.5$

$$weight\ of\ urea = 0.5mol \times 60\frac{g}{mol} = 30g$$

3. Correct option: (C)

$\Delta T_b\ for\ 1\ m\ urea\ solution = 0.54^\circ C$

$\Rightarrow 0.54^\circ C = K_b \times molality$

$\Rightarrow K_b = 0.54^\circ C kg\ mol^{-1}$

$$\frac{\Delta p}{p^o} = \frac{w_2 m_1}{m_2 w_1}$$

Where,

w_2 =mass of solute

w_1 = mass of solvent

m_1 = Molar mass of solute

m_2 = Molar mass of solvent

$$\frac{w_2}{m_2 w_1} = \frac{\Delta_p}{p^o m_1} = \frac{0.018}{18}$$

$$\Delta T_b(glucose) = K_b m = \frac{1000 K_b w_2}{m_2 w_1} = \frac{1000 \times 0.54 \times 0.018}{18} = 0.54^{\circ} C$$

4. Correct option: (B)

Vapour pressure of pure solvent (P^0) is more than the vapour pressure of solution (P). Tendency of the solvent particles to go into the vapour phase slightly decreases. Hence vapour pressure of the pure solvent is more than that of the solution.

5. Correct option: (A)

Boiling point of pure solvent is less than boiling point of solution.

When a non-volatile solute is dissolved in a pure solvent, its vapour pressure is decreased. The difference of boiling points of the solution and pure solvent is called elevation in boiling point.

6. Correct option: (C)

Mole fraction of solute can be written as,

$$x_2 = \frac{-\Delta_p}{p_1^*} = \frac{3000 - 2985}{3000} = 0.005$$

$$x_2 = \frac{m_2 / M_2}{\frac{m_1}{M_1} + \frac{m_2}{M_2}} = \frac{m_2 / M_2}{m_1 / M_1}$$

$$0.005 = \frac{5 / M_2}{100 / 18}$$

$$M_2 = 180 g mol^{-1}$$

7. Correct option: (A)

$K_b = 0.52 Kkg\ mol^{-1}$,

$molality\ = 0.1m$

$\Delta T_b = K_b m = 0.52 \times 0.1 = 0.052K$

$\left(\Delta T_b = 0.052K = 0.052C^0\right)$

$$= (T_b)H_2O + \Delta T_b$$

Hence, boiling point of solution = $= 100^0 C + 0.052C^0$

$$= 100.052^0 C$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

8. Correct option: (C)

Comparing the given options:A) 0.1 mole of glucose: Glucose ($C_6H_{12}O_6$) dissociates into one particle, so it has the same number of solute particles as sucrose.

B) 0.1 mole of urea: Urea ($CO\left(NH_2\right)_2$) does not dissociate in water, so it also has the same number of solute particles as sucrose.

C) Pure water: Pure water has no solute particles, so it will have a lower boiling point compared to a solution containing sucrose.

D) 0.01 mole of sucrose: This option has a lower amount of sucrose compared to the others, so it will have a lower elevation in boiling point and thus a lower boiling point compared to 0.1 mole of sucrose.

9. Correct option: (A)

The more volatile liquid evaporates fast as compared to the less volatile liquid at a low temperature because the volume increases with respect to temperature so it has a low boiling point.

10. Correct option: (A, C, D)

A) Liquid with lower boiling point is volatile and have higher vapour pressure.

This statement is generally correct. A lower boiling point typically indicates that a substance is more volatile, meaning it has a higher tendency to vaporize at a given temperature. Substances with lower boiling points tend to have higher vapor pressures because they transition from liquid to vapor more readily.

C) Addition of non-volatile solute lowers the vapor pressure of solvent.

This statement is correct. According to Raoult's Law, the vapor pressure of a solvent decreases when a non-volatile solute is added to it. The decrease in vapor pressure is proportional to the mole fraction of the solute in the solution.

D) Addition of non-volatile solute lowers the boiling point of solution.

This statement is also correct. The boiling point of a solution is elevated when a non-volatile solute is added to the solvent. This phenomenon is known as boiling point elevation and is a colligative property. The extent of the elevation depends on the molality of the solute particles in the solution.

11. Correct option: (B, C, D)

A) Colligative properties of dilute solutions are directly proportional to the number of solute particles rather than the molecular mass of the solute. Thus, statement A is incorrect.

B) The molal elevation constant (also known as the ebullioscopy constant) is indeed characteristic of a particular solvent. It represents the extent to which the boiling point of a solvent is elevated when a non-volatile solute is dissolved in it. Statement B is correct.

C) Vapour pressure of a liquid generally increases with an increase in temperature, as higher temperatures provide more thermal energy to the molecules, enabling more molecules to overcome the intermolecular forces holding them in the liquid phase and enter the vapor phase. Statement C is correct.

D) A liquid with a lower boiling point tends to be more volatile because it vaporizes more readily at a given temperature compared to a liquid with a higher boiling point. This statement is correct.

12. Correct option: (B)

The equation for Raoult's Law is: $P_{solution} = x_{solvent} \cdot P^{*}_{solvent}$

$P_{solution}$ vapor pressure of the solution,

$x_{solvent}$ mole fraction of the solvent,

$P^{*}_{solvent}$ vapor pressure of the pure solvent.

$P_{\text{solution}} = 20\,\text{mm Hg}$

$P^{\text{å}}_{\text{solvent}} = 25\,\text{mm Hg}$

$$x_{solvent} = \frac{P^{*}_{solvent}}{P_{solution}}$$

$$x_{solvent} = \frac{20mm\,Hg}{25mm\,Hg} = 0.8$$

Mole fraction of water is 0.8.

$x_{solvent} + x_{solute} = 1$

$0.8 + x_{solute} = 1$

$x_{solute} = 1 - 0.8 = 0.2$

COMPREHENSION TYPE

Passage-I:

13. Correct option: (C)

According to Raoult's law, Relative lowering of vapour pressure is equal to mole fraction of solute.

$$X_B = \frac{m \cdot M_A}{1000 + m \cdot M_A}$$

$\left(X_B = 0.0125\right)$

m=Molarity,

$$0.0125 = \frac{m \cdot 18}{1000 + m \cdot 18}$$

$12.5 + 0.225 = 18m$

$17.775m = 12.5$

$$m = \frac{17.775}{12.5} = 0.70$$

14. Correct option: (A)

5% aqueous solution refers to 5 g of solute in 100 g of solution.

Mass of solution (water)=100-5=95g

Vapour pressure of solution at 373 K = 745 mm

Vapour pressure of water at 373 K 760 mm

By Roults law,

$$\frac{p_A^0 - P_S}{P_S} = \frac{n_B}{n_A} = \frac{W_B}{M_B} \times \frac{M_A}{W_A}$$

$$\frac{(760mm - 745mm)}{(745mm)} = \frac{\%5g \times (18gmol^{-1})}{M_B \times (95g)}$$

$$\frac{15}{745} = \frac{(5g) \times (18gmol^{-1})}{M_B \times (95g)}$$

$$M_B = \frac{(5g) \times (18gmol^{-1}) \times 745}{(95g) \times 15} = 47gmol^{-1}$$

15. Correct option: (A)

Vapour pressure of water $(P^\circ) = 17.51mm$

Lowering of vapour pressure $(P^\circ - P_s) = 0.0614mm$

$(P^\circ) = 17.51mm$

$(P^\circ - P_s) = 0.0614mm$

$$\frac{(P^\circ - P_s)}{P^\circ} = \frac{0.0614}{17.51} = 0.00351$$

MATRIX MATCHING TYPE

16. Correct Matching is: a) Q b) PRS c) T d) Q

a) $\text{RLVP} = \frac{W_2}{M_2} \times \frac{M_1}{W_1}$

W1 and M1 are the mass and molar masses of solvent and W2 and M2 are the mass and molar masses of solute.

W1 and M1 are the mass and molar masses of solvent and W2 and M2 are the mass and molar masses of solute.

W_1 and M_1 are the mass and molar masses of solvent and W_2 and M_2 are the mass and molar masses of solute.

The expression

b) $\Delta T_b = T_b^0 - T_b$.

The expression $\Delta T_b = T_b^0 - T_b$ represents the boiling point elevation, where:

ΔT_b is the boiling point elevation,

T_b^0 is the boiling point of the pure solvent,

T_b is the boiling point of the solution.

This formula shows the difference between the boiling point of the pure solvent and the boiling point of the solution. It quantifies the increase in the boiling point due to the presence of a non-volatile solute in the solution.

c) Lowering of vapour pressure $= P^0 - P_s$

P is the vapour pressure of the solvent and P_s is the vapour pressure of the solution, then the lowering of vapour pressure can be written as $P^0 - P_s$.

d) Mole fraction of solute which contain non-volatile in dilute solution - $K_b \times \frac{W_2}{M_2} \times \frac{1000}{W_1}$

INTEGER TYPE

17. Correct Answer is:

Relative lowering of vapour pressure in dilute form:

$$\frac{p^\circ - p_s}{p^\circ} = X_B \left(X_B = Molefractionofsolute \right)$$

$$X_B = \frac{760 - 690}{760} = \frac{70}{760} = 0.0921$$

$$X_B = 0.92 \times 10^2$$

18. Correct Answer is:

Elevation in boiling point for the solution can be derived by following formula,

$\Delta T_b = T_b - T_b^0$

$\Delta T_b = 100.52 - 100$

$\Delta T_b = 0.52$

T_b is the boiling point of the solution.

T_b^0 is the boiling point of water.

19. Correct Answer is:

The boiling point of pure water is 373K at 1atmosphere.

This is the standard atmospheric pressure at sea level under normal conditions. At this pressure, water boils at 100^0C or 373K.

Osmosis and Osmotic Pressure

LEVEL-I

1. Correct option: (D)

The factor ΔT_f /ΔK_f represents molality (option D).

ΔT_f = represents the freezing point depression, which is the difference between the freezing point of the pure solvent and the freezing point of the solution.

ΔK_f = cryoscopic constant or the molal freezing point depression constant, which is specific to the solvent.

Molality (m) is defined as the number of moles of solute per kilogram of solvent.

2. Correct option: (C)

$\Delta T_f = k_f \times m$

$\Delta T_f = 1.86 \times 0.05$

$\Delta T_f = 0.093$

$\Delta T_f = 0 - 0.093 = -0.093$

3. Correct option: (A)

$\Delta T_f = -k_f \times m$

$molality\ of\ solution = (\frac{342}{342}) = 1$

$\Delta T_f = -1.86 \times \frac{(\frac{342}{342})}{1}$

$\Delta T_f = -1.86^0C$

4. Correct option: (D)

Depression in freezing point is given $\Delta T_f = -K_f \times m$, where K_f is the cryoscopic constant and m is the molality of the solution.

Elevation in boiling point formula can be written as $\Delta T_b = K_b \times m$,

K_b =ebullioscopy constant and m is the molality of the solution.

$$-\frac{\Delta T_f}{K_f} = \frac{\Delta T_b}{K_b}$$

$$\frac{0.186}{1.86} = \frac{x}{0.512}$$

$$x = \frac{0.186 \times 0.512}{1.86}$$

$$x = 0.0512^0C$$

5. Correct option: (A)

Freezing point of the solution can be written as below,

$$\Delta T_f = K_f \times \frac{w_B \times 1000}{m_B \times w_A}$$

$$\Delta T_f = 1.86 \times \frac{17 \times 1000}{46 \times 1000} = 0.69$$

$$Freezing\ point\ of\ solution = 0 - 0.69 = -0.69^{o}C$$

6. Correct option: (A)

In osmotic pressure, the number of solute particles plays a significant role. Urea, glucose, and sucrose are all non-electrolytes that do not dissociate into ions in solution, so they exert osmotic pressure proportional to the number of molecules present.

Comparing the options:

A 5% (w/v) solution of urea (A) would have the highest osmotic pressure among the 5% solutions because urea is a smaller molecule compared to glucose and sucrose, so it would have a higher number of particles per unit mass.

Among the 5% solutions, sucrose (B) would have the lowest osmotic pressure because it is a larger molecule compared to urea and glucose, and thus has fewer particles per unit mass.

A 10% (w/v) solution (D) would generally have a higher osmotic pressure than any of the 5% solutions because it has a higher concentration of solute particles.

7. Correct option: (D)

$$\pi = CRT$$

$$\pi = osmatic\ pressure = -2.5\ atm$$

$$T = 25^{0}C = 297K$$

$$R = Gas\ constant = 0.0821 L.atm K^{-1} mol^{-1}$$

$$\Rightarrow 2.5 = C \times 0.0821 \times 297$$

$$C = \frac{2.5}{0.0821 \times 297} = 0.1025$$

8. Correct option: (C)

$$\pi \times \frac{63}{1000} = \frac{3}{180} \times 0.0821 \times 288$$

$$\pi = 6.25 atm$$

9. Correct option: (D)

The osmotic pressure of a 1 molar aqueous glucose solution at 27°C can be calculated using the formula: $\pi = i \times M \times R \times T$

Where:

π = *osmotic pressure*

i = *van't Hoff factor* $\left(\text{for glucose, which is a non-electrolyte}, i = 1\right)$

M = *molarity of the solution* $\left(1\,M\right)$

R = *ideal gas constant* $\left(0.0821\ \frac{L}{atm} / \left(K \cdot mol\right)\right)$

T = *temperature in Kelvin* $\left(27°C = 300\,K\right)$

Substituting the values, we get:

$$\pi = 1 \times 1 \times 0.0821 \times 300 = 24.6 atm$$

10. Correct option: (C)

As we know πV=nRT, so $\pi \alpha V, T$ is constant this statement is wrong because temperature is not constant in the above expression.

11. Correct option: (A)

A semi-permeable membrane is chemically composed of copper ferrocyanide. This is supported by the chemical properties of the compound, which enable it to act as a semi-permeable membrane, allowing only certain substances to pass through while blocking others.

12. Correct option: (A)

Osmotic pressure is defined as the pressure required to prevent the flow of solvent across a semipermeable membrane from a region of lower solute concentration to a region of higher solute concentration. It is expressed in units of pressure, such as atmospheres (atm), pascals (Pa), or millimetres of mercury (mmHg). The most commonly used unit of osmotic pressure is atm, which is equivalent to 101.325 kPa or 760 mmHg.

13. Correct option: (B)

$$\pi V = nRT \Rightarrow \pi = \frac{n}{V}(RT) \Rightarrow \pi = CRT$$

The osmotic pressure of a dilute solution is given by the formula: π=CRT

$\pi =$ *osmotic pressure*

$c =$ *concentration of the solute in moles per liter* (M)

$R =$ *ideal gas constant* $\left(0.0821\, L\ atm/(K\ mol)\right)$

$T =$ *temperature in Kelvin* (K)

14. Correct option: (A)

Normal saline solution, which is a 0.9% (w/v) solution of sodium chloride (NaCl) in water, is isotonic with blood. This means that the concentration of solutes in normal saline is similar to that of blood, allowing for minimal osmotic pressure differences across cell membranes when it is used for medical purposes, such as intravenous injections.

15. Correct option: (A)

Isotonic solutions have the same osmotic pressure, meaning they exert the same pressure on a semi-permeable membrane and therefore do not cause osmotic movement of water into or out of cells. This property is important in medical contexts, especially when administering intravenous fluids or in cell culture techniques.

16. Correct option: (B)

Osmosis is the spontaneous flow of solvent molecules (usually water) from a region of lower solute concentration to a region of higher solute concentration through a semi-permeable membrane.

The correct option is:

B) The solvent from a solution of lower concentration to one of higher concentration.

17. Correct option: (C)

Osmotic pressure is directly proportional to the number of solute particles in a solution. As the number of solute molecules increases, the osmotic pressure of the solution also increases. This relationship is described by the van't Hoff equation for ideal solutions.

18. Correct option: (A)

Isotonic solutions have the same osmotic pressure as the cytoplasm of the cells. In an isotonic solution, there is no net movement of water into or out of the cells, so the cells maintain their normal shape and volume.

19. Correct option: (A)

The osmotic pressure (π) of a solution can be calculated using the formula πV = nRT,

$\pi =$ *osmotic pressure*

V = *volume of the solution in liters*

n = *moles of the solute*

$R =$ *ideal gas constant* $\left(0.0821\ L\ atm/\left(K\ mol\right)\right)$

$T =$ *temperature in Kelvin* $\left(K\right)$

$$\pi = \frac{nRT}{V}$$

$$\pi = \frac{0.1\times 0.0821\times 273}{1} = 2.243atm$$

20. Correct option is

Osmotic pressure π = CRT,

$$\pi = \frac{5}{100}\times 0.0821\times (150+273)$$

$$\pi = \frac{5}{100}\times 0.0821\times 423$$

$$\pi = 3.4atm$$

LEVEL-II

1. Correct option: (B)

Osmotic pressure is directly proportional to the concentration of solute particles in a solution, regardless of whether they are ions or molecules. This is a fundamental principle of osmosis and is described by the van't Hoff equation.

2. Correct option: (A)

Osmotic pressure is particularly useful for determining the molar mass of large molecules like proteins and polymers because it depends on the total number of solute particles in solution, which in turn relates to the molar mass of the solute. By measuring the osmotic pressure of a solution containing a known concentration of the protein or polymer, one can calculate the molar mass of the solute using the van't Hoff equation.

3. Correct option: (B)

The depression of the freezing point occurs when the solution is formed, and it's the liquid solvent that experiences the freezing point depression due to the presence of the solute particles. The solute particles disrupt the formation of the solid solvent lattice, thus lowering the freezing point.

4. Correct option: (D)

$$\Delta Tf = \frac{1000\times K_f \times w}{M\times W}$$

$$M = \frac{1000\times K_f \times w}{W\times \Delta T}$$

$K_f = 1.863 Kkgmol^{-1}$

$w = 13.6g$

$W = 20g$

$\Delta T = 3.7 \circ C$

$$M = \frac{1000 \times 1.863 \times 13.6}{20 \times 3.7} = 342.39$$

5. Correct option: (A)

Depression in freezing point is given by:

$$\Delta T = \frac{1000 \times K' f \times w}{W.m}$$

Molar mass of ethylene glycol m = 62g.

$$\Delta T = \frac{1000 \times K' f \times w}{W.m}$$

$$9.3 = \frac{1000 \times 1.86 \times 50}{62 \times W}$$

$W = 161.29$

Remaining water out of 200g will be separated as ice.

$Iceseparated = 200 - 161.29 = 38.71g$

6. Correct option: (B)

$Weight\ of\ solution = 100 + 8 = 108g$

$$Volume\ of\ solution = \frac{108}{1.025} \times 10^{-3} Litre$$

$$\pi = \frac{n}{V} RT$$

$$\pi = \frac{W \times R \times T}{M \times V}$$

$$\pi = \frac{8 \times 0.0821 \times 300 \times 1.025}{5 \times 108 \times 10^{-3}}$$

$\pi = 374$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

7. Correct option: (A)

Osmotic pressure is directly proportional to the molar concentration of the solute in the solution. This relationship is defined by the van't Hoff equation and is a fundamental principle of colligative properties.

8. Correct option: (A)

In the depression of freezing point experiment, it is found that the vapor pressure of the solution is less than that of the pure solvent.

Therefore, the correct answer is:

A) The vapor pressure of the solution is less than that of the pure solvent.

This is a well-established principle in the study of colligative properties, where the addition of a non-volatile solute to a solvent decreases the vapor pressure of the solution compared to that of the pure solvent. This leads to a lowering of the freezing point, which is the basis of the depression of freezing point experiment.

9. Correct option: (A)

Colligative properties depend on the number of solute particles present in a solution rather than the identity of the solute particles. When a non-volatile solute, such as a solid, is dissolved in a volatile liquid, the resulting solution exhibits colligative properties like changes in boiling point, freezing point, vapor pressure, and osmotic pressure.

10. Correct option: (B)

An isotonic solution refers to a solution where the osmotic pressure across a semi-permeable membrane is equal on both sides. This condition implies that the concentration of solutes (solute and solvent) is the same inside and outside the cell. While the solute and solvent may be different, what matters for isotonicity is the equality of osmotic pressure.

11. Correct option: (A, B, C)

The correct statements are:

A) For the same solution, elevation in boiling point = depression in freezing point.

B) The freezing point of water is depressed by the addition of glucose.

C) The elevation in boiling point is due to decrease in vapor pressure.

These statements are supported by the principles of colligative properties and the behaviour of solutions. The elevation in boiling point is directly related to the depression in freezing point for the same solution, and the freezing point of a solvent is indeed depressed by the addition of a solute, such as glucose.

12. Correct option: (C)

Isotonic solution has the same molarity.

13. Correct option: (A)

In hypertonic solution osmotic pressure is higher, so volume also increases.

14. Correct option: (A)

Increase in volume A meaning solvent moves from B to A.

Lower concentration of solute to higher concentration of solute. Check whether $C_A > C_B$

$20 gin 100 ml$

$$C_A = \frac{20}{276} \times \frac{1000}{100} = 0.7246$$

$10 grin 100 ml$

$$C_B = \frac{10}{276} \times \frac{1000}{100} = 0.3623$$

$C_A > C_B$

Passage-II:

15. Correct option: (A)

$$\Delta T_f = K_f \times m$$

$$m = \frac{\Delta T_f}{K_f}$$

$$\Delta T_f = 273 - 272.4 = 0.6k$$

$$K_f = 0.86 K kgmol^{-1}$$

$$m = \frac{0.6}{1.86} = 0.322$$

16. Correct option: (B)

Molal elevation constant of water = $0.512 Kkgmol^{-1}$

$\Delta T_f = K_f \times m$

$\Delta T_f = 273 - 272.4 = 0.6K$

$0.6K = 1.86 \times m$

$m = 0.322 mol/kg (molality)$

$\Delta T_b = K_b \times m$

$\Delta T_b = 0.512 \times 0.322 = 0.165K$

$Boiling point = 373 + 0.165 = 373.165K$

ΔT_b = elevation in boiling point,

K_b =Molal elevation constant,

m= molality.

17. Correct option: (D)

$\Delta T_f = K_f m$

$m = \dfrac{\Delta T_f}{K_f}$

$m = \dfrac{0.6K}{1.86 Kkgmol^{-1}}$

$m = 0.32m$

MATRIX MATCHING TYPE

18. Correct Matching is:

a) *Hypertonic – Q*) *One solution has higher osmotic pressure than the second solution.*

b) *Isotonic – P*) *Solution having the same osmotic pressure.*

c) *Reverse osmosis – S*) *Desalination of water*.

d) *Depression in freezing point – R*) *Colligative property*

INTEGER TYPE

19. Correct Answer is:

$\pi_1 = 200 mm;$

$T_1 = 283$

$\pi_1 = 105.3 mm,$

$T_2 = 298$

$\pi = \dfrac{n}{V}(RT)$

At T_1, $200 = \dfrac{n}{V_1} R \times 283 \rightarrow I$

At T_2, $50 = \dfrac{n}{V_2} R \times 298 \rightarrow II$

$\dfrac{I}{II} \Rightarrow \dfrac{200}{105.3} = \dfrac{V_2}{V_1} \times \dfrac{283}{298} \Rightarrow V_2 = 2V_1$

20. Correct Answer is:

$$\pi = CRT = \frac{(\frac{W_2}{Mw_2})RT}{V}$$

$W_2 = 40g$

$Mw_2 = 246$

Given $T = 27^0 C = 300k$

$V = 1L$

$$\pi = \left(\frac{40}{246}\right) \times 0.082 \times 300 = 4atm$$

Abnormal Molar Masses

LEVEL-I

1. Correct option: (C)

The osmotic pressure is given by the formula:

$\Pi = i \cdot M \cdot R \cdot T$

$i =$ *=van't Hoff factor,*

$M =$ *molarity of the solution,*

$R =$ *gas constant,*

$T =$ *temperature in Kelvin.*

The van't Hoff factor (i) represents the number of particles the salt dissociates into when it dissolves. For ionic compounds, such as salts, the van't Hoff factor is calculated based on the number of ions produced in solution.

$AgNO_3 (i = 2)$

$Na_2SO_4 (i = 3)$

$(NH_4)_3PO_4 (i = 4)$

$MgCl_2 (i = 3)$

$(NH_4)_3PO_4$ has the highest value of (i= 4).

2. Correct option: (D)

$KCl \rightarrow i - value = 2$

$BaCl_2 \rightarrow i - value = 3$

$FeCl_3 \rightarrow i - value = 4$

$Fe_2(SO_4)_3 \rightarrow i - value = 6$

Compound with maximum value of i exhibit maximum elevation in boiling point.

3. Correct option: (B)

The boiling point of a substance is the temperature at which the vapor pressure of the substance equals atmospheric pressure. A solution containing a non-volatile solute, having a lower vapor pressure than the pure solvent, must be at a higher temperature before its vapor pressure equals atmospheric pressure and it boils. Thus, the boiling point of a solution containing a non-volatile solute is higher than that of the pure solvent. This effect is called boiling point elevation.

4. Correct option: (D)

The freezing point of a 1% aqueous solution of calcium nitrate will be below 0°C. The presence of a solute (non-volatile) lowers the vapor pressure of a liquid, thereby lowering the freezing point. In the case of the 1% solution of calcium nitrate, the freezing point will be decreased, resulting in a temperature below 0°C.

5. Correct option: (C)

The correct relationship between the osmotic pressure is $P_1 > P_2$. CH_3COOH is a weak electrolyte whereas KNO_3 is 100% ionized. Here molality of KNO_3 is P_1 and molality of CH_3COOH is P_2.

6. Correct option: (C)

The colligative properties of solutions depend only on the number of solute particles, not the nature of the solute itself.

For example, when KCl dissolves in water, it dissociates into two ions (K^+ & Cl^-), while sugar (such as sucrose) does not dissociate into ions.

Therefore, the ratio of the value of any colligative property for a KCl solution to that of a sugar solution would be the ratio of the number of particles in KCl solution to the number of particles in sugar solution.

Since KCl dissociates into two ions and sugar does not dissociate, the ratio would be:

$$Ratio = \frac{2(\ for\ KCl\ )}{1(\ for\ sugar\ )} = 2$$

7. Correct option: (A)

The freezing point depression constant (K_f) depends on the nature of the solvent and not the solute. For a given solvent, K_f remains constant regardless of the solute.

In this case, K_f for water is a constant value, and it is independent of the solute used to determine it. The value given is, $K_f = 1.86^0 \frac{C}{mol}$, is constant for water.Therefore, the value of K_f for water calculated for NaCl solution will be the same as for the glucose solution, which is $K_f = 1.86^0 \frac{C}{mol}$.

8. Correct option: (A)

$NaCl \rightarrow Na^+ + Cl^-$ NaCl completely dissociates into sodium and chlorine ions respectively.

AAs we know molecular weight of NaCl is $58.44 \approx 58.5$

Normal molecular weight of electrolyte >Experimental molecular weight.

NaCl furnishes two ions on complete dissociation. so, i=2

$$i = \frac{M_{Theory}}{M_{observed}}$$

$$2 = \frac{M_{Theory}}{M_{observed}}$$

$$M_{Theory} = 2M_{observed}$$

$$M_{observed} < M_{Theory}$$

9. Correct option: (B)

The van't Hoff factor (i) is a measure of the extent of dissociation of a solute in a solution. For NaCl (sodium chloride), it dissociates completely into two ions: one sodium ion (Na^+) and one chloride ion (Cl^-). Hence, for NaCl, the van't Hoff factor is 2.

10. Correct option: (A)

According to modified equation of elevation of boiling point,

$\Delta T_f = iK_f m$

ΔT_f is directly proportional to i and m.

$0.005mMgSO_4 \Rightarrow (2)(0.005m) = 0.01m$

$0.005mC_2H_5OH \Rightarrow (1)(0.005m) = 0.005m$

$0.005mMgI_2 \Rightarrow (3)(0.005m) = 0.015m$

$0.005mNaCl \Rightarrow (2)(0.01m) = 0.02m$

0.01 NaCl having the highest value of has maximum ΔT_f value.

11. Correct option: (D)

The boiling point of a solution depends on the concentration of solute particles in the solvent. The more solute particles present in the solution, the higher the boiling point will be.

Let's analyze the options:

A) **1% glucose in water**: Glucose is a non-electrolyte, meaning it does not dissociate into ions when dissolved in water. Thus, it will only contribute one particle per molecule dissolved.

B) **1% sucrose in water**: Sucrose, like glucose, is also a non-electrolyte. It also contributes only one particle per molecule dissolved.

C) **1% NaCl in water**: NaCl (sodium chloride) dissociates completely into Na^+ and Cl^- ions when dissolved in water, resulting in two ions per formula unit. Therefore, it will contribute more solute particles compared to glucose or sucrose.

D) **1%** $CaCl_2$ **in water**: Calcium chloride ($CaCl_2$) dissociates into three ions when dissolved in water, one Ca^{+2} ion and two Cl^- ions. Therefore, it will contribute the highest number of solute particles among the options provided.

12. Correct option: (C)

The freezing point depression depends on the number of particles in solution. The greater the number of particles, the more significant the freezing point depression.

Let's analyze the options:

A) 1 molal NaCl solution: NaCl dissociates into Na+ and Cl- ions, so it yields 2 particles per formula unit in solution.

B) 1 molal KCl solution: KCl also dissociates into K+ and Cl- ions, so like NaCl, it also yields 2 particles per formula unit in solution.

C) 1 molal $CaCl_2$ solution: $CaCl_2$ dissociates into Ca^{+2} and $2Cl^-$ ions, so it yields 3 particles per formula unit in solution.

D) 1 molal urea solution: Urea, $(NH_2)_2 CO$, does not dissociate into ions in solution; it remains as individual molecules. Therefore, it yields only one particle per molecule in solution.

13. Correct option: (D)

The Van't Hoff factor (i) represents the number of particles into which a compound dissociates in solution.

$BaCl_2$ dissociates into one Ba^{+2} ion and two Cl^- ions when dissolved in water. Therefore, it contributes a total of 3 ions.

The formula for calculating the Van't Hoff factor (i) is:

$i = 1 + (n-1) \times \alpha$

n is the number of particles the solute breaks into (in this case, 3 for $BaCl_2$).

α is the degree of dissociation.

$i = 1 + (3-1) \times 1$

$i = 1 + 2 \times 1$

$i = 1 + 2$

$i = 3$

14. Correct option: (A)

Osmotic pressure (π) is directly proportional to the concentration of solute particles in a solution. Equimolar solutions contain the same number of moles of solute per unit volume.

Let's analyze the number of particles each solute forms:

$BaCl_2$ dissociates into three particles per formula unit: one Ba^{+2} ion and two Cl^- ions.

NaCl dissociates into two particles per formula unit: one Na^+ ion and one Cl^- ion.

Glucose does not dissociate into ions and remains as a single particle in solution.

Given this information, we can rank the osmotic pressure:

$BaCl_2$ solution will have the highest osmotic pressure because it produces the highest number of particles.

NaCl solution will have the next highest osmotic pressure because it produces two particles per formula unit.

Glucose solution will have the lowest osmotic pressure because it remains as a single particle.

15. Correct option: (D)

The solution with the maximum freezing point is D) 1 molar of urea solution. Urea is a covalent compound that does not dissociate into ions in solution. The freezing point depression is directly proportional to the number of particles into which the solute dissociates. Since urea does not dissociate, it will have the highest freezing point among the given options.

16. Correct option: (C)

The Van't Hoff factor (i) represents the number of particles into which a compound dissociates in a solution. For ionic compounds that dissociate into ions, such as $Ca(NO_3)_2$, the Van't Hoff factor is determined by the number of ions produced upon dissociation.

$Ca(NO_3)_2$ dissociates into one Ca^{2+} ion and two NO_3^- ions. Therefore, the total number of ions produced per formula unit of $Ca(NO_3)_2$ is 3.

Thus, the Van't Hoff factor (i) of a very dilute solution of $Ca(NO_3)_2$ is 3.

17. Correct option: (D)

Lowering of vapour pressure is directly proportional to the vant Hoff factor i. For $BaCl_2$ i = 3 maximum. Thus $BaCl_2$ has highest value relative lowering of vapour pressure.

18. Correct option: (C)

To determine which solution will have the highest boiling point at 1 atm pressure, we need to consider the colligative properties of the solutions, which depend on the number of solute particles present in the solution.

Let's analyze the options:

A) 0.1 M NaCl: NaCl dissociates into Na^+ and Cl^- ions in solution, resulting in two particles per formula unit.

B) 0.1 M sucrose: Sucrose does not dissociate into ions in solution; it remains as individual molecules, contributing one particle per molecule dissolved.

C) 0.1 M BaCl2: BaCl2 dissociates into one Ba^{2+} ion and two Cl^- ions in solution, resulting in three particles per formula unit.

D) 0.1 M glucose: Glucose does not dissociate into ions in solution; it remains as individual molecules, contributing one particle per molecule dissolved.

The boiling point elevation depends on the number of solute particles in the solution. The more particles, the greater the boiling point elevation.

19. Correct option: (C)

The ratio of freezing point depression value of 0.01 M solutions of urea, common salt, and Na_2SO_4 is 1:2:3. The depression in freezing point is directly proportional to the number of particles dissolved in the solution. Urea is a covalent compound that does not dissociate into ions in water, while NaCl dissociates into two ions (Na^+ and Cl^-) and Na2SO4 dissociates into three ions ($2Na^+$ and SO_4^{-2}). Therefore, the ratio of the depression in freezing point for 0.01 M solutions of urea, NaCl, and Na_2SO_4 is 1:2:3.

20. Correct option: (B)

The dissociation of the weak electrolyte is $\frac{\text{i-1}}{n-1}$

LEVEL-II

1. Correct option: (B)

To determine the increasing order of boiling points for the given solutions, we need to consider the colligative properties, which depend on the number of solute particles present in the solution.

Let's analyze the options:

0.001 M NaCl: NaCl dissociates into Na^+ and Cl^- ions in solution, providing two particles per formula unit.

0.001 M urea: Urea $\left(CO\left(NH_2\right)_2\right)$ does not dissociate into ions in solution; it remains as individual molecules, contributing one particle per molecule dissolved.

0.001 M $MgCl_2$: $MgCl_2$ dissociates into one Mg^{2+} ion and two Cl^- ions in solution, providing three particles per formula unit.

Since the boiling point elevation is directly proportional to the number of solute particles, the solution with the highest number of particles will exhibit the greatest increase in boiling point.

Therefore, the correct order of increasing boiling points is:

C) 2 (urea) < 1 (NaCl) < 3 ($MgCl_2$).

2. Correct option: (A)

Freezing point is higher if ΔT_f is lower (depends on m_{eff}).

$0.1\,M\ Ba_3(PO_4)_2 < 0.1\,M\ Na_2SO_4 < 0.1\,M\ ethanol$

3. Correct option: (A)

To determine the increasing order of osmotic pressure for the given solutions, we need to consider the colligative properties, which depend on the number of solute particles present in the solution.

Let's analyze the options:

0.1 M glucose: Glucose does not dissociate into ions in solution; it remains as individual molecules, contributing one particle per molecule dissolved.

0.1 M urea solution: Urea ($CO\left(NH_2\right)_2$) does not dissociate into ions in solution; it remains as individual molecules, contributing one particle per molecule dissolved.

0.1 M common salt (NaCl): NaCl dissociates into Na^+ and Cl^- ions in solution, providing two particles per formula unit.

Since osmotic pressure is directly proportional to the number of solute particles, the solution with the highest number of particles will exhibit the highest osmotic pressure.

Therefore, the correct order of increasing osmotic pressure is:

$3\left(0.1\,M\ common\ salt\right) < 1\left(0.1\,M\ glucose\right) < 2\left(0.1\,M\ urea\ solution\right)$

4. Correct option: (B)

To determine the increasing order of elevation in boiling points for the given solutions, we need to consider the colligative properties, which depend on the number of solute particles present in the solution.

Let's analyze the options:

0.1 M glucose: Glucose does not dissociate into ions in solution; it remains as individual molecules, contributing one particle per molecule dissolved.

0.1 M common salt (NaCl): NaCl dissociates into Na+ and Cl- ions in solution, providing two particles per formula unit.

0.1 M $MgCl_2$: $MgCl_2$ dissociates into one Mg^{2+} ion and two Cl^- ions in solution, providing three particles per formula unit.

Since the boiling point elevation is directly proportional to the number of solute particles, the solution with the highest number of particles will exhibit the greatest increase in boiling point.

5. Correct option: (B)

$$i = \frac{Calculated.MW}{Observed.MW}$$

$$i = \frac{94}{161.84} = 0.58$$

$$\alpha = \frac{1-i}{1-n}$$

$$\alpha = \frac{1-0.58}{1-0.5} = \frac{0.42}{0.5}$$

$$\alpha\% = \frac{0.42}{0.5} \times 100 = 84\%$$

6. Correct option: (D)

$Hg_2Cl_2\,(n=3) \Leftrightarrow Hg_2^{+2} + 2Cl$

$$\alpha = \frac{i-1}{n-1}$$

$$0.8 = \frac{i-1}{3-1}$$

$$i = 2.6$$

7. Correct option: (B)

$$Total\ moles = 1-\alpha+4\alpha+\alpha = 1+4\alpha$$

$$i = \frac{Total\ moles\ at\ equilibrium}{Initial\ moles}$$

$$i = \frac{1+4\alpha}{1} = 1+4\alpha\,(\alpha = 70\% = 0.7)$$

$$i = 1+4\times 0.7 = 3.8$$

8. Correct option: (B)

The expression for the osmotic pressure is

$$\Pi = iCRT$$

$$\Pi(NaCl) = \Pi(glucose)$$

$$i(NaCl)C(NaCl) = i(glucose)C(glucose)$$

$$i(NaCl)\times\frac{1.2}{58.5} = 1\times\frac{7.2}{180}$$

$$i(NaCl) = 1.95$$

α be the degree of ionization of NaCl,

$$\alpha = \frac{i-1}{n-1} = \frac{1.95-1}{2-1} = 0.95$$

9. Correct option: (B)

$$W_B = 2g, K_f = 4.9KKgmol^{-1}$$

$$W_A = 25g, \Delta T_f = 1.62K$$

$$\Delta T_f = K_f \times \frac{W_B}{M_B}\times\frac{1000}{W_A}$$

$$M_B = \frac{4.9\times 2\times 1000}{1.62\times 25} = 241.98g/mol$$

$$2C_6H_5COOH \square \quad (C_6H_5COOH)_2$$

X is the degree of association, (1−x) mole of benzoic acid left undissociated & corresponding x/2 as associated moles of C_6H_5COOH at equilibrium.

$1-x+\frac{x}{2}=1-\frac{x}{2}=i$

$i=\frac{Normalmolecularmass}{Abnormalmolecularmass}$

$1-\frac{x}{2}=\frac{122}{241.98}$

$\frac{x}{2}=1-\frac{122}{241.98}$

$x=0.992=99.2\%$

10. Correct option: (C)

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct option: (B, C)

The expression to determine van't Hoff factor of a solution is given by:

$i=1+(n-1)\alpha$

$n=$ *number of ions formed after dissociation*

$\alpha=$ *Degree of dissociationwhere,*

$i=$ *van't Hoff factor*

$For\ NaCl; n=2\ so\ i=1+\alpha$

$For\ Ca(NO_3)_2; n=3\ so\ i=1+2\alpha$

$For\ K_4[Fe(CN)_6]; n=5\ so\ i=1+4\alpha$

$For\ (NH_4)_3PO_4; n=4\ so\ i=1+3\alpha$

12. Correct option: (A)

The freezing point depression depends on the number of solute particles in the solution. Since all options are 0.1 M solutions, we need to consider the Van't Hoff factor for each solute.A) Potassium sulfate (K_2SO_4) dissociates into 3 ions in solution: $2K^+$ ions and 1 SO_4^{-2} ion. So, the Van't Hoff factor (i) for K_2SO_4 is 3.

B) Sodium chloride (NaCl) dissociates into 2 ions in solution: 1 Na^+ ion and 1 Cl^- ion. So, the Van't Hoff factor (i) for NaCl is 2.

C) Urea ($CO\left(NH_2\right)_2$) is a non-electrolyte, so it does not dissociate into ions in solution. Therefore, the Van't Hoff factor (i) for urea is 1.

D) Glucose ($C_6H_{12}O_6$) is a non-electrolyte, so it does not dissociate into ions in solution. Therefore, the Van't Hoff factor (i) for glucose is 1.

Potassium sulphate (K_2SO_4) with a Van't Hoff factor of 3 would produce the greatest depression in freezing point, and therefore it will have the lowest freezing point compared to the other 0.1 M solutions listed.

COMPREHENSITON TYPE

Passage-I:

13. Correct option: (D)

The Van't Hoff factor (i) is a measure of the number of particles into which a compound dissociates in solution. For acetic acid (CH_3COOH), which is a weak acid, it does not fully dissociate into ions in solution like a strong electrolyte would.

Theoretically, for a weak acid like acetic acid, the maximum Van't Hoff factor (i) would be greater than 1, indicating that it does not dissociate.

14. Correct option: (D)

The given information states that 50% of AB_3 dissociates into A^{+3} and B^- ions, and 50% associates to form the dimer $(AB_3)_2$. To calculate the Van't Hoff factor (i) for this system, we can use the expression:

$i = 1 + \alpha$

where α is the degree of dissociation. Since 50% of AB_3 dissociates, the degree of dissociation is 0.5. Therefore, the Van't Hoff factor is:

$i = 1 + 0.5 = 1.5$

The correct Van't Hoff factor is 1.5.

15. Correct option: (D)

The given molecular weight of the ionic compound is 133.5 g/mol, and the calculated molecular weight based on experiments is 76.286 g/mol. This means that only 56.3% (100% - 43.7%) of the compound dissociates into its constituent ions.

Since 50% of the compound associates to form the dimer $(AB_3)_2$, the remaining 56.3% dissociates into ions. The compound is an ionic compound, and it dissociates into cations and anions. Therefore, the number of ions produced on complete dissociation is:

$2 \times number\ of\ cations + 1 \times number\ of\ anions = 2 \times 1 + 1 \times 1 = 3$

MATRIX MATCHING TYPE

16. Correct Matching is: a) P) Q c) S d) R

a) $AlCl_3$ if α=0.8. $AlCl_3$ dissociates into Al^{+3} and $3Cl^-$ ions. Therefore, the Van't Hoff factor can be calculated as i=1+α×3=1+0.8×3=3.4. So, the correct match is i=3.4.

b) $BaCl_2$ dissociates into Ba^{2+} and $2Cl^-$ ions. Therefore, the Van't Hoff factor can be calculated as $i = 1 + \alpha \times 2 = 1 + 0.9 \times 2 = 2.8$. So, the correct match for b is Q with i=2.8.

c) $Na_3(PO_4)$ if $\alpha = 0.9\ matches\ with R) i = 3.8$

d) $K_4Fe(CN)_6$ if $\alpha = 0.7\ matches\ with R) i = 3.4$

INTEGER TYPE

17. Correct Answer is:

For urea, i = 1, while for $BaCl_2 (i = 3)$,therefore, the rations of any colligative property for $BaCl_2$ solution to urea solution is 3.

18. Correct Answer is:

The Van't Hoff factor (i) for a solute that does not dissociate or associate in solutions is equal to 1.

The Van't Hoff factor represents the ratio of the observed colligative property to the value expected from the number of solute particles in the solution. For a non-dissociating or non-associating solute, the number of solute particles in solution remains the same as the number of formula units dissolved, hence the Van't Hoff factor is 1.

Examples of solutes that do not dissociate or associate include most non-electrolytes such as glucose, sucrose, urea, and many organic compounds.

19. Correct Answer is:

$\Delta T_f = 273 - 269.28$

$\Delta T_f = 3.87K$

$3.72 = iK_f m$

$3.72 = i \times 86 \times 1$

$i = 2$

Complex produce 2 ions.

20. Correct Answer is:

Elevation of boiling point can be written as $\Delta T_b = iK_b m$

For $NaCl\ solution, i = 2$.

So, for NaCl, $\Delta T_b = 2$